United States Patent	12410436
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Gu; Bingnan et al.

Adeno-associated virus (AAV) producer cell lines

Abstract

The present disclosure relates to nucleic acids encoding helper genes and adeno-associated virus (AAV) genes, under the control of inducible promoters. The disclosure also relates to a mammalian cell line for producing AAV as well as methods of using the mammalian cells for producing AAVs.

Inventors: **Gu; Bingnan (Pearland, TX), Li; Gang (Pearland, TX), Tripp; Caitlin (Pearland, TX)**

Applicant: **LONZA HOUSTON, INC.** (Pearland, TX)

Family ID: **95485197**

Assignee: **LONZA HOUSTON, INC. (Pearland, TX)**

Appl. No.: **18/498577**

Filed: **October 31, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20250136999 A1	May. 01, 2025

Publication Classification

Int. Cl.: **C12N15/63** (20060101); **C07K14/005** (20060101); **C12N15/85** (20060101)

U.S. Cl.:

CPC **C12N15/635** (20130101); **C07K14/005** (20130101); **C12N15/85** (20130101); C12N2710/16222 (20130101); C12N2760/14122 (20130101); C12N2830/003 (20130101)

Field of Classification Search

CPC: C12N (15/635); C12N (15/85); C12N (2710/16222); C12N (2760/14122); C12N (2830/003); C07K (14/005)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5622856	12/1996	Natsoulis	N/A	N/A
2021/0062161	12/2020	Vink	N/A	C07K 14/005

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2006083800	12/2005	WO	A61P 25/28
2015/162211	12/2014	WO	N/A
2018/136566	12/2017	WO	N/A
2018/192982	12/2017	WO	N/A

OTHER PUBLICATIONS

Mi, Jing, et al. "H1 Rna polymerase III promoter-driven expression of an RNA aptamer leads to high-level inhibition of intracellular protein activity." Nucleic acids research 34.12 (2006): 3577-3584. (Year: 2006). cited by examiner

Chu et al., "SV40 DNA Transfection of Cells in Suspension: Analysis of the Efficiency of Transcription and Translation of T Antigen," Gene 13(2): 197-202 (1981). cited by applicant

Clark et al., "Cell Lines for the Production of Recombinant Adeno-Associated Virus," Human Gene Therapy 6:1329-1341 (1995). cited by applicant

Bianchi et al., "A potent enhancer element in the 5′-UTR intron is crucial for transcriptional regulation of the human ubiquitin C gene," Gene. 448(1):88-101 (2009). cited by applicant

Deuschle et al., "Tetracycline-Reversible Silencing of Eukaryotic Promoters," Mol. Cell Biol. 15(4): 1907-1914 (1995). cited by applicant

Farris and Pintel, "Improved splicing of Adeno-associated viral (AAV) capsid protein-supplying pre-mRNAs leads to increased recombinant AAV vector production," Hum Gene Ther. 19(12): 1421-1427 (2008). cited by applicant

Feng et al., "The VP35 protein of Ebola virus inhibits the antiviral effect mediated by double-stranded RNA-dependent protein kinase PKR," J Virol. 81(1):182-192 (2007). cited by applicant

Graham et al., "A New Technique for the Assay of Infectivity of Human Adenovirus 5 DNA," Virology, 52(2):456-457 (1973). cited by applicant

Guido et al., "Human bocavirus: Current knowledge and future challenges," World J. Gateroenterol 22:8684-8697. cited by applicant

Huang et al., "Adenovirus early region 4 encodes two gene products with redundant effects in lytic infection," Journal of Virology 63(6):2605-2615 (1989). cited by applicant

Ishibashi et al., "Adenoviruses of animals," The Adenoviruses, Ginsberg, ed., Plenum Press, New York, N.Y., pp. 497-562 (1984). cited by applicant

Kotin, R. M., "Prospects for the Use of Adeno-Associated Virus as a Vector for Human Gene Therapy," Human Gene Therapy 5:793-801 (1994). cited by applicant

Langland et al., "Inhibition of PKR by RNA and DNA viruses," Virus Res. 119(1):100-110 (2006). cited by applicant

Li et al., "Role for Highly Regulated rep Gene Expression in Adeno-Associated Virus Vector Production," Journal of Virology 71:5236-5243 (1997). cited by applicant

Machitani et al., "Development of an adenovirus vector lacking the expression of virus-associates RNAs," Journal of Controlled Release 154(3):285-289 (2011). cited by applicant

Mullick et al., "The cumate gene-switch: a system for regulated expression in mammalian cells," BMC Biotechnology 6:43 (1-18) (2006). cited by applicant

Muzyczka, N., "Use of Adeno-Associated Virus as a General Transduction Vector for Mammalian Cells," Current Topics in Microbiol. and Immunol. 158:97-129 (1992). cited by applicant

Nakamura et al., "Development of packaging cell lines for generation of adeno-associated virus vectors by lentiviral gene transfer of trans-complementary components," Eur. J. Haematol. 73:285-294 (2004). cited by applicant

Robert et al., “Manufacturing of Recombinant Adeno-Associated Viruses Using Mammalian Expression Platforms,” Biotechnology Journal 12(1600193): 1-16 (2017). cited by applicant
Smith and Katin, “An adeno-associated virus (AAV) initiator protein, Rep 78, catalyzes the cleavage and ligation of single-stranded AAV ori DNA,” J. Viral. 74(7):3122-3129 (2002). cited by applicant
Strauss, “Adenovirus infections in humans,” In The Adenoviruses, Ginsberg, ed., Plenum Press, New York, N.Y., pp. 451-596 (1984). cited by applicant
Szulc et al., “A versatile tool for conditional gene expression and knockdown,” Nature Methods 3:109-116 (2006). cited by applicant
Tetracycline On, “Tetracycline (Tet) Inducible Expression”, Addgene. Retrieved from https://www.addgene.org/collections/tetracycline/ on Jun. 2, 2023. cited by applicant
Tordo et al., “A Novel Adeno-Associated Virus Capsid with Enhanced Neurotropism Corrects a Lysosomal Transmembrane Enzyme Deficiency,” Brain ePub 141(7): 2014-2031 (2018). cited by applicant
Urabe et al., “Insect Cells as a Factory to Produce Adeno-Associated Virus Type 2 Vectors,” Human Gene Therapy 13: 193501943 (2002). cited by applicant
Wiederschain et al., “Single-vector inducible lentiviral RNAi system for oncology target validation,” Cell Cycle 8:498-504 (2009). cited by applicant
Yao et al., “Tetracycline Repressor, tetR, rather than the tetR-Mammalian Cell Transcription Factor Fusion Derivatives, Regulates Inducible Gene Expression in Mammalian Cells,” Human Gene Therapy 9:1939-1950 (1998). cited by applicant
Zhou et al., “A New Vector System with Inducible E2a Cell Line for Production of Higher Titer and Safer Adenoviral Vectors,” Virology 275(2):348-357 (2000). cited by applicant

---

Primary Examiner: Poliakova-Georgantas; Ekaterina

Assistant Examiner: McKillop; John Charles

Attorney, Agent or Firm: MEDLER FERRO WOODHOUSE & MILLS PLLC

Background/Summary

REFERENCE TO SEQUENCE LISTING

(1) The application contains a Sequence Listing which has been submitted electronically in .XML format and is hereby incorporated by reference in its entirety. Said .XML copy, created on Oct. 30, 2023, is named “0132-0318US1.xml” and is 513,568 bytes in size. The sequence listing contained in this .XML file is part of the specification and is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

(2) The present disclosure provides mammalian cell lines for producing adeno-associated virus (AAV). The cells suitably include isolated nucleic acids encoding helper genes and AAV genes, under the control of inducible promoters. The disclosure also relates to methods of producing AAVs using the mammalian cells and nucleic acids described herein.

BACKGROUND

(3) Recombinant adeno-associated virus (rAAV) has emerged as an important gene therapy vector for broad clinical application due to its high safety profile and strong potency for delivering functional genes into a variety of tissues with long-term therapeutic benefits. Multiple AAV-based gene therapies have been approved by the US Food and Drug Administration (FDA) and European Medicines Agency (EMA). With 25 viral-vector therapeutics in late-stage development and approximately 120 in Phase II trials as of February 2022, the number of approved therapies is only expected to grow.

(4) However, the challenges of current AAV manufacturing processes include low productivity and scalability, low batch-to-batch reproducibility, high cost and complex supply chain issues, which significantly limit the access to the gene delivery platform for the treatment of a variety of human diseases. Additionally, constitutive expression of both helper and Rep genes required for AAV production can be cytotoxic. AAV stable producer cell lines (PCLs) and vectors that are designed to tightly control the

expression of helper and Rep genes can overcome these limitations and bolster large-scale current good manufacturing practice (cGMP) manufacturing for AAV gene therapy.

SUMMARY OF THE INVENTION

(5) In some embodiments, the present disclosure is directed to an isolated nucleic acid molecule operably comprising a first inducible promoter with two tetracycline operator sequences (a TetO.sub.2 sequence), an E2A gene under control of the first inducible promoter, a viral associated (VA) non-coding RNA under control of a second inducible promoter, a third inducible promoter with a TetO.sub.2 sequence, an E4 gene under control of the third inducible promoter, and an antibiotic resistance gene.

(6) In further embodiments, the disclosure provides an isolated nucleic acid molecule operably comprising, a first inducible promoter with a TetO.sub.2 sequence, an E2A gene under control of the first inducible promoter, a viral associated (VA) non-coding RNA under control of a second inducible promoter, a third inducible promoter with a TetO.sub.2 sequence, an E4 gene under control of the third inducible promoter, an expression cassette encoding a protein ortholog of the VA RNA and a peptide protein tag under the control of a depressible promoter comprising a human Cytomegalovirus (CMV) promoter and a TetO.sub.2 sequence, a termination sequence, and an antibiotic resistance gene.

(7) In still further embodiments, the present disclosure is directed to an isolated nucleic acid molecule operably comprising a plasmid encoding a first inducible promoter with a TetO.sub.2 sequence, a Rep78 gene, including a silenced p19 promoter located within the Rep78 gene coding region, wherein the silenced p19 promoter comprises mutations in SP1, TATA-1, and/or TATA-2 sites, a second inducible promoter and a TetO.sub.2 sequence, a Rep52 gene under control of the second inducible promoter, termination sequence and an antibiotic resistance gene.

(8) In still further embodiments, the present disclosure is directed to a mammalian cell for producing an adeno-associated virus (AAV), comprising any of isolated nucleic acid molecules disclosed herein integrated into its genome. In some embodiments, the mammalian cell further comprises a nucleic acid encoding a transcriptional repression domain in frame with a nucleic acid encoding a tetracycline repressor protein. In some embodiments, the mammalian cell comprises a nucleic acid molecule encoding a gene of interest.

(9) In still further embodiments, the disclosure provides a method for producing an adeno-associated virus (AAV), comprising inducing production of the genomically integrated nucleic acid sequence of any of the mammalian cells disclosed herein, culturing the mammalian cell, and harvesting the AAV.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. **1** shows a schematic of the use of the Tet-On inducible system utilizing the tetracycline repressor protein to repress and activate expression of a protein.

(2) FIGS. **2**A and **2**B show an exemplary TetR and TetR-KRAB fusion system.

(3) FIGS. **3**A-**3**I show exemplary nucleic acid molecules for production of helper genes in accordance with embodiments hereof.

(4) FIGS. **4**A and **4**B show exemplary nucleic acid molecules for production of helper genes in accordance with embodiments hereof.

(5) FIGS. **5**A-**5**P show exemplary nucleic acid molecules for production of Rep genes in accordance with embodiments hereof.

(6) FIGS. **6**A-**6**E show exemplary nucleic acid molecules for production of Cap genes and genes of interest in accordance with embodiments hereof.

(7) FIG. **7** shows the results of AAV titers using the Helper, Rep and Cap-GOI vectors, in accordance with embodiments hereof.

(8) FIG. **8** shows a Western Blot image of the expression of helper, Rep and Cap proteins in accordance with embodiments hereof.

(9) FIG. **9** shows the results of AAV production in 3 Liter single-use bioreactors in accordance with embodiments hereof.

(10) FIG. **10** shows comparative results of AAV production of PCL clones.

(11) FIG. **11** shows the results of cell-based infectivity and potency assay of AAVs produced in accordance

with embodiments hereof.

DETAILED DESCRIPTION OF THE INVENTION

(12) Unless otherwise defined herein, scientific and technical terms used in the present disclosure shall have the meanings that are commonly understood by one of ordinary skill in the art. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. The use of the word “a” or “an” when used in conjunction with the term “comprising” in the claims and/or the specification may mean “one,” but it is also consistent with the meaning of “one or more,” “at least one,” and “one or more than one.”

(13) Throughout this application, the term “about” is used to indicate that a value includes the inherent variation of error for the method/device being employed to determine the value. Typically, the term is meant to encompass approximately or less than 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19% or 20% variability depending on the situation.

(14) The use of the term “or” in the claims is used to mean “and/or” unless explicitly indicated to refer only to alternatives or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.”

(15) As used in this specification and claim(s), the words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited, elements or method steps. It is contemplated that any embodiment discussed in this specification can be implemented with respect to any method, system, host cells, expression vectors, and/or composition of the invention. Furthermore, compositions, systems, cells, and/or nucleic acids of the invention can be used to achieve any of the methods as described herein.

(16) The use of the term “for example” and its corresponding abbreviation “e.g.” (whether italicized or not) means that the specific terms recited are representative examples and embodiments of the disclosure that are not intended to be limited to the specific examples referenced or cited unless explicitly stated otherwise.

(17) As used herein, “between” is a range inclusive of the ends of the range. For example, a number between x and y explicitly includes the numbers x and y, and any numbers that fall within x and y.

(18) As used herein, the term “adeno-associated virus (AAV)” refers to a small sized, replicative-defective nonenveloped virus containing a single stranded DNA of the family Parvoviridae and the genus Dependoparvovirus. Over 10 adeno-associated virus serotypes have been identified so far, with serotype AAV2 being the best characterized. Other non-limiting examples of AAV serotypes are ANC80, AAV1, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV10, and AAV11. In addition to these serotypes, AAV pseudotypes have been developed. An AAV pseudotype contains the capsid of a first serotype and the genome of a second serotype (e.g., the pseudotype AAV2/5 would correspond to an AAV with the genome of serotype AAV2 and the capsid of AAV5).

(19) As referred to herein, the term “adenovirus” refers to a nonenveloped virus with an icosahedral nucleocapsid containing a double stranded DNA of the family Adenoviridae. Over 50 adenoviral subtypes have been isolated from humans and many additional subtypes have been isolated from other mammals and birds. Birds. See, e.g., Ishibashi et al., “Adenoviruses of animals,” In The Adenoviruses, Ginsberg, ed., Plenum Press, New York, N.Y., pp. 497-562 (1984); Strauss, “Adenovirus infections in humans,” In The Adenoviruses, Ginsberg, ed., Plenum Press, New York, N.Y., pp. 451-596 (1984). These subtypes belong to the family Adenoviridae, which is currently divided into two genera, namely Mastadenovirus and Aviadenovirus. All adenoviruses are morphologically and structurally similar. In humans, however, adenoviruses show diverging immunological properties and are, therefore, divided into serotypes. Two human serotypes of adenovirus, namely AV2 and AV5, have been studied intensively and have provided the majority of general information about adenoviruses.

(20) Adeno-associated virus (AAV) has emerged as the vector of choice for gene therapy in over 120 clinical trials worldwide. The fast-growing demand of recombinant AAV requires highly efficient and robust manufacturing platforms. However, current methods for AAV production, including transient transfection and helper virus systems, are extremely costly and lab-intensive. Thus, described herein are isolated nucleic acid molecules for producing an adeno-associated virus (AAV) and plasmid/helper virus-free AAV producer cell lines, and methods of use thereof, that provide efficient AAV manufacturing for a long-term solution at significantly reduced cost. The AAV producer cell line described herein represents a

next generation platform for both clinical and commercial AAV manufacturing.

(21) As used herein, a “vector” or “expression vector” refers to a nucleic acid replicon, such as a plasmid, phage, virus, or cosmid, to which another nucleic acid segment may be attached to bring about the replication and/or expression of the attached nucleic acid segment in a host cell. The term “vector” includes episomal (e.g., plasmids) and non episomal vectors. The term “vector” may also include synthetic vectors. Non-limiting examples of vectors include baculovirus vectors, bacteriophage vectors, plasmids, phagemids, cosmids, fosmids, bacterial artificial chromosomes, viral vectors (e.g. viral vectors based on vaccinia virus, poliovirus, adenovirus, adeno-associated virus, SV40, herpes simplex virus, and the like), P1-based artificial chromosomes, yeast plasmids, yeast artificial chromosomes, and any other vectors specific for specific hosts of interest (e.g., mammalian cells such as the E1 complementary producer cells described herein, including but not limited to HEK293 cells, PER.C6 cells, CAP® cells, and derivatives thereof). Vectors may be introduced into the desired host cells, e.g., HEK293 cells, PER.C6 cells, CAP® cells, or derivatives thereof, by well-known methods, including, but not limited to, transfection, transduction, cell fusion, and lipofection. Vectors can comprise various regulatory elements including, e.g., constitutive and inducible promoters, transcription enhancers, transcription terminators, and the like.

(22) As used herein an “isolated nucleic acid molecule” includes vectors and plasmids that can contain the isolated nucleic acid molecule, as well as similar structures where the isolated nucleic acid molecule can be manipulated, stored, shipped, and ultimately utilized in various cell transfection systems. The isolated nucleic acid molecules described herein can be used for production of AAVs as described herein, but can also be utilized in various non-AAV producing cell lines (including transient transfection systems). The isolated nucleic acid molecules described herein suitably further include various additional elements and sequences as required to allow for use in the cellular systems, including mammalian cells, described herein.

(23) A “nucleic acid,” “nucleic acid molecule,” or “oligonucleotide” means a polymeric compound comprising covalently linked nucleotides. The term “nucleic acid” includes polyribonucleic acid (RNA) and polydeoxyribonucleic acid (DNA), both of which may be single- or double-stranded. DNA includes, but is not limited to, complimentary DNA (cDNA), genomic DNA, plasmid or vector DNA, and synthetic DNA. RNA includes, but is not limited to, mRNA, tRNA, rRNA, snRNA, microRNA, miRNA, or MIRNA.

(24) A “gene” refers to an assembly of nucleotides that encode a polypeptide, and includes cDNA and genomic DNA nucleic acid molecules. “Gene” also refers to a nucleic acid fragment that can act as a regulatory sequence preceding (5′ non-coding sequences) and following (3′ non-coding sequences) the coding sequence. In some embodiments, genes are integrated with multiple copies. In some embodiments, genes are integrated at predefined copy numbers.

(25) As referred to herein, the term “heterologous gene” or “HG” as it relates to nucleic acid sequences such as a coding sequence or a control sequence, denotes a nucleic acid sequence, e.g., a gene, that is not normally joined together, and/or are not normally associated with a particular cell. In some embodiments, a heterologous gene is a construct where the coding sequence itself is not found in nature (e.g., synthetic sequences having codons different from the native gene). Allelic variation or naturally occurring mutational events do not give rise to heterologous DNA, as used herein.

(26) In the various embodiments described herein, the nucleic acid molecules are capable of encoding the various genes. That is the nucleic acid molecules, when transcribed, produce mRNA for the genes described herein, which is then translated to the desired or required proteins.

(27) The various nucleic acid molecules encoding the various genes described herein are suitably under control of a promoter. As used herein “under control” refers to a gene being regulated by a “promoter,” “promoter sequence,” or “promoter region,” which refers to a DNA regulatory region/sequence capable of binding RNA polymerase and initiating transcription of a downstream coding or non-coding gene sequence. In other words, the promoter and the gene are in operable combination or operably linked. As referred to herein, the terms “in operable combination”, “in operable order”, “operably linked” and “operably” refer to the linkage of nucleic acid sequences in such a manner that a promoter capable of directing the transcription of a given gene and/or the synthesis of a desired protein molecule is produced.

(28) In some examples of the present disclosure, the promoter sequence includes the transcription initiation site and extends upstream to include the minimum number of bases or elements necessary to initiate transcription at levels detectable above background. In some embodiments, the promoter sequence includes a transcription initiation site, as well as protein binding domains responsible for the binding of RNA polymerase. Eukaryotic promoters will often, but not always, contain “TATA” boxes and “CAT” boxes.

Various promoters, including inducible promoters, may be used to drive the gene expression, e.g., in the host cell or vectors of the present disclosure. In some embodiments, the promoter is not a leaky promoter, i.e., the promoter is not constitutively expressing any of the gene products as described herein. In other embodiments as described herein, the promoter is a constitutive promoter, which initiates mRNA synthesis independent of the influence of an external regulation.

(29) Suitably, the promoters used to control the transcription of the various genes for producing the AAVs described herein are derepressible promoters. As used herein, a "derepressible promoter" refers to a structure that includes an inducible or a functional promoter and additional elements or sequences capable of binding to a repressor element to cause repression of the inducible or functional promoter. "Repression" refers to the decrease or inhibition of the initiation of transcription of a downstream coding or non-coding gene sequence by a promoter. A "repressor element" refers to a protein or polypeptide that is capable of binding to a promoter (or near a promoter) so as to decrease or inhibit the activity of the promoter. A repressor element can interact with a substrate or binding partner of the repressor element, such that the repressor element undergoes a conformation change. This conformation change in the repressor element takes away the ability of the repressor element to decrease or inhibit the promoter, resulting in the "derepression" of the promoter, thereby allowing the promoter to proceed with the initiation of transcription.

(30) In some embodiments, the derepressible promoter comprises an inducible or functional promoter and one or more tetracycline operator sequences (TetO.sub.2). An "inducible promoter" regulates (e.g., activates or inactivates) transcriptional activity of a nucleic acid to which it is operably linked when the promoter is influenced by or contacted by a corresponding regulatory protein or other external stimuli such as chemicals, stress, or biotic stimuli. A "functional promoter" refers to a promoter, that absent the action of the repressor element, would be capable of initiation transcription. Various inducible and functional promoters that can be used in the practice of the present invention are known in the art, and include for example, PCMV, PHI, P19, P5, P40 and promoters of Adenovirus helper genes (e.g., E1A, E1B, E2A, E4Orf6, and VA).

(31) The term "Tet operator sequence", "Tet operator sequence motif", "Tet operator", "TetO.sub.2" or "TetO" as used herein is intended to encompass all classes of Tet operator sequences, e.g., TetO(A), TetO(B), TetO(C), TetO(D), TetO(E), TetO(G), TetO(H), TetO(J) and TetO(Z). The nucleotide sequences of Tet repressors of members of the A, B, C, D, E, G, H, J and Z classes, and their corresponding Tet operator sequences are well known in the art, see, for example, Waters 1983, Null. Acids Res 11:6089-6105, Hillen 1983, Nucl. Acids Res. 11:525-539, Postle 1984, Nucl. Acids Res. 12:4849-4863, Unger 1984, Gene 31:103-108, Unger 1984, Nucl Acids Res. 12:7693-7703 and Tovar 1988, Mol. Gen. Genet. 215:76-80, which are incorporated herewith by reference with respect to the specifically disclosed Tet operator sequences and in their entireties. Tet operator sequences are also disclosed in U.S. Pat. No. 5,464,758.

(32) Exemplary repressor elements and their corresponding binding partners that can be used as derepressible promoters are known in the art, and include systems such as the cumate gene-switch system (CuO operator, CymR repressor and cumate binding partner) (see, e.g., Mullick et al., "The cumate gene-switch: a system for regulated expression in mammalian cells," *BMC Biotechnology* 6:43 (1-18) (2006), the disclosure of which is incorporated by reference herein in its entirety, including the disclosure of the derepressible promoter system described therein) and the TetO/TetR system described herein (see, e.g., Yao et al., "Tetracycline Repressor, tetR, rather than the tetR-Mammalian Cell Transcription Factor Fusion Derivatives, Regulates Inducible Gene Expression in Mammalian Cells," Human Gene Therapy 9:1939-1950 (1998), the disclosure of which is incorporated by reference herein in its entirety).

(33) VA (virus-associated) RNA is a non-coding type found in adenoviruses. It plays a role in the regulation of translation. There are two copies of this RNA called VAI or RNA I VA and VAII or RNA II VA. The two RNA VA genes are distinct genes in the adenovirus genome. RNA I VA is the main species with RNA II VA expressed at a lower level. Neither transcript is polyadenylated and both are transcribed by PolIII.

(34) As described herein, the nucleic acid molecules suitably encode viral helper gene. Viral helper genes include various adenoviral virus genes, herpes virus genes and bocavirus genes (see, e.g., Guido et al., "Human bocavirus: Current knowledge and future challenges," World J. Gateroenterol 22:8684-8697, the disclosure of which is incorporated by reference herein in its entirety). In exemplary embodiments, the viral helper gene is an adenovirus helper gene. As referred to herein, the term "adenovirus helper gene" or "AV helper gene" refers to a gene that is composed of one or more nucleic acid sequences derived from one or

more adenovirus subtypes or serotypes that contributes to Adeno-associated virus replication and packaging. In some embodiments, the Adenovirus helper gene is E1A, E1B, E2A, E4 (including E4Orf6), VA, or a combination thereof or any other adenovirus helper gene. In exemplary embodiments, the adenovirus helper gene comprises both E2A and E4Orf6 genes.

(35) In still further embodiments, provided herein is an isolated nucleic acid molecule operably comprising a first inducible promoter and two tetracycline operator sequences (a TetO.sub.2 sequence), an E2A gene under control of the first inducible promoter, a viral associated (VA) non-coding RNA under control of a second inducible promoter, a third inducible promoter and a TetO.sub.2 sequence, an E4 gene under control of the third inducible promoter, and an antibiotic resistance gene.

(36) In other embodiments, the present disclosure provides an isolated nucleic acid molecule operably comprising, a first inducible promoter and a TetO.sub.2 sequence, an E2A gene under control of the first inducible promoter, a viral associated (VA) non-coding RNA under control of a second inducible promoter, a third inducible promoter and a TetO.sub.2 sequence, an E4 gene under control of the third inducible promoter, an expression cassette encoding a protein ortholog of the VA RNA and a peptide protein tag under the control of a depressible promoter comprising a human Cytomegalovirus (CMV) promoter and a TetO.sub.2 sequence, a termination sequence, and an antibiotic resistance gene.

(37) As described herein, in exemplary embodiments, the E2A gene that is encoded by the nucleic acid molecules is suitably under the control of a first inducible promoter that is native to the E2A gene. In some embodiments, the VA non-coding RNA that is encoded by the nucleic acid molecules is suitably under the control of a second inducible promoter that is native to the VA non-coding RNA. In some embodiments, the second inducible promoter is an H1 promoter. In some embodiments, the E4 gene that is encoded by the nucleic acid molecules is suitably under the control of a third inducible promoter that is native to the E4 gene.

(38) As used herein, protein orthologs refer to proteins that have the same specificity in different organisms, e.g., bind the same ligand and similar DNA sites in related genomes. Hence, orthologous proteins carry the same or similar specificity determining residues. In some embodiments, the isolated nucleic acid molecule described herein comprises a protein ortholog of the of the VA RNA. In some embodiments, the protein ortholog is an influenza non-structural 1 protein (NS1), Ebola viral protein 35 (VP35), orthoreovirus sigma 3 protein (σ3), group C rotavirus non-structural protein 3 (NSP3), vaccinia virus E3L protein (E3L), herpes simplex virus type 1 US11 protein (US11), Epstein-Barr virus SM protein (SM), baculovirus PK2 protein (PK2), hepatitis C virus non-structural protein 5A protein (NS5A), human herpes virus-8 protein vIRF-2 (vIRF-2), human immunodeficiency virus protein Tat, vaccinia virus K3L protein (K3L), Herpes Simplex virus protein γ134.5/ICP34.5, or human papilloma virus-18 E6 protein (E6). In some embodiments, the protein ortholog is VP35. In some embodiments, the protein ortholog is SM.

(39) Alternatively, or in addition, the protein ortholog of the VA RNA of any isolated nucleic acid molecule of the present invention may be operably linked to a peptide tag to assist in its isolation and detection, or to assist in the identification and expression of the encoded protein. A variety of peptide tags can be used in the context of the present invention, including PK tags, FLAG tags, MYC tags, hemagglutinin (HA) tags and polyhistidine tags. Peptide tags can be diagnosed by immunodetection tests, which use anti-tag antibodies. In some embodiments, the peptide tag is located individually at the N-terminus of the protein ortholog of the VA RNA. In some embodiments, the isolated nucleic acid molecule comprises a peptide protein tag. In some embodiments, the peptide protein tag is an HA tag. In some embodiments, the peptide protein tag is a FLAG-tag. In some embodiments, the FLAG-tag is an N-terminal flag tag.

(40) As used herein, the role of the termination sequence is to define the end of a transcriptional unit (such as a gene) and initiate the process of releasing the newly synthesized RNA from the transcription machinery. Terminators are found downstream of the gene to be transcribed, and typically occur directly after any 3′ regulatory elements, such as the polyadenylation or poly(A) signal. In some embodiments, any of the nucleic acid molecules disclosed herein comprise a termination sequence. In some embodiments, the termination sequence is a bovine growth hormone polyadenylation (bgh-PolyA) signal.

(41) As referred to herein, a “reporter gene” is a gene whose expression confers a phenotype upon a cell that can be easily identified and measured. In some embodiments, the reporter gene comprises a fluorescent protein gene. In some embodiments, the reporter gene comprises a selection gene.

(42) In some embodiments, the isolated nucleic acid molecules comprise a selection gene. As referred to

herein, the term “selection gene” refers to the use of a gene which encodes an enzymatic activity that confers the ability to grow in medium lacking what would otherwise be an essential nutrient; in addition, a selection gene may confer resistance to an antibiotic or drug upon the cell in which the selection gene is expressed. A selection gene may be used to confer a particular phenotype upon a host cell. When a host cell must express a selection gene to grow in selective medium, the gene is said to be a positive selection gene. A selection gene can also be used to select against host cells containing a particular gene; a selection gene used in this manner is referred to as a negative selection gene. In some embodiments, the selection gene is an antibiotic resistance gene. In some embodiments, the antibiotic resistance gene is a puromycin resistance gene. In some embodiments, the antibiotic resistance gene is a blasticidin resistance gene. In some embodiments, the antibiotic resistance gene is a hygromycin resistance gene. In some embodiments, the antibiotic resistance gene is a zeocin resistance gene.

(43) In exemplary embodiments, the nucleic acid molecules include two inverted terminal repeat (ITR) sequences. As known in the art, these ITR sequences (i.e., AAV2 ITR) are single stranded sequence of nucleotides, followed downstream by its reverse compliment. ITR sequences represent the minimal sequence required for replication, rescue, packaging and integration of the AAV genome. Suitably, these ITR sequences flank a gene of interest. Thus, in embodiments, the nucleic acid molecules further encode a gene of interest. This gene of interest can be, for example, a reporter gene, a selection gene, or a gene of therapeutic interest, for example. In some embodiments, the nucleic acid sequence is flanked on both the 5′ and 3′ ends by transposon-specific inverted terminal repeats (ITR).

(44) As used herein, “core insulator” refers to a gene boundary element that blocks the interaction between an enhancer and a promoter. The presence of the core insulator between the enhancer and the promoter allows the insulator to block subsequent interactions. The core insulator can determine the set of genes that the enhancer can influence. Insulators are required when two adjacent genes on a chromosome have very different transcriptional patterns and it is necessary to induce or suppress one mechanism without inhibiting the adjacent genes. In some embodiments, any of the nucleic acid sequences disclosed herein comprise a core insulator sequence inserted downstream of the ITR on the 3′ end and upstream of the ITR on the 5′ end. In some embodiments, the nucleic acid molecule comprises a core insulator sequence inserted between the termination sequence and the antibiotic resistance gene.

(45) As referred to herein, the term “Rep” gene refers to the art-recognized region of the AAV genome which encodes the replication proteins of the virus which are collectively required for replicating the viral genome, or functional homologues thereof such as the human herpesvirus 6 (HHV-6) rep gene which is also known to mediate AAV2 DNA replication. Thus, the Rep coding region can include the genes encoding for AAV Rep78 and Rep68 (the “long forms of Rep”), and Rep52 and Rep40 (the “short forms of Rep”), or functional homologues thereof. The Rep coding region, as used herein, can be derived from any viral serotype, such as the AAV serotypes described herein. The region need not include all wild-type genes but may be altered, (e.g., by insertion, deletion or substitution of nucleotides), so long as the Rep genes present provide for sufficient integration functions when expressed in a suitable target cell. See, e.g., Muzyczka, N., Current Topics in Microbiol. and Immunol. 158:97-129 (1992); and Kotin, R. M., Human Gene Therapy 5:793-801 (1994).

(46) In still further embodiments, provided herein is an isolated nucleic acid molecule comprising a plasmid encoding a first inducible promoter and a TetO.sub.2 sequence, a Rep78 gene, including a silenced p19 promoter located within the Rep78 gene coding region, wherein the silenced p19 promoter comprises mutations in SP1, TATA-1, and/or TATA-2 sites, a second inducible promoter and a TetO.sub.2 sequence, a Rep52 gene under control of the second inducible promoter, termination sequence and an antibiotic resistance gene.

(47) The terms “sequence identity” or “% identity” in the context of nucleic acid sequences described herein refer to the percentage of residues in the compared sequences that are the same when the sequences are aligned over a specified comparison window. A comparison window can be a segment of at least 10 to over 1000 residues in which the sequences can be aligned and compared. Methods of alignment for determination of sequence identity are well-known and can be performed using publicly available databases such as BLAST (blast.ncbi.nlm.nih.gov/Blast. CGI.).

(48) The Kozak consensus sequence, Kozak consensus or Kozak sequence, is known as a sequence which occurs on eukaryotic mRNA and has the consensus (gcc)gccRccAUGG, where R is a purine (adenine or guanine) three bases upstream of the start codon (AUG), which is followed by another “G.” In some

embodiments, the isolated nucleic acid molecules disclosed herein comprise a Kozak consensus sequence. In some embodiments, the nucleic acid molecules comprise a consensus sequence that has at least about 70%, at least about 80%, at least about 90% sequence identity, or more identity to the Kozak consensus sequence. In some embodiments, the isolated nucleic acid molecule includes a Kozak consensus sequence after the gene cassette encoding one or more proteins of interest is inserted into the vector, e.g., at the restriction site downstream of the promoter. For example, the isolated nucleic acid molecule can include a nucleotide sequence of GCCGCCATG, where the ATG is the start codon of the protein of interest. In some embodiments, the isolated nucleic acid molecule comprises a nucleotide sequence of GCGGCCGCCATG, where the ATG is the start codon of the protein of interest. In some embodiments, the Rep78 gene of the isolated nucleic acid molecule further comprises a Kozak consensus sequence. In some embodiments, the Rep52 gene of the isolated nucleic acid molecule further comprises a Kozak consensus sequence.
(49) To control the production of Rep proteins more effectively, non-canonical start codons or non-canonical translational initiation codons can be used. In some embodiments, the Rep78 gene of the isolated nucleic acid molecules further comprises a non-canonical start codon. In some embodiments, the Rep52 gene of the isolated nucleic acid molecule further comprises a non-canonical start codon. In some embodiments, the non-canonical translation initiation codon is GTG, CTG, ACG or TTG. In some embodiments, the non-canonical translation initiation codon is CTG.
(50) Non-limiting examples of gene modifications that can be used in the present invention include codon optimization aimed at, for example, modifying a non-CTG leucine codon to CTG, or modifying a non-AAG lysine codon to AAG. Another example of a nucleic acid codon-optimizing modification is to increase GC content. To increase the vector stability and Rep52 expression, the DNA sequence of coding region of Rep52 was optimized for human cell codon usage while retaining the same protein sequence, which significantly reduced the DNA sequence identity of the Rep52 coding sequence to the Rep78 coding sequence, from 100% sequence identity to 80.1% sequence identity. In some embodiments, the Rep52 of any of the nucleic acid molecules disclosed herein is codon optimized. In some embodiments, the VP1 gene of any of the nucleic acid molecules disclosed herein, excluding the first 439 bp of the coding region, is optimized.
(51) In further embodiments, the first inducible promoter of the nucleic acid molecules disclosed herein is a human cytomegalovirus (CMV) immediate early enhancer-promoter, CMV immediate early enhancer fused chicken β-actin promoter (CAG), human ubiquitin C (UbC) promoter or Rous sarcoma virus (RSV) promoter. In some embodiments, the first inducible promoter is an RSV promoter. In some embodiments, the first inducible promoter is an UbC promoter. As described herein, the use of an artificial intron allows for removal of an inducible promoter following induction of the promoter and prior to the producing the AAV. In some embodiments, the nucleic acid molecule further comprises an intron of the UbC promoter inserted upstream of the UbC promoter.
(52) In some embodiments, the second inducible promoter of the nucleic acid molecules disclosed herein is human CMV immediate early enhancer-promoter, CMV immediate early enhancer fused chicken β-actin promoter (CAG), human ubiquitin C (UbC) promoter or Rous sarcoma virus (RSV) promoter. In some embodiments, the second inducible promoter is a CAG promoter. In some embodiments, the second inducible promoter is a CMV promoter.
(53) As referred to herein, the term “Cap” gene refers to the art-recognized region of the AAV genome which encodes the capsid proteins of the virus. Illustrative (non-limiting) examples of these capsid proteins are the AAV capsid proteins VP1, VP2, and VP3. Cap genes used in this disclosure can come from any AAV serotype or a combination of AAV serotypes.
(54) In still further embodiments, the present disclosure provides an isolated nucleic acid molecule operably comprising a first inducible promoter and a TetO.sub.2 sequence, a VP1 gene, a second inducible promoter and a TetO.sub.2 sequence, a VP2 gene and a VP3 gene, and an antibiotic resistance gene. In some embodiments, the nucleic acid sequence comprises a core insulator sequence inserted between the 1) VP1 gene and 2) the second inducible promoter and TetO.sub.2 sequence. In some embodiments, the nucleic acid sequence comprises a core insulator sequence inserted between 1) the VP2 and VP3 gene and 2) the antibiotic resistance gene.
(55) As referred to herein, the term “gene of interest”, or “gene of therapeutic interest” or “GOI” is used to describe a heterologous gene. The GOI of the present disclosure can comprise any desired gene that encodes a protein that is defective or missing from a target cell genome or that encodes a non-native protein

having a desired biological or therapeutic effect (e.g., an antiviral function), or the sequence can correspond to a molecule having an antisense or ribozyme function. Representative (non-limiting) examples of suitable genes of therapeutic interest include those used for the treatment of inflammatory diseases, autoimmune, chronic and infectious diseases, including such disorders as AIDS, cancer, neurological diseases, cardiovascular disease, hypercholestemia; various blood disorders including various anemias, thalassemias and hemophilia; genetic defects such as cystic fibrosis, Gaucher's Disease, adenosine deaminase (ADA) deficiency, emphysema, etc. Several antisense oligonucleotides (e.g., short oligonucleotides complementary to sequences around the translational initiation site (AUG codon) of an mRNA) that are useful in antisense therapy for cancer and for viral diseases have been described in the art and are also examples of suitable genes of therapeutic interest.

(56) In further embodiments, the nucleic acid molecule comprises a gene of interest (GOI) downstream of the VP2 and VP3 gene and upstream of the antibiotic resistance gene. In some embodiments, the nucleic acid molecule comprises a core insulator sequence inserted between the VP3 gene and the GOI. In some embodiments, the nucleic acid molecule comprises a core insulator sequence inserted between the GOI and the antibiotic resistance gene. In some embodiments, the nucleic acid molecule comprises a core insulator sequence inserted between the VP3 gene and the GOI, and between the GOI and the antibiotic resistance gene. In some embodiments, the GOI is a therapeutic gene.

(57) In some embodiments, the present disclosure provides a mammalian cell for producing an adeno-associated virus (AAV), comprising any of the isolated nucleic acid molecules disclosed herein integrated into its genome. In some embodiments, the mammalian cell further comprises a nucleic acid encoding a transcriptional repression domain in frame with a nucleic acid encoding a tetracycline repressor protein. In some embodiments, the mammalian cell disclosed herein further comprises a nucleic acid molecule encoding a gene of interest.

(58) The AAV producer cells described herein provide a long-term and cost-effective solution for large scale AAV manufacturing. As constitutive expression of either helper or Rep proteins can be cytotoxic, the strategies described herein allow for control of their expression by engineered, derepressible promoters.

(59) As used herein, the term “mammalian cell” includes cells from any member of the order Mammalia, such as, for example, human cells, mouse cells, rat cells, monkey cells, hamster cells, and the like. In some embodiments, the cell is a mouse cell, a human cell, a Chinese hamster ovary (CHO) cell, a CHOK1 cell, a CHO-DXB11 cell, a CHO-DG44 cell, a CHOK1SV cell including all variants (e.g., POTELLIGENT®, Lonza, Slough, UK), a CHOK1SV GS-KO (glutamine synthetase knockout) cell including all variants (e.g., XCEED™ Lonza, Slough, UK). In some embodiments, the mammalian cell is a Chinese hamster ovary (CHO) cell. In some embodiments, the mammalian cell is a human cell.

(60) Exemplary human cells include human embryonic kidney (HEK) cells, such as HEK293, a HeLa cell, or a HT1080 cell. As used herein, the term “human embryonic kidney 293 (HEK293) cell” refers to a cell line originally derived from human embryonic kidney cells and containing approximately 4.5 kb of Ad5 genome. Variants of HEK293 cells have been developed and include, e.g., HEK293S, HEK293T, HEK293F, HEK293FT, HEK293FTM, HEK293SG, HEK293SGGD, HEK293H, HEK293E, HEK293MSR, and HEK293A. The term “HEK293 cell” as used herein encompasses all HEK293 cell variants and derivatives, including but not limited to the variants described herein. See, e.g., Yuan et al., “The Scattered Twelve Tribes of HEK293,” Biomed Pharmacol J 2018; 11 (2). In some embodiments, the human cell is an HEK cell. In some embodiments, the HEK cell is an HEK293 cell.

(61) As described herein, suitably the mammalian cells are mammalian cell cultures, and in embodiments, can be suspension cultures. As described herein, the use of suspension cell cultures allows for increased scalability and production of AAV. Adherent cultures refer to cells that are grown on a substrate surface, for example a plastic plate, dish or other suitable cell culture growth platform, and may be anchorage dependent. Suspension cultures refer to cells that can be maintained in, for example, culture flasks or large suspension vats, which allows for a large surface area for gas and nutrient exchange. Suspension cell cultures often utilize a stirring or agitation mechanism to provide appropriate mixing. Media and conditions for maintaining cells in suspension are generally known in the art. An exemplary suspension cell culture includes human HEK293 clonal cells.

(62) “Transfection” as used herein means the introduction of an exogenous nucleic acid molecule, including a vector, into a cell. A “transfected” cell comprises an exogenous nucleic acid molecule inside the cell and a “transformed” cell is one in which the exogenous nucleic acid molecule within the cell induces a

phenotypic change in the cell. The transfected nucleic acid molecule can be integrated into the host cell's genomic DNA and/or can be maintained by the cell, temporarily or for a prolonged period of time, extra-chromosomally. Host cells or organisms that express exogenous nucleic acid molecules or fragments are referred to as “recombinant,” “transformed,” or “transgenic” organisms. A number of transfection techniques are generally known in the art. See, e.g., Graham et al., Virology, 52:456 (1973); Sambrook et al., Molecular Cloning, a laboratory manual, Cold Spring Harbor Laboratories, New York (1989); Davis et al., Basic Methods in Molecular Biology, Elsevier (1986); and Chu et al., Gene 13:197 (1981). Such techniques can be used to introduce one or more exogenous DNA moieties, such as an AAV vector cassette, AAV helper constructs, and other nucleic acid molecules, into suitable host cells.
(63) Various methods of transfecting the mammalian cells with the isolated nucleic acid molecules described herein (i.e., vectors), are known in the art and include various chemical and physical methods, for example, electroporation, cell injection, calcium phosphate exposure, liposome or polymer-based carrier systems, etc.
(64) In still further embodiments, the present disclosure provides a method for producing an adeno-associated virus (AAV), comprising inducing production of the genomically integrated nucleic acid sequence of any of the mammalian cells disclosed herein, culturing the mammalian cell, and harvesting the AAV.
(65) As discussed herein, the methods of producing and the mammalian cells of the present disclosure advantageously provide a high titer of the AAV produced by the mammalian cells. In some embodiments, the amount of the AAV produced is about 10.sup.10 to about 10.sup.11 viral genomes per mL (vg/mL). In some embodiments, the amount of the AAV produced is at least about 10.sup.10 viral genomes per mL (vg/mL). In some embodiments, the amount of the AAV produced is at least about 10.sup.11 viral genomes per mL (vg/mL).
(66) In some embodiments, the mammalian cells provided herein are substantially free of helper virus. As referred to herein, a “helper virus” is any non-AAV virus that is added to enable the replication and packaging of adeno-associated virus. Representative (non-limiting) examples of helper viruses are adenovirus and herpes virus. In some embodiments, the term substantially free of helper virus refers to a cell that has fewer than 100, fewer than 10, or fewer than 1 helper virus per cell. In some embodiments, the term substantially free of helper virus refers to a cell in which no helper viruses are present or to a population of cells in which no helper viruses are present using detection methods known to those skilled in the art. In some embodiments, no wild-type helper virus is in the cell. In some embodiments, the term wild-type virus refers to any complete-non-AAV virus that can replicate in the cell independently of any other virus.
(67) The methods of producing the AAVs can be used in a continuous manufacturing system. In exemplary embodiments, the use of a suspension cell culture allows for the production of large volumes of AAV, with high productivity and prolonged culture conditions to allow for multiple harvests of AAV for each batch of starting cells.
(68) Production methods can utilize any suitable reactor(s) including but not limited to stirred tank, airlift, fiber, microfiber, hollow fiber, ceramic matrix, fluidized bed, fixed bed, and/or spouted bed bioreactors. As used herein, “reactor” can include a fermenter or fermentation unit, or any other reaction vessel and the term “reactor” is used interchangeably with “fermenter.” The term fermenter or fermentation refers to both microbial and mammalian cultures. For example, in some aspects, an example bioreactor unit can perform one or more, or all, of the following: feeding of nutrients and/or carbon sources, injection of suitable gas (e.g., oxygen), inlet and outlet flow of fermentation or cell culture medium, separation of gas and liquid phases, maintenance of temperature, maintenance of oxygen and CO.sub.2 levels, maintenance of pH level, agitation (e.g., stirring), and/or cleaning/sterilizing. Example reactor units, such as a fermentation unit, may contain multiple reactors within the unit, for example the unit can have 1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, or 100, or more bioreactors in each unit and/or a facility may contain multiple units having a single or multiple reactors within the facility. In various embodiments, the bioreactor can be suitable for batch, semi fed-batch, fed-batch, perfusion, and/or a continuous fermentation processes. Any suitable reactor diameter can be used. In embodiments, the bioreactor can have a volume between about 100 mL and about 50,000 L. Non-limiting examples include a volume of 100 mL, 250 mL, 500 mL, 750 mL, 1 liter, 2 liters, 3 liters, 4 liters, 5 liters, 6 liters, 7 liters, 8 liters, 9 liters, 10 liters, 15 liters, 20 liters, 25 liters, 30 liters, 40 liters, 50 liters, 60 liters, 70 liters, 80 liters, 90 liters, 100 liters, 150 liters, 200 liters, 250

liters, 300 liters, 350 liters, 400 liters, 450 liters, 500 liters, 550 liters, 600 liters, 650 liters, 700 liters, 750 liters, 800 liters, 850 liters, 900 liters, 950 liters, 1000 liters, 1500 liters, 2000 liters, 2500 liters, 3000 liters, 3500 liters, 4000 liters, 4500 liters, 5000 liters, 6000 liters, 7000 liters, 8000 liters, 9000 liters, 10,000 liters, 15,000 liters, 20,000 liters, and/or 50,000 liters. Additionally, suitable reactors can be multi-use, single-use, disposable, or non-disposable and can be formed of any suitable material including metal alloys such as stainless steel (e.g., 316L or any other suitable stainless steel) and Inconel, plastics, and/or glass.

ADDITIONAL EXEMPLARY EMBODIMENTS

(69) Embodiment 1 is an isolated nucleic acid molecule operably comprising a first inducible promoter and two tetracycline operator sequences (a TetO.sub.2 sequence), an E2A gene under control of the first inducible promoter, a viral associated (VA) non-coding RNA under control of a second inducible promoter, a third inducible promoter and a TetO.sub.2 sequence, an E4 gene under control of the third inducible promoter, and an antibiotic resistance gene. Embodiment 2 includes the nucleic acid molecule of embodiment 1, wherein the first inducible promoter is native to the E2A gene. Embodiment 3 includes the nucleic acid molecule of embodiment 1 or 2, wherein the second inducible promoter is a promoter native to the VA non-coding RNA. Embodiment 4 includes the nucleic acid molecule of any one of embodiments 1 to 3, wherein the second inducible promoter is an H1 promoter. Embodiment 5 includes the nucleic acid molecule of any one of embodiments 1 to 4, wherein the third inducible promoter is native to the E4 gene. Embodiment 6 includes the nucleic acid molecule of any one of embodiments 1 to 5, wherein the antibiotic resistance gene is a puromycin resistance gene. Embodiment 7 includes the nucleic acid molecule of any one of embodiments 1 to 6, wherein the nucleic acid sequence comprises a core insulator sequence inserted between the E4 gene and the antibiotic resistance gene. Embodiment 8 includes the nucleic acid molecule of any one of embodiments 1 to 7, wherein the nucleic acid sequence is flanked on both the 5′ and 3′ ends by transposon-specific inverted terminal repeats (ITR). Embodiment 9 includes the nucleic acid molecule of embodiment 8, wherein the nucleic acid sequence comprises a core insulator sequence inserted downstream of the ITR on the 3′ end and upstream of the ITR on the 5′ end. Embodiment 10 includes the nucleic acid molecule of any one of embodiments 1 to 9, further comprising multiple copies of VA non-coding RNA. Embodiment 11 is an isolated nucleic acid molecule operably comprising a first inducible promoter and a TetO.sub.2 sequence, an E2A gene under control of the first inducible promoter, a viral associated (VA) non-coding RNA under control of a second inducible promoter, a third inducible promoter and a TetO.sub.2 sequence, an E4 gene under control of the third inducible promoter, an expression cassette encoding a protein ortholog of the VA RNA and a peptide protein tag under the control of a depressible promoter comprising a human Cytomegalovirus (CMV) promoter and a TetO.sub.2 sequence, a termination sequence, and an antibiotic resistance gene. Embodiment 12 includes the nucleic acid molecule of embodiment 11, wherein the first inducible promoter is native to the E2A gene. Embodiment 13 includes the nucleic acid molecule of embodiment 11 or 12, wherein the second inducible promoter is a promoter native to the VA non-coding RNA. Embodiment 14 includes the nucleic acid molecule of any one of embodiments 11 to 13, wherein the second inducible promoter is an H1 promoter. Embodiment 15 includes the isolated nucleic acid molecule of any one of embodiments 12 to 14, wherein the third inducible promoter is native to the E4 gene. Embodiment 16 includes the isolated nucleic acid molecule of any one of embodiments 11 to 15, wherein the protein ortholog is an influenza non-structural 1 protein (NS1), Ebola viral protein 35 (VP35), orthoreovirus sigma 3 protein (σ3), group C rotavirus non-structural protein 3 (NSP3), vaccinia virus E3L protein (E3L), herpes simplex virus type 1 US11 protein (US11), Epstein-Barr virus SM protein (SM), baculovirus PK2 protein (PK2), hepatitis C virus non-structural protein 5A protein (NS5A), human herpes virus-8 protein vIRF-2 (vIRF-2), human immunodeficiency virus protein Tat, vaccinia virus K3L protein (K3L), Herpes Simplex virus protein γ134.5/ICP34.5, and human papilloma virus-18 E6 protein (E6). Embodiment 17 includes the isolated nucleic acid molecule of embodiment 16, wherein the protein ortholog is VP35. Embodiment 18 includes the isolated nucleic acid molecule of embodiment 16, wherein the protein ortholog is SM. Embodiment 19 includes the isolated nucleic acid molecule of any one of embodiments 11 to 18, wherein the peptide protein tag is a FLAG-tag. Embodiment 20 includes the isolated nucleic acid molecule of embodiment 19, wherein the FLAG-tag is a N-terminal Flag tag. Embodiment 21 includes the isolated nucleic acid molecule of any one of embodiments 11 to 20, wherein the termination sequence is a bovine growth hormone polyadenylation (bgh-PolyA) signal. Embodiment 22 includes the nucleic acid molecule of any one of embodiments 11 to 21, wherein the antibiotic resistance gene is a puromycin resistance gene. Embodiment 23 includes the nucleic acid

molecule of any one of embodiments 11 to 22, wherein the nucleic acid sequence comprises a core insulator sequence inserted between the E4 gene and the expression cassette. Embodiment 24 includes the nucleic acid molecule of any one of embodiments 11 to 23, wherein the nucleic acid sequence is flanked on both the 5′ and 3′ ends by transposon-specific inverted terminal repeats (ITR). Embodiment 25 includes the nucleic acid molecule of embodiment 24, wherein the nucleic acid sequence comprises a core insulator sequence inserted downstream of the ITR on the 3′ end and upstream of the ITR on the 5′ end. Embodiment 26 is an isolated nucleic acid molecule operably comprising a plasmid encoding a first inducible promoter and a TetO.sub.2 sequence, a Rep78 gene, including a silenced p19 promoter located within the Rep78 gene coding region, wherein the silenced p19 promoter comprises mutations in SP1, TATA-1, and/or TATA-2 sites, a second inducible promoter and a TetO.sub.2 sequence, a Rep52 gene under control of the second inducible promoter, termination sequence and, an antibiotic resistance gene. Embodiment 27 includes the isolated nucleic acid molecule of embodiment 26, wherein the Rep78 gene further comprises a Kozak consensus sequence or a non-canonical start codon. Embodiment 28 includes the nucleic acid molecule of embodiment 26, wherein the Rep52 gene further comprises a Kozak consensus sequence or a non-canonical start codon. Embodiment 29 includes the isolated nucleic acid molecule of any one of embodiments 26 to 28, wherein the first inducible promoter is human cytomegalovirus (CMV) immediate early enhancer-promoter, CMV immediate early enhancer fused chicken β-actin promoter (CAG), human ubiquitin C (UbC) promoter or Rous sarcoma virus (RSV) promoter. Embodiment 30 includes the isolated nucleic acid molecule of embodiment 29, wherein the first inducible promoter is an RSV promoter. Embodiment 31 includes the isolated nucleic acid molecule of any one of embodiments 26 to 30, wherein the second inducible promoter is human CMV immediate early enhancer-promoter, CMV immediate early enhancer fused chicken β-actin promoter (CAG), human ubiquitin C (UbC) promoter or Rous sarcoma virus (RSV) promoter. Embodiment 32 includes the isolated nucleic acid molecule of embodiment 31, wherein the second inducible promoter is a CAG promoter. Embodiment 33 includes the isolated nucleic acid molecule of any one of embodiments 26 to 32, wherein the termination sequence is a bovine growth hormone polyadenylation (bgh-PolyA) signal. Embodiment 34 includes the nucleic acid molecule of any one of embodiments 26 to 33, wherein the antibiotic resistance gene is a blasticidin resistance gene. Embodiment 35 includes the nucleic acid molecule of any one of embodiments 26 to 34, wherein the nucleic acid sequence comprises a core insulator sequence inserted between the termination sequence and the antibiotic resistance gene. Embodiment 36 includes the nucleic acid molecule of any one of embodiments 26 to 35, wherein the Rep52 is codon optimized. Embodiment 37 includes the nucleic acid molecule of any one of embodiments 26 to 36, wherein the nucleic acid sequence is flanked on both the 5′ and 3′ ends by transposon-specific inverted terminal repeats (ITR). Embodiment 38 includes the nucleic acid molecule of embodiment 37, wherein the nucleic acid sequence comprises a core insulator sequence inserted downstream of the ITR on the 3′ end and upstream of the ITR on the 5′ end. Embodiment 39 is an isolated nucleic acid molecule operably comprising a first inducible promoter and a TetO.sub.2 sequence, a VP1 gene, a second inducible promoter and a TetO.sub.2 sequence, a VP2 gene and a VP3 gene, and an antibiotic resistance gene. Embodiment 40 includes the isolated nucleic acid molecule of embodiment 39, wherein the VP1 gene, excluding the first 439 bp of the coding region, is optimized. Embodiment 41 includes the isolated nucleic acid molecule of embodiment 39 or 40, wherein the first inducible promoter is human cytomegalovirus (CMV) immediate early enhancer-promoter, CMV immediate early enhancer fused chicken β-actin promoter (CAG), human ubiquitin C (UbC) promoter or Rous sarcoma virus (RSV) promoter. Embodiment 42 includes the isolated nucleic acid molecule of embodiment 41, wherein the first inducible promoter is an UbC promoter. Embodiment 43 includes the isolated nucleic acid molecule of embodiment 42, further comprising an intron of the UbC promoter inserted upstream of the UbC promoter. Embodiment 44 includes the isolated nucleic acid molecule of any one of embodiments 37 to 43, wherein the second inducible promoter is human CMV immediate early enhancer-promoter, CMV immediate early enhancer fused chicken β-actin promoter (CAG), human ubiquitin C (UbC) promoter or Rous sarcoma virus (RSV) promoter. Embodiment 45 includes the isolated nucleic acid molecule of embodiment 44, wherein the second inducible promoter is a CMV promoter. Embodiment 46 includes the isolated nucleic acid molecule of any one of embodiments 39 to 45, further comprising a gene of interest (GOI) downstream of the VP3 gene and upstream of the antibiotic resistance gene. Embodiment 47 includes the isolated nucleic acid molecule of embodiment 46, wherein the nucleic acid molecule comprises a core insulator sequence inserted between the VP3 gene and the GOI and between the GOI and the antibiotic resistance gene. Embodiment 48 includes the isolated nucleic acid molecule of embodiment 47, wherein the GOI is a therapeutic gene. Embodiment 49 includes the nucleic acid molecule of any one of embodiments 39 to 48, wherein the nucleic acid sequence comprises a core insulator sequence inserted between the VP1 gene and the second inducible promoter and TetO.sub.2 sequence. Embodiment 50 includes the nucleic acid molecule of any one of embodiments 39 to 49, wherein the nucleic acid sequence comprises a core insulator sequence inserted between the VP2 and VP3 gene of d) and the antibiotic resistance gene of e). Embodiment 51 includes the nucleic acid molecule of any one of embodiments 39 to 50, wherein the nucleic acid sequence is flanked on both the 5′ and 3′ ends by transposon-specific inverted terminal repeats (ITR). Embodiment 52 includes the nucleic acid molecule of embodiment 51, wherein the nucleic acid sequence comprises a core insulator sequence inserted downstream of the ITR on the 3′ end and upstream of the ITR on the 5′ end. Embodiment 53 includes the mammalian cell for producing an adeno-associated virus (AAV), comprising the isolated nucleic acid molecule of any one of embodiments 1 to 52 integrated into its genome. Embodiment 54 includes the mammalian cell of embodiment 53, further comprising a nucleic acid encoding a transcriptional repression domain in frame with a nucleic acid encoding a tetracycline repressor protein. Embodiment 55 includes the mammalian cell of embodiment 53 or 54, wherein the mammalian cell is a mammalian cell culture. Embodiment 56 includes the mammalian cell any one of embodiments 53 to 55, wherein the mammalian cell is a Chinese hamster ovary (CHO) cell. Embodiment 57 includes the mammalian cell of any one of embodiments 53 to 55, wherein the mammalian cell is a human cell. Embodiment 58 includes the mammalian of embodiment 57, wherein the human cell is a human embryonic kidney (HEK) cell. Embodiment 59 includes the mammalian cell of embodiment 58, wherein the HEK cell is an HEK293 cell. Embodiment 60 includes the mammalian cell of any one of embodiments 53 to 59, further comprising a nucleic acid molecule encoding a gene of interest. Embodiment 61 includes a method for producing an adeno-associated virus (AAV), comprising inducing production of the genomically integrated nucleic acid sequence of any of the mammalian cells of embodiments 53 to 60, culturing the mammalian cell, and harvesting the AAV. Embodiment 62 includes the method of embodiment 61, wherein an amount of the AAV produced is about 10.sup.10 to about 10.sup.11 viral genomes per mL (vg/mL). Embodiment 63 includes the method of embodiment 62, wherein an amount of the AAV produced is at least about 10.sup.10 viral genomes per mL (vg/mL).

EXAMPLES

Example 1: Preparation of HEK293 Stable Clonal Cells for TetR-KRAB Expression

(70) One of the original Tet-On inducible systems utilizes the tetracycline repressor protein (TetR) to silence the human cytomegalovirus (CMV) promoter with an insertion of two tetracycline operator sequences (TetO.sub.2) between the TATA box and transcriptional start site (TSS) (Yao et al., “Tetracycline repressor, tetR, rather than the tetR-mammalian cell transcription factor fusion derivatives, regulates inducible gene expression in mammalian cells,” Hum Gene Ther. 9 (13): 1939-50 (1998)). In the absence of doxycycline, transcription initiation is blocked by the binding of TetR on the TetO.sub.2 sites. When doxycycline is added into the medium, it competitively binds to TetR and changes its conformation. This leads to the release of TetR and derepression/activation of the CMV promoter and results in induced gene expression (See FIG. **1**). However, the high level of leaky expression of toxic proteins before induction in the original system poses a major challenge for the stability of the generated inducible cell lines and limits its application for an AAV producer cell line.

(71) Improvements of the Tet-On inducible system were implemented to minimize the leakiness level. First, an enhanced version of TetR was applied as described in Suzlc et al., “A versatile tool for conditional gene expression and knockdown,” Nat Methods 3 (2): 109-16 (2006). Briefly, a strong repressive domain of KRAB was fused in-frame to the C-terminus of original TetR, which dramatically improved its repressive activity and minimized basal gene expression before induction. An SV40 nuclear localization signal (NLS) was inserted as well to facilitate the nuclear entry of the larger TetR-KRAB fusion protein (See FIGS. **2**A and **2**B). Second, the expression cassette was optimized for high level expression of TetR-KRAB protein, including the strong human CMV promoter to drive a high level of mRNA expression, the consensus Kozak sequence to boost the initiation of the protein translation, and the rabbit beta-globin intron to increase mRNA level. The expression cassette was further protected on both ends by two chicken hypersensitive site-4 (cHS4) extended core chromatin insulators, which can prevent undesired silencing of the CMV promoter from the spreading of neighboring heterochromatin. An SV40 promoter driven Zeocin antibiotic resistance gene was included for stable selection. Third, a stable TetR-KRAB overexpression

clonal HEK293 cell line, C20, was established using this optimized construct by plasmid random integration. When compared to a commercially available TetR stable HEK293 cell line in a transient transfection test, the C20 cell line reduced the leaky production of AAV by approximately 200-fold.

(72) Sequence Listings

(73) Any nucleic acid and amino acid sequences listed herein or in the accompanying sequence listing are shown using standard letter abbreviations for nucleotide bases and amino acids, as defined in 37 C.F.R. § 1.822. In at least some cases, only one strand of each nucleic acid sequence is shown, but the complementary strand is understood as included by any reference to the displayed strand.

(74) The sequence of the pcDNA-TetR-Ins vector is provided below:

(75) TABLE-US-00001 pcDNA-TetR-Ins (7147 bp) (SEQ ID NO: 1)

gacggatcgggagatctgagctcacggggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgcta
gggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagccc
gggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaa
aagctttaggctgaaagagagatttagaatgacagaatcatagaacggcctgggttgcaaaggagcacagtgctcatccagatccaacccc
ctgctatgtgcagggtcatcaaccagcagcccaggctgcccagagccacatccagcctggccttgaatgcctgcaggcccgatcccctat
ggtcgactctcagtacaatctgctctgatgccgcatagttaagccagtatctgctccctgcttgtgtgttggaggtcgctgagtagtgcgcgag
caaaatttaagctacaacaaggcaaggcttgaccgacaattgcatgaagaatctgcttagggttaggcgttttgcgctgcttcgcgatgtacg
ggccagatatacgcgttgacattgattattgactagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcg
ttacataacttacggtaaatggcccgcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgcc
aatagggactttccattgacgtcaatgggtggactatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccc
cctattgacgtcaatgacggtaaatggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtatt
agtcatcgctattaccatggtgatgcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccacccc
attgacgtcaatgggagtttgttttggcaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggt
aggcgtgtacggtgggaggtctatataagcagagctctctggctaactagagaacccactgcttactggcttatcgaaattaatacgactcac
tatagggagacccaagctggctagcgtttaaacttaagctttctgtgagtttggggacccttgattgttctttctttttcgctattgtaaaattc
atgttatatggagggggcaaagttttcagggtgttgtttagaatgggaagatgtcccttgtatcaccatggaccctcatgataattttgtttctt
tcactttctactctgttgacaaccattgtctcctcttattttcttttcattttctgtaactttttcgttaaactttagcttgcatttgtaacgaa
tttttaaattcacttttgtttatttgtcagattgtaagtactttctctaatcacttttttttcaaggcaatcagggtatattatattgtacttca
gcacagttttagagaacaattgttataattaaatgataaggtagaatatttctgcatataaattctggctggcgtggaaatattcttattggtag
aaacaactacatcctggtcatcatcctgcctttctctttatggttacaatgatatacactgtttgagatgaggataaaatactctgagtccaaac
cgggcccctctgctaaccatgttcatgccttcttctttttcctacagctcctgggcaacgtgctggttattgtgctgtctcatcattttggcaaa
gaattgtaatacgactcactatagggcgagccaccatggctagattagataaaagtaaagtgattaacagcgcattagagctgcttaatgaggtc
ggaatcgaaggtttaacaacccgtaaactcgcccagaagctaggtgtagagcagcctacattgtattggcatgtaaaaaataagcgggctttgct
cgacgccttagccattgagatgttagataggcaccatactcacttttgccctttagaaggggaaagctggcaagatttttttacgtaataacgcta
aaagttttagatgtgctttactaagtcatcgcgatggagcaaaagtacatttaggtacacggcctacagaaaaacagtatgaaactctcgaaaat
caattagcctttttatgccaacaaggtttttcactagagaatgccttatatgcactcagcgccgtggggcattttactttaggttgcgtattgga
agatcaagagcatcaagtcgctaaagaagaaagggaaacacctactactgatagtatgccgccattattacgacaagctatcgaattatttgatc
accaaggtgcagagccagccttcttattcggccttgaattgatcatatgcggattagaaaaacaacttaaatgtgaaagtgggtccccaaaaaag
aagagaaaggtcgacggcggtggttcagtttaagcgtacagcgggatccactagtccagtgtggtggaattctgcagatatccagcacagtggcg
gccgctcgagtctagagggcccgtttaaacccgctgatcagcctcgactgtgccttctagttgccagccatctgttgtttgcccctcccccgtg
ccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgcattgtctgagtaggtgtcattctattct
ggggggtggggtggggcaggacagcaagggggaggattgggaagacaatagcaggcatgctggggatgcggtgggctctatggcttctgaggcgg
aaagaaccagctggggctctagggggtatccccacgcgccctgtagcggcgcattaagcgcggcgggtgtggtggttacgcgcagcgtgaccgc
tacacttgccagcgccctagcgcccgctcctttcgctttcttcccttcctttctcgccacgttcgccggctttccccgtcaagctctaaatcggg
gcatccctttagggttccgatttagtgctttacggcacctcgaccccaaaaaacttgattagggtgatggttcacgtagtgggccatcgccctga
tagacggtttttcgccctttgacgttggagtccacgttctttaatagtggactcttgttccaaactggaacaacactcaaccctatctcggtcta
ttcttttgatttataagggattttggggatttcggcctattggttaaaaaatgagctgatttaacaaaaatttaacgcgaattaattctgtggaa
tgtgtgtcagttagggtgtggaaagtccccaggctccccaggcaggcagaagtatgcaaagcatgcatctcaattagtcagcaaccaggtgtgga
aagtccccaggctccccagcaggcagaagtatgcaaagcatgcatctcaattagtcagcaaccatagtcccgcccctaactccgcccatcccgc
ccctaactccgcccagttccgcccattctccgccccatggctgactaattttttttatttatgcagaggccgaggccgcctctgcctctgagcta
ttccagaagtagtgaggaggcttttttggaggcctaggcttttgcaaaaagctcccgggagcttgtatatccattttcggatctgatcagcacgt
gttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaactaaaccatggccaagttgaccagtgccgttccggt
gctcaccgcgcgcgacgtcgccggagcggtcgagttctggaccgaccggctcgggttctcccgggacttcgtggaggacgacttcgccggtgt
ggtccgggacgacgtgaccctgttcatcagcgcggtccaggaccaggtggtgccggacaacaccctggcctgggtgtgggtgcgcggc

ctggacgagctgtacgccgagtggtcggaggtcgtgtccacgaacttccgggacgcctccgggccggccatgaccgagatcggcgagc
agccgtgggggcgggagttcgccctgcgcgacccggccggcaactgcgtgcacttcgtggccgaggagcaggactgacacgtgctac
gagatttcgattccaccgccgccttctatgaaaggttgggcttcggaatcgttttccgggacgccggctggatgatcctccagcgcggggat
ctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagc
atttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtataccgtcgacctctagctagagcttggcgt
aatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaagcctggggt
gcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcattaatgaatcgg
ccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggctgcggcgagcgg
tatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagagagctcacggggacagcccccccca
aagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctcccccc
gcatccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctct
ttgagcctgcagacacctgggggggatacggggaaaaagctttaggctgaaagagagatttagaatgacagaatcatagaacggcctgggt
tgcaaaggagcacagtgctcatccagatccaaccccctgctatgtgcagggtcatcaaccagcagcccaggctgcccagagccacatcca
gcctggccttgaatgcctgcaggacatgtgagcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttcca
taggctccgcccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgt
ttccccctggaagctccctcgtgcgctctcctgttccgaccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgct
ttctcaatgctcacgctgtaggtatctcagttcggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccg
ctgcgccttatccggtaactatcgtcttgagtccaacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagca
gagcgaggtatgtaggcggtgctacagagttcttgaagtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgc
tgaagccagttaccttcggaaaaagagttggtagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagca
gattacgcgcagaaaaaaaggatctcaagaagatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggat
tttggtcatgagattatcaaaaaggatcttcacctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaactt
ggtctgacagttaccaatgcttaatcagtgaggcacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtag
ataactacgatacgggagggcttaccatctggccccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataa
ccagccagccggaagggccgagcgcagaagtggtcctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagt
agttcgccagttaatagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccgg
ttcccaacgatcaaggcgagttacatgatcccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttg
gccgcagtgttatcactcatggttatggcagcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtact
caaccaagtcattctgagaatagtgtatgcggcgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaacttta
aaagtgctcatcattggaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcac
ccaactgatcttcagcatcttttactttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcg
acacggaaatgttgaatactcatactcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatg
tatttagaaaaataaacaaataggggttccgcgcacatttccccgaaaagtgccacctgacgtc

(76) The sequence of the pcDNA-TetR-KRAB-Ins vector is provided below:

(77) TABLE-US-00002 pcDNA-TetR-KRAB-Ins (7493 bp) (SEQ ID NO: 2)

gacggatcgggagatctgagctcacggggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagggggcag
cagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggcacgg
ggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaagctttag
gctgaaagagagatttagaatgacagaatcatagaacggcctgggttgcaaaggagcacagtgctcatccagatccaaccccctgctatgt
gcagggtcatcaaccagcagcccaggctgcccagagccacatccagcctggccttgaatgcctgcaggcccgatcccctatggtcgactc
tcagtacaatctgctctgatgccgcatagttaagccagtatctgctccctgcttgtgtgttggaggtcgctgagtagtgcgcgagcaaaatttaa
gctacaacaaggcaaggcttgaccgacaattgcatgaagaatctgcttagggttaggcgttttgcgctgcttcgcgatgtacgggccagatat
acgcgttgacattgattattgactagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgttacataact
tacggtaaatggcccgcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggact
ttccattgacgtcaatgggtggactatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgt
caatgacggtaaatggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgcta
ttaccatggtgatgcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaa
tgggagtttgttttggcaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtac
ggtgggaggtctatataagcagagctctctggctaactagagaacccactgcttactggcttatcgaaattaatacgactcactatagggaga
cccaagctggctagcgtttaaacttaagctttctgtgagtttggggacccttgattgttctttctttttcgctattgtaaaattcatgttatatg
gagggggcaaagttttcagggtgttgtttagaatgggaagatgtcccttgtatcaccatggaccctcatgataattttgtttctttcactttcta
ctctgttgacaaccattgtctcctcttattttcttttcattttctgtaactttttcgttaaactttagcttgcatttgtaacgaatttttaaatt
cacttttgtttatttgtcagattgtaagtactttctctaatcacttttttttcaaggcaatcagggtatattatattgtacttcagcacagtttt
agagaacaattgttataattaaatgataaggtagaatatttctgcatataaattctggctggcgtggaaatattcttattggtagaaacaactac

atcctggtcatcatcctgcctttctctttatggttacaatgatatacactgtttgagatgaggataaaatactctgagtccaaaccgggcccctc
tgctaaccatgttcatgccttcttctttttcctacagctcctgggcaacgtgctggttattgtgctgtctcatcattttggcaaagaattgtaat
acgactcactatagggcgagccaccatggctagattagataaaagtaaagtgattaacagcgcattagagctgcttaatgaggtcggaatcgaa
ggtttaacaacccgtaaactcgcccagaagctaggtgtagagcagcctacattgtattggcatgtaaaaaataagcgggctttgctcgacgcctt
agccattgagatgttagataggcaccatactcacttttgccctttagaaggggaaagctggcaagattttttacgtaataacgctaaaagtttta
gatgtgctttactaagtcatcgcgatggagcaaaagtacatttaggtacacggcctacagaaaaacagtatgaaactctcgaaaatcaattagc
ctttttatgccaacaaggtttttcactagagaatgccttatatgcactcagcgccgtggggcattttactttaggttgcgtattggaagatcaag
agcatcaagtcgctaaagaagaaagggaaacacctactactgatagtatgccgccattattacgacaagctatcgaattatttgatcaccaaggt
gcagagccagccttcttattcggccttgaattgatcatatgcggattagaaaaacaacttaaatgtgaaagtgggtccccaaaaaagaagagaa
aggtcgacggcggtggtgctttgtctcctcagcactctgctgtcactcaaggaagtatcatcaagaacaaggagggcatggatgctaagtcact
aactgcctggtcccggacactggtgaccttcaaggatgtatttgtggacttcaccagggaggagtggaagctgctggacactgctcagcagatc
gtgtacagaaatgtgatgctggagaactataagaacctggtttccttgggttatcagcttactaagccagatgtgatcctccggttggagaagg
gagaagagccctggctggtggagagagaaattcaccaagagacccatcctgattcagagactgcatttgaaatcaaatcatcagtttaagcgtac
agcggggatccactagtccagtgtggtggaattctgcagatatccagcacagtggcggccgctcgagtctagagggcccgtttaaacccgctgat
cagcctcgactgtgccttctagttgccagccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctt
tcctaataaaatgaggaaattgcatcgcattgtctgagtaggtgtcattctattctggggggtggggtggggcaggacagcaagggggaggattgg
gaagacaatagcaggcatgctggggatgcggtgggctctatggcttctgaggcggaaagaaccagctggggctctagggggtatccccacgcgcc
ctgtagcggcgcattaagcgcggcgggtgtggtggttacgcgcagcgtgaccgctacacttgccagcgccctagcgcccgctcctttcgcttt
cttcccttcctttctcgccacgttcgccggctttccccgtcaagctctaaatcgggggcatccctttagggttccgatttagtgctttacggcacc
tcgaccccaaaaaacttgattagggtgatggttcacgtagtgggccatcgccctgatagacggtttttcgccctttgacgttggagtccacgttc
tttaatagtggactcttgttccaaactggaacaacactcaaccctatctcggtctattcttttgatttataagggattttggggatttcggccta
ttggttaaaaaatgagctgatttaacaaaaatttaacgcgaattaattctgtggaatgtgtgtcagttagggtgtggaaagtccccaggctcccc
aggcaggcagaagtatgcaaagcatgcatctcaattagtcagcaaccaggtgtggaaagtccccaggctccccagcaggcagaagtatgcaaag
catgcatctcaattagtcagcaaccatagtcccgcccctaactccgcccatcccgcccctaactccgcccagttccgcccattctccgcccca
tggctgactaattttttttatttatgcagaggccgaggccgcctctgcctctgagctattccagaagtagtgaggaggcttttttggaggcctag
gcttttgcaaaaagctcccgggagcttgtatatccattttcggatctgatcagcacgtgttgacaattaatcatcggcatagtatatcggcatag
tataatacgacaaggtgaggaactaaaccatggccaagttgaccagtgccgttccggtgctcaccgcgcgcgacgtcgccggagcggtcg
agttctggaccgaccggctcgggttctcccgggacttcgtggaggacgacttcgccggtgtggtccgggacgacgtgaccctgttcatcag
cgcggtccaggaccaggtggtgccggacaacaccctggcctgggtgtgggtgcgcggcctggacgagctgtacgccgagtggtcgga
ggtcgtgtccacgaacttccgggacgcctccgggccggccatgaccgagatcggcgagcagccgtgggggcgggagttcgccctgcg
cgacccggccggcaactgcgtgcacttcgtggccgaggagcaggactgacacgtgctacgagatttcgattccaccgccgccttctatga
aaggttgggcttcggaatcgttttccgggacgccggctggatgatcctccagcgcggggatctcatgctggagttcttcgcccaccccaact
tgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttg
tccaaactcatcaatgtatcttatcatgtctgtataccgtcgacctctagctagagcttggcgtaatcatggtcatagctgtttcctgtgtgaa
attgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaagcctggggtgcctaatgagtgagctaactcacattaatt
gcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtat
tgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacg
gttatccacagaatcaggggataacgcaggaaagagagctcacggggacagccccccccaaagcccccagggatgtaattacgtccct
cccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggg
gacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggata
cggggaaaaagctttaggctgaaagagagatttagaatgacagaatcatagaacggcctgggttgcaaaggagcacagtgctcatccaga
tccaacccccctgctatgtgcagggtcatcaaccagcagcccaggctgcccagagccacatccagcctggccttgaatgcctgcaggacat
gtgagcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatc
acaaaaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgct
ctcctgttccgaccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcaatgctcacgctgtaggtatctc
agttcggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtctt
gagtccaacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctaca
gagttcttgaagtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaaga
gttggtagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatct
caagaagatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggat
cttcacctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatca
gtgaggcacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttacca
tctggccccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagc
gcagaagtggtcctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgc

aacgttgttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttac atgatcccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggtta tggcagcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgt atgcggcgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgtt cttcggggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttact ttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcatac tcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaacaaataggg gttccgcgcacatttccccgaaaagtgccacctgacgtc

Example 2: Design and Validation of Derepressible Helper Genes

(78) pPBBG-iHelper2.0-HA Vector

(79) To drive the expression of Adenovirus Helper Genes E2A and E4Orf6, two tetracycline operator sequences (TetO.sub.2) were inserted between the TATA box and transcriptional start site (TSS) in the native promoters of the E2A and E4 genes. Expression of the VA RNA is driven by its native promoter or H1 promoter with or without TetO.sub.2 insertion. The VA gene promoter is embedded in the gene body of the non-coding RNA, and the TetO.sub.2 sequences were added upstream of the VA gene. To facilitate the detection of E4orf6 protein expression, an HA tag was fused in frame to the N-terminus of E4orf6 coding sequence. The expression cassettes of inducible E2A, E4 and VA as one gene block was flanked by core insulator sequences, long-range regulatory sequences to prevent the activation or silencing of the gene promoters from neighboring distal enhancers or spreading heterochromatin, respectively. The sequence was fully synthesized and subcloned into a piggyBac transposon vector between the transposon-specific ITRs on 5′ and 3′ ends. To allow the antibiotic selection of the recombinant transposon in cells, the puromycin antibiotic resistance gene driven by a CMV promoter was also included and flanked by core insulators inside the ITRs, which resulted in the pPBBG-iHelper2.0-HA vector (See FIG. **3**A).

(80) To enhance the VA RNA expression of the pPBBG-iHelper2.0-HA vector design, a series of modifications were applied to the VA cassettes in the pPBBG-iHelper2.0-HA vector design.

(81) In the iHelper2.1 vector, both the TetO.sub.2 and VAII regions originally in the pPBBG-iHelper2.0-HA vector design were removed (see FIG. **3**B). In the iHelper2.2 vector, the VAI cassette in iHelper2.1 was relocated between E2A and E4 cassettes to achieve inducibility from neighboring TetO.sub.2 sequences (See FIG. **3**C). In the iHelper2.3 vector, the constitutive H1 promoter was added upstream of VAI cassette in the iHelper2.1 to increase the VA expression (see FIG. **3**D). In iHelper2.4 vector, the VAI and H1 promoter in iHelper2.3 was relocated between E2A and E4 cassettes (see FIG. **3**E). To further augment the VA expression, the VA cassette was increased from a single copy in the iHelper 2.1, 2.2, 2.3 and 2.4 vectors to four tandem repeats in the iHelper 2.5, 2.6, 2.7 and 2.8, accordingly. The addition of further promoters (H1) in various locations was also investigated. (See FIGS. **3**F-**3**I).

(82) To further enhance the helper function, functional protein orthologs to VA RNA were screened to enhance AAV production. These protein-encoding genes are derived from different viruses with reported function as inhibitors of protein kinase R (PKR), a key function of VA RNA (Langland et al., “Inhibition of PKR by RNA and DNA viruses,” Virus Res. 119 (1): 100-10 (2006) and Feng et al., “The VP35 protein of Ebola virus inhibits the antiviral effect mediated by double-stranded RNA-dependent protein kinase PKR,” J Virol. 81 (1): 182-92 (2007). Two candidates boosted AAV production, VP35 from Ebola virus and SM from Epstein-Barr virus. To incorporate the protein orthologs into the iHelper2.2 vector, a VP35 or SM gene was fused in frame to an N-terminal Flag tag, and placed under an inducible human CMV promoter with an insertion of TetO.sub.2, followed by a BGH Poly(A) signal. The expression cassette was then inserted into the region between E4 and puromycin resistance gene in the iHelper2.2, flanked by core insulators. The iHelper3.1 vector with a VP35 gene is shown in FIG. **4**A and the iHelper3.2 vector with a SM gene is shown in FIG. **4**B.

(83) Sequence Listings

(84) The sequence of the pPBBG-iHelper2.0-HA vector is provided below:

(85) TABLE-US-00003 pPBBG-iHelper2.0-HA (16,549 bp) (SEQ ID NO: 3)
ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA

AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttacccccacccttgccgtctg
cgccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgct
cccgtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctctg
caaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtact
tgtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttg
ccgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctgggga
tggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgcttttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc
cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtacccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagatacccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg

agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacgggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaataaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccgggcgttttagggcggagtaacttgcatgtattgggaattgtag
ttttttaaaatgggaagtgacgtatcgtgggaaaacggaagtgaagatttgaggaagttgtgggttttttggctttcgtttctgggcgtaggtt
cgcgtgcggttttctgggtgtttttgtggactttaaccgttacgtcatttttagtcctatatatactcgctctgtaCTCTCCCTATCAGT
GATAGAGATCTCCCTATCAGTGATAGAGATCGcttggccctttttacactgtgactgattgagctggtgccgtgt
cgagtggtgtttttaataggtttttttactggtaaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggt
gctattttgcctaggcaggagggtttttcaggtgtttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgtc
tctacgcctgcgggtatgtattcccccgggctatttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttaca
ttatgactccggacatgaccgaggaactgtcggtggtgctttttaatcacggtgaccagttttttttacggtcacgccggcatggccgtagtccgt
cttatgcttataagggttgtttttcctgttgtaagacaggcttctaatgtttaaatgtttttttttttgttattttattttgtgtttaatgcagg
aacccgcagacatgtttgagagaaaaatggtgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacgatgtg
cttgctttttgcgcgaggctttgcctgatttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggctacgct
ggttagcatagctccgagtatgcgtgtcataatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgcagacc
tgcacgattatgttcagctggccctgcgaagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctgtgag
gaacctgaatttttgcaatcatgattcgctgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatattcgg
gatttgcttagagacatattgataaggtggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcaccctga
agggtttagcctttacgtccacttggacgtgagggcagtttgccttttggaagccattgtgcaacatcttacaaatgccattatctgttctttgg
ctgtagagtttgaccacgccaccggaggggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaaaaaaa
aacatggttcttccagctcttcccgctcctcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggtggtgga
tgttatcagggcagcggcgcatgaaggagtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactacaactact
acacagagcgagctaagcgacgagaccggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATACGATGT
TCCAGATTACGCTGGAGGAGGCGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatct
cggttgtctcggcgcactccgtacagtagggatcgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgct
gcccgaatgtaacactttgacaatgcacaacgtgagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgtt
ccctgggatatggttctgacgcgggaggagcttgtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacg
agcatgatgatccatggttacgagtcctgggctctccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttgg

ccagctggtttaggatggtggtggatggcgccatgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagag
gtaatgtttatgtccagcgtgtttatgaggggtcgccacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccat
gagctttggatacagcgccttgcactgtgggattttgaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgc
gctgctgtgcccggaggacaaggcgtctcatgctgcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggac
ggagcggcggcggcagcagtttattcgcgcgctgctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcg
tggacttccccttcgccgcccgttgagcaaccgcaagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgag
ctgcccggggagtttattaatatcactgatgagcgtttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatg
atatgatgctttttaaggccagccggggagaaaggactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgag
tttgattaaggtacggtgatcaatataagctatgtggtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaata
aacacgttgaaacataacatgcaacaggttcacgattctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtgg
tgtattttccactttcccaggaccatgtaaaagacatagagtaagtgcttacctcgctagtttctgtggattcactagaatcgatgtaggatgtt
gcccctcctgacgcggtaggagaagggggagggtgccctgcatgtctgccgctgctcttgctcttgccgctgctgaggagggggggcgcatctg
ccgcagcaccggatgcatctgggaaaagcaaaaaaggggctcgtccctgtttccggaggaatttgcaagcggggtcttgcatgacgggg
aggcaaacccccgttcgccgcagtccggccggcccgagactcgaaccgggggtcctgcgactcaacccttggaaaataaccctccggct
acagggagcgagccacttaatgctttcgctttccagcctaaccgcttacgccgcgcgcggccagtggccaaaaaagctagcgcagcagcc
gccgcgcctggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggggtaaccgcatgg
atcacggcggacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgc
ccgcttacaCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggctctcctt
ttgcacggtctagagcgtcaacgactgcgcacgcctcaccggccagagcgtcccgaccatggagcactttttgccgctgcgcaacatctgg
aaccgcgtccgcgactttccgcgcgcctccaccaccgccgccggcatcacctggatgtccaggtacatctacggattacgtcgacgtttaa
accatatgatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaaggccag
caaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttccactttggccgcggctcgagTGAGCTATTCCAGAAGT
AGTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATGCCC
GGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTAA
TTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGTC
CGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGGG
GAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGCA
GACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGTT
GGGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTA
GGCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAGATATACGCGTTGACATTG
ATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATAT
ATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAAC
GACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGG
ACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTA
CATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGG
CCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTAC
ATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGTACATCAAT
GGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCATTGACGTC
AATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTTCCAAAATGTCGTAACAAC
TCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTCTATATAAG
CAGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCC
TGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAG
CCTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgaccgagtacaagcccacggtgcgcctcgc
cacccgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactaccccgccacgcgccacaccgtcgatccggac
cgccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacatcggcaaggtgtgggtcgcggacgacg
gcgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccgagatcggcccgcgcatggccgagttga
gcggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggcccaaggagcccgcgtggttcctggccaccgt
cggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagtggaggcggccgagcgcgccggggtgc
ccgccttcctggagacctccgcgccccgcaacctcccattctacgagcggctcggcttcaccgtcaccgccgacgtcgaggtgcccgaag
gaccgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATCTCATGCTGGAGTTCTTCG
CCCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCA
CAAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACT
CATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGCAgg
TGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGG

AATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCC
AGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTG
CCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGG
AGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAG
GGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGCCGC
CACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTAATA
CGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTATCG
ATAAGCTTGATATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAGTAG
AACATGAAATAACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAAAA
GATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACACT
TACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCCTA
AATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAATG
CATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGATC
ATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCAT
CGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAGT
TTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATGA
TGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCCA
CCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCGA
AGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGTG
CAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATCC
TGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGCG
CTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAATT
CCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAGT
GAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCT
GTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGTA
TTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGCG
GCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGG
ATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAA
AAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACAA
AAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGG
CGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCGG
ATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGT
AGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACCC
CCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCCG
GTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGC
GAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACAC
TAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAG
AGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGT
TTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATCT
TTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCA
TGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTTA
AATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCA
GTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCCC
CGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAAT
GATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAG
CCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCTA
TTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAACG
TTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCATT
CAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAAA
AGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGTT
ATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAG
ATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCG
GCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCA
GAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGA

TCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTC
AGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGC
CGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAATACTCAT

(86) The sequence of the pPBBG-iHelper2.1-HA vector is provided below:

(87) TABLE-US-00004 pPBBG-iHelper2.1-HA (15,878 bp) (SEQ ID NO: 4)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttacccccacccttgccgtctg
cgccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgct
cccgtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctctg
caaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtact
tgtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttg
ccgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctgggga
tggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgctttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc

cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtaccccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagataccccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctaccccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacgggggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaaaaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccgggcgttttagggcggagtaacttgcatgtattgggaattgtag
ttttttaaaatgggaagtgacgtatcgtgggaaaacggaagtgaagatttgaggaagttgtgggtttttttggctttcgtttctgggcgtaggtt
cgcgtgcggttttctgggtgtttttttgtggactttaaccgttacgtcatttttagtcctatatatactcgctctgtaCTCTCCCTATCAGT
GATAGAGATCTCCCTATCAGTGATAGAGATCGcttggccctttttacactgtgactgattgagctggtgccgtgt
cgagtggtgtttttaataggtttttttactggtaaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggt
gctattttgcctaggcaggagggtttttcaggtgtttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgtc
tctacgcctgcgggtatgtattcccccgggctatttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttaca
ttatgactccggacatgaccgaggaactgtcggtggtgctttttaatcacggtgaccagttttttttacggtcacgccggcatggccgtagtccg
tcttatgcttataagggttgtttttcctgttgtaagacaggcttctaatgtttaaatgtttttttttttgttattttattttgtgtttaatgcag
gaacccgcagacatgtttgagagaaaaatggtgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacgatgt
gcttgctttttttgcgcgaggctttgcctgatttttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggctacgc
tggttagcatagctccgagtatgcgtgtcataatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgcaga
cctgcacgattatgttcagctggccctgcgaagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctgtg
aggaacctgaatttttgcaatcatgattcgctgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatattc
gggatttgcttagagacatattgataaggtggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcaccct

gaagggtttagcctttacgtccacttggacgtgagggcagtttgccttttggaagccattgtgcaacatcttacaaatgccattatctgttcttt
ggctgtagagtttgaccacgccaccggaggggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaaaa
aaaacatggttcttccagctcttcccgctcctcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggtggtg
gatgttatcagggcagcggcgcatgaaggagtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactacaacta
ctacacagagcgagctaagcgacgagaccggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATACGAT
GTTCCAGATTACGCTGGAGGAGGCGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatct
cggttgtctcggcgcactccgtacagtagggatcgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgct
gcccgaatgtaacactttgacaatgcacaacgtgagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgtt
ccctgggatatggttctgacgcgggaggagcttgtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacg
agcatgatgatccatggttacgagtcctgggctctccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttgg
ccagctggtttaggatggtggtggatggcgccatgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagag
gtaatgtttatgtccagcgtgtttatgaggggtcgccacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccat
gagctttggatacagcgccttgcactgtgggattttgaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgc
gctgctgtgcccggaggacaaggcgtctcatgctgcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggac
ggagcggcggcggcagcagtttattcgcgcgctgctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcg
tggacttccccttcgccgcccgttgagcaaccgcaagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgag
ctgcccggggagtttattaatatcactgatgagcgtttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatg
atatgatgctttttaaggccagccggggagaaaggactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgag
tttgattaaggtacggtgatcaatataagctatgtggtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaata
aacacgttgaaacataacatgcaacaggttcacgattctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtgg
tgtattttccactttcccaggaccatgtaaaagacatagagtaagtgcttacctcgctagtttctgtggattcactagaaccagtggccaaaaaa
gctagcgcagcagccgccgcgcctggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgg
gcggtaaccgcatggatcacggcggacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccag
accacggaagagtgcccgcttacaggctctccttttgcacggtctagagcgtcaacgactgcgcacgtccactttggccgcggctcgagT
GAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAG
CTCCGGATCGATGCCCGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAA
GCCCCCAGGGATGTAATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCC
GGGGCTCCGCTCCGGTCCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGG
GACAGCCCGGGCACGGGGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCG
CTGCTCTTTGAGCCTGCAGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCC
AGCTTCCCACAATAAGTTGGGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGA
GGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAG
ATATACGCGTTGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTC
ATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCC
GCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCC
CATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTA
AACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGA
CGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGA
CTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGG
TTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGT
CTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTT
CCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACG
GTGGGAGGTCTATATAAGCAGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTA
GTGAACCGTCAGATCGCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGA
CACCGGGACCGATCCAGCCTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgac
cgagtacaagcccacggtgcgcctcgccacccgcgacgacgtccccagggccgtacgcaccctcgccgccggttcgccgactacccc
gccacgcgccacaccgtcgatccggaccgccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacat
cggcaaggtgtgggtcgcggacgacggcgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccg
agatcggcccgcgcatggccgagttgagcggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggccca
aggagcccgcgtggttcctggccaccgtcggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagt
ggaggcggccgagcgcgccggggtgcccgccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcacc
gtcaccgccgacgtcgaggtgcccgaaggaccgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATC
TCATGCTGGAGTTCTTCGCCCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAA
ATAAAGCAATAGCATCACAAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAG

TTGTGGTTTGTCCAAACTCATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAg
gATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCA
GCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTT
TTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGA
GGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGC
TCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTG
CCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGT
CCCTCTAGAGCGGCCGCCACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGG
GTTAATTAGATCTTAATACGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCT
CGAGGTCGACGGTATCGATAAGCTTGATATCTATAACAAGAAAATATATATATAATA
AGTTATCACGTAAGTAGAACATGAAATAACAATATAATTATCGTATGAGTTAAATCT
TAAAAGTCACGTAAAAGATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTT
CAAAATCAGTGACACTTACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGG
CGACTGAGATGTCCTAAATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAG
CAATATTTCAAGAATGCATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCT
AGCTGCATCAGGATCATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCT
GGCGCTATCTGGGCATCGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAA
GCGGCATGGAAAGAGTTTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAA
ACGCACGTTTACCATGATGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGA
ACTGTTCGTTCAGGCCACCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTT
CCGGATGGTCAGCCCGAAGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGA
ACTGCCGTGCCGGTGTGCAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCA
GCGAGGACGGGTATCCTGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAG
TTACCCGGCGGGCGCGCTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATT
GTTATCCGCTCACAATTCCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCC
TGGGGTGCCTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCT
TTCCAGTCGGGAAACCTGTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGG
GAGAGGCGGTTTGCGTATTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCG
CTCGGTCGTTCGGCTGCGGCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTT
ATCCACAGAATCAGGGGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAA
AAGGCCAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCC
CCTGACGAGCATCACAAAAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGG
ACTATAAAGATACCAGGCGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCC
GACCCTGCCGCTTACCGGATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCT
TTCTCATAGCTCACGCTGTAGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTG
GGCTGTGTGCACGAACCCCCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTAT
CGTCTTGAGTCCAACCCGGTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGT
AACAGGATTAGCAGAGCGAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTG
GCCTAACTACGGCTACACTAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCC
AGTTACCTTCGGAAAAAGAGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTG
GTAGCGGTGGTTTTTTTGTTTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTC
AAGAAGATCCTTTGATCTTTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCAC
GTTAAGGGATTTTGGTCATGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAA
ATTAAAAATGAAGTTTTAAATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACA
GTTACCAATGCTTAATCAGTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATC
CATAGTTGCCTGACTCCCCGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATC
TGGCCCCAGTGCTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATC
AGCAATAAACCAGCCAGCCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTAT
CCGCCTCCATCCAGTCTATTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAG
TTAATAGTTTGCGCAACGTTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTC
GTTTGGTATGGCTTCATTCAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATC
CCCCATGTTGTGCAAAAAAGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAG
TAAGTTGGCCGCAGTGTTATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACT
GTCATGCCATCCGTAAGATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCT

GAGAATAGTGTATGCGGCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAAT
ACCGCGCCACATAGCAGAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGG
CGAAAACTCTCAAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGT
GCACCCAACTGATCTTCAGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAA
ACAGGAAGGCAAAATGCCGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAA TACTCAT

(88) The sequence of the pPBBG-iHelper2.2-HA vector is provided below:

(89) TABLE-US-00005 pPBBG-iHelper2.2-HA (15,878 bp) (SEQ ID NO: 5)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttacccccacccttgccgtctg
cgccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgct
cccgtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctctg
caaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtact
tgtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttg
ccgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctgggga
tggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgctttttgggggcgcgcgggg

aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc
cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcggggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtacccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagataccccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacgggggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaaaaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccccagtggccaaaaaagctagcgcagcagccgccgcgcctg
gaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcgg
acggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacag
gctctccttttgcacggtctagagcgtcaacgactgcgcacggggcgttttagggcggagtaacttgcatgtattgggaattgtagttttttaa
aatgggaagtgacgtatcgtgggaaaacggaagtgaagatttgaggaagttgtgggttttttggctttcgtttctgggcgtaggttcgcgtgcg
gttttctgggtgtttttgtggactttaaccgttacgtcatttttagtcctatatatactcgctctgtaCTCTCCCTATCAGTGATAG
AGATCTCCCTATCAGTGATAGAGATCGcttggccctttttacactgtgactgattgagctggtgccgtgtcgagtggt
gtttttaataggtttttttactggtaaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggtgctatttt
gcctaggcaggagggtttttcaggtgtttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgtctctacgcc
tgcgggtatgtattcccccgggctatttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttacattatgact
ccggacatgaccgaggaactgtcggtggtgctttttaatcacggtgaccagttttttttacggtcacgccggcatggccgtagtccgtcttatgct
tataagggttgtttttcctgttgtaagacaggcttctaatgtttaaatgttttttttttgttattttattttgtgtttaatgcaggaacccgca
gacatgtttgagagaaaaatggtgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacgatgtgcttgcttt

tttgcgcgaggctttgcctgatttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggctacgctggttagca
tagctccgagtatgcgtgtcataatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgcagacctgcacgat
tatgttcagctggccctgcgaagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctgtgaggaacctga
atttttgcaatcatgattcgctgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatattcgggatttgct
tagagacatattgataaggtggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcaccctgaagggttta
gcctttacgtccacttggacgtgagggcagtttgccttttggaagccattgtgcaacatcttacaaatgccattatctgttctttggctgtagag
tttgaccacgccaccggaggggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaaaaaaaaacatggt
tcttccagctcttcccgctcctcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggtggtggatgttatca
gggcagcggcgcatgaaggagtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactacaactactacacagag
cgagctaagcgacgagaccggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATACGATGTTCCAGA
TTACGCTGGAGGAGGCGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatctcggttgtct
cggcgcactccgtacagtagggatcgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgctgcccgaat
gtaacactttgacaatgcacaacgtgagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgttccctggga
tatggttctgacgcgggaggagcttgtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacgagcatgatg
atccatggttacgagtcctgggctctccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttggccagctggt
ttaggatggtggtggatggcgccatgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagaggtaatgtttat
gtccagcgtgtttatgaggggtcgccacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccatgagctttgga
tacagcgccttgcactgtgggattttgaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgcgctgctgtgc
ccggaggacaaggcgtctcatgctgcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggacggagcggc
ggcggcagcagtttattcgcgcgctgctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcgtggacttcc
ccttcgccgcccgttgagcaaccgcaagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgagctgcccgg
ggagtttattaatatcactgatgagcgtttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatgatatgatgc
tttttaaggccagccggggagaaaggactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgagtttgattaa
ggtacggtgatcaatataagctatgtggtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaataaacacgttg
aaacataacatgcaacaggttcacgattctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtggtgtattttc
cactttcccaggaccatgtaaaagacatagagtaagtgcttacctcgctagtttctgtggattcactagaatccactttggccgcggctcgagT
GAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAG
CTCCGGATCGATGCCCGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAA
GCCCCCAGGGATGTAATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCC
GGGGCTCCGCTCCGGTCCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGG
GACAGCCCGGGCACGGGGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCG
CTGCTCTTTGAGCCTGCAGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCC
AGCTTCCCACAATAAGTTGGGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGA
GGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAG
ATATACGCGTTGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTC
ATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCC
GCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCC
CATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTA
AACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGA
CGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGA
CTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGG
TTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGT
CTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTT
CCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACG
GTGGGAGGTCTATATAAGCAGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTA
GTGAACCGTCAGATCGCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGA
CACCGGGACCGATCCAGCCTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgac
cgagtacaagcccacggtgcgcctcgccacccgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactacccc
gccacgcgccacaccgtcgatccggaccgccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacat
cggcaaggtgtgggtcgcggacgacggcgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccg
agatcggcccgcgcatggccgagttgagcggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggccca
aggagcccgcgtggttcctggccaccgtcggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagt
ggaggcggccgagcgcgccggggtgcccgccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcacc
gtcaccgccgacgtcgaggtgcccgaaggaccgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATC

TCATGCTGGAGTTCTTCGCCCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAA
ATAAAGCAATAGCATCACAAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAG
TTGTGGTTTGTCCAAACTCATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAg
gATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCA
GCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTT
TTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGA
GGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGC
TCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTG
CCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGT
CCCTCTAGAGCGGCCGCCACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGG
GTTAATTAGATCTTAATACGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCT
CGAGGTCGACGGTATCGATAAGCTTGATATCTATAACAAGAAAATATATATATAATA
AGTTATCACGTAAGTAGAACATGAAATAACAATATAATTATCGTATGAGTTAAATCT
TAAAAGTCACGTAAAAGATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTT
CAAAATCAGTGACACTTACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGG
CGACTGAGATGTCCTAAATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAG
CAATATTTCAAGAATGCATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCT
AGCTGCATCAGGATCATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCT
GGCGCTATCTGGGCATCGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAA
GCGGCATGGAAAGAGTTTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAA
ACGCACGTTTACCATGATGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGA
ACTGTTCGTTCAGGCCACCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTT
CCGGATGGTCAGCCCGAAGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGA
ACTGCCGTGCCGGTGTGCAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCA
GCGAGGACGGGTATCCTGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAG
TTACCCGGCGGGCGCGCTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATT
GTTATCCGCTCACAATTCCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCC
TGGGGTGCCTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCT
TTCCAGTCGGGAAACCTGTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGG
GAGAGGCGGTTTGCGTATTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCG
CTCGGTCGTTCGGCTGCGGCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTT
ATCCACAGAATCAGGGGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAA
AAGGCCAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCC
CCTGACGAGCATCACAAAAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGG
ACTATAAAGATACCAGGCGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCC
GACCCTGCCGCTTACCGGATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCT
TTCTCATAGCTCACGCTGTAGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTG
GGCTGTGTGCACGAACCCCCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTAT
CGTCTTGAGTCCAACCCGGTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGT
AACAGGATTAGCAGAGCGAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTG
GCCTAACTACGGCTACACTAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCC
AGTTACCTTCGGAAAAAGAGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTG
GTAGCGGTGGTTTTTTTGTTTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTC
AAGAAGATCCTTTGATCTTTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCAC
GTTAAGGGATTTTGGTCATGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAA
ATTAAAAATGAAGTTTTAAATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACA
GTTACCAATGCTTAATCAGTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATC
CATAGTTGCCTGACTCCCCGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATC
TGGCCCCAGTGCTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATC
AGCAATAAACCAGCCAGCCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTAT
CCGCCTCCATCCAGTCTATTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAG
TTAATAGTTTGCGCAACGTTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTC
GTTTGGTATGGCTTCATTCAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATC
CCCCATGTTGTGCAAAAAAGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAG

TAAGTTGGCCGCAGTGTTATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACT
GTCATGCCATCCGTAAGATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCT
GAGAATAGTGTATGCGGCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAAT
ACCGCGCCACATAGCAGAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGG
CGAAAACTCTCAAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGT
GCACCCAACTGATCTTCAGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAA
ACAGGAAGGCAAAATGCCGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAA TACTCAT

(90) The sequence of the pPBBG-iHelper2.3-HA vector is provided below:

(91) TABLE-US-00006 pPBBG-iHelper2.3-HA (15,992 bp) (SEQ ID NO: 6)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttacccccacccttgccgtctg
cgccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgct
cccgtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctctg
caaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtact
tgtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttg
ccgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctgggga

tggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgctttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc
cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtacccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagatacccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacgggggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaataaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccgggcgttttagggcggagtaacttgcatgtattgggaattgtag
ttttttaaaatgggaagtgacgtatcgtgggaaaacggaagtgaagatttgaggaagttgtgggtttttggctttcgtttctgggcgtaggtt
cgcgtgcggttttctgggtgtttttgtggactttaaccgttacgtcatttttagtcctatatatactcgctctgtaCTCTCCCTATCAGT
GATAGAGATCTCCCTATCAGTGATAGAGATCGcttggccctttttacactgtgactgattgagctggtgccgtgt
cgagtggtgtttttaataggtttttttactggtaaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggt
gctattttgcctaggcaggagggtttttcaggtgtttatgtgtttttctctcctattaatttgttatacctcctatgggggctgtaatgttgt
ctctacgcctgcgggtatgtattcccccgggctatttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttac
attatgactccggacatgaccgaggaactgtcggtggtgctttttaatcacggtgaccagtttttttacggtcacgccggcatggccgtagtcc
gtcttatgcttataagggttgtttttcctgttgtaagacaggcttctaatgtttaaatgtttttttttttgttattttattttgtgtttaatgca
ggaacccgcagacatgtttgagagaaaaatggtgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacgatg
tgcttgctttttgcgcgaggctttgcctgatttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggctac

gctggttagcatagctccgagtatgcgtgtcataatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgcag
acctgcacgattatgttcagctggccctgcgaagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctgt
gaggaacctgaatttttgcaatcatgattcgctgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatatt
cgggatttgcttagagacatattgataaggtggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcaccc
tgaagggtttagcctttacgtccacttggacgtgagggcagtttgccttttggaagccattgtgcaacatcttacaaatgccattatctgttctt
tggctgtagagtttgaccacgccaccggaggggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaaa
aaaaaacatggttcttccagctcttcccgctcctcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggtgg
tggatgttatcagggcagcggcgcatgaaggagtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactacaa
ctactacacagagcgagctaagcgacgagaccggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATAC
GATGTTCCAGATTACGCTGGAGGAGGCGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatct
cggttgtctcggcgcactccgtacagtagggatcgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgct
gcccgaatgtaacactttgacaatgcacaacgtgagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgtt
ccctgggatatggttctgacgcgggaggagcttgtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacg
agcatgatgatccatggttacgagtcctgggctctccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttgg
ccagctggtttaggatggtggtggatggcgccatgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagag
gtaatgtttatgtccagcgtgtttatgaggggtcgccacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccat
gagctttggatacagcgccttgcactgtgggattttgaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgc
gctgctgtgcccggaggacaaggcgtctcatgctgcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggac
ggagcggcggcggcagcagtttattcgcgcgctgctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcg
tggacttccccttcgccgcccgttgagcaaccgcaagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgag
ctgcccggggagtttattaatatcactgatgagcgtttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatg
atatgatgctttttaaggccagccggggagaaaggactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgag
tttgattaaggtacggtgatcaatataagctatgtggtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaata
aacacgttgaaacataacatgcaacaggttcacgattctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtgg
tgtattttccactttcccaggaccatgtaaaagacatagagtaagtgcttacctcgctagtttctgtggattcactagaaccagtggccaaaaaa
gctagcgcagcagccgccgcgcctggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgg
gcggtaaccgcatggatcacggcggacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccag
accacggaagagtgcccgcttacagGTGGTCTCATACAGAACTTATAAGATTCCCAAATCCAAAGA
CATTTCACGTTTATGGTGATTTCCCAGAACACATAGCGACATGCAAATATTGCAGGG
CGCCACTCCCCTGTCCgctctccttttgcacggtctagagcgtcaacgactgcgcacgtccactttggccgcggctcgagTG
AGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGC
TCCGGATCGATGCCCGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAG
CCCCCAGGGATGTAATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCG
GGGCTCCGCTCCGGTCCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGG
ACAGCCCGGGCACGGGGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCT
GCTCTTTGAGCCTGCAGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAG
CTTCCCACAATAAGTTGGGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGAGG
CTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAGAT
ATACGCGTTGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCAT
TAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGC
CTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCA
TAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAA
CTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACG
TCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACT
TTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTT
TTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCT
CCACCCCATTGACGTCAATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTTCC
AAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGT
GGGAGGTCTATATAAGCAGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTAGT
GAACCGTCAGATCGCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACA
CCGGGACCGATCCAGCCTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgaccga
gtacaagcccacggtgcgcctcgccacccgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactaccccgcc
acgcgccacaccgtcgatccggaccgccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacatcgg
caaggtgtgggtcgcggacgacggcgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccgaga

tcggcccgcgcatggccgagttgagcggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggcccaagg
agcccgcgtggttcctggccaccgtcggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagtgga
ggcggccgagcgcgccggggtgcccgccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcaccgtca
ccgccgacgtcgaggtgcccgaaggaccgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATCTC
ATGCTGGAGTTCTTCGCCCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAAAT
AAAGCAATAGCATCACAAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAGTT
GTGGTTTGTCCAAACTCATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAggA
TCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAG
CCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTTT
TTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGA
GGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGC
TCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTG
CCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGT
CCCTCTAGAGCGGCCGCCACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGG
GTTAATTAGATCTTAATACGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCT
CGAGGTCGACGGTATCGATAAGCTTGATATCTATAACAAGAAAATATATATATAATA
AGTTATCACGTAAGTAGAACATGAAATAACAATATAATTATCGTATGAGTTAAATCT
TAAAAGTCACGTAAAAGATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTT
CAAAATCAGTGACACTTACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGG
CGACTGAGATGTCCTAAATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAG
CAATATTTCAAGAATGCATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCT
AGCTGCATCAGGATCATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCT
GGCGCTATCTGGGCATCGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAA
GCGGCATGGAAAGAGTTTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAA
ACGCACGTTTACCATGATGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGA
ACTGTTCGTTCAGGCCACCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTT
CCGGATGGTCAGCCCGAAGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGA
ACTGCCGTGCCGGTGTGCAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCA
GCGAGGACGGGTATCCTGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAG
TTACCCGGCGGGCGCGCTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATT
GTTATCCGCTCACAATTCCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCC
TGGGGTGCCTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCT
TTCCAGTCGGGAAACCTGTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGG
GAGAGGCGGTTTGCGTATTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCG
CTCGGTCGTTCGGCTGCGGCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTT
ATCCACAGAATCAGGGGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAA
AAGGCCAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCC
CCTGACGAGCATCACAAAAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGG
ACTATAAAGATACCAGGCGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCC
GACCCTGCCGCTTACCGGATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCT
TTCTCATAGCTCACGCTGTAGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTG
GGCTGTGTGCACGAACCCCCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTAT
CGTCTTGAGTCCAACCCGGTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGT
AACAGGATTAGCAGAGCGAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTG
GCCTAACTACGGCTACACTAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCC
AGTTACCTTCGGAAAAAGAGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTG
GTAGCGGTGGTTTTTTTGTTTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTC
AAGAAGATCCTTTGATCTTTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCAC
GTTAAGGGATTTTGGTCATGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAA
ATTAAAAATGAAGTTTTAAATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACA
GTTACCAATGCTTAATCAGTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATC
CATAGTTGCCTGACTCCCCGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATC
TGGCCCCAGTGCTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATC
AGCAATAAACCAGCCAGCCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTAT

CCGCCTCCATCCAGTCTATTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAG
TTAATAGTTTGCGCAACGTTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTC
GTTTGGTATGGCTTCATTCAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATC
CCCCATGTTGTGCAAAAAAGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAG
TAAGTTGGCCGCAGTGTTATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACT
GTCATGCCATCCGTAAGATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCT
GAGAATAGTGTATGCGGCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAAT
ACCGCGCCACATAGCAGAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGG
CGAAAACTCTCAAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGT
GCACCCAACTGATCTTCAGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAA
ACAGGAAGGCAAAATGCCGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAA TACTCAT
(92) The sequence of the pPBBG-iHelper2.4-HA vector is provided below:
(93) TABLE-US-00007 pPBBG-iHelper2.4-HA (15,992 bp) (SEQ ID NO: 7)
ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttacccccacccttgccgtctg
cgccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgc
tcccgtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctct
gcaaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc

ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtact
tgtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttg
ccgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctgggga
tggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgcttttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc
cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtacccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagataccccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacggggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaaaaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccccagtggccaaaaaagctagcgcagcagccgccgcgcctg
gaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcgg
acggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacag
GTGGTCTCATACAGAACTTATAAGATTCCCAAATCCAAAGACATTTCACGTTTATGG
TGATTTCCCAGAACACATAGCGACATGCAAATATTGCAGGGCGCCACTCCCCTGTCC
gctctccttttgcacggtctagagcgtcaacgactgcgcacggggcgttttagggcggagtaacttgcatgtattgggaattgtagttttttaa
aatgggaagtgacgtatcgtgggaaaacggaagtgaagatttgaggaagttgtgggtttttggctttcgtttctgggcgtaggttcgcgtgcg

gttttctgggtgtttttgtggactttaaccgttacgtcatttttagtcctatatatactcgctctgtaCTCTCCCTATCAGTGATAG
AGATCTCCCTATCAGTGATAGAGATCGcttggccctttttacactgtgactgattgagctggtgccgtgtcgagtggt
gtttttaataggtttttttactggtaaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggtgctatttt
gcctaggcaggagggtttttcaggtgtttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgtctctacgcc
tgcgggtatgtattcccccgggctatttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttacattatgact
ccggacatgaccgaggaactgtcggtggtgctttttaatcacggtgaccagttttttttacggtcacgccggcatggccgtagtccgtcttatgct
tataagggttgtttttcctgttgtaagacaggcttctaatgtttaaatgttttttttttgttattttattttgtgtttaatgcaggaacccgca
gacatgtttgagagaaaaatggtgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacgatgtgcttgcttt
tttgcgcgaggctttgcctgatttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggctacgctggttagca
tagctccgagtatgcgtgtcataatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgcagacctgcacgat
tatgttcagctggccctgcgaagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctgtgaggaacctga
atttttgcaatcatgattcgctgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatattcgggatttgct
tagagacatattgataaggtggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcaccctgaagggttta
gcctttacgtccacttggacgtgagggcagtttgccttttggaagccattgtgcaacatcttacaaatgccattatctgttctttggctgtagag
tttgaccacgccaccggaggggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaaaaaaaaacatggt
tcttccagctcttcccgctcctcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggtggtggatgttatca
gggcagcggcgcatgaaggagtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactacaactactacacagag
cgagctaagcgacgagaccggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATACGATGTTCCAGA
TTACGCTGGAGGAGGCGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatctcggttgtct
cggcgcactccgtacagtagggatcgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgctgcccgaat
gtaacactttgacaatgcacaacgtgagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgttccctggga
tatggttctgacgcgggaggagcttgtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacgagcatgatg
atccatggttacgagtcctgggctctccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttggccagctggt
ttaggatggtggtggatggcgccatgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagaggtaatgtttat
gtccagcgtgtttatgaggggtcgccacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccatgagctttgga
tacagcgccttgcactgtgggattttgaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgcgctgctgtgc
ccggaggacaaggcgtctcatgctgcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggacggagcggc
ggcggcagcagtttattcgcgcgctgctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcgtggacttcc
ccttcgccgcccgttgagcaaccgcaagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgagctgcccgg
ggagtttattaatatcactgatgagcgtttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatgatatgatgc
tttttaaggccagccggggagaaaggactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgagtttgattaa
ggtacggtgatcaatataagctatgtggtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaataaacacgttg
aaacataacatgcaacaggttcacgattctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtggtgtattttc
cactttcccaggaccatgtaaaagacatagagtaagtgcttacctcgctagtttctgtggattcactagaatccactttggccgcggctcgagT
GAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAG
CTCCGGATCGATGCCCGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAA
GCCCCCAGGGATGTAATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCC
GGGGCTCCGCTCCGGTCCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGG
GACAGCCCGGGCACGGGGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCG
CTGCTCTTTGAGCCTGCAGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCC
AGCTTCCCACAATAAGTTGGGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGA
GGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAG
ATATACGCGTTGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTC
ATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCC
GCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCC
CATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTA
AACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGA
CGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGA
CTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGG
TTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGT
CTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTT
CCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACG
GTGGGAGGTCTATATAAGCAGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTA
GTGAACCGTCAGATCGCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGA

CACCGGGACCGATCCAGCCTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgac
cgagtacaagcccacggtgcgcctcgccacccgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactacccc
gccacgcgccacaccgtcgatccggaccgccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacat
cggcaaggtgtgggtcgcggacgacggcgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccg
agatcggcccgcgcatggccgagttgagcggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggccca
aggagcccgcgtggttcctggccaccgtcggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagt
ggaggcggccgagcgcgccggggtgcccgccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcacc
gtcaccgccgacgtcgaggtgcccgaaggaccgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATC
TCATGCTGGAGTTCTTCGCCCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAA
ATAAAGCAATAGCATCACAAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAG
TTGTGGTTTGTCCAAACTCATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAg
gATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCA
GCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTT
TTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGA
GGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGC
TCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTG
CCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGT
CCCTCTAGAGCGGCCGCCACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGG
GTTAATTAGATCTTAATACGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCT
CGAGGTCGACGGTATCGATAAGCTTGATATCTATAACAAGAAAATATATATATAATA
AGTTATCACGTAAGTAGAACATGAAATAACAATATAATTATCGTATGAGTTAAATCT
TAAAAGTCACGTAAAAGATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTT
CAAAATCAGTGACACTTACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGG
CGACTGAGATGTCCTAAATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAG
CAATATTTCAAGAATGCATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCT
AGCTGCATCAGGATCATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCT
GGCGCTATCTGGGCATCGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAA
GCGGCATGGAAAGAGTTTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAA
ACGCACGTTTACCATGATGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGA
ACTGTTCGTTCAGGCCACCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTT
CCGGATGGTCAGCCCGAAGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGA
ACTGCCGTGCCGGTGTGCAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCA
GCGAGGACGGGTATCCTGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAG
TTACCCGGCGGGCGCGCTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATT
GTTATCCGCTCACAATTCCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCC
TGGGGTGCCTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCT
TTCCAGTCGGGAAACCTGTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGG
GAGAGGCGGTTTGCGTATTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCG
CTCGGTCGTTCGGCTGCGGCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTT
ATCCACAGAATCAGGGGATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAA
AAGGCCAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCC
CCTGACGAGCATCACAAAAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGG
ACTATAAAGATACCAGGCGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCC
GACCCTGCCGCTTACCGGATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCT
TTCTCATAGCTCACGCTGTAGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTG
GGCTGTGTGCACGAACCCCCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTAT
CGTCTTGAGTCCAACCCGGTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGT
AACAGGATTAGCAGAGCGAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTG
GCCTAACTACGGCTACACTAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCC
AGTTACCTTCGGAAAAAGAGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTG
GTAGCGGTGGTTTTTTTGTTTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTC
AAGAAGATCCTTTGATCTTTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCAC
GTTAAGGGATTTTGGTCATGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAA
ATTAAAAATGAAGTTTTAAATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACA

GTTACCAATGCTTAATCAGTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATC
CATAGTTGCCTGACTCCCCGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATC
TGGCCCCAGTGCTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATC
AGCAATAAACCAGCCAGCCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTAT
CCGCCTCCATCCAGTCTATTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAG
TTAATAGTTTGCGCAACGTTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTC
GTTTGGTATGGCTTCATTCAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATC
CCCCATGTTGTGCAAAAAAGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAG
TAAGTTGGCCGCAGTGTTATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACT
GTCATGCCATCCGTAAGATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCT
GAGAATAGTGTATGCGGCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAAT
ACCGCGCCACATAGCAGAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGG
CGAAAACTCTCAAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGT
GCACCCAACTGATCTTCAGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAA
ACAGGAAGGCAAAATGCCGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAA TACTCAT

(94) The sequence of the pPBBG-iHelper2.5-HA vector is provided below:

(95) TABLE-US-00008 pPBBG-iHelper2.5-HA (16,658 bp) (SEQ ID NO: 8)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttaccccccacccttgccgtctg
cgccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa

actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgc
tcccgtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctct
gcaaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtact
tgtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttg
ccgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctgggga
tggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgcttttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc
cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtacccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagatacccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacgggggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaataaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccgggcgttttagggcggagtaacttgcatgtattgggaattgtag
ttttttaaaatgggaagtgacgtatcgtgggaaaacggaagtgaagatttgaggaagttgtgggtttttggctttcgtttctgggcgtaggtt
cgcgtgcggttttctgggtgtttttgtggactttaaccgttacgtcatttttagtcctatatatactcgctctgtaCTCTCCCTATCAGT

GATAGAGATCTCCCTATCAGTGATAGAGATCGcttggccctttttacactgtgactgattgagctggtgccgtgt
cgagtggtgtttttaataggtttttttactggtaaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggt
gctattttgcctaggcaggagggtttttcaggtgtttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgtc
tctacgcctgcgggtatgtattcccccgggctatttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttaca
ttatgactccggacatgaccgaggaactgtcggtggtgctttttaatcacggtgaccagttttttttacggtcacgccggcatggccgtagtccgt
cttatgcttataagggttgtttttcctgttgtaagacaggcttctaatgtttaaatgtttttttttttgttattttattttgtgtttaatgcagg
aacccgcagacatgtttgagagaaaaatggtgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacgatgtg
cttgctttttttgcgcgaggctttgcctgatttttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggctacgct
ggttagcatagctccgagtatgcgtgtcataatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgcagac
ctgcacgattatgttcagctggccctgcgaagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctgtg
aggaacctgaatttttgcaatcatgattcgctgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatattc
gggatttgcttagagacatattgataaggtggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcaccct
gaagggtttagcctttacgtccacttggacgtgagggcagtttgccttttggaagccattgtgcaacatcttacaaatgccattatctgttcttt
ggctgtagagtttgaccacgccaccggaggggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaaaaa
aaaacatggttcttccagctcttcccgctcctcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggtggt
ggatgttatcagggcagcggcgcatgaaggagtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactacaact
actacacagagcgagctaagcgacgagaccggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATACGA
TGTTCCAGATTACGCTGGAGGAGGCGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatct
cggttgtctcggcgcactccgtacagtagggatcgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgct
gcccgaatgtaacactttgacaatgcacaacgtgagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgtt
ccctgggatatggttctgacgcgggaggagcttgtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacg
agcatgatgatccatggttacgagtcctgggctctccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttgg
ccagctggtttaggatggtggtggatggcgccatgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagag
gtaatgtttatgtccagcgtgtttatgaggggtcgccacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccat
gagctttggatacagcgccttgcactgtgggattttgaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgc
gctgctgtgcccggaggacaaggcgtctcatgctgcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggac
ggagcggcggcggcagcagtttattcgcgcgctgctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcg
tggacttccccttcgccgcccgttgagcaaccgcaagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgag
ctgcccggggagtttattaatatcactgatgagcgtttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatg
atatgatgctttttaaggccagccggggagaaaggactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgag
tttgattaaggtacggtgatcaatataagctatgtggtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaata
aacacgttgaaacataacatgcaacaggttcacgattctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtgg
tgtattttccactttcccaggaccatgtaaaagacatagagtaagtgcttacctcgctagtttctgtggattcactagaaccagtggccaaaaaa
gctagcgcagcagccgccgcgcctggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgg
gcggtaaccgcatggatcacggcggacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccag
accacggaagagtgcccgcttacaggctctccttttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgc
agcagccgccgcgcctggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaac
cgcatggatcacggcggacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacgga
agagtgcccgcttacaggctctccttttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccg
ccgcgcctggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggggtaaccgcatggat
cacggcggacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcc
cgcttacaggctctccttttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccgccgcgcct
ggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcg
gacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttaca
ggctctccttttgcacggtctagagcgtcaacgactgcgcacgtccactttggccgcggctcgagTGAGCTATTCCAGAAGT
AGTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATGCCC
GGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTAA
TTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGTC
CGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGGG
GAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGCA
GACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGTT
GGGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTA
GGCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAGATATACGCGTTGACATTG
ATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATAT

ATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAAC
GACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGG
ACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTA
CATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGG
CCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTAC
ATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGTACATCAAT
GGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCATTGACGTC
AATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTTCCAAAATGTCGTAACAAC
TCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTCTATATAAG
CAGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCC
TGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAG
CCTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgaccgagtacaagcccacggtgcgcctcgc
cacccgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactaccccgccacgcgccacaccgtcgatccggac
cgccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacatcggcaaggtgtgggtcgcggacgacg
gcgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccgagatcggcccgcgcatggccgagttga
gcggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggcccaaggagcccgcgtggttcctggccaccgt
cggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagtggaggcggccgagcgcgccggggtgc
ccgccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcaccgtcaccgccgacgtcgaggtgcccgaag
gaccgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATCTCATGCTGGAGTTCTTCG
CCCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCA
CAAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACT
CATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGCAgg
TGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGG
AATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCC
AGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTG
CCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGG
AGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAG
GGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGCCGC
CACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTAATA
CGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTATCG
ATAAGCTTGATATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAGTAG
AACATGAAATAACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAAAA
GATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACACT
TACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCCTA
AATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAATG
CATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGATC
ATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCAT
CGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAGT
TTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATGA
TGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCCA
CCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCGA
AGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGTG
CAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATCC
TGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGCG
CTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAATT
CCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAGT
GAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCT
GTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGTA
TTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGCG
GCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGG
ATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAA
AAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACAA
AAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGG
CGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCGG

ATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGT
AGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACCC
CCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCCG
GTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGC
GAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACAC
TAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAG
AGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGT
TTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATCT
TTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCA
TGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTTA
AATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCA
GTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCCC
CGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAAT
GATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAG
CCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCTA
TTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAACG
TTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCATT
CAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAAA
AGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGTT
ATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAG
ATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCG
GCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCA
GAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGA
TCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTC
AGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGC
CGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAATACTCAT
(96) The sequence of the pPBBG-iHelper2.6-HA vector is provided below:
(97) TABLE-US-00009 pPBBG-iHelper2.6-HA (16,658 bp) (SEQ ID NO: 9)
ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG

GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttacccccacccttgccgtctg
cgccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgct
cccgtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctctg
caaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtact
tgtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttg
ccgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctgggga
tggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgcttttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc
cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtacccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagatacccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctccctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacgggggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc

agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaataaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccccagtggccaaaaaagctagcgcagcagccgccgcgcctg
gaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcgg
acggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacag
gctctccttttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccgccgcgcctggaaggaa
gccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcggacggccg
gatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacaggctctcctt
ttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccgccgcgcctggaaggaagccaaaa
ggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcggacggccggatccgg
ggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacaggctctccttttgcacg
gtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccgccgcgcctggaaggaagccaaaaggagcg
ctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcggacggccggatccggggttcga
accccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacaggctctccttttgcacggtctagag
cgtcaacgactgcgcacggggcgttttagggcggagtaacttgcatgtattgggaattgtagttttttttaaaatgggaagtgacgtatcgtggg
aaaacggaagtgaagatttgaggaagttgtgggtttttttggctttcgtttctgggcgtaggttcgcgtgcggttttctgggtgtttttttgtggac
tttaaccgttacgtcatttttttagtcctatatatactcgctctgtaCTCTCCCTATCAGTGATAGAGATCTCCCTATCA
GTGATAGAGATCGcttggccctttttacactgtgactgattgagctggtgccgtgtcgagtggtgtttttttaataggtttttttactggt
aaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggtgctattttgcctaggcaggagggtttttcaggt
gtttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgtctctacgcctgcgggtatgtattcccccgggcta
tttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttacattatgactccggacatgaccgaggaactgtcg
gtggtgctttttaatcacggtgaccagtttttttacggtcacgccggcatggccgtagtccgtcttatgcttataagggttgtttttcctgttg
taagacaggcttctaatgtttaaatgtttttttttttgttattttattttgtgtttaatgcaggaacccgcagacatgtttgagagaaaaatgg
tgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacgatgtgcttgctttttttgcgcgaggctttgcctga
ttttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggctacgctggttagcatagctccgagtatgcgtgtca
taatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgcagacctgcacgattatgttcagctggccctgcg
aagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctgtgaggaacctgaatttttgcaatcatgattcg
ctgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatattcgggatttgcttagagacatattgataaggt
ggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcaccctgaagggtttagcctttacgtccacttggac
gtgagggcagtttgccttttggaagccattgtgcaacatcttacaaatgccattatctgttctttggctgtagagtttgaccacgccaccggagg
ggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaaaaaaaaaacatggttcttccagctcttcccgctc
ctcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggtggtggatgttatcagggcagcggcgcatgaagg
agtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactacaactactacacagagcgagctaagcgacgagacc
ggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATACGATGTTCCAGATTACGCTGGAGGAGG
CGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatctcggttgtctcggcgcactccgtacagtagggat
cgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgctgcccgaatgtaacactttgacaatgcacaacgt
gagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgttccctgggatatggttctgacgcgggaggagctt
gtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacgagcatgatgatccatggttacgagtcctgggctc
tccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttggccagctggtttaggatggtggtggatggcgcca
tgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagaggtaatgtttatgtccagcgtgtttatgaggggtcgc
cacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccatgagctttggatacagcgccttgcactgtgggatttt
gaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgcgctgctgtgcccggaggacaaggcgtctcatgct
gcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggacggagcggcggcggcagcagtttattcgcgcgct
gctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcgtggacttccccttcgccgcccgttgagcaaccgc
aagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgagctgcccggggagtttattaatatcactgatgagcg
tttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatgatatgatgctttttaaggccagccggggagaaag
gactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgagtttgattaaggtacggtgatcaatataagctatgtg
gtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaataaacacgttgaaacataacatgcaacaggttcacga

ttctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtggtgtattttccactttcccaggaccatgtaaaagac
atagagtaagtgcttacctcgctagtttctgtggattcactagaatccactttggccgcggctcgagTGAGCTATTCCAGAAGT
AGTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATGCCC
GGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTAA
TTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGTC
CGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGGG
GAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGCA
GACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGTT
GGGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTA
GGCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAGATATACGCGTTGACATTG
ATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATAT
ATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAAC
GACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGG
ACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTA
CATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGG
CCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTAC
ATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGTACATCAAT
GGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCATTGACGTC
AATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTTCCAAAATGTCGTAACAAC
TCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTCTATATAAG
CAGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCC
TGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAG
CCTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgaccgagtacaagcccacggtgcgcctcgc
cacccgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactaccccgccacgcgccacaccgtcgatccggac
cgccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacatcggcaaggtgtgggtcgcggacgacg
gcgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccgagatcggcccgcgcatggccgagttga
gcggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggcccaaggagcccgcgtggttcctggccaccgt
cggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagtggaggcggccgagcgcgccggggtgc
ccgccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcaccgtcaccgccgacgtcgaggtgcccgaag
gaccgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATCTCATGCTGGAGTTCTTCG
CCCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCA
CAAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACT
CATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGCAgg
TGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGG
AATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCC
AGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTG
CCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGGG
AGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAG
GGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGCCGC
CACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTAATA
CGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTATCG
ATAAGCTTGATATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAGTAG
AACATGAAATAACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAAAA
GATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACACT
TACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCCTA
AATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAATG
CATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGATC
ATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCAT
CGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAGT
TTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATGA
TGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCCA
CCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCGA
AGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGTG
CAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATCC

TGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGCG
CTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAATT
CCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAGT
GAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCT
GTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGTA
TTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGCG
GCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGG
ATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAA
AAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACAA
AAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGG
CGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCGG
ATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGT
AGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACCC
CCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCCG
GTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGC
GAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACAC
TAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAG
AGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGT
TTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATCT
TTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCA
TGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTTA
AATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCA
GTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCCC
CGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAAT
GATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAG
CCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCTA
TTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAACG
TTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCATT
CAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAAA
AGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGTT
ATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAG
ATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCG
GCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCA
GAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGA
TCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTC
AGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGC
CGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAATACTCAT

(98) The sequence of the pPBBG-iHelper2.7-HA vector is provided below:

(99) TABLE-US-00010 pPBBG-iHelper2.7-HA (17,114 bp) (SEQ ID NO: 10)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT

CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttaccccccacccttgccgtctg
cgccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgc
tcccgtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctct
gcaaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtact
tgtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttg
ccgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctgggga
tggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgcttttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc
cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtacccccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagatacccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgaccccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg

gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacggggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaataaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccgggcgttttagggcggagtaacttgcatgtattgggaattgtag
tttttttaaaatgggaagtgacgtatcgtgggaaaacggaagtgaagatttgaggaagttgtgggtttttttggctttcgtttctgggcgtaggtt
cgcgtgcggttttctgggtgtttttttgtggactttaaccgttacgtcattttttagtcctatatatactcgctctgtaCTCTCCCTATCAGT
GATAGAGATCTCCCTATCAGTGATAGAGATCGcttggccctttttacactgtgactgattgagctggtgccgtgt
cgagtggtgttttttaataggtttttttactggtaaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggt
gctattttgcctaggcaggagggtttttcaggtgtttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgt
ctctacgcctgcgggtatgtattcccccgggctatttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttac
attatgactccggacatgaccgaggaactgtcggtggtgctttttaatcacggtgaccagttttttttacggtcacgccggcatggccgtagtcc
gtcttatgcttataagggttgtttttcctgttgtaagacaggcttctaatgtttaaatgtttttttttttgttattttattttgtgtttaatgc
aggaacccgcagacatgtttgagagaaaaatggtgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacga
tgtgcttgctttttttgcgcgaggctttgcctgatttttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggcta
cgctggttagcatagctccgagtatgcgtgtcataatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgca
gacctgcacgattatgttcagctggccctgcgaagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctg
tgaggaacctgaatttttgcaatcatgattcgctgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatat
tcgggatttgcttagagacatattgataaggtggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcacc
ctgaagggtttagcctttacgtccacttggacgtgagggcagtttgccttttggaagccattgtgcaacatcttacaaatgccattatctgttc
tttggctgtagagtttgaccacgccaccggaggggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaa
aaaaaaacatggttcttccagctcttcccgctcctcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggt
ggtggatgttatcagggcagcggcgcatgaaggagtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactaca
actactacacagagcgagctaagcgacgagaccggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATA
CGATGTTCCAGATTACGCTGGAGGAGGCGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatct
cggttgtctcggcgcactccgtacagtagggatcgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgct
gcccgaatgtaacactttgacaatgcacaacgtgagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgtt
ccctgggatatggttctgacgcgggaggagcttgtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacg
agcatgatgatccatggttacgagtcctgggctctccactgtcattgttccagtcccggttccctgcagtgcatagccgggggcaggttttgg
ccagctggtttaggatggtggtggatggcgccatgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagag
gtaatgtttatgtccagcgtgtttatgaggggtcgccacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccat
gagctttggatacagcgccttgcactgtgggattttgaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgc
gctgctgtgcccggaggacaaggcgtctcatgctgcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggac
ggagcggcggcggcagcagtttattcgcgcgctgctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcg
tggacttccccttcgccgcccgttgagcaaccgcaagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgag
ctgcccggggagtttattaatatcactgatgagcgtttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatg
atatgatgctttttaaggccagccggggagaaaggactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgag
tttgattaaggtacggtgatcaatataagctatgtggtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaata
aacacgttgaaacataacatgcaacaggttcacgattctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtgg

tgtattttccactttcccaggaccatgtaaaagacatagagtaagtgcttacctcgctagtttctgtggattcactagaaccagtggccaaaaaa
gctagcgcagcagccgccgcgcctggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgg
gcggtaaccgcatggatcacggcggacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccag
accacggaagagtgcccgcttacagGTGGTCTCATACAGAACTTATAAGATTCCCAAATCCAAAGA
CATTTCACGTTTATGGTGATTTCCCAGAACACATAGCGACATGCAAATATTGCAGGG
CGCCACTCCCCTGTCCgctctccttttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgca
gcagccgccgcgcctggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaacc
gcatggatcacggcggacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaa
gagtgcccgcttacagGTGGTCTCATACAGAACTTATAAGATTCCCAAATCCAAAGACATTTC
ACGTTTATGGTGATTTCCCAGAACACATAGCGACATGCAAATATTGCAGGGCGCCAC
TCCCCTGTCCgctctccttttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccgcc
gcgcctggaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatca
cggcggacggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccg
cttacagGTGGTCTCATACAGAACTTATAAGATTCCCAAATCCAAAGACATTTCACGTTT
ATGGTGATTTCCCAGAACACATAGCGACATGCAAATATTGCAGGGCGCCACTCCCCT
GTCCgctctccttttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccgccgcgcctgg
aaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggggtaaccgcatggatcacggcgga
cggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacagG
TGGTCTCATACAGAACTTATAAGATTCCCAAATCCAAAGACATTTCACGTTTATGGT
GATTTCCCAGAACACATAGCGACATGCAAATATTGCAGGGCGCCACTCCCCTGTCCg
ctctccttttgcacggtctagagcgtcaacgactgcgcacgtccactttggccgcggctcgagTGAGCTATTCCAGAAGTA
GTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATGCCCG
GGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTAAT
TACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGTCC
GGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGGGG
AAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGCAG
ACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGTTG
GGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTAG
GCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAGATATACGCGTTGACATTGA
TTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATATA
TGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACG
ACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGA
CTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTAC
ATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGGC
CCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTACA
TCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGTACATCAATG
GGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCATTGACGTCA
ATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTTCCAAAATGTCGTAACAACT
CCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTCTATATAAGC
AGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCT
GGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGC
CTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgaccgagtacaagcccacggtgcgcctcgcc
acccgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactaccccgccacgcgccacaccgtcgatccggacc
gccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacatcggcaaggtgtgggtcgcggacgacgg
cgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccgagatcggcccgcgcatggccgagttgag
cggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggcccaaggagcccgcgtggttcctggccaccgtc
ggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagtggaggcggccgagcgcgccggggtgcc
cgccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcaccgtcaccgccgacgtcgaggtgcccgaagg
accgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATCTCATGCTGGAGTTCTTCGC
CCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCAC
AAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACTC
ATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGCAggT
GTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGGA
ATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCCA

GGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTGC
CACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGGG
AGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAG
GGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGCCGC
CACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTAATA
CGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTATCG
ATAAGCTTGATATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAGTAG
AACATGAAATAACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAAAA
GATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACACT
TACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCCTA
AATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAATG
CATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGATC
ATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCAT
CGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAGT
TTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATGA
TGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCCA
CCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCGA
AGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGTG
CAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATCC
TGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGCG
CTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAATT
CCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAGT
GAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCT
GTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGTA
TTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGCG
GCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGG
ATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAA
AAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACAA
AAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGG
CGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCGG
ATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGT
AGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACCC
CCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCCG
GTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGC
GAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACAC
TAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAG
AGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGT
TTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATCT
TTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCA
TGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTTA
AATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCA
GTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCCC
CGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAAT
GATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAG
CCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCTA
TTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAACG
TTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCATT
CAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAAA
AGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGTT
ATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAG
ATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCG
GCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCA
GAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGA
TCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTC

AGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGC
CGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAATACTCAT

(100) The sequence of the pPBBG-iHelper2.8-HA vector is provided below:

(101) TABLE-US-00011 pPBBG-iHelper2.8-HA (17,114 bp) (SEQ ID NO: 11)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttacccccacccttgccgtctgcg
ccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgctccc
gtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctctg
caaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtactt
gtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttgc
cgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctggggat
ggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgcttttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc
cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa

gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtacccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagatacccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaaggacgggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaataaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccccagtggccaaaaaagctagcgcagcagccgccgcgcctg
gaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcgg
acggccggatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacag
GTGGTCTCATACAGAACTTATAAGATTCCCAAATCCAAAGACATTTCACGTTTATGG
TGATTTCCCAGAACACATAGCGACATGCAAATATTGCAGGGCGCCACTCCCCTGTCC
gctctccttttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccgccgcgcctggaaggaa
gccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcggacggccg
gatccggggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacagGTGGT
CTCATACAGAACTTATAAGATTCCCAAATCCAAAGACATTTCACGTTTATGGTGATT
TCCCAGAACACATAGCGACATGCAAATATTGCAGGGCGCCACTCCCCTGTCCgctctcctt
ttgcacggtctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccgccgcgcctggaaggaagccaaaa
ggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcggacggccggatccgg
ggttcgaaccccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacagGTGGTCTCAT
ACAGAACTTATAAGATTCCCAAATCCAAAGACATTTCACGTTTATGGTGATTTCCCA
GAACACATAGCGACATGCAAATATTGCAGGGCGCCACTCCCCTGTCCgctctccttttgcacgg
tctagagcgtcaacgactgcgcacgccagtggccaaaaaagctagcgcagcagccgccgcgcctggaaggaagccaaaaggagcgct

cccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcggacggccggatccggggttcgaac
cccggtcgtccgccatgatacccttgcgaatttatccaccagaccacggaagagtgcccgcttacagGTGGTCTCATACAGA
ACTTATAAGATTCCCAAATCCAAAGACATTTCACGTTTATGGTGATTTCCCAGAACA
CATAGCGACATGCAAATATTGCAGGGCGCCACTCCCCTGTCCgctctccttttgcacggtctagagc
gtcaacgactgcgcacggggcgttttagggcggagtaacttgcatgtattgggaattgtagtttttttaaaatgggaagtgacgtatcgtggga
aaacggaagtgaagatttgaggaagttgtgggtttttttggctttcgtttctgggcgtaggttcgcgtgcggttttctgggtgtttttgtggacttta
accgttacgtcatttttttagtcctatatatactcgctctgtaCTCTCCCTATCAGTGATAGAGATCTCCCTATCAG
TGATAGAGATCGcttggccctttttacactgtgactgattgagctggtgccgtgtcgagtggtgtttttttaataggttttttttactggtaa
ggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggtgctattttgcctaggcaggagggtttttcaggtgt
ttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgtctctacgcctgcgggtatgtattcccccgggctatttcg
gtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttacattatgactccggacatgaccgaggaactgtcggtggtgctt
tttaatcacggtgaccagttttttttacggtcacgccggcatggccgtagtccgtcttatgcttataagggttgtttttcctgttgtaagacaggcttc
taatgtttaaatgtttttttttttgttattttattttgtgtttaatgcaggaacccgcagacatgtttgagagaaaaaatggtgtctttttctgtggtg
gttccggaacttacctgcctttatctgcatgagcatgactacgatgtgcttgctttttttgcgcgaggctttgcctgatttttttgagcagcaccttgca
ttttatatcgccgcccatgcaacaagcttacataggggctacgctggttagcatagctccgagtatgcgtgtcataatcagtgtgggttcttttgtca
tggttcctggcggggaagtggccgcgctggtccgtgcagacctgcacgattatgttcagctggccctgcgaagggacctacgggatcgcggt
atttttgttaatgttccgcttttgaatcttatacaggtctgtgaggaacctgaatttttgcaatcatgattcgctgcttgaggctgaaggtggagggc
gctctggagcagatttttacaatggccggacttaatattcgggatttgcttagagacatattgataaggtggcgagatgaaaattatttgggcat
ggttgaaggtgctggaatgtttatagaggagattcaccctgaagggtttagcctttacgtccacttggacgtgagggcagtttgccttttggaa
gccattgtgcaacatcttacaaatgccattatctgttctttggctgtagagtttgaccacgccaccggaggggagcgcgttcacttaatagatct
tcattttgaggttttggataatcttttggaataaaaaaaaaaaaaacatggttcttccagctcttcccgctcctcccgtgtgtgactcgcagaacga
atgtgtaggttggctgggtgtggcttattctgcggtggtggatgttatcagggcagcggcgcatgaaggagtttacatagaacccgaagcca
gggggcgcctggatgctttgagagagtggatatactacaactactacacagagcgagctaagcgacgagaccggagacgcagatctgttt
gtcacgcccgcacctggttttgcttcaggaaatatgTACCCATACGATGTTCCAGATTACGCTGGAGGAGG
CGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatctcggttgtctcggcgcactccgtacagtagggat
cgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgctgcccgaatgtaacactttgacaatgcacaacgt
gagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgttccctgggatatggttctgacgcgggaggagctt
gtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacgagcatgatgatccatggttacgagtcctgggctc
tccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttggccagctggtttaggatggtggtggatggcgcca
tgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagaggtaatgtttatgtccagcgtgtttatgaggggtcgc
cacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccatgagctttggatacagcgccttgcactgtgggatttt
gaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgcgctgctgtgcccggaggacaaggcgtctcatgct
gcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggacggagcggcggcggcagcagtttattcgcgcgct
gctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcgtggacttccccttcgccgcccgttgagcaaccgc
aagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgagctgcccggggagtttattaatatcactgatgagcg
tttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatgatatgatgctttttaaggccagccggggagaaag
gactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgagtttgattaaggtacggtgatcaatataagctatgtg
gtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaataaacacgttgaaacataacatgcaacaggttcacga
ttctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtggtgtattttccactttcccaggaccatgtaaaagaca
tagagtaagtgcttacctcgctagtttctgtggattcactagaatccactttggccgcggctcgagTGAGCTATTCCAGAAGT
AGTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATGCCC
GGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTAA
TTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGTC
CGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGGG
GAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGCA
GACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGTT
GGGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTA
GGCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAGATATACGCGTTGACATTG
ATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATAT
ATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAAC
GACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGG
ACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTA
CATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGG
CCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTAC

ATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGTACATCAAT
GGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCATTGACGTC
AATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTTCCAAAATGTCGTAACAAC
TCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTCTATATAAG
CAGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCC
TGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAG
CCTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgaccgagtacaagcccacggtgcgcctcgc
cacccgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactaccccgccacgcgccacaccgtcgatccggac
cgccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacatcggcaaggtgtgggtcgcggacgacg
gcgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccgagatcggcccgcgcatggccgagttga
gcggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggcccaaggagcccgcgtggttcctggccaccgt
cggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagtggaggcggccgagcgcgccggggtgc
ccgccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcaccgtcaccgccgacgtcgaggtgcccgaag
gaccgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATCTCATGCTGGAGTTCTTCG
CCCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCA
CAAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACT
CATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGCAgg
TGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGG
AATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCC
AGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTG
CCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGGG
AGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAG
GGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGCCGC
CACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTAATA
CGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTATCG
ATAAGCTTGATATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAGTAG
AACATGAAATAACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAAAA
GATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACACT
TACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCCTA
AATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAATG
CATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGATC
ATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCAT
CGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAGT
TTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATGA
TGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCCA
CCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCGA
AGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGTG
CAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATCC
TGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGCG
CTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAATT
CCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAGT
GAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCT
GTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGTA
TTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGCG
GCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGG
ATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAA
AAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACAA
AAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGG
CGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCGG
ATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGT
AGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACCC
CCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCCG
GTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGC
GAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACAC

TAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAG
AGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGT
TTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATCT
TTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCA
TGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTTA
AATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCA
GTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCCC
CGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAAT
GATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAG
CCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCTA
TTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAACG
TTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCATT
CAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAAA
AGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGTT
ATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAG
ATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCG
GCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCA
GAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGA
TCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTC
AGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGC
CGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAATACTCAT

(102) The sequence of the pPBBG-iHelper3.1-HA vector is provided below:

(103) TABLE-US-00012 pPBBG-iHelper3.1-HA (18,311 bp) (SEQ ID NO: 12)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttacccccacccttgccgtctgcg

ccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgctccc
gtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctctg
caaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtactt
gtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttgc
cgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctggggat
ggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgcttttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc
cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcaccccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtaccccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagatacccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacgggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaaaaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt

accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggggctattaccacacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccccagtggccaaaaaagctagcgcagcagccgccgcgcctg
gaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggcggtaaccgcatggatcacggcgg
acggccggatccggggttcgaaccccggtcgtccgccatgataccttgcgaatttatccaccagaccacggaagagtgcccgcttacag
gctctccttttgcacggtctagagcgtcaacgactgcgcacggggcgttttagggcggagtaacttgcatgtattgggaattgtagttttttaa
aatgggaagtgacgtatcgtgggaaaacggaagtgaagatttgaggaagttgtgggtttttggctttcgtttctgggcgtaggttcgcgtgcg
gttttctgggtgtttttgtggactttaaccgttacgtcatttttagtcctatatatactcgctctgtaCTCTCCCTATCAGTGATAG
AGATCTCCCTATCAGTGATAGAGATCGcttggcccttttacactgtgactgattgagctggtgccgtgtcgagtggt
gtttttaataggtttttttactggtaaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggtgctattttgcc
taggcaggagggtttttcaggtgtttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgtctctacgcctgcggg
tatgtattcccccgggctatttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttacattatgactccggacatg
accgaggaactgtcggtggtgctttttaatcacggtgaccagttttttttacggtcacgccggcatggccgtagtccgtcttatgcttataagggtt
gtttttcctgttgtaagacaggcttctaatgtttaaatgtttttttttttgttattttattttgtgtttaatgcaggaacccgcagacatgtttgaga
gaaaaatggtgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacgatgtgcttgctttttttgcgcgaggctttg
cctgatttttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggctacgctggttagcatagctccgagtatgcgtgt
cataatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgcagacctgcacgattatgttcagctggccctgc
gaagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctgtgaggaacctgaatttttgcaatcatgattcgc
tgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatattcgggatttgcttagagacatattgataaggt
ggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcaccctgaagggtttagcctttacgtccacttgga
cgtgagggcagtttgcctttttggaagccattgtgcaacatcttacaaatgccattatctgttctttggctgtagagtttgaccacgccaccggag
gggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaaaaaaaaaacatggttcttccagctcttcccgctcc
tcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggtggtggatgttatcagggcagcggcgcatgaag
gagtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactacaactactacacagagcgagctaagcgac
gagaccggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATACGATGTTCCAGA
TTACGCTGGAGGAGGCGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatctcggttgtct
cggcgcactccgtacagtagggatcgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgctgcccgaat
gtaacactttgacaatgcacaacgtgagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgttccctggga
tatggttctgacgcgggaggagcttgtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacgagcatgatg
atccatggttacgagtcctgggctctccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttggccagctggt
ttaggatggtggtggatggcgccatgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagaggtaatgtttat
gtccagcgtgtttatgaggggtcgccacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccatgagctttgga
tacagcgccttgcactgtgggattttgaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgcgctgctgtgc
ccggaggacaaggcgtctcatgctgcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggacggagcggc
ggcggcagcagtttattcgcgcgctgctgcagcaccaccgccctatcctgatgcacgattatgactctaccccccatgtaggcgtggacttcc
ccttcgccgcccgttgagcaaccgcaagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgagctgcccgg
ggagtttattaatatcactgatgagcgtttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatgatatgatgc
tttttaaggccagccggggagaaaggactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgagtttgattaa
ggtacggtgatcaatataagctatgtggtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaataaacacgttg
aaacataacatgcaacaggttcacgattctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtggtgtattttc
cactttcccaggaccatgtaaaagacatagagtaagtgcttacctcgctagtttctgtggattcactagaatccactttggccgcggctcgagg
agggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgct
ccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcct
ctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttggg
tgaattttggctcattcctcctttctataggattgaggtcagagcGACATTGATTATTGACTAGTTATTAATAGTA
ATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACT
TACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAAT
AATGACGTATGTTCCCATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGT
GGAGTATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAG
TACGCCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTA
CATGACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATT
ACCATGGTGATGCGGTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCA
CGGGGATTTCCAAGTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGAACCAA

AATCAACGGGACTTTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGG
CGGTAGGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTCTCCCTATCAGTGATAG
AGATCTCCCTATCAGTGATAGAGATCGTCGACGAGCTCGTTTAGTGAACCGTCAGAT
CGCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATC
CAGCCTgaacgcgcagccgccatgGACTACAAAGACGATGACGACAAGGGAGGAGGCGGAG
GCacaactagaacaaagggcaggggccatactgcggccacgactcaaaacgacagaatgccaggccctgagctttcgggctggatctc
tgagcagctaatgaccggaagaattcctgtaagcgacatcttctgtgatattgagaacaatccaggattatgctacgcatcccaaatgcaaca
aacgaagccaaacccgaagacgcgcaacagtcaaacccaaacggacccaatttgcaatcatagttttgaggaggtagtacaaacattggct
tcattggctactgttgtgcaacaacaaaccatcgcatcagaatcattagaacaacgcattacgagtcttgagaatggtctaaagccagtttatg
atatggcaaaaacaatctcctcattgaacagggtttgtgctgagatggttgcaaaatatgatcttctggtgatgacaaccggtcgggcaacag
caaccgctgcggcaactgaggcttattgggccgaacatggtcaaccaccacctggaccatcactttatgaagaaagtgcgattcggggtaa
gattgaatctagagatgagaccgtccctcaaagtgttagggaggcattcaacaatctaaacagtaccacttcactaactgaggaaaattttgg
gaaacctgacatttcggcaaaggatttgagaaacattatgtatgatcacttgcctggttttggaactgctttccaccaattagtacaagtgatttgt
aaattgggaaaagatagcaactcattggacatcattcatgctgagttccaggccagcctggctgaaggagactctcctcaatgtgccctaatt
caaattacaaaaagagttccaatcttccaagatgctgctccacctgtcatccacatccgctctcgaggtgacattccccgagcttgccagaaa
agcttgcgtccagtcccaccatcgcccaagattgatcgaggttgggtatgtgtttttcagcttcaagatggtaaaacacttggactcaaaatttg
aGGATCCACTAGTCCAGTGTGGTGGAATTCTGCAGATATCCAGCACAGTGGCGGCCG
CTCGAGTCTAGAGGGCCCGTTTAAACCCGCTGATCAGCCTCGACTGTGCCTTCTAGT
TGCCAGCCATCTGTTGTTTGCCCCTCCCCCGTGCCTTCCTTGACCCTGGAAGGTGCCA
CTCCCACTGTCCTTTCCTAATAAAATGAGGAAATTGCATCGCATTGTCTGAGTAGGT
GTCATTCTATTCTGGGGGGTGGGGTGGGGCAGGACAGCAAGGGGGAGGATTGGGAA
GACAATAGCAGGCATGCTGGGGATGCGGTGGGCTCTATGGTGAGCTATTCCAGAAG
TAGTGAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCT
AGGCTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAGATATACGCGTTGACATT
GATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCAT
ATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCA
ACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAG
GGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAG
TACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAAT
GGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGT
ACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGTACATCA
ATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCATTGACG
TCAATGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTTCCAAAATGTCGTAACA
ACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTCTATATA
AGCAGAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCG
CCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCA
GCCTCCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgaccgagtacaagcccacggtgcgcctc
gccacccgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactaccccgccacgcgccacaccgtcgatccg
gaccgccacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacatcggcaaggtgtgggtcgcggacg
acggcgccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccgagatcggcccgcgcatggccgag
ttgagcggttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggcccaaggagcccgcgtggttcctggcca
ccgtcggcgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagtggaggcggccgagcgcgccggg
gtgcccgccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcaccgtcaccgccgacgtcgaggtgccc
gaaggaccgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATCTCATGCTGGAGTTCTT
CGCCCACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCAT
CACAAATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAA
CTCATCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGC
AggTGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCC

GGAATTGACTGGATTCCTTTTTTAGGGCCATTGGTATGGCTTTTTCCCCGTATCCCC
CCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCG
TGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGG
GGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGG
AGGGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGC
CGCCACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTA
ATACGACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTA
TCGATAAGCTTGATATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAG
TAGAACATGAAATAACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAA
AAGATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACA
CTTACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCC
TAAATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAA
TGCATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGAT
CATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCA
TCGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAG
TTTGCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATG
ATGATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCC
ACCTGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCG
AAGCGCATCAGCAACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGT
GCAGATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATC
CTGGCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGC
GCTTGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAAT
TCCACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAG
TGAGCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACC
TGTCGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGT
ATTGGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGC
GGCGAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGG
GATAACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTA
AAAAGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACA
AAAATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAG
GCGTTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCG
GATACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTG
TAGGTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACC
CCCCGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCC
GGTAAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAG
CGAGGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACA
CTAGAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAA
GAGTTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTG
TTTGCAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATC
TTTTCTACGGGGTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTC
ATGAGATTATCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTT
AAATCAATCTAAAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATC
AGTGAGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCC
CCGTCGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAA
TGATACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCA
GCCGGAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCT
ATTAATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAAC
GTTGTTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCAT
TCAGCTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAA
AAGCGGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGT
TATCACTCATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAG
ATGCTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCG
GCGACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCA
GAACTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGA

TCTTACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTC
AGCATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGC
CGCAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAATACTCAT

(104) The sequence of the pPBBG-iHelper3.2-HA vector is provided below:

(105) TABLE-US-00013 pPBBG-iHelper3.2-HA (18,728 bp) (SEQ ID NO: 13)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCggtacccaactccatgcttaacagtccccaggtacagcccaccctgcgtcgcaaccaggaaca
gctctacagcttcctggagcgccactcgccctacttccgcagccacagtgcgcagattaggagcgccacttctttttgtcacttgaaaaacat
gtaaaaataatgtactaggagacactttcaataaaggcaaatgtttttatttgtacactctcgggtgattatttacccccacccttgccgtctgcg
ccgtttaaaaatcaaaggggttctgccgcgcatcgctatgcgccactggcagggacacgttgcgatactggtgtttagtgctccacttaaact
caggcacaaccatccgcggcagctcggtgaagttttcactccacaggctgcgcaccatcaccaacgcgtttagcaggtcgggcgccgata
tcttgaagtcgcagttggggcctccgccctgcgcgcgcgagttgcgatacacagggttgcagcactggaacactatcagcgccgggtggt
gcacgctggccagcacgctcttgtcggagatcagatccgcgtccaggtcctccgcgttgctcagggcgaacggagtcaactttggtagctg
ccttcccaaaaagggtgcatgcccaggctttgagttgcactcgcaccgtagtggcatcagaaggtgaccgtgcccggtctgggcgttagga
tacagcgcctgcatgaaagccttgatctgcttaaaagccacctgagcctttgcgccttcagagaagaacatgccgcaagacttgccggaaa
actgattggccggacaggccgcgtcatgcacgcagcaccttgcgtcggtgttggagatctgcaccacatttcggccccaccggttcttcacg
atcttggccttgctagactgctccttcagcgcgcgctgcccgttttcgctcgtcacatccatttcaatcacgtgctccttatttatcataatgctccc
gtgtagacacttaagctcgccttcgatctcagcgcagcggtgcagccacaacgcgcagcccgtgggctcgtggtgcttgtaggttacctctg
caaacgactgcaggtacgcctgcaggaatcgccccatcatcgtcacaaaggtcttgttgctggtgaaggtcagctgcaacccgcggtgctc
ctcgtttagccaggtcttgcatacggccgccagagcttccacttggtcaggcagtagcttgaagtttgcctttagatcgttatccacgtggtactt
gtccatcaacgcgcgcgcagcctccatgcccttctcccacgcagacacgatcggcaggctcagcgggtttatcaccgtgctttcactttccg
cttcactggactcttccttttcctcttgcgtccgcataccccgcgccactgggtcgtcttcattcagccgccgcaccgtgcgcttacctcccttgc
cgtgcttgattagcaccggtgggttgctgaaacccaccatttgtagcgccacatcttctctttcttcctcgctgtccacgatcacctctggggat
ggcgggcgctcgggcttgggagaggggcgcttctttttctttttggacgcaatggccaaatccgccgtcgaggtcgatggccgcgggctgg
gtgtgcgcggcaccagcgcatcttgtgacgagtcttcttcgtcctcggactcgagacgccgcctcagccgcttttttgggggcgcgcgggg
aggcggcggcgacggcgacggggacgacacgtcctccatggttggtggacgtcgcgccgcaccgcgtccgcgctcgggggtggtttc
gcgctgctcctcttcccgactggccatttccttctcctataggcagaaaaagatcatggagtcagtcgagaaggaggacagcctaaccgccc

cctttgagttcgccaccaccgcctccaccgatgccgccaacgcgcctaccaccttccccgtcgaggcacccccgcttgaggaggaggaa
gtgattatcgagcaggacccaggttttgtaagcgaagacgacgaggatcgctcagtaccaacagaggataaaaagcaagaccaggacga
cgcagaggcaaacgaggaacaagtcgggcgggggggaccaaaggcatggcgactacctagatgtgggagacgacgtgctgttgaagca
tctgcagcgccagtgcgccattatctgcgacgcgttgcaagagcgcagcgatgtgcccctcgccatagcggatgtcagccttgcctacgaa
cgccacctgttctcaccgcgcgtacccccaaacgccaagaaaacggcacatgcgagcccaacccgcgcctcaacttctaccccgtatttg
ccgtgccagaggtgcttgccacctatcacatctttttccaaaactgcaagatacccctatcctgccgtgccaaccgcagccgagcggacaag
cagctggccttgcggcagggcgctgtcatacctgatatcgcctcgctcgacgaagtgccaaaaatctttgagggtcttggacgcgacgaga
aacgcgcggcaaacgctctgcaacaagaaaacagcgaaaatgaaagtcactgtggagtgctggtggaacttgagggtgacaacgcgcg
cctagccgtgctgaaacgcagcatcgaggtcacccactttgcctacccggcacttaacctacccccaaggttatgagcacagtcatgagc
gagctgatcgtgcgccgtgcacgacccctggagagggatgcaaacttgcaagaacaaaccgaggagggcctacccgcagttggcgatg
agcagctggcgcgctggcttgagacgcgcgagcctgccgacttggaggagcgacgcaagctaatgatggccgcagtgcttgttaccgtg
gagcttgagtgcatgcagcggttctttgctgacccggagatgcagcgcaagctagaggaaacgttgcactacacctttcgccagggctacg
tgcgccaggcctgcaaaatttccaacgtggagctctgcaacctggtctcctaccttggaattttgcacgaaaaccgcctcgggcaaaacgtg
cttcattccacgctcaagggcgaggcgcgccgcgactacgtccgcgactgcgtttacttatttctgtgctacacctggcaaacggccatggg
cgtgtggcagcaatgcctggaggagcgcaacctaaaggagctgcagaagctgctaaagcaaaacttgaaggacctatggacggccttca
acgagcgctccgtggccgcgcacctggcggacattatcttccccgaacgcctgcttaaaaccctgcaacagggtctgccagacttcaccag
tcaaagcatgttgcaaaactttaggaactttatcctagagcgttcaggaattctgcccgccacctgctgtgcgcttcctagcgactttgtgccca
ttaagtaccgtgaatgccctccgccgctttggggtcactgctaccttctgcagctagccaactaccttgcctaccactccgacatcatggaaga
cgtgagcggtgacggcctactggagtgtcactgtcgctgcaacctatgcaccccgcaccgctccctggtctgcaattcgcaactgcttagcg
aaagtcaaattatcggtacctttgagctgcagggtccctcgcctgacgaaaagtccgcggctccggggttgaaactcactccggggctgtg
gacgtcCGATCTCTATCACTGATAGGGAGATCTCTATCACTGATAGGGAGAGggcttaccttcg
caaatttgtacctgaggactaccacgcccacgagattaggttctacgaagaccaatcccgcccgccaaatgcggagcttaccgcctgcgtc
attacccagggccacatccttggccaattgcaagccatcaacaaagcccgccaagagtttctgctacgaaagggacgggggggtttacctgg
acccccagtccggcgaggagctcaacccaatcccccgccgccgcagccctatcagcagccgcgggcccttgcttcccaggatggcac
ccaaaaagaagctgcagctgccgccgccgccacccacggacgaggaggaatactgggacagtcaggcagaggaggttttggacgagg
aggaggagatgatggaagactgggacagcctagacgaagcttccgaggccgaagaggtgtcagacgaaacaccgtcaccctcggtcgc
attcccctcgccggcgccccagaaattggcaaccgttcccagcatcgctacaacctccgctcctcaggcgccgccggcactgcctgttcgc
cgacccaaccgtagatgggacaccactggaaccagggccggtaagtctaagcagccgccgccgttagcccaagagcaacaacagcgc
caaggctaccgctcgtggcgcgggcacaagaacgccatagttgcttgcttgcaagactgtgggggcaacatctccttcgcccgccgctttct
tctctaccatcacggcgtggccttcccccgtaacatcctgcattactaccgtcatctctacagcccctactgcaccggcggcagcggcagcg
gcagcaacagcagcggtcacacagaagcaaaggcgaccggatagcaagactctgacaaagcccaagaaatccacagcggcggcagc
agcaggaggaggagcgctgcgtctggcgcccaacgaacccgtatcgacccgcgagcttagaaataggatttttcccactctgtatgctatat
ttcaacaaagcaggggccaagaacaagagctgaaaataaaaaacaggtctctgcgctccctcacccgcagctgcctgtatcacaaaagcg
aagatcagcttcggcgcacgctggaagacgcggaggctctcttcagcaaCGATCTCTATCACTGATAGGGAGATC
TCTATCACTGATAGGGAGAGatactgcgcgctgactcttaaggactagtttcgcgccctttctcaaatttaagcgcgaaaa
ctacgtcatctccagcggccacacccggcgccagcacctgtcgtcagcgccattatgagcaaggaaattcccacgccctacatgtggagtt
accagccacaaatgggacttgcggctggagctgcccaagactactcaacccgaataaactacatgagcgcgggaccccacatgatatccc
gggtcaacggaatccgcgcccaccgaaaccgaattctcctcgaacaggcggctattaccaccacacctcgtaataaccttaatccccgtagt
tggcccgctgccctggtgtaccaggaaagtcccgctcccaccactgtggtacttcccagagacgcccaggccgaagttcagatgactaact
caggggcgcagcttgcgggcggctttcgtcacagggtgcggtcgcccccagtggccaaaaaagctagcgcagcagccgccgcgcctg
gaaggaagccaaaaggagcgctcccccgttgtctgacgtcgcacacctgggttcgacacgcgggggtaaccgcatggatcacggcgg
acggccggatccggggttcgaaccccggtcgtccgccatgataccccttgcgaatttatccaccagaccacggaagagtgcccgcttacag
gctctccttttgcacggtctagagcgtcaacgactgcgcacggggcgttttagggcggagtaacttgcatgtattgggaattgtagtttttttaa
aatgggaagtgacgtatcgtgggaaaacggaagtgaagatttgaggaagttgtgggtttttttggctttcgtttctgggcgtaggttcgcgtgcg
gttttctgggtgtttttttgtggactttaaccgttacgtcattttttagtcctatatatactcgctctgtaCTCTCCCTATCAGTGATAG
AGATCTCCCTATCAGTGATAGAGATCGcttggcccttttttacactgtgactgattgagctggtgccgtgtcgagtggt
gtttttaataggtttttttactggtaaggctgactgttatggctgccgctgtggaagcgctgtatgttgttctggagcgggagggtgctattttgcc
taggcaggagggtttttcaggtgtttatgtgtttttctctcctattaattttgttatacctcctatgggggctgtaatgttgtctctacgcctgcggg
tatgtattcccccgggctatttcggtcgctttttagcactgaccgatgttaaccaacctgatgtgtttaccgagtcttacattatgactccggacatg
accgaggaactgtcggtggtgctttttaatcacggtgaccagttttttttacggtcacgccggcatggccgtagtccgtcttatgcttataagggtt
gtttttcctgttgtaagacaggcttctaatgtttaaatgtttttttttttgttattttattttgtgtttaatgcaggaacccgcagacatgtttgaga
gaaaaatggtgtctttttctgtggtggttccggaacttacctgcctttatctgcatgagcatgactacgatgtgcttgctttttttgcgcgaggctttg
cctgattttttgagcagcaccttgcattttatatcgccgcccatgcaacaagcttacataggggctacgctggttagcatagctccgagtatgcgtgt
cataatcagtgtgggttcttttgtcatggttcctggcggggaagtggccgcgctggtccgtgcagacctgcacgattatgttcagctggccctgc

gaagggacctacgggatcgcggtatttttgttaatgttccgcttttgaatcttatacaggtctgtgaggaacctgaatttttgcaatcatgattcgc
tgcttgaggctgaaggtggagggcgctctggagcagatttttacaatggccggacttaatattcgggatttgcttagagacatattgataaggt
ggcgagatgaaaattatttgggcatggttgaaggtgctggaatgtttatagaggagattcaccctgaagggtttagcctttacgtccacttgga
cgtgagggcagtttgccttttggaagccattgtgcaacatcttacaaatgccattatctgttctttggctgtagagtttgaccacgccaccggag
gggagcgcgttcacttaatagatcttcattttgaggttttggataatcttttggaataaaaaaaaaaaaacatggttcttccagctcttcccgctcc
tcccgtgtgtgactcgcagaacgaatgtgtaggttggctgggtgtggcttattctgcggtggtggatgttatcagggcagcggcgcatgaag
gagtttacatagaacccgaagccagggggcgcctggatgctttgagagagtggatatactacaactactacacagagcgagctaagcgac
gagaccggagacgcagatctgtttgtcacgcccgcacctggttttgcttcaggaaatatgTACCCATACGATGTTCCAGA
TTACGCTGGAGGAGGCGGAGGCactacgtccggcgttccatttggcatgacactacgaccaacacgatctcggttgtct
cggcgcactccgtacagtagggatcgcctacctccttttgagacagagacccgcgctaccatactggaggatcatccgctgctgcccgaat
gtaacactttgacaatgcacaacgtgagttacgtgcgaggtcttccctgcagtgtgggatttacgctgattcaggaatgggttgttccctggga
tatggttctgacgcgggaggagcttgtaatcctgaggaagtgtatgcacgtgtgcctgtgttgtgccaacattgatatcatgacgagcatgatg
atccatggttacgagtcctgggctctccactgtcattgttccagtcccggttccctgcagtgcatagccggcgggcaggttttggccagctggt
ttaggatggtggtggatggcgccatgtttaatcagaggtttatatggtaccgggaggtggtgaattacaacatgccaaaagaggtaatgtttat
gtccagcgtgtttatgaggggtcgccacttaatctacctgcgcttgtggtatgatggccacgtgggttctgtggtccccgccatgagctttgga
tacagcgccttgcactgtgggattttgaacaatattgtggtgctgtgctgcagttactgtgctgatttaagtgagatcagggtgcgctgctgtgc
ccggaggacaaggcgtctcatgctgcgggcggtgcgaatcatcgctgaggagaccactgccatgttgtattcctgcaggacggagcggc
ggcggcagcagtttattcgcgcgctgctgcagcaccaccgccctatcctgatgcacgattatgactctacccccatgtaggcgtggacttcc
ccttcgccgcccgttgagcaaccgcaagttggacagcagcctgtggctcagcagctggacagcgacatgaacttaagcgagctgcccgg
ggagtttattaatatcactgatgagcgtttggctcgacaggaaaccgtgtggaatataacacctaagaatatgtctgttacccatgatatgatgc
tttttaaggccagccggggagaaaggactgtgtactctgtgtgttgggagggaggtggcaggttgaatactagggttctgtgagtttgattaa
ggtacggtgatcaatataagctatgtggtggtggggctatactactgaatgaaaaatgacttgaaattttctgcaattgaaaaataaacacgttg
aaacataacatgcaacaggttcacgattctttattcctgggcaatgtaggagaaggtgtaagagttggtagcaaaagtttcagtggtgtattttc
cactttcccaggaccatgtaaaagacatagagtaagtgcttacctcgctagtttctgtggattcactagaatccactttggccgcggctcgagg
agggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgct
ccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcct
ctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttggg
tgaattttggctcattcctcctttctataggattgaggtcagagcGACATTGATTATTGACTAGTTATTAATAGTA
ATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACT
TACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAAT
AATGACGTATGTTCCCATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGT
GGAGTATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAG
TACGCCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTA
CATGACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATT
ACCATGGTGATGCGGTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCA
CGGGGATTTCCAAGTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGAACCAA
AATCAACGGGACTTTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGG
CGGTAGGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTCTCCCTATCAGTGATAG
AGATCTCCCTATCAGTGATAGAGATCGTCGACGAGCTCGTTTAGTGAACCGTCAGAT
CGCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATC
CAGCCTgaacgcgcagccgccatgGACTACAAAGACGATGACGACAAGGGAGGAGGCGGAG
GCgttccttctcagagactctcccgaactagcagcatttcctccaacgaggatcccgcagagagccacattctggaactcgaagcggtctc
agacaccaacacagactgcgatctggaccctatggagggcagcgaagaacactccacagatggagagatttcatcctcagaggaggagg
atgaagatccaactccggcccacgccatacctgcacggccctccagcgtggtcataacccctacctcggcatcgtttgtgattcccagaaag
aagtgggacctacaggacaagacagtcacattgcatcgctcacccctgtgcagggacgaggacgagaaggaggagactggcaactcct
cttacaccagaggccacaaaaggcgacgcggagaggtccatggctgcaccgatgaaagttatggcaagcgccgacacctgcccccggg
agcaagagcgcccagagccccaagggcccccagggtgcctagagcaccgaggtctccaagagctccccgaagcaacagagcaacca
gaggtccccggtcagaatctcgaggggccggcaggagcacaaggaagcaggcgaggcaagaacgcagccagaggcccctgccgaa
caaaccgtggtttgacatgagtctggttaagcctgtctccaagattacatttgtcaccttgcccagcccccctggcctctctgaccctagagccc
atccaagacccgttcctacagtcgatgctggcggtggccgcccatccagagattggagcctggcagaaagtgcaacccagacacgagct
gcgcaggagctacaagacactacgtgagtttttcaccaagtcaaccaacaaggacacatggctggatgcacgcatgcaggcgatccagaa
cgcggggctctgcaccctggtggccatgctagaagagaccatcttttggctccaggagatcacctaccacggcgacctgcccctagctccc
gcggaagacatcctcctggcctgcgccatgagtcttagcaaggtgatcctgaccaagctcaaagagctggcaccctgcttccttcctaacac
gcgagactacaactttgtgaagcaactcttctacatcacctgtgccacggcccgtcaaaacaaggtggtggagaccctgagcagctcatatg

tgaagcagcccctctgtctcttggcagcatatgcggcagtagccccagcctacattaacgccaactgcagacggagacacgatgaagttga
attcctgggccactacatcaagaattacaaccctggcacgctaagctcccttttgacagaggccgtggagactcacacacgtgactgccgaa
gtgcatcatgcagccgacttgtcagggccattctctctccgggcactgggtcactaggactgtttttgttcctggattaaatcaataaGGAT
CCACTAGTCCAGTGTGGTGGAATTCTGCAGATATCCAGCACAGTGGCGGCCGCTCGA
GTCTAGAGGGCCCGTTTAAACCCGCTGATCAGCCTCGACTGTGCCTTCTAGTTGCCA
GCCATCTGTTGTTTGCCCCTCCCCCGTGCCTTCCTTGACCCTGGAAGGTGCCACTCCC
ACTGTCCTTTCCTAATAAAATGAGGAAATTGCATCGCATTGTCTGAGTAGGTGTCAT
TCTATTCTGGGGGGTGGGGTGGGGCAGGACAGCAAGGGGGAGGATTGGGAAGACA
ATAGCAGGCATGCTGGGGATGCGGTGGGCTCTATGGTGAGCTATTCCAGAAGTAGT
GAAGAGGCTTTTTTGGAGGCCTAGGCTTTTGCAAAAAGCTCCGGATCGATGCCCGGG
GGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTAATTA
CGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGTCCG
GCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGGGGA
AGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGCAGA
CACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGTTGG
GTGAATTTTGGCTGAGCTATTCCAGAAGTAGTGAAGAGGCTTTTTTGGAGGCCTAGG
CTTTTGCAAAAAGCTCCGGATCGATCATATATGGCAGATATACGCGTTGACATTGAT
TATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATAGCCCATATAT
GGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGA
CCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGAC
TTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTACA
TCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGGCC
CGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTACAT
CTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGTACATCAATGG
GCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCATTGACGTCAA
TGGGAGTTTGTTTTGGCACCAAAATCAACGGGACTTTCCAAAATGTCGTAACAACTC
CGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTCTATATAAGCA
GAGCTCTCTGGCTAACTATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTG
GAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCT
CCGGACTCTAGCGTTTAAACTTAAGCTTGCCACCatgaccgagtacaagcccacggtgcgcctcgccacc
cgcgacgacgtccccagggccgtacgcaccctcgccgccgcgttcgccgactaccccgccacgcgccacaccgtcgatccggaccgc
cacatcgagcgggtcaccgagctgcaagaactcttcctcacgcgcgtcgggctcgacatcggcaaggtgtgggtcgcggacgacggcg
ccgcggtggcggtctggaccacgccggagagcgtcgaagcgggggcggtgttcgccgagatcggcccgcgcatggccgagttgagcg
gttcccggctggccgcgcagcaacagatggaaggcctcctggcgccgcaccggcccaaggagcccgcgtggttcctggccaccgtcgg
cgtctcgcccgaccaccagggcaagggtctgggcagcgccgtcgtgctccccggagtggaggcggccgagcgcgccggggtgcccg
ccttcctggagacctccgcgccccgcaacctccccttctacgagcggctcggcttcaccgtcaccgccgacgtcgaggtgcccgaaggac
cgcgcacctggtgcatgacccgcaagcccggtgcctgaAGCGCGGGGATCTCATGCTGGAGTTCTTCGCC
CACCCCAACTTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCACA
AATTTCACAAATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACTCA
TCAATGTATCTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGCAggTG
TCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGGAA
TTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCCAG
GTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTGCC
ACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGGGA
GCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAGG
GACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGCCGCC
ACCGCGGTGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTAATAC
GACTCACTATAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTATCGA
TAAGCTTGATATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAGTAGA
ACATGAAATAACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAAAAG
ATAATCATGCGTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACACTT
ACCGCATTGACAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCCTAA
ATGCACAGCGACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAATGC
ATGCGTCAATTTTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGATCAT

ATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCATCG
GGGAGGAAGAAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAGTTT
GCCGAGGATGACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATGATG
ATTCGGGAAGGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCCACC
TGGGATACCAGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCGAAG
CGCATCAGCAACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGTGCA
GATTAATGACAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATCCTG
GCTGGATGCCGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGCGCT
TGGCGTAATCATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAATTCC
ACACAACATACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAGTGA
GCTAACTCACATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCTGT
CGTGCCAGCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGTATT
GGGCGCTCTTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGCGGC
GAGCGGTATCAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGGAT
AACGCAGGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAAAA
AGGCCGCGTTGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACAAAA
ATCGACGCTCAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGGCG
TTTCCCCCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCGGAT
ACCTGTCCGCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGTAG
GTATCTCAGTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACCCCC
CGTTCAGCCCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCCGGT
AAGACACGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGCGA
GGTATGTAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACACTA
GAAGGACAGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAGAG
TTGGTAGCTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGTTTG
CAAGCAGCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATCTTTT
CTACGGGGTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCATGA
GATTATCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTTAAAT
CAATCTAAAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCAGTG
AGGCACCTATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCCCCGT
CGTGTAGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAATGAT
ACCGCGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAGCCG
GAAGGGCCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCTATTA
ATTGTTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAACGTTG
TTGCCATTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCATTCAG
CTCCGGTTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAAAAGC
GGTTAGCTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGTTATC
ACTCATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAGATG
CTTTTCTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCGGCG
ACCGAGTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCAGAA
CTTTAAAAGTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGATCT
TACCGCTGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTCAG
CATCTTTTACTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGCCG
CAAAAAAGGGAATAAGGGCGACACGGAAATGTTGAATACTCAT

Example 3: Design and Validation of Inducible Rep Genes

(106) Producing inducible Rep proteins can be challenging due to the critical maintenance of the ratio of Rep78 and Rep52 protein expression needed for high titer AAV quality and quantity. In addition, the p19 promoter required for Rep52 expression is embedded in the coding region of Rep78, which is difficult for direct promoter engineering. To overcome these challenges, two strategies were developed.

(107) First, the expression cassettes of Rep78 and Rep52 were split from overlapped genes into two separate ones, each of which was driven by one of the following viral or non-viral promoters with diverse strength, including human cytomegalovirus (CMV) immediate early enhancer-promoter, CMV immediate early enhancer fused chicken β-actin promoter (CAG), human ubiquitin C (UbC) promoter and Rous sarcoma virus (RSV) promoter. To convert these constitutive promoters into inducible versions, two

tetracycline operator sequences (TetO.sub.2) were inserted between the TATA box and transcriptional start site (TSS).

(108) The original p19 promoter in the Rep78 open reading frame (ORF) was silenced by changing six nucleotides in three core regulatory elements required for p19 activity (SP1, TATA-1, and TATA-2 sites). These changes did not alter the Rep78 protein sequence. To increase vector stability and Rep52 expression, the DNA sequence of coding region of Rep52 was optimized for human cell codon usage while retaining the same protein sequence, which significantly reduced the sequence identity of the Rep52 sequence with the Rep78 sequence, from 100% to 80.1%.

(109) To improve the expression of Rep genes, a Kozak sequence or non-canonical start codon ACG was applied to enhance or reduce the protein expression, respectively. Sixteen inducible Rep vectors were designed in piggyBac transposon vectors with core insulators flanking the Rep78 and Rep52 cassettes, as well as an expression cassette for human PGK promoter driven blasticidin antibiotic resistance gene (BSD) (See FIG. **5**A-**5**P). A summary of the inducible promoter designs for Rep genes are shown in Table 1.

(110) TABLE-US-00014 TABLE 1 iCMV/ iCMV/ iCMV no Kozak ACG iCAG iUbC iRSV PB- Rep78 x hiRep-1 Rep52 x PB- Rep78 x hiRep-2 Rep52 x PB- Rep78 x hiRep-3 Rep52 x PB- Rep78 x hiRep-4 Rep52 x PB- Rep78 x hiRep-5 Rep52 x PB- Rep78 x hiRep-6 Rep52 x PB- Rep78 x hiRep-7 Rep52 x PB- Rep78 x hiRep-8 Rep52 x PB- Rep78 x hiRep-9 Rep52 x PB- Rep78 x hiRep-10 Rep52 x PB- Rep78 x hiRep-11 Rep52 x PB- Rep78 x hiRep-12 Rep52 x PB- Rep78 x hiRep-13 Rep52 x PB- Rep78 x hiRep-14 Rep52 x PB- Rep78 x hiRep-15 Rep52 x PB- Rep78 x hiRep-16 Rep52 x

(111) The sequence of the PB-hiRep-1 #(11,866 bp) vector is provided below:

(112) TABLE-US-00015 PB-hiRep-1# (11,866 bp) (SEQ ID NO: 14)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaac
ttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgct
agggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tgacattgattattgactagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggta
aatggcccgcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggactttccatt
gacgtcaatgggtggagtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatga
cggtaaatggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccat
ggtgatgcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggag
tttgttttggaaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtggg
aggtctatataagcagagctctccctatcagtgatagagatctccctatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagat
cgcctggagacgccatccacgctgttttgacctccatagaagacaccgggaccgatccagcctgaacgcgcagccgccaatggatgccg
ggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtggccgagaag
gaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagcgcgactttc
tgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgcacgtgctc
gtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcgggatcga
gccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggggggaacaaggtggtggatgagtgctacatcccca
attacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcacggagcgt
aaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatgcgccggt
gatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtggatccagg

aggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaagattatgag
cctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaactaaacg
ggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgggcctgca
actaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactttcccttca
acgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattctcggagga
agcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaacatgtgcgc
cgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctggatcatg
actttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattctacgtcaaa
aagggtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcagccatcga
cgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgc
agacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaatctcaac
ccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgcgatctggt
caatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaa
gaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaat
aaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaagcccc
cagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccc
cgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcct
gcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctatag
gattgaggtcagagcGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCAT
TAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGC
CTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCA
TAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAA
CTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACG
TCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACT
TTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTT
TTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCT
CCACCCCATTGACGTCAATGGGAGTTTGTTTTGGAACCAAAATCAACGGGACTTTCC
AAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGT
GGGAGGTCTATATAAGCAGAGCTCTCCCTATCAGTGATAGAGATCTCCCTATCAGTG
ATAGAGATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGCCATC
CACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTatcccaattctgatgcgc
cggtgatcagatcaaaaacttcagccaggtacgccaccatggaactggtcggatggctggtggataagggcatcacaagcgagaagcaat
ggatccaggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgccggcaa
aatcatgagcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataagatcctg
gaactgaacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatctggctg
ttcggacctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgag
aacttccccttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggcta
tcctgggaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagcaataccaa
catgtgcgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacg
gctggatcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtggaacacga
gttctacgtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgt
ggcccagcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacc
tgatgctgtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatgcttccct
gtgtccgagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcc
tgtaccgcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgc
cagccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatc
gcattgtctgagtaggtgtcattctattctggggggtggggtggggcaggacagcaagggggaggattgggaagacaatagcaggcatgc
tggggatgcggtgggctctatggaccttttttagggcccattggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagca
gcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggg
gagcgccggaccggagcggagccccgggcggctcgctgctgcccctagcgggggggacgtaattacatccctgggggctttggg
ggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcact
atagggcgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgcgcctttt
ccaaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcaca
ttcttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcggga

aggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccaggga
gcaatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaagggg
cggtgcgggaggcggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacg
tcggcagtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatcagcac
gtgttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaag
aatccaccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctctagcga
cggccgcatcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagct
ggcaacctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgc
atcctgggatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggag
ggctaaagcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcac
aaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacc
taggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctt
ttttagggcccattggtatggctttttccccgtatccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatc
ccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcggagcccc
gggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgc
caccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccc
cctcgaggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatatatatataataagttatcacgtaagtag
aacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagt
tcaaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacggatt
cgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatc
atatcgtcgggtctttttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgag
gttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaag
gtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccggatgg
tcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggtgcggc
gctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgggcgcg
cttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaag
cctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcatta
atgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggc
tgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaa
ggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcga
cgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccga
ccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtag
gtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtccaaccc
ggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagttcttgaa
gtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctc
ttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatc
ctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttcacctag
atccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcacct
atctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggccccagt
gctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcagaagtggt
cctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgcc
attgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatcccccat
gttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggcagcactg
cataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccg
agttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcggggcga
aaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtt
tctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat
(113) The sequence of the PB-hiRep-2 #vector is provided below:
(114) TABLE-US-00016 PB-hiRep-2# (11,861 bp) (SEQ ID NO: 15)
actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga

acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaa
cttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgc
tagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tgacattgattattgactagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggta
aatggcccgcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggactttccatt
gacgtcaatgggtggagtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatga
cggtaaatggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccat
ggtgatgcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggag
tttgttttggaaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtggg
aggtctatataagcagagctctccctatcagtgatagagatctccctatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagat
cgcctggagacgccatccacgctgttttgacctccatagaagacaccgggaccgatccagcctgaacgcgcagccgccacgccggggtt
ttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtggccgagaaggaat
gggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagcgcgactttctgac
ggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgcacgtgctcgtgg
aaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcgggatcgagccg
actttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaacaaggtggtggatgagtgctacatccccaattac
ttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcacggagcgtaaac
ggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatgcgccggtgatc
agatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtggatccaggagga
ccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaagattatgagcctga
ctaaaaccgcccccgactacctgggggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaactaaacgggtac
gatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgggcctgcaactac
cgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactttcccttcaacgac
tgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattctcggaggaagcaa
ggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaacatgtgcgccgtgat
tgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctggatcatgactttg
ggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattctacgtcaaaaagg
gtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcagccatcgacgtc
agacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgcagac
aatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaatctcaacccgt
ttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgcgatctggtcaat
gtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaa
cagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaag
catttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaagcccccagg
gatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgag
ccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcag
acacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattg
aggtcagagcGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAG
TTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTG
GCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAG
TAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTG
CCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCA
ATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTC
CTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTG
GCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCA

CCCCATTGACGTCAATGGGAGTTTGTTTTGGAACCAAAATCAACGGGACTTTCCAAA
ATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGG
AGGTCTATATAAGCAGAGCTCTCCCTATCAGTGATAGAGATCTCCCTATCAGTGATA
GAGATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGCCATCCA
CGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTatcccaattctgatgcgccggt
gatcagatcaaaaacttcagccaggtacgccaccatggaactggtcggatggctggtggataagggcatcacaagcgagaagcaatggat
ccaggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgccggcaaaatca
tgagcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataagatcctggaac
tgaacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatctggctgttcg
gacctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgagaact
tccccttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggctatcct
gggaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagcaataccaacat
gtgcgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacggct
ggatcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtggaacacgagtt
ctacgtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgtgg
cccagcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacctg
atgctgtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatgcttccctgt
gtccgagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcctg
taccgcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgcca
gccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgc
attgtctgagtaggtgtcattctattctggggggtggggggggcaggacagcaagggggaggattgggaagacaatagcaggcatgctg
gggatgcggtgggctctatgggaccttttttagggcccattggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagcag
cgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggg
agcgccggaccggagcggagccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggg
gggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcacta
tagggcgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgcgcctttttc
caaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacatt
cttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaa
ggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggag
caatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggc
ggtgcgggaggcggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgt
cggcagtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacg
tgttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaaga
atccaccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctctagcgac
ggccgcatcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagctg
gcaacctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgca
tcctgggatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggag
ggctaaagcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcac
aaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatac
ctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctt
ttttagggcccattggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatc
ccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcggagcccc
gggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgc
caccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccc
cctcgaggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatatatatataataagttatcacgtaagtag
aacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagt
tcaaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacggatt
cgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatc
atatcgtcgggtcttttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgag
gttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaag
gtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccggatgg
tcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggtgcggc
gctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgggcgcg
cttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaag

cctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcatta
atgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggc
tgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaa
ggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcga
cgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccga
ccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtag
gtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtccaaccc
ggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagttcttgaa
gtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctc
ttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatc
ctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttcacctag
atccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcacct
atctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggccccag
tgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcagaagtggt
cctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgcc
attgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatcccccat
gttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggcagcactg
cataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccg
agttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcggggcga
aaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtt
tctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat

(115) The sequence of the PB-hiRep-3 #vector is provided below:

(116) TABLE-US-00017 PB-hiRep-3# (11,580 bp) (SEQ ID NO: 16)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaac
ttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgct
agggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tggcctccgcgccgggttttggcgcctcccgcgggcgcccccctcctcacggcgagcgctgccacgtcagacgaagggcgcagcgagc
gtcctgatccttccgcccggacgctcaggacagcggcccgctgctcataagactcggccttagaaccccagtatcagcagaaggacatttt
aggacgggacttgggtgactctagggcactggttttctttccagagagcggaacaggcgaggaaaagtagtcccttctcggcgattctgcg
gagggatctccgtggggcggtgaacgccgatgattatataaggacgcgccggtccctatcagtgatagagatctccctatcagtgatagag
agtgtggcacagctagttccgtcgcagccgggatttgggtcgcggttcttgtttgtggatcgctgtgatcgtcacttgacgaacgcgcagccg
ccatgccggggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgacagctttgtgaactggggg
ccgagaaggaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcaccccctgaccgtggccgagaagctgcagc
gcgactttctgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgc
acgtgctcgtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcg
ggatcgagccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaacaaggtggtggatgagtgctac
atccccaattacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcac
ggagcgtaaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatg
cgccggtgatcagatcaaaaacttcagccaggtacatggagctggtcgggggctcgtggacaaggggattacctcggagaagcagtgg
atccaggaggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaaga

ttatgagcctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaa
ctaaacgggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgg
gcctgcaactaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactt
tcccttcaacgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattct
cggaggaagcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaaca
tgtgcgccgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctg
gatcatgactttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattcta
cgtcaaaaagggtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcag
ccatcgacgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtt
tccctgcagacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcaga
atctcaacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgc
gatctggtcaatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagt
ccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaattt
cacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccca
aagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctcccccg
catccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctcttt
gagcctgcagacacctgggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctccttt
ctataggattgaggtcagagcGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGG
TCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGC
CCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTT
CCCATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGG
TAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATT
GACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGG
GACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGC
GGTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAA
GTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGAACCAAAATCAACGGGACT
TTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTAC
GGTGGGAGGTCTATATAAGCAGAGCTCTCCCTATCAGTGATAGAGATCTCCCTATCA
GTGATAGAGATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGC
CATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTatcccaattctg
atgcgccggtgatcagatcaaaaacttcagccaggtacgccaccatggaactggtcggatggctggtggataagggcatcacaagcgag
aagcaatggatccaggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgc
cggcaaaatcatgagcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataa
gatcctggaactgaacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccat
ctggctgttcggacctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggaca
aacgagaacttccccttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgcc
aaggctatcctgggaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctaccсctgtgatcgtgaccagc
aataccaacatgtgcgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctca
ccagacggctggatcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtgg
aacacgagttctacgtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcggg
aaagcgtggcccagcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggc
atgaacctgatgctgtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaat
gcttccctgtgtccgagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgcc
agacgcctgtaccgcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgcctt
ctagttgccagccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaa
attgcatcgcattgtctgagtaggtgtcattctattctggggggtggggtggggcaggacagcaagggggaggattgggaagacaatagca
ggcatgctggggatgcggtgggctctatgggacctttttttagggcccattggtatggctttttccccgtatccccccaggtgtctgcaggctca
aagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggat
gcgggggggagcgccggaccggagcggagccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggg
gctttggggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatac
gactcactatagggcgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggtt
gcgccttttccaaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggt
ctcgcacattcttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaa
gtcgggaaggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgc

cagggagcaatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccggg
aaggggcggtgcgggaggcggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggag
cgcacgtcggcagtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgat
cagcacgtgttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctc
aagaagaatccaccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctc
tagcgacggccgcatcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcg
gcagctggcaacctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctc
gatctgcatcctgggatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgt
gtgggagggctaaagcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaata
gcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctg
atgtatacctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactg
gattccttttttagggcccattggtatggctttttccccgtatccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaa
gcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggggagcgccggaccggagcgga
gccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttggggggggggctgtccctctagagcg
gccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccggg
ccccccctcgaggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatatatatatataataagttatcacgt
aagtagaacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgt
tatagttcaaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcga
cggattcgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatca
ggatcatatcgtcgggtcttttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccg
cgaggttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgg
gaaggtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccg
gatggtcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggt
gcggcgctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgg
gcgcgcttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagt
gtaaagcctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagc
tgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtc
gttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtga
gcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaa
aaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcct
gttccgaccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttc
ggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtc
caacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagtt
cttgaagtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttgg
tagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaaga
agatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttca
cctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgag
gcacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggc
cccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcaga
agtggtcctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgtt
gttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatc
ccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggca
gcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcgg
cgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcg
gggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcac
cagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat

(117) The sequence of the PB-hiRep-4 #vector is provided below:

(118) TABLE-US-00018 PB-hiRep-4# (11,702 bp) (SEQ ID NO: 17)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc

acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaa
cttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccg
ctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tctgctccctgcttgtgtgttggaggtcgctgagtagtgcgcgagcaaaatttaagctacaacaaggcaaggcttgaccgacaattgcatgaa
gaatctgcttagggttaggcgttttgcgctgcttcgcgatgtacgggccagatatacgcgtatctgaggggactagggtgtgtttaggcgaaa
agcggggcttcggttgtacgcggttaggagtcccctcaggatatagtagtttcgcttttgcatagggagggggaaatgtagtcttatgcaatac
acttgtagtcttgcaacatggtaacgatgagttagcaacatgccttacaaggagagaaaaagcaccgtgcatgccgattggtggaagtaagg
tggtacgatcgtgccttattaggaaggcaacagacaggtctgacatggattggacgaaccactgaattccgcattgcagagataattgtattta
agtgcctagctccctatcagtgatagagatctccctatcagtgatagagatcgatacaataaacgccatttgaccattcaccacattggtgtgc
accgaacgcgcagccgccatgccggggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgaca
gctttgtgaactgggtggccgagaaggaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgt
ggccgagaagctgcagcgcgactttctgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaaggga
gagagctacttccacatgcacgtgctcgtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaact
gattcagagaatttaccgcgggatcgagccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaaca
aggtggtggatgagtgctacatccccaattacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacct
cagcgcctgtttgaatctcacggagcgtaaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagaga
atcagaatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggat
tacctcggagaagcagtggatccaggaggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgcc
ttggacaatgcgggaaagattatgagcctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaat
cggatttataaaattttggaactaaacgggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagagg
aacaccatctggctgtttgggcctgcaactaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaa
actggaccaatgagaactttcccttcaacgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtgg
agtcggccaaagccattctcggaggaagcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatc
gtcacctccaacaccaacatgtgcgccgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaat
ttgaactcacccgccgtctggatcatgactttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggtt
gaggtggagcatgaattctacgtcaaaaagggtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacggg
tgcgcgagtcagttgcgcagccatcgacgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgt
gggcatgaatctgatgctgtttccctgcagacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgttta
gagtgctttcccgtgtcagaatctcaacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgc
cagacgcttgcactgcctgcgatctggtcaatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcagg
tgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaata
aagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactgg
attgagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgct
ccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcct
ctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttggg
tgaattttggctcattcctcctttctataggattgaggtcagagcGACATTGATTATTGACTAGTTATTAATAGTA
ATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACT
TACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAAT
AATGACGTATGTTCCCATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGT
GGAGTATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAG
TACGCCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTA
CATGACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATT
ACCATGGTGATGCGGTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCA
CGGGGATTTCCAAGTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGAACCAA
AATCAACGGGACTTTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGG
CGGTAGGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTCTCCCTATCAGTGATAG

AGATCTCCCTATCAGTGATAGAGATCGTCGACGAGCTCGTTTAGTGAACCGTCAGAT
CGCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATC
CAGCCTatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacgccaccatggaactggtcggatggctggtgga
taagggcatcacaagcgagaagcaatggatccaggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatc
aaggctgctctggacaacgccggcaaaatcatgagcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatat
cagcagcaacagaatctataagatcctggaactgaacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaa
gttcggcaagcggaacaccatctggctgttcggacctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgccttt
ctacggctgcgtgaactggacaaacgagaacttccccttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgac
agctaaggtggtggaatctgccaaggctatcctgggaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccc
taccccctgtgatcgtgaccagcaataccaacatgtgcgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcagga
ccggatgttcaagtttgagctcaccagacggctggatcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggcc
aaagaccacgttgttgaggtggaacacgagttctacgtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatca
gcgagcctaagagagtgcgggaaagcgtggcccagcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaa
caagtgcagccgccacgtgggcatgaacctgatgctgtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacaca
cggccagaaagattgcctggaatgcttccctgtgtccgagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatc
caccacatcatgggcaaggtgccagacgcctgtaccgcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttg
tgatgggaattctgtgctgtgccttctagttgccagccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccact
gtcctttcctaataaaatgaggaaattgcatcgcattgtctgagtaggtgtcattctattctggggggtggggggggcaggacagcaagggg
gaggattgggaagacaatagcaggcatgctggggatgcggtgggctctatgggaccttttttagggcccattggtatggctttttccccgtat
ccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccc
cgcacgctgccggctcggggatgcggggggagcgccggaccggagcggagccccgggcggctcgctgctgccccctagcgggggga
gggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttta
gtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctag
ggggtatccccggggttggggttgcgccttttccaaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaa
cgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccgg
cgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcacta
gtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggc
gcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgt
tccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccggga
gcttgtatatccattttcggatctgatcagcacgtgttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaac
gccaccatggccaagcctttgtctcaagaagaatccaccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagacta
cagcgtcgccagcgcagctctctctagcgacggccgcatcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtg
gtgctgggcactgctgctgctgcggcagctggcaacctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcg
gacggtgccgacaggtgcttctcgatctgcatcctgggatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattc
gtgaattgctgccctctggttatgtgtgggagggctaaagcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagctt
ataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatg
tatcttatcatgtctgtagctgatgtatacctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagcc
aggtttagccccggaattgactggattcctttttttagggcccattggtatggctttttccccgtatccccccaggtgtctgcaggctcaaagagc
agcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggg
ggagcgccggaccggagcggagccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgg
gggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactca
ctatagggcgaattgggtaccgggccccccctcgaggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaa
aatatatatataataagttatcacgtaagtagaacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatc
atgcgtcattttgactcacgcggtcgttatagttcaaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgac
tgagatgtcctaaatgcacagcgacggattcgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatc
tttctagggttaatctagctgcatcaggatcatatcgtcgggtctttttttccggctcagtcatcgcccaagctggcgctatctgggcatcgggga
ggaagaagcccgtgccttttcccgcgaggttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaa
cgcacgtttaccatgatgattcgggaaggtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcg
cggcttttccggacacagttccggatggtcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccgg
tgtgcagattaatgacagcggtgcggcgctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggat
accccgtgagttacccggcgggcgcgcttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaaca
tacgagccggaagcataaagtgtaaagcctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagt
cgggaaacctgtcgtgccagctgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgct
cactgactcgctgcgctcggtcgttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggat

aacgcaggaaagaacatgtgagcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccg
cccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctgg
aagctccctcgtgcgctctcctgttccgaccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagct
cacgctgtaggtatctcagttcggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttat
ccggtaactatcgtcttgagtccaacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggta
tgtaggcggtgctacagagttcttgaagtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagtt
accttcggaaaaagagttggtagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgc
agaaaaaaaggatctcaagaagatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatga
gattatcaaaaaggatcttcacctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagtt
accaatgcttaatcagtgaggcacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatac
gggagggcttaccatctggccccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagcc
ggaagggccgagcgcagaagtggtcctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgcc
agttaatagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacg
atcaaggcgagttacatgatcccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagt
gttatcactcatggttatggcagcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtc
attctgagaatagtgtatgcggcgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctc
atcattggaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgat
cttcagcatcttttactttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaa
atgttgaatactcat

(119) The sequence of the PB-hiRep-5 #vector is provided below:

(120) TABLE-US-00019 PB-hiRep-5# (11,865 bp) (SEQ ID NO: 18)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaac
ttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgct
agggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtccccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATA
GCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGAC
CGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGC
CAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACT
TGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACG
GTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTT
GGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGT
ACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCA
TTGACGTCAATGGGAGTTTGTTTTGGAACCAAAATCAACGGGACTTTCCAAAATGTC
GTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTC
TATATAAGCAGAGCTCTCCCTATCAGTGATAGAGATCTCCCTATCAGTGATAGAGAT
CGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGCCATCCACGCTGT
TTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTgAACGCGCAGCCGCCaatgg
ATGccggggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtgg
ccgagaaggaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagc
gcgactttctgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgc

acgtgctcgtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcg
ggatcgagccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaacaaggtggtggatgagtgctac
atccccaattacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcac
ggagcgtaaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatg
cgccggtgatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtgg
atccaggaggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaaga
ttatgagcctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaa
ctaaacgggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgg
gcctgcaactaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactt
tcccttcaacgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattct
cggaggaagcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaaca
tgtgcgccgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctg
gatcatgactttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattcta
cgtcaaaaagggtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcag
ccatcgacgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtt
tccctgcagacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcaga
atctcaacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgc
gatctggtcaatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagt
ccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaattt
cacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagccccccccca
aagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccg
catccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctcttt
gagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctccttt
ctataggattgaggtcagagcgacattgattattgactagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttcc
gcgttacataacttacggtaaatggcccgcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaac
gccaatagggactttccattgacgtcaatgggtggagtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacg
ccccctattgacgtcaatgacggtaaatggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgt
attagtcatcgctattaccatggtgatgcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccacc
ccattgacgtcaatgggagtttgttttggaaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcg
gtaggcgtgtacggtgggaggtctatataagcagagctctccctatcagtgatagagatctccctatcagtgatagagatcgtcgacgagctc
gtttagtgaaccgtcagatcgcctggagacgccatccacgctgttttgacctccatagaagacaccgggaccgatccagcctatcccaattct
gatgcgccggtgatcagatcaaaaacttcagccaggtacaatggatggaactggtcggatggctggtggataagggcatcacaagcgaga
agcaatggatccaggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgcc
ggcaaaatcatgagcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataag
atcctggaactgaacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatc
tggctgttcggacctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaa
acgagaacttccccttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgcca
aggctatcctgggaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagca
ataccaacatgtgcgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcac
cagacggctggatcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtgga
acacgagttctacgtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcggga
aagcgtggcccagcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggca
tgaacctgatgctgtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatg
cttccctgtgtccgagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgcca
gacgcctgtaccgcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttct
agttgccagccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaatt
gcatcgcattgtctgagtaggtgtcattctattctggggggtggggggggcaggacagcaagggggaggattgggaagacaatagcagg
catgctggggatgcggtgggctctatgggacctttttttagggcccattggtatggctttttccccgtatcccccccaggtgtctgcaggctcaaa
gagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgc
ggggggagcgccggaccggagcggagccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggc
tttggggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacga
ctcactatagggcgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgc
gccttttccaaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtct
cgcacattcttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagt

cgggaaggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgcca
gggagcaatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaa
ggggcggtgcgggaggcggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagc
gcacgtcggcagtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatc
agcacgtgttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctca
agaagaatccaccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctct
agcgacggccgcatcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcg
gcagctggcaacctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctc
gatctgcatcctgggatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgt
gtgggagggctaaagcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaata
gcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctga
tgtatacctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactg
gattcctttttttagggcccattggtatggctttttccccgtatccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaa
gcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcgga
gccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcg
gccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccggg
ccccccctcgaggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatatatatatataataagttatcacgt
aagtagaacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgt
tatagttcaaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcga
cggattcgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatca
ggatcatatcgtcgggtcttttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccg
cgaggttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgg
gaaggtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccg
gatggtcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggt
gcggcgctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgg
gcgcgcttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagt
gtaaagcctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagc
tgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtc
gttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtga
gcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaa
aaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcct
gttccgaccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttc
ggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtc
caacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagtt
cttgaagtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttgg
tagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaaga
agatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttca
cctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgagg
cacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggcc
ccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcaga
agtggtcctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgtt
gttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatc
ccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggca
gcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcgg
cgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcg
gggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcac
cagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat

(121) The sequence of the PB-hiRep-6 #vector is provided below:

(122) TABLE-US-00020 PB-hiRep-6# (11,860 bp) (SEQ ID NO: 19)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga

acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactt
acatactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaa
cttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgc
tagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATA
GCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGAC
CGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGC
CAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACT
TGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACG
GTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTT
GGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGT
ACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCA
TTGACGTCAATGGGAGTTTGTTTTGGAACCAAAATCAACGGGACTTTCCAAAATGTC
GTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTC
TATATAAGCAGAGCTCTCCCTATCAGTGATAGAGATCTCCCTATCAGTGATAGAGAT
CGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGCCATCCACGCTGT
TTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTgAACGCGCAGCCGCCacgccg
gggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtggccgagaag
gaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagcgcgactttc
tgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgcacgtgctc
gtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcgggatcga
gccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaacaaggtggtggatgagtgctacatcccca
attacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcacggagcgt
aaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatgcgccggt
gatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtggatccagg
aggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaagattatgag
cctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaactaaacg
ggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgggcctgca
actaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactttcccttca
acgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattctcggagga
agcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaacatgtgcgc
cgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctggatcatg
actttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattctacgtcaaa
aagggtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcagccatcga
cgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgc
agacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaatctcaac
ccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgcgatctggt
caatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaa
gaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaat
aaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaagcccc
cagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccc
cgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcct
gcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctatag
gattgaggtcagagcgacattgattattgactagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgtta
ataacttacggtaaatggcccgcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaa
ctagggactttccattgacgtcaatgggtggagtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccc

tattgacgtcaatgacggtaaatggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagt
catcgctattaccatggtgatgcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccatt
gacgtcaatgggagtttgttttggaaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtag
gcgtgtacggtgggaggtctatataagcagagctctccctatcagtgatagagatctccctatcagtgatagagatcgtcgacgagctcgttt
agtgaaccgtcagatcgcctggagacgccatccacgctgttttgacctccatagaagacaccgggaccgatccagcctatcccaattctgat
gcgccggtgatcagatcaaaaacttcagccaggtacaatggatggaactggtcggatggctggtggataagggcatcacaagcgagaag
caatggatccaggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgccgg
caaaatcatgagcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataagat
cctggaactgaacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatctg
gctgttcggacctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaa
cgagaacttccccttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgccaa
ggctatcctgggaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagcaat
accaacatgtgcgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcacca
gacggctggatcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtggaac
acgagttctacgtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcgggaaa
gcgtggcccagcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggcatg
aacctgatgctgtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatgct
tccctgtgtccgagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgccaga
cgcctgtaccgcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctag
ttgccagccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgc
atcgcattgtctgagtaggtgtcattctattctggggggtggggtggggcaggacagcaagggggaggattgggaagacaatagcaggca
tgctggggatgcggtgggctctatgggacctttttagggcccattggtatggcttttccccgtatcccccaggtgtctgcaggctcaaaga
gcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgg
ggggagcgccggaccggagcggagccccgggcggctcgctgctgcccccctagcgggggagggacgtaattacatccctgggggcttt
gggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgact
cactatagggcgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgcg
ccttttccaaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctc
gcacattcttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtc
gggaaggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgcca
gggagcaatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaa
ggggcggtgcgggaggcggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagc
gcacgtcggcagtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatc
agcacgtgttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctca
agaagaatccaccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctct
agcgacggccgcatcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcg
gcagctggcaacctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctc
gatctgcatcctgggatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgt
gtgggagggctaaagcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaata
gcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctga
tgtatacctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactg
gattcctttttagggcccattggtatggcttttccccgtatcccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaa
gcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggagcgccggaccggagcgga
gccccgggcggctcgctgctgcccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcg
gccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccggg
cccccccctcgaggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatatatatataataagttatcacgt
aagtagaacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgt
tatagttcaaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcga
cggattcgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatca
ggatcatatcgtcgggtctttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccg
cgaggttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgg
gaaggtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccg
gatggtcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggt
gcggcgctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgg
gcgcgcttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagt

gtaaagcctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagc
tgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtc
gttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtga
gcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaa
aaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcct
gttccgaccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttc
ggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtc
caacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagtt
cttgaagtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttgg
tagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaaga
agatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttca
cctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgagg
cacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggcc
ccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcaga
agtggtcctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgtt
gttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatc
ccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggca
gcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcgg
cgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcg
gggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcac
cagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat
(123) The sequence of the PB-hiRep-7 #vector is provided below:
(124) TABLE-US-00021 PB-hiRep-7# (11,579 bp) (SEQ ID NO: 20)
actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaa
cttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccg
ctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tggcctccgcgccgggttttggcgcctcccgcgggcgcccccctcctcacggcgagcgctgccacgtcagacgaagggcgcagcgagc
gtcctgatccttccgcccggacgctcaggacagcggcccgctgctcataagactcggccttagaaccccagtatcagcagaaggacatttt
aggacgggacttgggtgactctagggcactggttttctttccagagagcggaacaggcgaggaaaagtagtcccttctcggcgattctgcg
gagggatctccgtggggcggtgaacgccgatgattatataaggacgcgccggTCCCTATCAGTGATAGAGAtcTCC
CTATCAGTGATAGAGAgtgtggcacagctagttccgtcgcagccgggatttgggtcgcggttcttgtttgtggatcgctgtgat
cgtcacttgacGAACGCGCAGCCGCCATGccggggttttacgagattgtgattaaggtccccagcgaccttgacgagcatc
tgcccggcatttctgacagctttgtgaactgggtggccgagaaggaatgggagttgccgccagattctgacatggatctgaatctgattgagc
aggcacccctgaccgtggccgagaagctgcagcgcgactttctgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgt
gcaatttgagaagggagagagctacttccacatgcacgtgctcgtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtc
agattcgcgaaaaactgattcagagaatttaccgcgggatcgagccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgc
cggaggcgggaacaaggtggtggatgagtgctacatccccaattacttgctccccaaaacccagcctgagctccaatgggcatggaccaa
catggaacagtacctcagcgcctgtttgaatctcacggagcgtaaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcagga
gcagaacaaagagaatcagaatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctc
gtggacaaggggattacctcggagaagcagtggatccaggaggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcc

caaatcaaggctgccttggacaatgcgggaaagattatgagcctgactaaaaccgcccccgactacctggtgggccagcagcccgtgga
ggacatttccagcaatcggatttataaaattttggaactaaacgggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaa
aagttcggcaagaggaacaccatctggctgtttgggcctgcaactaccgggaagaccaacatcgcggaggccatagcccacactgtgcc
cttctacgggtgcgtaaactggaccaatgagaactttcccttcaacgactgtgtcgacaagatggtgatctggtgggaggaggggaagatg
accgccaaggtcgtggagtcggccaaagccattctcggaggaagcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatag
acccgactcccgtgatcgtcacctccaacaccaacatgtgcgccgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgca
agaccggatgttcaaatttgaactcacccgccgtctggatcatgactttgggaaggtcaccaagcaggaagtcaaagactttttccggtggg
caaaggatcacgtggttgaggtggagcatgaattctacgtcaaaaagggggagccaagaaaagacccgcccccagtgacgcagatata
agtgagcccaaacgggtgcgcgagtcagttgcgcagccatcgacgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaa
caaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgcagacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacg
gacagaaagactgtttagagtgctttcccgtgtcagaatctcaacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcat
atcatgggaaaggtgccagacgcttgcactgcctgcgatctggtcaatgtggatttggatgactgcatctttgaacaataaaatggctaggat
ccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagc
ttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaa
tgtatcttattgactggattgagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagcc
gcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggc
acgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagctt
cccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagcgacattgattattgactagttattaatagtaatca
attacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggtaaatggcccgcctggctgaccgcccaacgaccccc
gcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggactttccattgacgtcaatgggtggagtatttacggtaaactgc
ccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatgacggtaaatggcccgcctggcattatgcccagta
catgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccatggtgatgcggttttggcagtacatcaatgggcg
tggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggagtttgttttggaaccaaaatcaacgggactttcca
aaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtgggaggtctatataagcagagctctccctatcagtga
tagagatctccctatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagatcgcctggagacgccatccacgctgttttgacct
ccatagaagacaccgggaccgatccagcctatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacaatggatggaa
ctggtcggatggctggtggataagggcatcacaagcgagaagcaatggatccaggaggaccaggcctcatatatttcttttaacgccgcta
gcaattccagaagccagatcaaggctgctctggacaacgccggcaaaatcatgagcctgaccaagaccgcccctgactacctggtggga
cagcagcctgtggaagatatcagcagcaacagaatctataagatcctggaactgaacggctacgacccccagtacgccgcctccgtgttcc
tgggctgggctacaaagaagttcggcaagcggaacaccatctggctgttcggacctgccaccacaggcaaaaccaatatcgccgaggcc
atcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgagaacttcccattcaacgactgtgtggacaagatggtgatctggtg
ggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggctatcctgggaggctctaaggtcagggtggatcagaagtgtaaaa
gcagcgcccagattgaccctacccctgtgatcgtgaccagcaataccaacatgtgcgccgtgatcgacggcaacagcaccaccttcgagc
atcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacggctggatcacgacttcggcaaggtgaccaagcaggaggtga
aggatttcttcagatgggccaaagaccacgttgttgaggtggaacacgagttctacgtgaagaagggcggcgccaagaaaagacccgcc
cctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgtggcccagcccagcacatctgatgctgaggccagcatcaact
acgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacctgatgctgtttccctgcagacagtgtgaaagaatgaaccaga
attctaatatctgctttacacacggccagaaagattgcctggaatgcttccctgtgtccgagagccaaccagtgtctgtggtgaaaaaggccta
ccagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcctgtaccgcctgcgacctggtcaacgtggacctggacgactg
catcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgccagccatctgttgtttgcccctcccccgtgccttccttgaccc
tggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgcattgtctgagtaggtgtcattctattctggggggggggtg
gggcaggacagcaagggggaggattgggaagacaatagcaggcatgctggggatgcggtgggctctatgggaccttttttagggcccat
tggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttc
cccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggagcgccggaccggagcggagccccgggcggctcgct
gctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtgg
agctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctgaggcggaa
agaaccagctggggctctagggggtatccccggggttggggttgcgccttttccaaggcagccctgggtttgcgcagggacgcggctgct
ctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcacccggatcttcgccgc
tacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgccggacgtgacaaac
ggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccgcgatgggctgtggccaa
tagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtggggcggtagtgtgggcc
ctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgaccgaatcaccgacctctc
tccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacgtgttgacaattaatcatcggcatagtatatcggcatagtata
atacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaagaatccaccctcattgaaagagcaacggctacaatcaacag

catccccatctctgaagactacagcgtcgccagcgcagctctctctagcgacggccgcatcttcactggtgtcaatgtatatcattttactggg
ggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagctggcaacctgacttgtatcgtcgcgatcggaaatgagaaca
ggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgcatcctgggatcaaagccatagtgaaggacagtgatggac
agccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggagggctaaagcgcggggatctcatgctggagttcttcgccc
accccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagt
tgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacctaggatccggccggcctgcaggtgtcctcacaggaacga
agtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctttttagggcccattggtatggctttttccccgtatccccc
caggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgct
gccggctcggggatgcgggggggagcgccggaccggagcggagccccgggcggctcgctgctgccccctagcgggggagggacgta
attacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggtt
aattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctcgaggtcgacggtatcctcgaggtcgacggtatcgata
agcttgatatctataacaagaaaatatatatatataataagttatcacgtaagtagaacatgaaataacaatataattatcgtatgagttaaatctta
aaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagttcaaaatcagtgacacttaccgcattgacaagcacgcctcac
gggagctccaagcggcgactgagatgtcctaaatgcacagcgacggattcgcgctatttagaaagagagagcaatatttcaagaatgcatg
cgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatcatatcgtcgggtcttttttccggctcagtcatcgcccaagctg
gcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgaggttgaagcggcatggaaagagtttgccgaggatgactgctgc
tgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaaggtgtggccatgcacgcctttaacggtgaactgttcgttcaggcc
acctgggataccagttcgtcgcggcttttccggacacagttccggatggtcagcccgaagcgcatcagcaacccgaacaataccggcgac
agccggaactgccgtgccggtgtgcagattaatgacagcggtgcggcgctgggatattacgtcagcgaggacgggtatcctggctggatg
ccgcagaaatggacatggataccccgtgagttacccgggggcgcgcttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatc
cgctcacaattccacacaacatacgagccggaagcataaagtgtaaagcctggggtgcctaatgagtgagctaactcacattaattgcgttg
cgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtattg
ggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacgg
ttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgc
tggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaag
ataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccgaccctgccgcttaccggatacctgtccgcctttctcccttcggga
agcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttc
agcccgaccgctgcgccttatccggtaactatcgtcttgagtccaacccggtaagacacgacttatcgccactggcagcagccactggtaa
caggattagcagagcgaggtatgtaggcggtgctacagagttcttgaagtggtggcctaactacggctacactagaaggacagtatttggta
tctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtt
tgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactc
acgttaagggattttggtcatgagattatcaaaaaggatcttcacctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatata
tgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccc
cgtcgtgtagataactacgatacgggagggcttaccatctggccccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttat
cagcaataaaccagccagccggaagggccgagcgcagaagtggtcctgcaactttatccgcctccatccagtctattaattgttgccggga
agctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttc
attcagctccggttcccaacgatcaaggcgagttacatgatcccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtc
agaagtaagttggccgcagtgttatcactcatggttatggcagcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgact
ggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacat
agcagaactttaaaagtgctcatcattggaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaac
ccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaag
ggaataagggcgacacggaaatgttgaatactcat

(125) The sequence of the PB-hiRep-8 #vector is provided below:

(126) TABLE-US-00022 PB-hiRep-8# (11,701 bp) (SEQ ID NO: 21)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa

tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaac
ttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgct
agggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tctgctccctgcttgtgtgttggaggtcgctgagtagtgcgcgagcaaaatttaagctacaacaaggcaaggcttgaccgacaattgcatgaa
gaatctgcttagggttaggcgttttgcgctgcttcgcgatgtacgggccagatatacgcgtatctgaggggactagggtgtgtttaggcgaaa
agcggggcttcggttgtacgcggttaggagtcccctcaggatatagtagtttcgcttttgcatagggagggggaaatgtagtcttatgcaatac
acttgtagtcttgcaacatggtaacgatgagttagcaacatgccttacaaggagagaaaaagcaccgtgcatgccgattggtggaagtaagg
tggtacgatcgtgccttattaggaaggcaacagacaggtctgacatggattggacgaaccactgaattccgcattgcagagataattgtattta
agtgcctagcTCCCTATCAGTGATAGAGAtcTCCCTATCAGTGATAGAGAtcgatacaataaacgccat
ttgaccattcaccacattggtgtgcaccGAACGCGCAGCCGCCATGccggggttttacgagattgtgattaaggtccccagc
gaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtggccgagaaggaatgggagttgccgccagattctgacatgg
atctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagcgcgactttctgacggaatggcgccgtgtgagtaaggccc
cggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgcacgtgctcgtggaaaccaccggggtgaaatccatggtttt
gggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcgggatcgagccgactttgccaaactggttcgcggtcacaa
agaccagaaatggcgccggaggcgggaacaaggtggtggatgagtgctacatccccaattacttgctccccaaaacccagcctgagctc
caatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcacggagcgtaaacggttggtggcgcagcatctgacgcac
gtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacatg
gagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtggatccaggaggaccaggcctcatacatctccttcaatgcg
gcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaagattatgagcctgactaaaaccgcccccgactacctggtg
ggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaactaaacgggtacgatccccaatatgcggcttccgtctttc
tgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgggcctgcaactaccgggaagaccaacatcgcggaggcc
atagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactttcccttcaacgactgtgtcgacaagatggtgatctggtgg
gaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattctcggaggaagcaaggtgcgcgtggaccagaaatgcaag
tcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaacatgtgcgccgtgattgacgggaactcaacgaccttcgaac
accagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctggatcatgactttgggaaggtcaccaagcaggaagtcaa
agactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattctacgtcaaaaagggtggagccaagaaaagacccgcccc
cagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcagccatcgacgtcagacgcggaagcttcgatcaactacg
cagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgcagacaatgcgagagaatgaatcagaattcaa
atatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaatctcaacccgtttctgtcgtcaaaaaggcgtatcagaaa
ctgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgcgatctggtcaatgtggatttggatgactgcatctttgaaca
ataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccgga
aaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtg
gtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctag
ggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccg
ggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaa
ggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagcgacattgattattgact
agttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggtaaatggcccgcctggctga
ccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggactttccattgacgtcaatgggtgga
gtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatgacggtaaatggcccgc
ctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccatggtgatgcggttttg
gcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggagtttgttttggaaccaaa
atcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtgggaggtctatataagcag
agctctccctatcagtgatagagatctccctatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagatcgcctggagacgcca
tccacgctgttttgacctccatagaagacaccgggaccgatccagcctatcccaattctgatgcgccggtgatcagatcaaaaacttcagcca
ggtacaatggatggaactggtcggatggctggtggataagggcatcacaagcgagaagcaatggatccaggaggaccaggcctcatata
tttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgccggcaaaatcatgagcctgaccaagaccgcccct
gactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataagatcctggaactgaacggctacgacccccagtac
gccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatctggctgttcggacctgccaccacaggcaaaacc
aatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgagaacttccccttcaacgactgtgtggaca
agatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggctatcctgggaggctctaaggtcagggtgg

atcagaagtgtaaaagcagcgcccagattgaccctaccctgtgatcgtgaccagcaataccaacatgtgcgccgtgatcgacggcaaca
gcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacggctggatcacgacttcggcaaggtgac
caagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtggaacacgagttctacgtgaagaagggcggcgcca
agaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgtggcccagcccagcacatctgatgctg
aggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacctgatgctgtttccctgcagacagtgtg
aaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatgcttccctgtgtccgagagccaaccagtgtctg
tggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcctgtaccgcctgcgacctggtcaacg
tggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgccagccatctgttgtttgcccctcccc
gtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgcattgtctgagtaggtgtcattctatt
ctggggggtggggtggggcaggacagcaagggggaggattgggaagacaatagcaggcatgctggggatgcggtgggctctatggga
cctttttagggcccattggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcg
atcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggggagcgccggaccggagcggagcc
ccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggcc
gccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccc
ccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgcgccttttccaaggcagccctgggtttgcgcag
ggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcacc
cggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgcc
ggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccgcgat
gggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtggggc
ggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgaccga
atcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacgtgttgacaattaatcatcggcatagtat
atcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaagaatccaccctcattgaaagagcaacgg
ctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctctagcgacggccgcatcttcactggtgtcaatgtat
atcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagctggcaacctgacttgtatcgtcgcgatcg
gaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgcatcctgggatcaaagccatagtgaagg
acagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggagggctaaagcgcggggatctcatgctg
gagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttca
ctgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacctaggatccggccggcctgcaggtgtcc
tcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctttttagggcccattggtatggcttttt
ccccgtatcccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggct
gtccccgcacgctgccggctcggggatgcgggggggagcgccggaccggagcggagccccgggcggctcgctgctgccccctagcgg
gggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttc
cctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctcgaggtcgacggtatcctcgaggt
cgacggtatcgataagcttgatatctataacaagaaaatatatatataataagttatcacgtaagtagaacatgaaataacaatataattatcgta
tgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagttcaaaatcagtgacacttaccgcattga
caagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacggattcgcgctatttagaaagagagagcaatat
ttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatcatatcgtcgggtctttttccggctcagt
catcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgaggttgaagcggcatggaaagagtttgccga
ggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaaggtgtggccatgcacgcctttaacggtgaac
tgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccggatggtcagcccgaagcgcatcagcaacccgaa
caataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggtgcggcgctgggatattacgtcagcgaggacgggt
atcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgggcgcgcttggcgtaatcatggtcatagctgtttcctg
tgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaagcctggggtgcctaatgagtgagctaactca
cattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcattaatgaatcggccaacgcgcggggagagg
cggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggctgcggcgagcggtatcagctcactcaaa
ggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaaggccagcaaaaggccaggaaccgtaaa
aaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtggcgaaacccg
acaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccgaccctgccgcttaccggatacctgtccgcc
tttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtaggtcgttcgctccaagctgggctgtgtgca
cgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtccaacccggtaagacacgacttatcgccactggcag
cagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagttcttgaagtggtggcctaactacggctacactagaag
gacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctcttgatccggcaaacaaaccaccgctggtag
cggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatcctttgatcttttctacggggtctgacgctcagt
ggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttcacctagatccttttaaattaaaaatgaagttttaaatcaa

tctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcacctatctcagcgatctgtctatttcgttcatccatag
ttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggccccagtgctgcaatgataccgcgagacccacgctcac
cggctccagatttatcagcaataaaccagccagccggaagggccgagcgcagaagtggtcctgcaactttatccgcctccatccagtctatt
aattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcacgctcgt
cgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatcccccatgttgtgcaaaaaagcggttagctccttcggt
cctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggcagcactgcataattctcttactgtcatgccatccgtaaga
tgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccgagttgctcttgcccggcgtcaatacgggata
ataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttgagat
ccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaa
tgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat

(127) The sequence of the PB-hiRep-9 #vector is provided below:

(128) TABLE-US-00023 PB-hiRep-9# (11, 861 bp) (SEQ ID NO: 22)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactt
acatactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaa
acttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccc
gctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTTCATA
GCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGAC
CGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGC
CAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACT
TGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACG
GTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTT
GGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGCAGT
ACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACCCCA
TTGACGTCAATGGGAGTTTGTTTTGGAACCAAAATCAACGGGACTTTCCAAAATGTC
GTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGGTC
TATATAAGCAGAGCTCTCCCTATCAGTGATAGAGATCTCCCTATCAGTGATAGAGAT
CGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGCCATCCACGCTGT
TTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTgAACGCGCAGCCGCCacgccg
gggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtggccgagaag
gaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagcgcgactttc
tgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgcacgtgctc
gtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcgggatcga
gccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggagggggaacaaggtggtggatgagtgctacatcccca
attacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcacggagcgt
aaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatgcgccggt
gatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtggatccagg
aggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaagattatgag
cctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaactaaacg
ggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgggcctgca
actaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactttcccttca

acgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattctcggagga
agcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaacatgtgcgc
cgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctggatcatg
actttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattctacgtcaaa
aagggtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcagccatcga
cgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgc
agacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaatctcaac
ccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgcgatctggt
caatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaa
gaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaat
aaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaagcccc
cagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccc
cgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcct
gcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctatag
gattgaggtcagagcgacattgattattgactagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgtta
cataacttacggtaaatggcccgcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaa
tagggactttccattgacgtcaatgggtggagtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccc
tattgacgtcaatgacggtaaatggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagt
catcgctattaccatggtgatgcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccatt
gacgtcaatgggagtttgttttggaaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggggtag
gcgtgtacggtgggaggtctatataagcagagctctccctatcagtgatagagatctccctatcagtgatagagatcgtcgacgagctcgttt
agtgaaccgtcagatcgcctggagacgccatccacgctgttttgacctccatagaagacaccgggaccgatccagcctatcccaattctgat
gcgccggtgatcagatcaaaaacttcagccaggtacgccaccAcGgaactggtcggatggctggtggataagggcatcacaagcgaga
agcaatggatccaggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgcc
ggcaaaatcatgagcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataag
atcctggaactgaacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatc
tggctgttcggacctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaa
acgagaacttccccttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgcca
aggctatcctgggaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagca
ataccaacatgtgcgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcac
cagacggctggatcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtgga
acacgagttctacgtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcggga
aagcgtggcccagcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggca
tgaacctgatgctgtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatg
cttccctgtgtccgagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgcca
gacgcctgtaccgcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttct
agttgccagccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaatt
gcatcgcattgtctgagtaggtgtcattctattctggggggtggggtggggcaggacagcaagggggaggattgggaagacaatagcagg
catgctggggatgcggtgggctctatggaccttttttagggcccattggtatggctttttccccgtatcccccaggtgtctgcaggctcaaa
gagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgc
ggggggagcgccggaccggagcggagccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggc
tttggggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacga
ctcactatagggcgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgc
gccttttccaaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtct
cgcacattcttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagt
cgggaaggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgcca
gggagcaatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaa
ggggcggtgcgggaggggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagc
gcacgtcggcagtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatc
agcacgtgttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctca
agaagaatccaccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctct
agcgacggccgcatcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcg
gcagctggcaacctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctc
gatctgcatcctgggatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgt

gtgggagggctaaagcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaata
gcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctga
tgtatacctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactg
gattcctttttttagggcccattggtatggctttttccccgtatccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaa
gcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggggagcgccggaccggagcgga
gccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttggggggggggctgtccctctagagcg
gccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccggg
ccccccctcgaggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatatatatatataataagttatcacgt
aagtagaacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgt
tatagttcaaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcga
cggattcgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatca
ggatcatatcgtcgggtcttttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccg
cgaggttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgg
gaaggtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccg
gatggtcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggt
gcggcgctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgg
gcgcgcttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagt
gtaaagcctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagc
tgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtc
gttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtga
gcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaa
aaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcct
gttccgaccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttc
ggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtc
caacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagtt
cttgaagtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttgg
tagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaaga
agatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttca
cctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgagg
cacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggcc
ccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcaga
agtggtcctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgtt
gttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatc
ccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggca
gcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcgg
cgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcg
gggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcac
cagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat

(129) The sequence of the PB-hiRep-10 #vector is provided below:

(130) TABLE-US-00024 PB-hiRep-10# (11,580 bp) (SEQ ID NO: 23)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactt
acatactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaa
acttttatcgaattcctgcagcccgggggatccactagttctagagggacagccccccccaaagcccccagggatgtaattacgtccctcccccg
ctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca

cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tggcctccgcgccgggttttggcgcctcccgcgggcgcccccctcctcacggcgagcgctgccacgtcagacgaagggcgcagcgagc
gtcctgatccttccgcccggacgctcaggacagcggcccgctgctcataagactcggccttagaaccccagtatcagcagaaggacatttt
aggacgggacttgggtgactctagggcactggttttctttccagagagcggaacaggcgaggaaaagtagtcccttctcggcgattctgcg
gagggatctccgtggggcggtgaacgccgatgattatataaggacgcgccggTCCCTATCAGTGATAGAGAtcTCC
CTATCAGTGATAGAGAgtgtggcacagctagttccgtcgcagccgggatttgggtcgcggttcttgtttgtggatcgctgtgat
cgtcacttgacGAACGCGCAGCCGCCATGccggggttttacgagattgtgattaaggtccccagcgaccttgacgagcatc
tgcccggcatttctgacagctttgtgaactgggtggccgagaaggaatgggagttgccgccagattctgacatggatctgaatctgattgagc
aggcacccctgaccgtggccgagaagctgcagcgcgactttctgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgt
gcaatttgagaagggagagagctacttccacatgcacgtgctcgtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtc
agattcgcgaaaaactgattcagagaatttaccgcgggatcgagccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgc
cggaggcgggaacaaggtggtggatgagtgctacatccccaattacttgctccccaaaacccagcctgagctccaatgggcatggaccaa
catggaacagtacctcagcgcctgtttgaatctcacggagcgtaaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcagga
gcagaacaaagagaatcagaatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctc
gtggacaaggggattacctcggagaagcagtggatccaggaggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcc
caaatcaaggctgccttggacaatgcgggaaagattatgagcctgactaaaaccgcccccgactacctggtgggccagcagcccgtgga
ggacatttccagcaatcggatttataaaattttggaactaaacgggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaa
aagttcggcaagaggaacaccatctggctgtttgggcctgcaactaccgggaagaccaacatcgcggaggccatagcccacactgtgcc
cttctacgggtgcgtaaactggaccaatgagaactttcccttcaacgactgtgtcgacaagatggtgatctggtgggaggaggggaagatg
accgccaaggtcgtggagtcggccaaagccattctcggaggaagcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatag
acccgactcccgtgatcgtcacctccaacaccaacatgtgcgccgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgca
agaccggatgttcaaatttgaactcacccgccgtctggatcatgactttgggaaggtcaccaagcaggaagtcaaagactttttccggtggg
caaaggatcacgtggttgaggtggagcatgaattctacgtcaaaaagggtggagccaagaaaagacccgcccccagtgacgcagatata
agtgagcccaaacgggtgcgcgagtcagttgcgcagccatcgacgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaa
caaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgcagacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacg
gacagaaagactgtttagagtgctttcccgtgtcagaatctcaacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcat
atcatgggaaaggtgccagacgcttgcactgcctgcgatctggtcaatgtggatttggatgactgcatctttgaacaataaaatggctaggat
ccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagc
ttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatca
atgtatcttattgactggattgagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagcc
gcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggc
acgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctggggggatacggggaaaaggcctccaaggccagctt
cccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagcgacattgattattgactagttattaatagtaatca
attacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggtaaatggcccgcctggctgaccgcccaacgaccccc
gcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggactttccattgacgtcaatgggtggagtatttacggtaaactgc
ccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatgacggtaaatggcccgcctggcattatgcccagta
catgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccatggtgatgcggttttggcagtacatcaatgggcg
tggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggagtttgttttggaaccaaaatcaacgggactttcca
aaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtgggaggtctatataagcagagctctccctatcagtga
tagagatctccctatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagatcgcctggagacgccatccacgctgttttgacct
ccatagaagacaccgggaccgatccagcctatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacgccaccAcG
gaactggtcggatggctggtggataagggcatcacaagcgagaagcaatggatccaggaggaccaggcctcatatatttcttttaacgccg
ctagcaattccagaagccagatcaaggctgctctggacaacgccggcaaaatcatgagcctgaccaagaccgcccctgactacctggtgg
gacagcagcctgtggaagatatcagcagcaacagaatctataagatcctggaactgaacggctacgacccccagtacgccgcctccgtgtt
cctgggctgggctacaaagaagttcggcaagcggaacaccatctggctgttcggacctgccaccacaggcaaaaccaatatcgccgagg
ccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgagaacttccccttcaacgactgtgtggacaagatggtgatctgg
tgggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggctatcctgggaggctctaaggtcagggtggatcagaagtgtaaa
agcagcgcccagattgaccctacccctgtgatcgtgaccagcaataccaacatgtgcgccgtgatcgacggcaacagcaccaccttcgag
catcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacggctggatcacgacttcggcaaggtgaccaagcaggaggtg
aaggatttcttcagatgggccaaagaccacgttgttgaggtggaacacgagttctacgtgaagaagggcggcgccaagaaaagacccgc
ccctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgtggcccagcccagcacatctgatgctgaggccagcatcaac
tacgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacctgatgctgtttccctgcagacagtgtgaaagaatgaaccag

aattctaatatctgctttacacacggccagaaagattgcctggaatgcttccctgtgtccgagagccaaccagtgtctgtggtgaaaaaggcct
accagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcctgtaccgcctgcgacctggtcaacgtggacctggacgact
gcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgccagccatctgttgtttgcccctcccccgtgccttccttgacc
ctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgcattgtctgagtaggtgtcattctattctggggggtggggtg
gggcaggacagcaagggggaggattgggaagacaatagcaggcatgctggggatgcggtgggctctatgggaccttttttagggcccat
tggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttc
cccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggagcgccggaccggagcggagccccgggcggctcgct
gctgcccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtgg
agctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctgaggcggaa
agaaccagctggggctctagggggtatccccggggttggggttgcgccttttccaaggcagccctgggtttgcgcagggacgcggctgct
ctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcacccggatcttcgccgc
taccctttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgccggacgtgacaaac
ggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccgcgatgggctgtggccaa
tagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtggggcggtagtgtgggcc
ctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgaccgaatcaccgacctctc
tccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacgtgttgacaattaatcatcggcatagtatatcggcatagtata
atacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaagaatccaccctcattgaaagagcaacggctacaatcaacag
catccccatctctgaagactacagcgtcgccagcgcagctctctctagcgacggccgcatcttcactggtgtcaatgtatatcattttactggg
ggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagctggcaacctgacttgtatcgtcgcgatcggaaatgagaaca
ggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgcatcctgggatcaaagccatagtgaaggacagtgatggac
agccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggagggctaaagcgcggggatctcatgctggagttcttcgccc
accccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagt
tgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacctaggatccggccggcctgcaggtgtcctcacaggaacgaa
gtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctttttagggcccattggtatggctttttccccgtatcccccc
aggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgct
gccggctcggggatgcgggggagcgccggaccggagcggagccccgggcggctcgctgctgcccctagcgggggagggacgta
attacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggtt
aattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctcgaggtcgacggtatcctcgaggtcgacggtatcgata
agcttgatatctataacaagaaaatatatatataataagttatcacgtaagtagaacatgaaataacaatataattatcgtatgagttaaatctta
aaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagttcaaaatcagtgacacttaccgcattgacaagcacgcctcac
gggagctccaagcggcgactgagatgtcctaaatgcacagcgacggattcgcgctatttagaaagagagagagcaatatttcaagaatgcatg
cgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatcatatcgtcgggtctttttttccggctcagtcatcgcccaagctg
gcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgaggttgaagcggcatggaaagagtttgccgaggatgactgctgc
tgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaaggtgtggccatgcacgcctttaacggtgaactgttcgttcaggcc
acctgggataccagttcgtcgcggcttttccggacacagttccggatggtcagcccgaagcgcatcagcaacccgaacaataccggcgac
agccggaactgccgtgccggtgtgcagattaatgacagcggtgcggcgctgggatattacgtcagcgaggacgggtatcctggctggatg
ccgcagaaatggacatggataccccgtgagttacccggcgggcgcgcttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatc
cgctcacaattccacacaacatacgagccggaagcataaagtgtaaagcctggggtgcctaatgagtgagctaactcacattaattgcgttg
cgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtattg
ggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacgg
ttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgc
tggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaag
ataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccgaccctgccgcttaccggatacctgtccgcctttctcccttcggga
agcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttc
agcccgaccgctgcgccttatccggtaactatcgtcttgagtccaacccggtaagacacgacttatcgccactggcagcagccactggtaa
caggattagcagagcgaggtatgtaggcggtgctacagagttcttgaagtggtggcctaactacggctacactagaaggacagtatttggta
tctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtt
tgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactc
acgttaagggattttggtcatgagattatcaaaaaggatcttcacctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatata
tgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccc
cgtcgtgtagataactacgatacgggagggcttaccatctggccccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttat
cagcaataaaccagccagccggaagggccgagcgcagaagtggtcctgcaactttatccgcctccatccagtctattaattgttgccggga
agctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttc
attcagctccggttcccaacgatcaaggcgagttacatgatcccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtc

agaagtaagttggccgcagtgttatcactcatggttatggcagcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgact
ggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacat
agcagaactttaaaagtgctcatcattggaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaac
ccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaag
ggaataagggcgacacggaaatgttgaatactcat

(131) The sequence of the PB-hiRep-11 #vector is provided below:

(132) TABLE-US-00025 PB-hiRep-11# (11,702 bp) (SEQ ID NO: 24)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagct
cattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacacttac
atactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaactt
ttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagg
gggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tctgctccctgcttgtgtgttggaggtcgctgagtagtgcgcgagcaaaatttaagctacaacaaggcaaggcttgaccgacaattgcatgaa
gaatctgcttagggttaggcgttttgcgctgcttcgcgatgtacgggccagatatacgcgtatctgaggggactagggtgtgtttaggcgaaa
agcggggcttcggttgtacgcggttaggagtcccctcaggatatagtagtttcgcttttgcatagggagggggaaatgtagtcttatgcaatac
acttgtagtcttgcaacatggtaacgatgagttagcaacatgccttacaaggagagaaaaagcaccgtgcatgccgattggtggaagtaagg
tggtacgatcgtgccttattaggaaggcaacagacaggtctgacatggattggacgaaccactgaattccgcattgcagagataattgtattta
agtgcctagcTCCCTATCAGTGATAGAGAtcTCCCTATCAGTGATAGAGAtcgatacaataaacgccat
ttgaccattcaccacattggtgtgcaccGAACGCGCAGCCGCCATGccggggttttacgagattgtgattaaggtccccagc
gaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtggccgagaaggaatgggagttgccgccagattctgacatgg
atctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagcgcgactttctgacggaatggcgccgtgtgagtaaggccc
cggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgcacgtgctcgtggaaaccaccggggtgaaatccatggtttt
gggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcgggatcgagccgactttgccaaactggttcgcggtcacaa
agaccagaaatggcgccggaggcgggaacaaggtggtggatgagtgctacatccccaattacttgctccccaaaacccagcctgagctc
caatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcacggagcgtaaacggttggtggcgcagcatctgacgcac
gtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacatg
gagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtggatccaggaggaccaggcctcatacatctccttcaatgcg
gcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaagattatgagcctgactaaaaccgcccccgactacctggtg
ggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaactaaacgggtacgatccccaatatgcggcttccgtctttc
tgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgggcctgcaactaccgggaagaccaacatcgcggaggcc
atagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactttcccttcaacgactgtgtcgacaagatggtgatctggtgg
gaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattctcggaggaagcaaggtgcgcgtggaccagaaatgcaag
tcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaacatgtgcgccgtgattgacgggaactcaacgaccttcgaac
accagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctggatcatgactttgggaaggtcaccaagcaggaagtcaa
agactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattctacgtcaaaaagggtggagccaagaaaagacccgcccc
cagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcagccatcgacgtcagacgcggaagcttcgatcaactacg
cagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgcagacaatgcgagagaatgaatcagaattcaa
atatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaatctcaacccgtttctgtcgtcaaaaaggcgtatcagaaa
ctgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgcgatctggtcaatgtggatttggatgactgcatctttgaaca
ataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccgga
aaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggt
ttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctag

ggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccg
ggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaa
ggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagcgacattgattattgact
agttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggtaaatggcccgcctggctga
ccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggactttccattgacgtcaatgggtgga
gtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatgacggtaaatggcccgc
ctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccatggtgatgcggttttgg
cagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggagtttgttttggaaccaaa
atcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtgggaggtctatataagcag
agctctccctatcagtgatagagatctccctatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagatcgcctggagacgcca
tccacgctgttttgacctccatagaagacaccgggaccgatccagcctatcccaattctgatgcgccggtgatcagatcaaaaacttcagcca
ggtacgccaccAcGgaactggtcggatggctggtggataagggcatcacaagcgagaagcaatggatccaggaggaccaggcctcat
atatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgccggcaaaatcatgagcctgaccaagaccgcc
cctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataagatcctggaactgaacggctacgacccccag
tacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatctggctgttcggacctgccaccacaggcaaa
accaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgagaacttccccttcaacgactgtgtgg
acaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggctatcctgggaggctctaaggtcagg
gtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagcaataccaacatgtgcgccgtgatcgacggc
aacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacggctggatcacgacttcggcaag
gtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtggaacacgagttctacgtgaagaagggcggc
gccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgtggcccagcccagcacatctgat
gctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacctgatgctgtttccctgcagacag
tgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatgcttccctgtgtccgagagccaaccagtg
tctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcctgtaccgcctgcgacctggtca
acgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgccagccatctgttgtttgcccctcc
cccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgcattgtctgagtaggtgtcattct
attctggggggtggggtggggcaggacagcaagggggaggattgggaagacaatagcaggcatgctggggatgcggtgggctctatg
ggaccttttttagggcccattggtatggctttttccccgtatccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaa
gcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcgga
gccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcg
gccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccggg
ccccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgcgccttttccaaggcagccctgggtttgcg
cagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtc
acccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgt
gccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccg
cgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtg
gggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttga
ccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacgtgttgacaattaatcatcggcat
agtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaagaatccaccctcattgaaagagca
acggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctctagcgacggccgcatcttcactggtgtcaa
tgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagctggcaacctgacttgtatcgtcgcg
atcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgcatcctgggatcaaagccatagtga
aggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggagggctaaagcgcggggatctcatg
ctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcattttttt
cactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacctaggatccggccggcctgcaggtgtcc
tcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattccttttttagggcccattggtatggcttttt
ccccgtatccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgg
gctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcggagccccgggggctcgctgctgccccctag
cgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtggagctccagctttt
gttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctcgaggtcgacggtatcctcga
ggtcgacggtatcgataagcttgatatctataacaagaaaatatatatataataagttatcacgtaagtagaacatgaaataacaatataattatc
gtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagttcaaaatcagtgacacttaccgcat
tgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacggattcgcgctatttagaaagagagagc
aatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatcatatcgtcgggtctttttttccggct

cagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgaggttgaagcggcatggaaagagttt
gccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaaggtgtggccatgcacgcctttaacgg
tgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccggatggtcagcccgaagcgcatcagcaacc
cgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggtgcggcgctgggatattacgtcagcgaggac
gggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgggcgcgcttggcgtaatcatggtcatagctgttt
cctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaagcctggggtgcctaatgagtgagcta
actcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcattaatgaatcggccaacgcgcgggga
gaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggctgcggcgagcggtatcagctcact
caaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaaggccagcaaaaggccaggaacc
gtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtggcgaaa
cccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccgaccctgccgcttaccggatacctgtc
cgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtaggtcgttcgctccaagctgggctgtg
tgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtccaacccggtaagacacgacttatcgccactg
gcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagttcttgaagtggtggcctaactacggctacacta
gaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctcttgatccggcaaacaaaccaccgctg
gtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatcctttgatcttttctacggggtctgacgct
cagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttcacctagatccttttaaattaaaaatgaagttttaaa
tcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcacctatctcagcgatctgtctatttcgttcatcca
tagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggccccagtgctgcaatgataccgcgagacccacgc
tcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcagaagtggtcctgcaactttatccgcctccatccagt
ctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcacgc
tcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatcccccatgttgtgcaaaaaagcggttagctcctt
cggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggcagcactgcataattctcttactgtcatgccatccgt
aagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccgagttgctcttgcccggcgtcaatacgg
gataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttg
agatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtttctgggtgagcaaaaacaggaaggc
aaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat

(133) The sequence of the PB-hiRep-12 #vector is provided below:

(134) TABLE-US-00026 PB-hiRep-12# (11,589 bp) (SEQ ID NO: 25)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagct
cattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacacttac
atactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaactt
ttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagg
gggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tgacattgattattgactagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggtaa
atggcccgcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggactttccatt
gacgtcaatgggtggagtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatga
cggtaaatggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccat
ggtgatgcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggag
tttgttttggaaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtggg
aggtctatataagcagagctctccctatcagtgatagagatctccctatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagat
cgcctggagacgccatccacgctgttttgacctccatagaagacaccgggaccgatccagcctgaacgcgcagccgccaatggatgccg

gggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtggccgagaag
gaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagcgcgactttc
tgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgcacgtgctc
gtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcgggatcga
gccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaacaaggtggtggatgagtgctacatcccca
attacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcacggagcgt
aaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatgcgccggt
gatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtggatccagg
aggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaagattatgag
cctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaactaaacg
ggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgggcctgca
actaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactttcccttca
acgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattctcggagga
agcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaacatgtgcgc
cgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctggatcatg
actttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattctacgtcaaa
aagggtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcagccatcga
cgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgc
agacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaatctcaac
ccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgcgatctggt
caatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaa
gaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaat
aaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaagcccc
cagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccc
cgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcct
gcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctatag
gattgaggtcagagcGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAAC
GACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGG
ACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTA
CATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGG
CCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTAC
ATCTACGTATTAGTCATCGCTATTACCATGGTCGAGGTGAGCCCCACGTTCTGCTTCA
CTCTCCCCATCTCCCCCCCCTCCCCACCCCCAATTTTGTATTTATTTATTTTTTAATTA
TTTTGTGCAGCGATGGGGGCGGGGGGGGGGGGGGGGCGCGCGCCAGGCGGGGCGG
GGCGGGGCGAGGGGCGGGGCGGGGCGAGGCGGAGAGGTGCGGCGGCAGCCAATCA
GAGCGGCGCGCTCCGAAAGTTTCCTTTTATGGCGAGGCGGCGGCGGCGGCGGCCCT
ATAAAAAGCGAAGCGTCCCTATCAGTGATAGAGAtcTCCCTATCAGTGATAGAGACG
CGGCGGGCGGGAGTCGCTGCGTTGCCTTCGCCCCGTGCCCCGCTCCGCGCCGCCTCG
CGCCGCCCGCCCCGGCTCTGACTGACCGCGTTACTCCCACAGatcccaattctgatgcgccggtgat
cagatcaaaaacttcagccaggtacgccaccatggaactggtcggatggctggtggataagggcatcacaagcgagaagcaatggatcc
aggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgccggcaaaatcatg
agcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataagatcctggaactg
aacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatctggctgttcgga
cctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgagaacttc
cccttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggctatcctgg
gaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagcaataccaacatgtg
cgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacggctgga
tcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtggaacacgagttctac
gtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgtggccca
gcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacctgatgct
gtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatgcttccctgtgtcc
gagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcctgtacc
gcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgccagcca
tctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgcattgt

ctgagtaggtgtcattctattctggggggtggggggggcaggacagcaagggggaggattgggaagacaatagcaggcatgctgggga
tgcggtgggctctatgggaccttttttagggcccattggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagcagcgaga
agcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggggagcgc
cggaccggagcggagccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggggg
ctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactataggg
cgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgcgccttttccaag
gcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttca
cgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttc
cttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatg
gcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgc
gggaggcggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggca
gtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacgtgttga
caattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaagaatcca
ccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctctagcgacggccg
catcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagctggcaac
ctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgcatcctgg
gatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggagggctaa
agcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttc
acaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacctaggatcc
ggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctttttagg
gcccattggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgc
caccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggggagcgccggaccggagcggagccccgggcgg
ctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgc
ggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctcga
ggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatatatatataataagttatcacgtaagtagaacatg
aaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagttcaaaat
cagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacggattcgcgct
atttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatcatatcgt
cgggtctttttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgaggttgaag
cggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaaggtgtggc
catgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccggatggtcagccc
gaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggtgcggcgctggga
tattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgggcgcgcttggcgt
aatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaagcctgggg
tgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcattaatgaatc
ggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggctgcggc
gagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaaggccag
caaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcgacgctca
agtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccgaccctgc
cgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtaggtcgtt
cgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtccaacccggtaag
acacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagttcttgaagtggtgg
cctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctcttgatccg
gcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatcctttgatctt
ttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttcacctagatccttttaa
attaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcacctatctcagcgat
ctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggccccagtgctgcaatga
taccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcagaagtggtcctgcaact
ttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgccattgctaca
ggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatcccccatgttgtgca
aaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggcagcactgcataattct
cttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccgagttgctct
tgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcggggcgaaaactctc
aaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtttctgggtg

agcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat
(135) The sequence of the PB-hiRep-13 #vector is provided below:
(136) TABLE-US-00027 PB-hiRep-13# (11, 854 bp) (SEQ ID NO: 26)
actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagct
cattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacacttac
atactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaactt
ttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagg
gggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tgacattgattattgactagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggtaa
atggcccgcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggactttccatt
gacgtcaatgggtggagtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatga
cggtaaatggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccat
ggtgatgcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggag
tttgttttggaaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtggg
aggtctatataagcagagctctccctatcagtgatagagatctccctatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagat
cgcctggagacgccatccacgctgttttgacctccatagaagacaccgggaccgatccagcctgaacgcgcagccgccacgccggggtt
ttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtggccgagaaggaat
gggagttgccgccagattctgacatggatctgaatctgattgagcaggcaccccctgaccgtggccgagaagctgcagcgcgactttctgac
ggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgcacgtgctcgtgg
aaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcgggatcgagccg
actttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaacaaggtggtggatgagtgctacatccccaattac
ttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcacggagcgtaaac
ggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatgcgccggtgatc
agatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtggatccaggagga
ccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaagattatgagcctga
ctaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaactaaacgggtac
gatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgggcctgcaactac
cgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactttcccttcaacgac
tgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattctcggaggaagcaa
ggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaacatgtgcgccgtgat
tgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctggatcatgactttg
ggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattctacgtcaaaaagg
gtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcagccatcgacgtc
agacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtttccctgcagac
aatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaatctcaacccgt
ttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgcgatctggtcaat
gtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaa
cagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaag
catttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaagcccccagg
gatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgag
ccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcag
acacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattg

aggtcagagcGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGAC
CCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGACT
TTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTACAT
CAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGGCCC
GCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAGTACATCT
ACGTATTAGTCATCGCTATTACCATGGTCGAGGTGAGCCCCACGTTCTGCTTCACTCT
CCCCATCTCCCCCCCCTCCCCACCCCCAATTTTGTATTTATTTATTTTTTAATTATTTT
GTGCAGCGATGGGGGCGGGGGGGGGGGGGGGGCGCGCGCCAGGCGGGGCGGGGCG
GGGCGAGGGGGGGGCGGGGCGAGGCGGAGAGGTGCGGCGGCAGCCAATCAGAGC
GGCGCGCTCCGAAAGTTTCCTTTTATGGCGAGGCGGCGGCGGCGGCGGCCCTATAA
AAAGCGAAGCGTCCCTATCAGTGATAGAGAtcTCCCTATCAGTGATAGAGACGCGGC
GGGCGGGAGTCGCTGCGTTGCCTTCGCCCCGTGCCCCGCTCCGCGCCGCCTCGCGCC
GCCCGCCCCGGCTCTGACTGACCGCGTTACTCCCACAGatcccaattctgatgcgccggtgatcagatc
aaaaacttcagccaggtacgccaccatggaactggtcggatggctggtggataagggcatcacaagcgagaagcaatggatccaggagg
accaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgccggcaaaatcatgagcctg
accaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataagatcctggaactgaacggc
tacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatctggctgttcggacctgcca
ccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgagaacttccccttcaa
cgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggctatcctgggaggctc
taaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagcaataccaacatgtgcgccgtg
atcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacggctggatcacgac
ttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtggaacacgagttctacgtgaaga
agggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgtggcccagcccagc
acatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacctgatgctgtttccct
gcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatgcttccctgtgtccgagagc
caaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcctgtaccgcctgc
gacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgccagccatctgttg
tttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgcattgtctgagta
ggtgtcattctattctggggggtggggggggcaggacagcaagggggaggattgggaagacaatagcaggcatgctggggatgcggt
gggctctatgggacctttttttagggcccattggtatggctttttccccgtatccccccaggtgtctgcaggctcaaagagcagcgagaagcgtt
cagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggggagcgccggac
cggagcggagccccgggcggctcgctgctgcccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtcc
ctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaatt
gggtaccgggcccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgcgccttttccaaggcagcc
ctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccg
ttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcgg
ttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcg
cgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggagg
cggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggct
ccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacgtgttgacaattaat
catcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaagaatccaccctcatt
gaaagagcaacggctacaatcaacagcatccccatctctgaagactacagegtegccagcgcagctctctctagcgacggccgcatcttca
ctggtgtcaatgtatatcattttactggggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagctggcaacctgacttg
tatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgcatcctgggatcaaa
gccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggagggctaaagcgcg
gggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaata
aagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacctaggatccggccggc
ctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctttttttagggcccatt
ggtatggctttttccccgtatccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttc
cccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcggagccccgggcggctcgct
gctgcccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtgg
agctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggcccccctcgaggtcg
acggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatatatatataataagttatcacgtaagtagaacatgaaata
acaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagttcaaaatcagtg

acacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacggattcgcgctatttag
aaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatcatatcgtcgggt
ctttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgaggttgaagcggca
tggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaaggtgtggccatgca
cgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccggatggtcagcccgaagc
gcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggtgcggcgctgggatattac
gtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgggcgcgcttggcgtaatcat
ggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaagcctggggtgccta
atgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcattaatgaatcggccaa
cgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggctgcggcgagcgg
tatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaaggccagcaaaag
gccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcgacgctcaagtcaga
ggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccgaccctgccgcttacc
ggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtaggtcgttcgctccaa
gctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtccaacccggtaagacacgact
tatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagttcttgaagtggtggcctaacta
cggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctcttgatccggcaaaca
aaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatcctttgatcttttctacgg
ggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttcacctagatccttttaaattaaaaa
tgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcacctatctcagcgatctgtctat
ttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggccccagtgctgcaatgataccgcg
agacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcagaagtggtcctgcaactttatccgc
ctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgccattgctacaggcatcgt
ggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatcccccatgttgtgcaaaaaagcg
gttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggcagcactgcataattctcttactgtca
tgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccgagttgctcttgcccggc
gtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcggggcgaaaactctcaaggatctt
accgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtttctgggtgagcaaaaa
caggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat

(137) The sequence of the PB-hiRep-14 #vector is provided below:

(138) TABLE-US-00028 PB-hiRep-14# (11,573 bp) (SEQ ID NO: 27)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagct
cattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaac
ttttatcgaattcctgcagcccgggggatccactagttctagagggacagccccccccaaagcccccagggatgtaattacgtccctcccccgct
agggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tggcctccgcgccgggttttggcgcctcccgcgggcgcccccctcctcacggcgagcgctgccacgtcagacgaagggcgcagcgagc
gtcctgatccttccgcccggacgctcaggacagcggcccgctgctcataagactcggccttagaaccccagtatcagcagaaggacatttt
aggacgggacttgggtgactctagggcactggttttctttccagagagcggaacaggcgaggaaaagtagtcccttctcggcgattctgcg
gagggatctccgtggggcggtgaacgccgatgattatataaggacgcgccggtccctatcagtgatagagatctccctatcagtgatagag
agtgtggcacagctagttccgtcgcagccgggatttgggtcgcggttcttgtttgtggatcgctgtgatcgtcacttgacgaacgcgcagccg
ccatgccggggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgacagctttgtgaactgggtgg

ccgagaaggaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagc
gcgactttctgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaagggagagagctacttccacatgc
acgtgctcgtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcg
ggatcgagccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaacaaggtggtggatgagtgctac
atccccaattacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacctcagcgcctgtttgaatctcac
ggagcgtaaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatg
cgccggtgatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggattacctcggagaagcagtgg
atccaggaggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaaga
ttatgagcctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaatcggatttataaaattttggaa
ctaaacgggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagaggaacaccatctggctgtttgg
gcctgcaactaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaactt
tcccttcaacgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtggagtcggccaaagccattct
cggaggaagcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaaca
tgtgcgccgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaatttgaactcacccgccgtctg
gatcatgactttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggttgaggtggagcatgaattcta
cgtcaaaaagggtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagttgcgcag
ccatcgacgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtt
tccctgcagacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcaga
atctcaacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgc
gatctggtcaatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagt
ccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaattt
cacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaa
agcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccg
catccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctcttt
gagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctccttt
ctataggattgaggtcagagcGCGTTACATAACTTACGGTAAATGGCCCGCCTGGCTGACCGCCC
AACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTAACGCCAATA
GGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCCCACTTGGCA
GTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAATGACGGTAAA
TGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCTACTTGGCAG
TACATCTACGTATTAGTCATCGCTATTACCATGGTCGAGGTGAGCCCCACGTTCTGCT
TCACTCTCCCCATCTCCCCCCCCTCCCCACCCCCAATTTTGTATTTATTTATTTTTTAA
TTATTTTGTGCAGCGATGGGGGCGGGGGGGGGGGGGGGGCGCGCGCCAGGCGGGGC
GGGGCGGGGCGAGGGGCGGGGCGGGGCGAGGCGGAGAGGTGCGGCGGCAGCCAAT
CAGAGCGGCGCGCTCCGAAAGTTTCCTTTTATGGCGAGGCGGCGGCGGCGGCGGCC
CTATAAAAAGCGAAGCGTCCCTATCAGTGATAGAGAtcTCCCTATCAGTGATAGAGA
CGCGGCGGGCGGGAGTCGCTGCGTTGCCTTCGCCCCGTGCCCCGCTCCGCGCCGCCT
CGCGCCGCCCGCCCCGGCTCTGACTGACCGCGTTACTCCCACAGatcccaattctgatgcgccggt
gatcagatcaaaaacttcagccaggtacgccaccatggaactggtcggatggctggtggataagggcatcacaagcgagaagcaatggat
ccaggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgccggcaaaatca
tgagcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataagatcctggaac
tgaacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatctggctgttcg
gacctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgagaact
tccccttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggctatcct
gggaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagcaataccaacat
gtgcgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacggct
ggatcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtggaacacgagtt
ctacgtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgtgg
cccagcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacctg
atgctgtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatgcttccctgt
gtccgagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcctg
taccgcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgcca
gccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgc
attgtctgagtaggtgtcattctattctggggggtggggggggcaggacagcaagggggaggattgggaagacaatagcaggcatgctg

gggatgcggtgggctctatgggaccttttttagggcccattggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagcag
cgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggg
agcgccggaccggagcggagccccgggcggctcgctgctgcccccctagcgggggagggacgtaattacatccctgggggctttgggg
gggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcacta
tagggcgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctaggggggtatccccggggttggggttgcgccttttc
caaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacatt
cttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaa
ggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggag
caatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggc
ggtgcgggaggggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgt
cggcagtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacg
tgttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaaga
atccaccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctctagcgac
ggccgcatcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagctg
gcaacctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgca
tcctgggatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggag
ggctaaagcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcac
aaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacct
aggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctt
ttttagggcccattggtatggctttttccccgtatcccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatc
ccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcggagcccc
gggcggctcgctgctgcccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgc
caccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccc
cctcgaggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatatatatatataataagttatcacgtaagtag
aacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagtt
caaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacggatt
cgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatc
atatcgtcgggtcttttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgag
gttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaag
gtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccggatgg
tcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggtgcggc
gctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgggcgcg
cttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaag
cctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcatta
atgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggc
tgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaa
ggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcga
cgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccga
ccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtag
gtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtccaaccc
ggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagttcttgaa
gtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctc
ttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatc
ctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttcacctag
atccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcaccta
tctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggccccagtg
ctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcagaagtggt
cctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgcc
attgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatcccccat
gttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggcagcactg
cataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccg
agttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcggggcga
aaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgttt
ctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat

(139) The sequence of the PB-hiRep-15 #vector is provided below:
(140) TABLE-US-00029 PB-hiRep-15# (11,695 bp) (SEQ ID NO: 28)
actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagct
cattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaact
tttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctag
ggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tctgctccctgcttgtgtgttggaggtcgctgagtagtgcgcgagcaaaatttaagctacaacaaggcaaggcttgaccgacaattgcatgaa
gaatctgcttagggttaggcgttttgcgctgcttcgcgatgtacgggccagatatacgcgtatctgaggggactagggtgtgtttaggcgaaa
agcggggcttcggttgtacgcggttaggagtcccctcaggatatagtagtttcgcttttgcatagggagggggaaatgtagtcttatgcaatac
acttgtagtcttgcaacatggtaacgatgagttagcaacatgccttacaaggagagaaaaagcaccgtgcatgccgattggtggaagtaagg
tggtacgatcgtgccttattaggaaggcaacagacaggtctgacatggattggacgaaccactgaattccgcattgcagagataattgtattta
agtgcctagctccctatcagtgatagagatctccctatcagtgatagagatcgatacaataaacgccatttgaccattcaccacattggtgtgc
accgaacgcgcagccgccatgccggggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgaca
gctttgtgaactgggtggccgagaaggaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgt
ggccgagaagctgcagcgcgactttctgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaaggga
gagagctacttccacatgcacgtgctcgtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaact
gattcagagaatttaccgcgggatcgagccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaaca
aggtggtggatgagtgctacatccccaattacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacct
cagcgcctgtttgaatctcacggagcgtaaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagaga
atcagaatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacatggagctggtcgggggctcgtggacaaggggat
tacctcggagaagcagtggatccaggaggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgcc
ttggacaatgcgggaaagattatgagcctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaat
cggatttataaaattttggaactaaacgggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagagg
aacaccatctggctgtttgggcctgcaactaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaa
actggaccaatgagaactttcccttcaacgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtgg
agtcggccaaagccattctcggaggaagcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatc
gtcacctccaacaccaacatgtgcgccgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaat
ttgaactcacccgccgtctggatcatgactttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggtt
gaggtggagcatgaattctacgtcaaaaagggtggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacggg
tgcgcgagtcagttgcgcagccatcgacgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgt
gggcatgaatctgatgctgtttccctgcagacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgttta
gagtgctttcccgtgtcagaatctcaacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgc
cagacgcttgcactgcctgcgatctggtcaatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcagg
tgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaata
aagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggat
tgagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgct
ccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcct
ctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttggg
tgaattttggctcattcctcctttctataggattgaggtcagagcGCGTTACATAACTTACGGTAAATGGCCCGCC
TGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCAT
AGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAAC

TGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGT
CAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTT
TCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTCGAGGTGAG
CCCCACGTTCTGCTTCACTCTCCCCATCTCCCCCCCCTCCCCACCCCCAATTTTGTATT
TATTTATTTTTTAATTATTTTGTGCAGCGATGGGGGCGGGGGGGGGGGGGGGGCGCG
CGCCAGGCGGGGCGGGGCGGGGCGAGGGGCGGGGCGGGGCGAGGCGGAGAGGTGC
GGCGGCAGCCAATCAGAGCGGCGCGCTCCGAAAGTTTCCTTTTATGGCGAGGCGGC
GGCGGCGGCGGCCCTATAAAAAGCGAAGCGTCCCTATCAGTGATAGAGAtcTCCCTA
TCAGTGATAGAGACGCGGCGGGCGGGAGTCGCTGCGTTGCCTTCGCCCCGTGCCCC
GCTCCGCGCCGCCTCGCGCCGCCCGCCCCGGCTCTGACTGACCGCGTTACTCCCACA
Gatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacgccaccatggaactggtcggatggctggtggataagggc
atcacaagcgagaagcaatggatccaggaggaccaggcctcatatatttcttttaacgccgctagcaattccagaagccagatcaaggctg
ctctggacaacgccggcaaaatcatgagcctgaccaagaccgcccctgactacctggtgggacagcagcctgtggaagatatcagcagc
aacagaatctataagatcctggaactgaacggctacgacccccagtacgccgcctccgtgttcctgggctgggctacaaagaagttcggca
agcggaacaccatctggctgttcggacctgccaccacaggcaaaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggct
gcgtgaactggacaaacgagaacttcccctttcaacgactgtgtggacaagatggtgatctggtgggaggaaggcaaaatgacagctaagg
tggtggaatctgccaaggctatcctgggaggctctaaggtcagggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgt
gatcgtgaccagcaataccaacatgtgcgccgtgatcgacggcaacagcaccaccttcgagcatcagcagcctctgcaggaccggatgtt
caagtttgagctcaccagacggctggatcacgacttcggcaaggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagacca
cgttgttgaggtggaacacgagttctacgtgaagaagggcggcgccaagaaaagacccgcccctagcgacgccgacatcagcgagcct
aagagagtgcgggaaagcgtggcccagcccagcacatctgatgctgaggccagcatcaactacgccgatagataccaaaacaagtgca
gccgccacgtgggcatgaacctgatgctgtttccctgcagacagtgtgaaagaatgaaccagaattctaatatctgctttacacacggccag
aaagattgcctggaatgcttccctgtgtccgagagccaaccagtgtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacat
catgggcaaggtgccagacgcctgtaccgcctgcgacctggtcaacgtggacctggacgactgcatcttcgagcagtgatttgtgatggga
attctgtgctgtgccttctagttgccagccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcc
taataaaatgaggaaattgcatcgcattgtctgagtaggtgtcattctattctggggggtggggtggggcaggacagcaagggggaggatt
gggaagacaatagcaggcatgctggggatgcggtgggctctatgggacctttttttagggcccattggtatggctttttccccgtatcccccca
ggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgc
tgccggctcggggatgcgggggggagcgccggaccggagcggagccccgggcggctcgctgctgccccctagcgggggagggacgt
aattacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgaggg
ttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctgaggcggaaagaaccagctggggctctagggggtat
ccccggggttggggttgcgccttttccaaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaacgcagc
ggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccggcgacgc
ttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcactagtaccct
cgcagacggacagcgccagggagcaatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggcgcgccg
agagcagcggccgggaaggggcggtgcgggaggcggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcat
tctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgaccgaatcaccgacctctctccccagaagctcccgggagcttgtat
atccattttcggatctgatcagcacgtgttgacaattaatcatcggcatagtatatcggcatagtataatacgacaaggtgaggaacgccacca
tggccaagcctttgtctcaagaagaatccaccctcattgaaagagcaacggctacaatcaacagcatccccatctctgaagactacagcgtc
gccagcgcagctctctctagcgacggccgcatcttcactggtgtcaatgtatatcattttactgggggaccttgtgcagaactcgtggtgctgg
gcactgctgctgctgcggcagctggcaacctgacttgtatcgtcgcgatcggaaatgagaacaggggcatcttgagcccctgcggacggt
gccgacaggtgcttctcgatctgcatcctgggatcaaagccatagtgaaggacagtgatggacagccgacggcagttgggattcgtgaatt
gctgccctctggttatgtgtgggagggctaaagcgcggggatctcatgctggagttcttcgcccaccccaacttgtttattgcagcttataatg
gttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatctt
atcatgtctgtagctgatgtatacctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggttt
agccccggaattgactggattcctttttttagggcccattggtatggctttttccccgtatcccccccaggtgtctgcaggctcaaagagcagcga
gaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcgggggggagc
gccggaccggagcggagccccgggggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttggggggg
ggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatag
ggcgaattgggtaccgggccccccctcgaggtcgacggtatcctcgaggtcgacggtatcgataagcttgatatctataacaagaaaatata
tatataataagttatcacgtaagtagaacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtc
attttgactcacgcggtcgttatagttcaaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcggcgactgagat
gtcctaaatgcacagcgacggattcgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagactatctttctag
ggttaatctagctgcatcaggatcatatcgtcgggtctttttttccggctcagtcatcgcccaagctggcgctatctgggcatcggggaggaag

aagcccgtgccttttcccgcgaggttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcgaaaacgcac
gtttaccatgatgattcgggaaggtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttcgtcgcggctt
ttccggacacagttccggatggtcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtgccggtgtgca
gattaatgacagcggtgcggcgctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacatggataccccg
tgagttacccggcgggcgcgcttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacacaacatacgag
ccggaagcataaagtgtaaagcctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttccagtcggga
aacctgtcgtgccagctgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcctcgctcactga
ctcgctgcgctcggtcgttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcaggggataacgca
ggaaagaacatgtgagcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggctccgcccccc
tgacgagcatcacaaaaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttccccctggaagctc
cctcgtgcgctctcctgttccgaccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctcatagctcacgct
gtaggtatctcagttcggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgccttatccggta
actatcgtcttgagtccaacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcgaggtatgtagg
cggtgctacagagttcttgaagtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagccagttaccttc
ggaaaaagagttggtagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaa
aaaaggatctcaagaagatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggtcatgagattat
caaaaaggatcttcacctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtctgacagttaccaatg
cttaatcagtgaggcacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtcgtgtagataactacgatacgggagg
gcttaccatctggccccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccagccagccggaa
gggccgagcgcagaagtggtcctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagttcgccagtta
atagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttcccaacgatcaa
ggcgagttacatgatcccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggccgcagtgttatc
actcatggttatggcagcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaaccaagtcattctga
gaatagtgtatgcggcgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaaagtgctcatcatt
ggaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcacccaactgatcttca
gcatcttttactttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacacggaaatgtt gaatactcat

(141) The sequence of the PB-hiRep-16 #vector is provided below:

(142) TABLE-US-00030 PB-hiRep-16# (11,421 bp) (SEQ ID NO: 29)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatcagc
tcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaacaag
agtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaat
caagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacactta
catactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaacaaaac
ttttatcgaattcctgcagcccgggggatccactagttctagagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgct
agggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggca
cggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgtgg
aatgtgtgtcagttagggtgtggaaagtccccgcgatcgctagcttaagcgctgatcaattggcgcgccgaattcgttatctgcagaattcggct
tctgctccctgcttgtgtgttggaggtcgctgagtagtgcgcgagcaaaatttaagctacaacaaggcaaggcttgaccgacaattgcatgaa
gaatctgcttagggttaggcgttttgcgctgcttcgcgatgtacgggccagatatacgcgtatctgaggggactagggtgtgtttaggcgaaa
agcggggcttcggttgtacgcggttaggagtcccctcaggatatagtagtttcgcttttgcatagggagggggaaatgtagtcttatgcaatac
acttgtagtcttgcaacatggtaacgatgagttagcaacatgccttacaaggagagaaaaagcaccgtgcatgccgattggtggaagtaagg
tggtacgatcgtgccttattaggaaggcaacagacaggtctgacatggattggacgaaccactgaattccgcattgcagagataattgtattta
agtgcctagctccctatcagtgatagagatctccctatcagtgatagagatcgatacaataaacgccatttgaccattcaccacattggtgtgc
accgaacgcgcagccgccatgccggggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggcatttctgaca
gctttgtgaactgggtggccgagaaggaatgggagttgccgccagattctgacatggatctgaatctgattgagcaggcacccctgaccgt
ggccgagaagctgcagcgcgactttctgacggaatggcgccgtgtgagtaaggccccggaggctcttttctttgtgcaatttgagaaggga

gagagctacttccacatgcacgtgctcgtggaaaccaccggggtgaaatccatggttttgggacgtttcctgagtcagattcgcgaaaaact
gattcagagaatttaccgcgggatcgagccgactttgccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaaca
aggtggtggatgagtgctacatccccaattacttgctccccaaaacccagcctgagctccaatgggcatggaccaacatggaacagtacct
cagcgcctgtttgaatctcacggagcgtaaacggttggtggcgcagcatctgacgcacgtgtcgcagacgcaggagcagaacaaagaga
atcagaatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggggat
tacctcggagaagcagtggatccaggaggaccaggcctcatacatctccttcaatgcggcctccaactcgcggtcccaaatcaaggctgcc
ttggacaatgcgggaaagattatgagcctgactaaaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaat
cggatttataaaattttggaactaaacgggtacgatccccaatatgcggcttccgtctttctgggatgggccacgaaaaagttcggcaagagg
aacaccatctggctgtttgggcctgcaactaccgggaagaccaacatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaa
actggaccaatgagaactttcccttcaacgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgccaaggtcgtgg
agtcggccaaagccattctcggaggaagcaaggtgcgcgtggaccagaaatgcaagtcctcggcccagatagacccgactcccgtgatc
gtcacctccaacaccaacatgtgcgccgtgattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaat
ttgaactcacccgccgtctggatcatgactttgggaaggtcaccaagcaggaagtcaaagactttttccggtgggcaaaggatcacgtggtt
gaggtggagcatgaattctacgtcaaaaagggggagccaagaaaagacccgcccccagtgacgcagatataagtgagcccaaacggg
tgcgcgagtcagttgcgcagccatcgacgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaatgttctcgtcacgt
gggcatgaatctgatgctgtttccctgcagacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgttta
gagtgctttcccgtgtcagaatctcaacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgc
cagacgcttgcactgcctgcgatctggtcaatgtggatttggatgactgcatctttgaacaataaaatggctaggatccggccggcctgcagg
tgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaata
aagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactgga
ttgagggacagcccccccccaaagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgct
ccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcct
ctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttggg
tgaattttggctcattcctcctttctataggattgaggtcagagcggcctccgcgccgggttttggcgcctcccgcgggcgcccccctcctcac
ggcgagcgctgccacgtcagacgaagggcgcagcgagcgtcctgatccttccgcccggacgctcaggacagcggcccgctgctcataa
gactcggccttagaaccccagtatcagcagaaggacattttaggacgggacttgggtgactctagggcactggttttctttccagagagcgg
aacaggcgaggaaaagtagtcccttctcggcgattctgcggagggatctccgtggggcggtgaacgccgatgattatataaggacgcgcc
ggTCCCTATCAGTGATAGAGAtcTCCCTATCAGTGATAGAGAgtgtggcacagctagttccgtcgcag
ccgggatttgggtcgcggttcttgtttgtggatcgctgtgatcgtcacttgacatcccaattctgatgcgccggtgatcagatcaaaaacttcag
ccaggtacgccaccatggaactggtcggatggctggtggataagggcatcacaagcgagaagcaatggatccaggaggaccaggcctc
atatatttcttttaacgccgctagcaattccagaagccagatcaaggctgctctggacaacgccggcaaaatcatgagcctgaccaagaccg
cccctgactacctggtgggacagcagcctgtggaagatatcagcagcaacagaatctataagatcctggaactgaacggctacgaccccc
agtacgccgcctccgtgttcctgggctgggctacaaagaagttcggcaagcggaacaccatctggctgttcggacctgccaccacaggca
aaaccaatatcgccgaggccatcgcccacaccgtgcctttctacggctgcgtgaactggacaaacgagaacttccccttcaacgactgtgtg
gacaagatggtgatctggtgggaggaaggcaaaatgacagctaaggtggtggaatctgccaaggctatcctgggaggctctaaggtcag
ggtggatcagaagtgtaaaagcagcgcccagattgaccctacccctgtgatcgtgaccagcaataccaacatgtgcgccgtgatcgacgg
caacagcaccaccttcgagcatcagcagcctctgcaggaccggatgttcaagtttgagctcaccagacggctggatcacgacttcggcaa
ggtgaccaagcaggaggtgaaggatttcttcagatgggccaaagaccacgttgttgaggtggaacacgagttctacgtgaagaagggcgg
cgccaagaaaagacccgcccctagcgacgccgacatcagcgagcctaagagagtgcgggaaagcgtggcccagcccagcacatctga
tgctgaggccagcatcaactacgccgatagataccaaaacaagtgcagccgccacgtgggcatgaacctgatgctgtttccctgcagaca
gtgtgaaagaatgaaccagaattctaatatctgctttacacacggccagaaagattgcctggaatgcttccctgtgtccgagagccaaccagt
gtctgtggtgaaaaaggcctaccagaagctgtgctacatccaccacatcatgggcaaggtgccagacgcctgtaccgcctgcgacctggtc
aacgtggacctggacgactgcatcttcgagcagtgatttgtgatgggaattctgtgctgtgccttctagttgccagccatctgttgtttgcccctc
ccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattgcatcgcattgtctgagtaggtgtcatt
ctattctggggggggggtggggcaggacagcaagggggaggattgggaagacaatagcaggcatgctggggatgcggtgggctctat
gggaccttttttagggcccattggtatggcttttttccccgtatcccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaa
agcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcgg
agccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagc
ggccgccaccgcggtggagctccagcttttgttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgg
gccccccctgaggcggaaagaaccagctggggctctagggggtatccccggggttggggttgcgccttttccaaggcagccctgggtttgc
gcagggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgt
cacccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcg
tgccggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccg
cgatgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtg

gggcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttga
ccgaatcaccgacctctctccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacgtgttgacaattaatcatcggcat
agtatatcggcatagtataatacgacaaggtgaggaacgccaccatggccaagcctttgtctcaagaagaatccaccctcattgaaagagca
acggctacaatcaacagcatccccatctctgaagactacagcgtcgccagcgcagctctctctagcgacggccgcatcttcactggtgtcaa
tgtatatcattttactgggggaccttgtgcagaactcgtggtgctgggcactgctgctgctgcggcagctggcaacctgacttgtatcgtcgcg
atcggaaatgagaacaggggcatcttgagcccctgcggacggtgccgacaggtgcttctcgatctgcatcctgggatcaaagccatagtga
aggacagtgatggacagccgacggcagttgggattcgtgaattgctgccctctggttatgtgtgggagggctaaagcgcggggatctcatg
ctggagttcttcgcccaccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttt
tcactgcattctagttgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacctaggatccggccggcctgcaggtgt
cctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctttttagggcccattggtatggctttt
tccccgtatcccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgg
gctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcggagccccgggggctcgctgctgccccctag
cgggggagggacgtaattacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtggagctccagctttt
gttccctttagtgagggttaattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctcgaggtcgacggtatcctcga
ggtcgacggtatcgataagcttgatatctataacaagaaaatatatatataataagttatcacgtaagtagaacatgaaataacaatataattatc
gtatgagttaaatcttaaaagtcacgtaaaagataatcatgcgtcattttgactcacgcggtcgttatagttcaaaatcagtgacacttaccgcat
tgacaagcacgcctcacgggagctccaagcggcgactgagatgtcctaaatgcacagcgacggattcgcgctatttagaaagagagagc
aatatttcaagaatgcatgcgtcaattttacgcagactatctttctagggttaatctagctgcatcaggatcatatcgtcgggtctttttttccggc
tcagtcatcgcccaagctggcgctatctgggcatcggggaggaagaagcccgtgccttttcccgcgaggttgaagcggcatggaaagagttt
gccgaggatgactgctgctgcattgacgttgagcgaaaacgcacgtttaccatgatgattcgggaaggtgtggccatgcacgcctttaacgg
tgaactgttcgttcaggccacctgggataccagttcgtcgcggcttttccggacacagttccggatggtcagcccgaagcgcatcagcaacc
cgaacaataccggcgacagccggaactgccgtgccggtgtgcagattaatgacagcggtgcggcgctgggatattacgtcagcgaggac
gggtatcctggctggatgccgcagaaatggacatggataccccgtgagttacccggcgggcgcgcttggcgtaatcatggtcatagctgttt
cctgtgtgaaattgttatccgctcacaattccacacaacatacgagccggaagcataaagtgtaaagcctggggtgcctaatgagtgagcta
actcacattaattgcgttgcgctcactgcccgctttccagtcgggaaacctgtcgtgccagctgcattaatgaatcggccaacgcgcgggga
gaggcggtttgcgtattgggcgctcttccgcttcctcgctcactgactcgctgcgctcggtcgttcggctgcggcgagcggtatcagctcact
caaaggcggtaatacggttatccacagaatcaggggataacgcaggaaagaacatgtgagcaaaaggccagcaaaaggccaggaacc
gtaaaaaggccgcgttgctggcgtttttccataggctccgcccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtggcgaaa
cccgacaggactataaagataccaggcgtttccccctggaagctccctcgtgcgctctcctgttccgaccctgccgcttaccggatacctgtc
cgcctttctcccttcgggaagcgtggcgctttctcatagctcacgctgtaggtatctcagttcggtgtaggtcgttcgctccaagctgggctgtg
tgcacgaaccccccgttcagcccgaccgctgcgccttatccggtaactatcgtcttgagtccaacccggtaagacacgacttatcgccactg
gcagcagccactggtaacaggattagcagagcgaggtatgtaggcggtgctacagagttcttgaagtggtggcctaactacggctacacta
gaaggacagtatttggtatctgcgctctgctgaagccagttaccttcggaaaaagagttggtagctcttgatccggcaaacaaaccaccgctg
gtagcggtggtttttttgtttgcaagcagcagattacgcgcagaaaaaaaggatctcaagaagatcctttgatcttttctacggggtctgacgct
cagtggaacgaaaactcacgttaagggattttggtcatgagattatcaaaaaggatcttcacctagatccttttaaattaaaaatgaagttttaaa
tcaatctaaagtatatatgagtaaacttggtctgacagttaccaatgcttaatcagtgaggcacctatctcagcgatctgtctatttcgttcatcc
atagttgcctgactccccgtcgtgtagataactacgatacgggagggcttaccatctggccccagtgctgcaatgataccgcgagacccacgc
tcaccggctccagatttatcagcaataaaccagccagccggaagggccgagcgcagaagtggtcctgcaactttatccgcctccatccagt
ctattaattgttgccgggaagctagagtaagtagttcgccagttaatagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcacgc
tcgtcgtttggtatggcttcattcagctccggttcccaacgatcaaggcgagttacatgatcccccatgttgtgcaaaaaagcggttagctcctt
cggtcctccgatcgttgtcagaagtaagttggccgcagtgttatcactcatggttatggcagcactgcataattctcttactgtcatgccatccgt
aagatgcttttctgtgactggtgagtactcaaccaagtcattctgagaatagtgtatgcggcgaccgagttgctcttgcccggcgtcaatacgg
gataataccgcgccacatagcagaactttaaaagtgctcatcattggaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttg
agatccagttcgatgtaacccactcgtgcacccaactgatcttcagcatcttttactttcaccagcgtttctgggtgagcaaaaacaggaaggc
aaaatgccgcaaaaaagggaataagggcgacacggaaatgttgaatactcat

Example 4: Design and Validation of Inducible Cap Genes and Gene of Interest

(143) Two strategies were implemented for the inducible Cap gene expression. In the first strategy, the Cap genes were split into two cassettes, one for the VP1 expression, and the other for VP2 and VP3. To reduce the DNA sequence homology between VP1 and VP2/3 genes in order to increase the vector stability, the DNA sequence of VP1 coding region was optimized for human cell codon usage without change of protein sequence. To maintain the protein coding sequence of membrane-associated accessory protein (MAAP), the first 439 bp of VP1 coding region that included the MAAP gene was excluded for codon optimization. The DNA sequence identity between the optimized VP1 and the original VP2/3 gene was reduced to 80.7%.

(144) An AAV ITR flanked transgene expression cassette, such as the GFP gene driven by CMV promoter,

was also included in the vector as a representative of a gene of interest (GOI) inserted into the vector. In addition, the third antibiotic resistance gene for hygromycin driven by human PGK promoter was included. All the gene cassettes were subcloned into the piggyBac transposon vectors, and flanked by core insulators.
(145) In the PB-iCapsplit-ITRGFP-hygro-1 vector, the previously described inducible CMV promoter was used to drive the VP1 gene expression without the Kozak sequence before ATG start codon of VP1, while the same promoter was also used for the expression of VP2 and VP3 gene expression by the native leader sequence (ggcgctaag) before the original ACG start codon (See FIG. **6**A). In the PB-iCapsplit-ITRGFP-hygro-2 vector, the VP1 cassette was modified with the native VP2 leader sequence followed by the ACG start codon, while no other changes compared to PB-iCapsplit-ITRGFP-hygro-1 vector (See FIG. **6**B). In the PB-iCapsplit-ITRGFP-hygro-3 vector, the promoter for VP1 cassette was switched to the inducible UbC promoter, and the Kozak sequence was added before the ATG start codon of VP1 gene, without other changes compared to PB-iCapsplit-ITRGFP-hygro-1 vector (See FIG. **6**C). To increase the expression of VP1 protein, the 814 bp UbC intron that was identified as an enhancer (Bianchi et al., "A potent enhancer element in the 5′-UTR intron is crucial for transcriptional regulation of the human ubiquitin C gene," Gene 448 (1): 88-101 (2009) was inserted before the inducible UbC promoter in the PB-iCapsplit-ITRGFP-hygro-3 vector, and resulted in the PB-iCapsplit-ITRGFP-hygro-4 vector (See FIG. **6**D).
(146) In the second strategy, the VP1, VP2 and VP3 genes remained in the original architecture in a single native DNA sequence. The strong inducible CMV promoter was applied to drive the Cap gene expression. To enhance the expression level of VP proteins, the original suboptimal splicing donor site for the VP genes was optimized to the conserved donor sequence (caggtaAGT from 2309 to 2317 bp) without affecting the ratios of VP proteins (Farris and Pintel, "Improved splicing of Adeno-associated viral (AAV) capsid protein-supplying pre-mRNAs leads to increased recombinant AAV vector production," Hum Gene Ther. 19 (12): 1421-1427 (2008). The pPBBG-iCap8cd-ITRGFP-hygro vector is shown in FIG. **6**E.
(147) The sequence of the PB-iCap8split-ITRGFP-hygro-1 vector is provided below:
(148) TABLE-US-00031 PB-iCap8split-ITRGFP-hygro-1 (17,121 bp) (SEQ ID NO: 30)
ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCAGATATACGCGTgacattgattattgactagttattaatagtaatcaattacggggtcatt

agttcatagcccatatatggagttccgcgttacataacttacggtaaatggcccgcctggctgaccgcccaacgacccccgcccattgacgt
caataatgacgtatgttcccatagtaacgccaatagggactttccattgacgtcaatgggtggagtatttacggtaaactgcccacttggcagt
acatcaagtgtatcatatgccaagtacgccccctattgacgtcaatgacggtaaatggcccgcctggcattatgcccagtacatgaccttatg
ggactttcctacttggcagtacatctacgtattagtcatcgctattaccatggtgatgcggttttggcagtacatcaatgggcgtggatagcggtt
tgactcacggggatttccaagtctccaccccattgacgtcaatgggagtttgttttggaaccaaaatcaacgggactttccaaaatgtcgtaac
aactccgccccattgacgcaaatgggcggtaggcgtgtacggtgggaggtctatataagcagagctctccctatcagtgatagagatctccc
tatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagatcgcctggagacgccatccacgctgttttgacctccatagaagaca
ccgggaccgatccagcctgaacgcgcagccgccaatggATGGCCGCTGATGGCTATCTGCCTGATTGGCT
GGAAGATAACCTGAGTGAGGGCATCCGGGAATGGTGGGCCCTGAAGCctggagccccgaag
cccaaagccaaccagcaaaagcaggacgacggccggggtctggtgcttcctggctacaagtacctcggacccttcaacggactcgacaa
gggggagcccgtcaacgcggcggacgcagcggccctcgagcacgacaaggcctacgaccagcagctgcaggcgggtgacaatccgt
acctgcggtataaccacgccgacgccgagtttcaggagcgtctgcaagaagatacgtcttttgggggcaacctcgggcgagcagtcttcca
ggccaagaagcgggttctcgaacctctcggtctggttgaggaaggcgctaagacggctcctggaaagaagaggccggtagAACCT
AGCCCCCAGCGATCTCCAGATTCTTCTACCGGCATCGGCAAAAAGGGCCAACAGCC
TGCGAGAAAGCGGCTGAACTTCGGCCAGACCGGAGACAGCGAGAGCGTGCCCGACC
CTCAGCCTCTGGGCGAGCCCCCTGCCGCCCCTTCTGGCGTGGGCCCCAACACCATGG
CCGCCGGCGGCGGAGCCCCTATGGCCGACAACAACGAGGGCGCCGACGGCGTGGG
AAGCAGCAGCGGCAATTGGCACTGCGACTCCACCTGGCTGGGCGACAGAGTGATCA
CCACATCTACAAGAACCTGGGCCCTGCCCACCTACAACAATCACCTGTACAAGCAG
ATCAGCAATGGAACCTCTGGAGGCGCCACCAACGATAACACCTACTTTGGCTACAG
CACCCCTTGGGGATATTTCGACTTCAACCGGTTCCACTGTCACTTCAGCCCTCGGGA
CTGGCAGCGTCTCATTAACAATAACTGGGGCTTCAGACCAAAGAGACTGAGCTTCA
AGCTGTTCAACATCCAGGTGAAGGAAGTGACCCAAAACGAGGGCACCAAGACCATC
GCCAATAACCTCACCTCCACAATCCAAGTGTTCACCGATTCCGAGTACCAGCTGCCT
TACGTGCTGGGCTCCGCCCACCAGGGCTGCCTGCCACCATTCCCCGCCGACGTGTTC
ATGATCCCTCAGTACGGCTACCTGACACTGAACAACGGATCTCAAGCAGTGGGCAG
AAGCTCCTTCTACTGTCTGGAGTACTTCCCTAGCCAGATGCTGAGGACAGGCAACAA
TTTCCAGTTCACCTACACATTCGAGGACGTCCCTTTTCACAGCTCTTATGCCCATAGC
CAGAGCCTGGATAGACTGATGAACCCTCTGATCGACCAGTATCTGTATTACCTGAGC
AGAACGCAAACAACAGGCGGCACCGCTAATACCCAGACCCTGGGTTTCAGCCAAGG
CGGCCCTAACACAATGGCCAATCAGGCCAAAAACTGGCTTCCTGGCCCCTGCTACA
GACAGCAAAGAGTGAGCACAACCACCGGCCAGAACAACAACTCTAACTTCGCCTGG
ACAGCCGGCACCAAATACCACCTGAACGGCAGAAACAGCCTGGCCAACCCTGGAAT
CGCTATGGCTACACACAAGGACGATGAGGAACGGTTCTTCCCCAGCAACGGCATCC
TGATCTTCGGCAAGCAGAACGCCGCTCGCGACAACGCCGACTACAGCGACGTGATG
CTGACCAGCGAGGAAGAAATCAAGACAACAAACCCGGTCGCCACCGAGGAATACG
GCATTGTGGCCGATAACCTGCAGCAGCAGAATACCGCCCCTCAGATCGGCACCGTG
AACTCCCAGGGAGCCCTGCCTGGCATGGTGTGGCAGAACAGAGATGTGTACCTGCA
GGGCCCTATCTGGGCCAAGATCCCCCACACCGACGGAAATTTCCATCCAAGCCCTCT
GATGGGCGGATTTGGCCTGAAGCACCCCCCTCCACAGATTCTCATCAAAAACACACC
TGTGCCAGCCGACCCTCCCACAACATTTAATCAGTCTAAGCTGAATAGCTTCATCAC
CCAGTACAGCACCGGCCAGGTGTCCGTGGAAATCGAGTGGGAGCTGCAGAAAGAGA
ACAGCAAGAGATGGAACCCTGAGATCCAGTACACCAGCAACTACTACAAAAGCACA
AGCGTGGACTTCGCCGTTAATACTGAGGGCGTCTACTCAGAGCCTCGGCCCATCGGC
ACTCGGTACCTGACCAGAAACCTGaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccct
aaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcaca
aataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaagc
ccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatc
cccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgag
cctgcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctat
aggattgaggtcagagcGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTC
ATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCC
GCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCC
CATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTA

AACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGA
CGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGA
CTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGG
TTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGT
CTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGAACCAAAATCAACGGGACTTT
CCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACG
GTGGGAGGTCTATATAAGCAGAGCTCTCCCTATCAGTGATAGAGATCTCCCTATCAG
TGATAGAGATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGCC
ATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTatcccaattctgat
gcgccggtgatcagatcaaaaacttcagccaggtacggcgctaagacggctcctggaaagaagaggccggtagagccatcaccccagc
gttctccagactcctctacgggcatcggcaagaaaggccaacagcccgccagaaaaagactcaattttggtcagactggcgactcagagtc
agttccagaccctcaacctctcggagaacctccagcagcgccctctggtgtgggacctaatacaatggctgcaggcggtggcgcaccaat
ggcagacaataacgaaggcgccgacggagtgggtagttcctcgggaaattggcattgcgattccacatggctgggcgacagagtcatcac
caccagcacccgaacctgggccctgcccacctacaacaaccacctctacaagcaaatctccaacgggacatcgggaggagccaccaac
gacaacacctacttcggctacagcaccccctgggggtattttgactttaacagattccactgccacttttcaccacgtgactggcagcgactca
tcaacaacaactggggattccggcccaagagactcagcttcaagctcttcaacatccaggtcaaggaggtcacgcagaatgaaggcacca
agaccatcgccaataacctcaccagcaccatccaggtgtttacggactcggagtaccagctgccgtacgttctcggctctgcccaccaggg
ctgcctgcctccgttcccggcggacgtgttcatgattccccagtacggctacctaacactcaacaacggtagtcaggccgtgggacgctcct
ccttctactgcctggaatactttccttcgcagatgctgagaaccggcaacaacttccagtttacttacaccttcgaggacgtgcctttccacagc
agctacgcccacagccagagcttggaccggctgatgaatcctctgattgaccagtacctgtactacttgtctcggactcaaacaacaggagg
cacggcaaatacgcagactctgggcttcagccaaggtgggcctaatacaatggccaatcaggcaaagaactggctgccaggaccctgtta
ccgccaacaacgcgtctcaacgacaaccgggcaaaacaacaatagcaactttgcctggactgctgggaccaaataccatctgaatggaag
aaattcattggctaatcctggcatcgctatggcaacacacaaagacgacgaggagcgtttttttcccagtaacgggatcctgatttttggcaaa
caaaatgctgccagagacaatgcggattacagcgatgtcatgctcaccagcgaggaagaaatcaaaaccactaaccctgtggctacagag
gaatacggtatcgtggcagataacttgcagcagcaaaacacggctcctcaaattggaactgtcaacagccaggggggccttacccggtatgg
tctggcagaaccgggacgtgtacctgcagggtcccatctgggccaagattcctcacacggacggcaacttccacccgtctccgctgatgg
gcggctttggcctgaaacatcctccgcctcagatcctgatcaagaacacgcctgtacctgcggatcctccgaccaccttcaaccagtcaaag
ctgaactctttcatcacgcaatacagcaccggacaggtcagcgtggaaattgaatgggagctgcagaaggaaaacagcaagcgctggaa
ccccgagatccagtacacctccaactactacaaatctacaagtgtggactttgctgttaatacagaaggcgtgtactctgaaccccgccccatt
ggcacccgttacctcacccgtaatctgtaaactagtttgcttgttaatcaataaaccgtttaattcgtttcagttgaGCGGCCGTCGAG
TCTAGAGGGCCCGTTTAAACCCGCTGATCAGCCTCGACTGTGCCTTCTAGTTGCCAG
CCATCTGTTGTTTGCCCCTCCCCCGTGCCTTCCTTGACCCTGGAAGGTGCCACTCCCA
CTGTCCTTTCCTAATAAAATGAGGAAATTGCATCGCATTGTCTGAGTAGGTGTCATT
CTATTCTGGGGGGTGGGGTGGGGCAGGACAGCAAGGGGGAGGATTGGGAAGACAA
TAGCAGGCATGCTGGGGATGCGGTGGGCTCTATGGCTTCTGAGGCGGAAAGAACCA
GCTGGGGCTCTAGGGGGTATCCCCGGCGCGCCGAATTCGTTAACAAGCTtTAATTAaC
GCgtATAcCTAggATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAG
AAACAGTGGCAGCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCA
TTGGTATGGCTTTTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGA
GAAGCGTTCAGAGGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCG
CACGCTGCCGGCTCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGG
CGGCTCGCTGCTGCCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTG
GGGGGGGGCTGTCCCTGGCCTCCAAGGCCAGCTTCCCACAATAAGTTGGGTGAATTT
TGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATGGGAATTCTGTG
GAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcAAACGCCAGCAACG
CGGCCTTTTTACGGTTCCTGGCCTTTTGCTGGCCTTTTGCTCACATGTCCTGCAGGCA
GCTGCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGA
CCTTTGGTCGCCCGGCCTCAGTGAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAAC
TCCATCACTAGGGGTTCCTGCGGCCGCACGCGTGGAGCTAGTTATTAATAGTAATCA
ATTACGGGGTCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACG
GTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATG
ACGTATGTTCCCATAGTAACGTCAATAGGGACTTTCCATTGACGTCAATGGGTGGAG
TATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACG
CCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATG

ACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCA
TGGTGATGCGGTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGG
GATTTCCAAGTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGCACCAAAATCA
ACGGGACTTTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTA
GGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTCGTTTAGTGAACCGTCAGATC
GCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCC
AGCCTCCGCGGATTCGAATCCCGGCCGGGAACGGTGCATTGGAACGCGGATTCCCC
GTGCCAAGAGTGACGTAAGTACCGCCTATAGAGTCTATAGGCCCACAAAAAATGCT
TTCTTCTTTTAATATACTTTTTTGTTTATCTTATTTCTAATACTTTCCCTAATCTCTTTC
TTTCAGGGCAATAATGATACAATGTATCATGCCTCTTTGCACCATTCTAAAGAATAA
CAGTGATAATTTCTGGGTTAAGGCAATAGCAATATTTCTGCATATAAATATTTCTGC
ATATAAATTGTAACTGATGTAAGAGGTTTCATATTGCTAATAGCAGCTACAATCCAG
CTACCATTCTGCTTTTATTTTATGGTTGGGATAAGGCTGGATTATTCTGAGTCCAAGC
TAGGCCCTTTTGCTAATCATGTTCATACCTCTTATCTTCCTCCCACAGCTCCTGGGCA
ACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTGGGATTCGAACATC
GATTGAATTCTGAATGGTGAGCAAGGGCGAGGAGCTGTTCACCGGGGTGGTGCCCA
TCCTGGTCGAGCTGGACGGCGACGTAAACGGCCACAAGTTCAGCGTGTCCGGCGAG
GGCGAGGGCGATGCCACCTACGGCAAGCTGACCCTGAAGTTCATCTGCACCACCGG
CAAGCTGCCCGTGCCCTGGCCCACCCTCGTGACCACCCTGACCTACGGCGTGCAGTG
CTTCAGCCGCTACCCCGACCACATGAAGCAGCACGACTTCTTCAAGTCCGCCATGCC
CGAAGGCTACGTCCAGGAGCGCACCATCTTCTTCAAGGACGACGGCAACTACAAGA
CCCGCGCCGAGGTGAAGTTCGAGGGCGACACCCTGGTGAACCGCATCGAGCTGAAG
GGCATCGACTTCAAGGAGGACGGCAACATCCTGGGGCACAAGCTGGAGTACAACTA
CAACAGCCACAACGTCTATATCATGGCCGACAAGCAGAAGAACGGCATCAAGGTGA
ACTTCAAGATCCGCCACAACATCGAGGACGGCAGCGTGCAGCTCGCCGACCACTAC
CAGCAGAACACCCCCATCGGCGACGGCCCCGTGCTGCTGCCCGACAACCACTACCT
GAGCACCCAGTCCGCCCTGAGCAAAGACCCCAACGAGAAGCGCGATCACATGGTCC
TGCTGGAGTTCGTGACCGCCGCCGGGATCACTCTCGGCATGGACGAGCTGTACAAGT
ACTCAGATCTCGAGCTCAAGTAGGGATCCTCTAGAGTCGACCTGCAGAAGCTTGCCT
CGAGCAGCGCTGCTCGAGAGATCTACGGGTGGCATCCCTGTGACCCCTCCCCAGTGC
CTCTCCTGGCCCTGGAAGTTGCCACTCCAGTGCCCACCAGCCTTGTCCTAATAAAAT
TAAGTTGCATCATTTTGTCTGACTAGGTGTCCTTCTATAATATTATGGGGTGGAGGG
GGGTGGTATGGAGCAAGGGGCAAGTTGGGAAGACAACCTGTAGGGCCTGCGGGGTC
TATTGGGAACCAAGCTGGAGTGCAGTGGCACAATCTTGGCTCACTGCAATCTCCGCC
TCCTGGGTTCAAGCGATTCTCCTGCCTCAGCCTCCCGAGTTGTTGGGATTCCAGGCAT
GCATGACCAGGCTCAGCTAATTTTTGTTTTTTTGGTAGAGACGGGGTTTCACCATATT
GGCCAGGCTGGTCTCCAACTCCTAATCTCAGGTGATCTACCCACCTTGGCCTCCCAA
ATTGCTGGGATTACAGGCGTGAACCACTGCTCCCTTCCCTGTCCTTCTGATTTTGTAG
GTAACCACGTGCGGACCGAGCGGCCGCAGGAACCCCTAGTGATGGAGTTGGCCACT
CCCTCTCTGCGCGCTCGCTCGCTCACTGAGGCCGGGCGACCAAAGGTCGCCCGACGC
CCGGGCTTTGCCCGGGCGGCCTCAGTGAGCGAGCGAGCGCGCAGCTGCCTGCAGGG
GCGCCTGATGCGGTATTTTCTCCTTACGCATCTGTGCGGTATTTCACACCGCATACGT
CgtaGCtGATcaATTgGCGCGCCGAATTCGTTAACAAGCTtTAATTAaCGCgtATAcCTAgg
ATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCA
GCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTT
TTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGA
GGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGC
TCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTG
CCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGT
CCCTGGCCTCCAAGGCCAGCTTCCCACAATAAGTTGGGTGAATTTTGGCTCATTCCT
CCTTTCTATAGGATTGAGGTCAGAGCggggttggggttgcgccttttccaaggcagccctgggtttgcgcagggac
gcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcacccgga
tcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgccggac
gtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccgcgatgggc

tgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtggggcggta
gtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgaccgaatca
ccgacctctctccccagAAGCTCCCGGGAGCTTGTATATCCATTTTCGGATCTGATCAGCACGT
GTTGACAATTAATCATCGGCATAGTATATCGGCATAGTATAATACGACAAGGTGAG
GAACgccaccATGAAAAAGCCTGAACTCACCGCGACGTCTGTCGAGAAGTTTCTGATC
GAAAAGTTCGACAGCGTCTCCGACCTGATGCAGCTCTCGGAGGGCGAAGAATCTCG
TGCTTTCAGCTTCGATGTAGGAGGGCGTGGATATGTCCTGCGGGTAAATAGCTGCGC
CGATGGTTTCTACAAAGATCGTTATGTTTATCGGCACTTTGCATCGGCCGCGCTCCCG
ATTCCGGAAGTGCTTGACATTGGGGAATTCAGCGAGAGCCTGACCTATTGCATCTCC
CGCCGTGCACAGGGTGTCACGTTGCAAGACCTGCCTGAAACCGAACTGCCCGCTGTT
CTGCAGCCGGTCGCGGAGGCCATGGATGCGATCGCTGCGGCCGATCTTAGCCAGAC
GAGCGGGTTCGGCCCATTCGGACCGCAAGGAATCGGTCAATACACTACATGGCGTG
ATTTCATATGCGCGATTGCTGATCCCCATGTGTATCACTGGCAAACTGTGATGGACG
ACACCGTCAGTGCGTCCGTCGCGCAGGCTCTCGATGAGCTGATGCTTTGGGCCGAGG
ACTGCCCCGAAGTCCGGCACCTCGTGCACGCGGATTTCGGCTCCAACAATGTCCTGA
CGGACAATGGCCGCATAACAGCGGTCATTGACTGGAGCGAGGCGATGTTCGGGGAT
TCCCAATACGAGGTCGCCAACATCTTCTTCTGGAGGCCGTGGTTGGCTTGTATGGAG
CAGCAGACGCGCTACTTCGAGCGGAGGCATCCGGAGCTTGCAGGATCGCCGCGGCT
CCGGGCGTATATGCTCCGCATTGGTCTTGACCAACTCTATCAGAGCTTGGTTGACGG
CAATTTCGATGATGCAGCTTGGGCGCAGGGTCGATGCGACGCAATCGTCCGATCCGG
AGCCGGGACTGTCGGGCGTACACAAATCGCCCGCAGAAGCGCGGCCGTCTGGACCG
ATGGCTGTGTAGAAGTACTCGCCGATAGTGGAAACCGACGCCCCAGCACTCGTCCG
AGGGCAAAGGAATAGAGCGCGGGGATCTCATGCTGGAGTTCTTCGCCCACCCCAAC
TTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCACAAATTTCACA
AATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACTCATCAATGTAT
CTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGCAggTGTCCTCACAG
GAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGGAATTGACTGGA
TTCCTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCCAGGTGTCTGCAG
GCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTGCCACCTTCCCCG
TGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGGGAGCGCCGGACC
GGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAGGGACGTAATTA
CATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGCCGCCACCGCGGTG
GAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTAATACGACTCACTA
TAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTATCGATAAGCTTGAT
ATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAGTAGAACATGAAAT
AACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAAAAGATAATCATGC
GTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACACTTACCGCATTGA
CAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCCTAAATGCACAGCG
ACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAATGCATGCGTCAATT
TTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGATCATATCGTCGGGT
CTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCATCGGGGAGGAAG
AAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAGTTTGCCGAGGAT
GACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATGATGATTCGGGAA
GGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCCACCTGGGATACC
AGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCGAAGCGCATCAGC
AACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGTGCAGATTAATGA
CAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATCCTGGCTGGATGC
CGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGCGCTTGGCGTAAT
CATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAATTCCACACAACAT
ACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAGTGAGCTAACTCA
CATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCTGTCGTGCCAGC
TGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGTATTGGGCGCTCTT
CCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGCGGCGAGCGGTAT
CAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGGATAACGCAGGA

AAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAAAAAGGCCGCGT
TGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACAAAAATCGACGCT
CAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGGCGTTTCCCCCT
GGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCGGATACCTGTCC
GCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGTAGGTATCTCA
GTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACCCCCCGTTCAGC
CCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCCGGTAAGACACG
ACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGCGAGGTATGTA
GGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACACTAGAAGGAC
AGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAGAGTTGGTAG
CTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGTTTGCAAGCA
GCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATCTTTTCTACGGG
GTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCATGAGATTATC
AAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTTAAATCAATCTA
AAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCAGTGAGGCACC
TATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCCCCGTCGTGTAG
ATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAATGATACCGCG
AGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAGCCGGAAGGG
CCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCTATTAATTGTT
GCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAACGTTGTTGCCA
TTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCATTCAGCTCCGG
TTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAAAAGCGGTTAG
CTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGTTATCACTCATG
GTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAGATGCTTTTCTG
TGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCGGCGACCGAGTT
GCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCAGAACTTTAAAA
GTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGATCTTACCGCTG
TTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTCAGCATCTTTTA
CTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGCCGCAAAAAAG
GGAATAAGGGCGACACGGAAATGTTGAATACTCAT

(149) The sequence of the PB-iCap8split-ITRGFP-hygro-2 vector is provided below:

(150) TABLE-US-00032 PB-iCap8split-ITRGFP-hygro-2 (17,119 bp) (SEQ ID NO: 31)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT

CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCAGATATACGCGTgacattgattattgactagttattaatagtaatcaattacggggtcatt
agttcatagcccatatatggagttccgcgttacataacttacggtaaatggcccgcctggctgaccgcccaacgacccccgcccattgacgt
caataatgacgtatgttcccatagtaacgccaatagggactttccattgacgtcaatgggtggagtatttacggtaaactgcccacttggcagt
acatcaagtgtatcatatgccaagtacgccccctattgacgtcaatgacggtaaatggcccgcctggcattatgcccagtacatgaccttatg
ggactttcctacttggcagtacatctacgtattagtcatcgctattaccatggtgatgcggttttggcagtacatcaatgggcgtggatagcgg
tttgactcacggggatttccaagtctccaccccattgacgtcaatgggagtttgttttggaaccaaaatcaacgggactttccaaaatgtcgta
acaactccgccccattgacgcaaatgggcggtaggcgtgtacggtgggaggtctatataagcagagctctccctatcagtgatagagatctccc
tatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagatcgcctggagacgccatccacgctgttttgacctccatagaagaca
ccgggaccgatccagcctgaacgcgcaggcgctaagACGGCCGCTGATGGCTATCTGCCTGATTGGCTG
GAAGATAACCTGAGTGAGGGCATCCGGGAATGGTGGGCCCTGAAGCctggagccccgaagc
ccaaagccaaccagcaaaagcaggacgacggccggggtctggtgcttcctggctacaagtacctcggacccttcaacggactcgacaag
ggggagcccgtcaacgcggcggacgcagcggccctcgagcacgacaaggcctacgaccagcagctgcaggcgggtgacaatccgta
cctgcggtataaccacgccgacgccgagtttcaggagcgtctgcaagaagatacgtcttttgggggcaacctcgggcgagcagtcttccag
gccaagaagcgggttctcgaacctctcggtctggttgaggaaggcgctaagacggctcctggaaagaagaggccggtagAACCTA
GCCCCCAGCGATCTCCAGATTCTTCTACCGGCATCGGCAAAAAGGGCCAACAGCCT
GCGAGAAAGCGGCTGAACTTCGGCCAGACCGGAGACAGCGAGAGCGTGCCCGACC
CTCAGCCTCTGGGCGAGCCCCCTGCCGCCCCTTCTGGCGTGGGCCCCAACACCATGG
CCGCCGGCGGCGGAGCCCCTATGGCCGACAACAACGAGGGCGCCGACGGCGTGGG
AAGCAGCAGCGGCAATTGGCACTGCGACTCCACCTGGCTGGGCGACAGAGTGATCA
CCACATCTACAAGAACCTGGGCCCTGCCCACCTACAACAATCACCTGTACAAGCAG
ATCAGCAATGGAACCTCTGGAGGCGCCACCAACGATAACACCTACTTTGGCTACAG
CACCCCTTGGGGATATTTCGACTTCAACCGGTTCCACTGTCACTTCAGCCCTCGGGA
CTGGCAGCGTCTCATTAACAATAACTGGGGCTTCAGACCAAAGAGACTGAGCTTCA
AGCTGTTCAACATCCAGGTGAAGGAAGTGACCCAAAACGAGGGCACCAAGACCATC
GCCAATAACCTCACCTCCACAATCCAAGTGTTCACCGATTCCGAGTACCAGCTGCCT
TACGTGCTGGGCTCCGCCCACCAGGGCTGCCTGCCACCATTCCCCGCCGACGTGTTC
ATGATCCCTCAGTACGGCTACCTGACACTGAACAACGGATCTCAAGCAGTGGGCAG
AAGCTCCTTCTACTGTCTGGAGTACTTCCCTAGCCAGATGCTGAGGACAGGCAACAA
TTTCCAGTTCACCTACACATTCGAGGACGTCCCTTTTCACAGCTCTTATGCCCATAGC
CAGAGCCTGGATAGACTGATGAACCCTCTGATCGACCAGTATCTGTATTACCTGAGC
AGAACGCAAACAACAGGCGGCACCGCTAATACCCAGACCCTGGGTTTCAGCCAAGG
CGGCCCTAACACAATGGCCAATCAGGCCAAAAACTGGCTTCCTGGCCCCTGCTACA
GACAGCAAAGAGTGAGCACAACCACCGGCCAGAACAACAACTCTAACTTCGCCTGG
ACAGCCGGCACCAAATACCACCTGAACGGCAGAAACAGCCTGGCCAACCCTGGAAT
CGCTATGGCTACACACAAGGACGATGAGGAACGGTTCTTCCCCAGCAACGGCATCC
TGATCTTCGGCAAGCAGAACGCCGCTCGCGACAACGCCGACTACAGCGACGTGATG
CTGACCAGCGAGGAAGAAATCAAGACAACAAACCCGGTCGCCACCGAGGAATACG
GCATTGTGGCCGATAACCTGCAGCAGCAGAATACCGCCCCTCAGATCGGCACCGTG
AACTCCCAGGGAGCCCTGCCTGGCATGGTGTGGCAGAACAGAGATGTGTACCTGCA
GGGCCCTATCTGGGCCAAGATCCCCCACACCGACGGAAATTTCCATCCAAGCCCTCT
GATGGGCGGATTTGGCCTGAAGCACCCCCCTCCACAGATTCTCATCAAAAACACACC
TGTGCCAGCCGACCCTCCCACAACATTTAATCAGTCTAAGCTGAATAGCTTCATCAC
CCAGTACAGCACCGGCCAGGTGTCCGTGGAAATCGAGTGGGAGCTGCAGAAAGAGA
ACAGCAAGAGATGGAACCCTGAGATCCAGTACACCAGCAACTACTACAAAAGCACA
AGCGTGGACTTCGCCGTTAATACTGAGGGCGTCTACTCAGAGCCTCGGCCCATCGGC
ACTCGGTACCTGACCAGAAACCTGaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccct
aaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcaca
aataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagcccccccccaaag

cccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatc
cccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgag
cctgcagacacctgggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctat
aggattgaggtcagagcGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGGTC
ATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCC
GCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCC
CATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTA
AACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGA
CGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGA
CTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGG
TTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGT
CTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGAACCAAAATCAACGGGACTTT
CCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACG
GTGGGAGGTCTATATAAGCAGAGCTCTCCCTATCAGTGATAGAGATCTCCCTATCAG
TGATAGAGATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGCC
ATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTatcccaattctgat
gcgccggtgatcagatcaaaaacttcagccaggtacggcgctaagacggctcctggaaagaagaggccggtagagccatcaccccagc
gttctccagactcctctacgggcatcggcaagaaaggccaacagcccgccagaaaaagactcaattttggtcagactggcgactcagagtc
agttccagaccctcaacctctcggagaacctccagcagcgccctctggtgtgggacctaatacaatggctgcaggcggtggcgcaccaat
ggcagacaataacgaaggcgccgacggagtgggtagttcctcgggaaattggcattgcgattccacatggctgggcgacagagtcatcac
caccagcacccgaacctgggccctgcccacctacaacaaccacctctacaagcaaatctccaacgggacatcgggaggagccaccaac
gacaacacctacttcggctacagcaccccctgggggtattttgactttaacagattccactgccacttttcaccacgtgactggcagcgactca
tcaacaacaactggggattccggcccaagagactcagcttcaagctcttcaacatccaggtcaaggaggtcacgcagaatgaaggcacca
agaccatcgccaataacctcaccagcaccatccaggtgtttacggactcggagtaccagctgccgtacgttctcggctctgcccaccaggg
ctgcctgcctccgttcccggcggacgtgttcatgattccccagtacggctacctaacactcaacaacggtagtcaggccgtgggacgctcct
ccttctactgcctggaatactttccttcgcagatgctgagaaccggcaacaacttccagtttacttacaccttcgaggacgtgcctttccacagc
agctacgcccacagccagagcttggaccggctgatgaatcctctgattgaccagtacctgtactacttgtctcggactcaaacaacaggagg
cacggcaaatacgcagactctgggcttcagccaaggtgggcctaatacaatggccaatcaggcaaagaactggctgccaggaccctgtta
ccgccaacaacgcgtctcaacgacaaccgggcaaaacaacaatagcaactttgcctggactgctgggaccaaataccatctgaatggaag
aaattcattggctaatcctggcatcgctatggcaacacacaaagacgacgaggagcgtttttttcccagtaacgggatcctgatttttggcaaa
caaaatgctgccagagacaatgcggattacagcgatgtcatgctcaccagcgaggaagaaatcaaaaccactaaccctgtggctacagag
gaatacggtatcgtggcagataacttgcagcagcaaaacacggctcctcaaattggaactgtcaacagccagggggccttacccggtatgg
tctggcagaaccgggacgtgtacctgcagggtcccatctgggccaagattcctcacacggacggcaacttccacccgtctccgctgatgg
gcggctttggcctgaaacatcctccgcctcagatcctgatcaagaacacgcctgtacctgcggatcctccgaccaccttcaaccagtcaaag
ctgaactctttcatcacgcaatacagcaccggacaggtcagcgtggaaattgaatgggagctgcagaaggaaaacagcaagcgctggaa
ccccgagatccagtacacctccaactactacaaatctacaagtgtggactttgctgttaatacagaaggcgtgtactctgaaccccgccccatt
ggcacccgttacctcacccgtaatctgtaaactagtttgcttgttaatcaataaaccgtttaattcgtttcagttgaGCGGCCGTCGAG
TCTAGAGGGCCCGTTTAAACCCGCTGATCAGCCTCGACTGTGCCTTCTAGTTGCCAG
CCATCTGTTGTTTGCCCCTCCCCCGTGCCTTCCTTGACCCTGGAAGGTGCCACTCCCA
CTGTCCTTTCCTAATAAAATGAGGAAATTGCATCGCATTGTCTGAGTAGGTGTCATT
CTATTCTGGGGGGTGGGGTGGGGCAGGACAGCAAGGGGGAGGATTGGGAAGACAA
TAGCAGGCATGCTGGGGATGCGGTGGGCTCTATGGCTTCTGAGGCGGAAAGAACCA
GCTGGGGCTCTAGGGGGTATCCCCGGCGCGCCGAATTCGTTAACAAGCTtTAATTAaC
GCgtATAcCTAggATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAG
AAACAGTGGCAGCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCA
TTGGTATGGCTTTTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGA
GAAGCGTTCAGAGGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCG
CACGCTGCCGGCTCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGG
CGGCTCGCTGCTGCCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTG
GGGGGGGGCTGTCCCTGGCCTCCAAGGCCAGCTTCCCACAATAAGTTGGGTGAATTT
TGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATGGGAATTCTGTG
GAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcAAACGCCAGCAACG
CGGCCTTTTTACGGTTCCTGGCCTTTTGCTGGCCTTTTGCTCACATGTCCTGCAGGCA
GCTGCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGA

CCTTTGGTCGCCCGGCCTCAGTGAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAAC
TCCATCACTAGGGGTTCCTGCGGCCGCACGCGTGGAGCTAGTTATTAATAGTAATCA
ATTACGGGGTCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACG
GTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATG
ACGTATGTTCCCATAGTAACGTCAATAGGGACTTTCCATTGACGTCAATGGGTGGAG
TATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACG
CCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATG
ACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCA
TGGTGATGCGGTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGG
GATTTCCAAGTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGCACCAAAATCA
ACGGGACTTTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTA
GGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTCGTTTAGTGAACCGTCAGATC
GCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCC
AGCCTCCGCGGATTCGAATCCCGGCCGGGAACGGTGCATTGGAACGCGGATTCCCC
GTGCCAAGAGTGACGTAAGTACCGCCTATAGAGTCTATAGGCCCACAAAAAATGCT
TTCTTCTTTTAATATACTTTTTTGTTTATCTTATTTCTAATACTTTCCCTAATCTCTTTC
TTTCAGGGCAATAATGATACAATGTATCATGCCTCTTTGCACCATTCTAAAGAATAA
CAGTGATAATTTCTGGGTTAAGGCAATAGCAATATTTCTGCATATAAATATTTCTGC
ATATAAATTGTAACTGATGTAAGAGGTTTCATATTGCTAATAGCAGCTACAATCCAG
CTACCATTCTGCTTTTATTTTATGGTTGGGATAAGGCTGGATTATTCTGAGTCCAAGC
TAGGCCCTTTTGCTAATCATGTTCATACCTCTTATCTTCCTCCCACAGCTCCTGGGCA
ACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTGGGATTCGAACATC
GATTGAATTCTGAATGGTGAGCAAGGGCGAGGAGCTGTTCACCGGGGTGGTGCCCA
TCCTGGTCGAGCTGGACGGCGACGTAAACGGCCACAAGTTCAGCGTGTCCGGCGAG
GGCGAGGGCGATGCCACCTACGGCAAGCTGACCCTGAAGTTCATCTGCACCACCGG
CAAGCTGCCCGTGCCCTGGCCCACCCTCGTGACCACCCTGACCTACGGCGTGCAGTG
CTTCAGCCGCTACCCCGACCACATGAAGCAGCACGACTTCTTCAAGTCCGCCATGCC
CGAAGGCTACGTCCAGGAGCGCACCATCTTCTTCAAGGACGACGGCAACTACAAGA
CCCGCGCCGAGGTGAAGTTCGAGGGCGACACCCTGGTGAACCGCATCGAGCTGAAG
GGCATCGACTTCAAGGAGGACGGCAACATCCTGGGGCACAAGCTGGAGTACAACTA
CAACAGCCACAACGTCTATATCATGGCCGACAAGCAGAAGAACGGCATCAAGGTGA
ACTTCAAGATCCGCCACAACATCGAGGACGGCAGCGTGCAGCTCGCCGACCACTAC
CAGCAGAACACCCCCATCGGCGACGGCCCCGTGCTGCTGCCCGACAACCACTACCT
GAGCACCCAGTCCGCCCTGAGCAAAGACCCCAACGAGAAGCGCGATCACATGGTCC
TGCTGGAGTTCGTGACCGCCGCCGGGATCACTCTCGGCATGGACGAGCTGTACAAGT
ACTCAGATCTCGAGCTCAAGTAGGGATCCTCTAGAGTCGACCTGCAGAAGCTTGCCT
CGAGCAGCGCTGCTCGAGAGATCTACGGGTGGCATCCCTGTGACCCCTCCCCAGTGC
CTCTCCTGGCCCTGGAAGTTGCCACTCCAGTGCCCACCAGCCTTGTCCTAATAAAAT
TAAGTTGCATCATTTTGTCTGACTAGGTGTCCTTCTATAATATTATGGGGGGAGGG
GGGTGGTATGGAGCAAGGGGCAAGTTGGGAAGACAACCTGTAGGGCCTGCGGGGTC
TATTGGGAACCAAGCTGGAGTGCAGTGGCACAATCTTGGCTCACTGCAATCTCCGCC
TCCTGGGTTCAAGCGATTCTCCTGCCTCAGCCTCCCGAGTTGTTGGGATTCCAGGCAT
GCATGACCAGGCTCAGCTAATTTTTGTTTTTTTGGTAGAGACGGGGTTTCACCATATT
GGCCAGGCTGGTCTCCAACTCCTAATCTCAGGTGATCTACCCACCTTGGCCTCCCAA
ATTGCTGGGATTACAGGCGTGAACCACTGCTCCCTTCCCTGTCCTTCTGATTTTGTAG
GTAACCACGTGCGGACCGAGCGGCCGCAGGAACCCCTAGTGATGGAGTTGGCCACT
CCCTCTCTGCGCGCTCGCTCGCTCACTGAGGCCGGGCGACCAAAGGTCGCCCGACGC
CCGGGCTTTGCCCGGGCGGCCTCAGTGAGCGAGCGAGCGCGCAGCTGCCTGCAGGG
GCGCCTGATGCGGTATTTTCTCCTTACGCATCTGTGCGGTATTTCACACCGCATACGT
CgtaGCtGATcaATTgGCGCGCCGAATTCGTTAACAAGCTtTAATTAaCGCgtATAcCTAgg
ATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCA
GCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTT
TTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGA
GGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGC

TCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTG
CCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGT
CCCTGGCCTCCAAGGCCAGCTTCCCACAATAAGTTGGGTGAATTTTGGCTCATTCCT
CCTTTCTATAGGATTGAGGTCAGAGCggggttggggttgcgccttttccaaggcagccctgggtttgcgcagggac
gcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcacccgga
tcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgccggac
gtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccgcgatgggc
tgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtggggcggta
gtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgaccgaatca
ccgacctctctccccagAAGCTCCCGGGAGCTTGTATATCCATTTTCGGATCTGATCAGCACGT
GTTGACAATTAATCATCGGCATAGTATATCGGCATAGTATAATACGACAAGGTGAG
GAACgccaccATGAAAAAGCCTGAACTCACCGCGACGTCTGTCGAGAAGTTTCTGATC
GAAAAGTTCGACAGCGTCTCCGACCTGATGCAGCTCTCGGAGGGCGAAGAATCTCG
TGCTTTCAGCTTCGATGTAGGAGGGCGTGGATATGTCCTGCGGGTAAATAGCTGCGC
CGATGGTTTCTACAAAGATCGTTATGTTTATCGGCACTTTGCATCGGCCGCGCTCCCG
ATTCCGGAAGTGCTTGACATTGGGGAATTCAGCGAGAGCCTGACCTATTGCATCTCC
CGCCGTGCACAGGGTGTCACGTTGCAAGACCTGCCTGAAACCGAACTGCCCGCTGTT
CTGCAGCCGGTCGCGGAGGCCATGGATGCGATCGCTGCGGCCGATCTTAGCCAGAC
GAGCGGGTTCGGCCCATTCGGACCGCAAGGAATCGGTCAATACACTACATGGCGTG
ATTTCATATGCGCGATTGCTGATCCCCATGTGTATCACTGGCAAACTGTGATGGACG
ACACCGTCAGTGCGTCCGTCGCGCAGGCTCTCGATGAGCTGATGCTTTGGGCCGAGG
ACTGCCCCGAAGTCCGGCACCTCGTGCACGCGGATTTCGGCTCCAACAATGTCCTGA
CGGACAATGGCCGCATAACAGCGGTCATTGACTGGAGCGAGGCGATGTTCGGGGAT
TCCCAATACGAGGTCGCCAACATCTTCTTCTGGAGGCCGTGGTTGGCTTGTATGGAG
CAGCAGACGCGCTACTTCGAGCGGAGGCATCCGGAGCTTGCAGGATCGCCGCGGCT
CCGGGCGTATATGCTCCGCATTGGTCTTGACCAACTCTATCAGAGCTTGGTTGACGG
CAATTTCGATGATGCAGCTTGGGCGCAGGGTCGATGCGACGCAATCGTCCGATCCGG
AGCCGGGACTGTCGGGCGTACACAAATCGCCCGCAGAAGCGCGGCCGTCTGGACCG
ATGGCTGTGTAGAAGTACTCGCCGATAGTGGAAACCGACGCCCCAGCACTCGTCCG
AGGGCAAAGGAATAGAGCGCGGGGATCTCATGCTGGAGTTCTTCGCCCACCCCAAC
TTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCACAAATTTCACA
AATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACTCATCAATGTAT
CTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGCAggTGTCCTCACAG
GAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGGAATTGACTGGA
TTCCTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCCAGGTGTCTGCAG
GCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTGCCACCTTCCCCG
TGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGGGAGCGCCGGACC
GGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAGGGACGTAATTA
CATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGCCGCCACCGCGGTG
GAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTAATACGACTCACTA
TAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTATCGATAAGCTTGAT
ATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAGTAGAACATGAAAT
AACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAAAAGATAATCATGC
GTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACACTTACCGCATTGA
CAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCCTAAATGCACAGCG
ACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAATGCATGCGTCAATT
TTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGATCATATCGTCGGGT
CTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCATCGGGGAGGAAG
AAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAGTTTGCCGAGGAT
GACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATGATGATTCGGGAA
GGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCCACCTGGGATACC
AGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCGAAGCGCATCAGC
AACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGTGCAGATTAATGA
CAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATCCTGGCTGGATGC

CGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGCGCTTGGCGTAAT
CATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAATTCCACACAACAT
ACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAGTGAGCTAACTCA
CATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCTGTCGTGCCAGC
TGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGTATTGGGCGCTCTT
CCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGCGGCGAGCGGTAT
CAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGGATAACGCAGGA
AAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAAAAAGGCCGCGT
TGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACAAAAATCGACGCT
CAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGGCGTTTCCCCCT
GGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCGGATACCTGTCC
GCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGTAGGTATCTCA
GTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACCCCCCGTTCAGC
CCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCCGGTAAGACACG
ACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGCGAGGTATGTA
GGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACACTAGAAGGAC
AGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAGAGTTGGTAG
CTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGTTTGCAAGCA
GCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATCTTTTCTACGGG
GTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCATGAGATTATC
AAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTTAAATCAATCTA
AAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCAGTGAGGCACC
TATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCCCCGTCGTGTAG
ATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAATGATACCGCG
AGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAGCCGGAAGGG
CCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCTATTAATTGTT
GCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAACGTTGTTGCCA
TTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCATTCAGCTCCGG
TTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAAAAGCGGTTAG
CTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGTTATCACTCATG
GTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAGATGCTTTTCTG
TGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCGGCGACCGAGTT
GCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCAGAACTTTAAAA
GTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGATCTTACCGCTG
TTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTCAGCATCTTTTA
CTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGCCGCAAAAAAG
GGAATAAGGGCGACACGGAAATGTTGAATACTCAT

(151) The sequence of the PB-iCap8split-ITRGFP-hygro-3 vector is provided below:

(152) TABLE-US-00033 PB-iCap8split-ITRGFP-hygro-3 (16,835 bp) (SEQ ID NO: 32)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT

CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCAGATATACGCGTggcctccgcgccgggttttggcgcctcccgcgggcgcccccctc
ctcacggcgagcgctgccacgtcagacgaagggcgcagcgagcgtcctgatccttccgcccggacgctcaggacagcggcccgctgct
cataagactcggccttagaaccccagtatcagcagaaggacattttaggacgggacttgggtgactctagggcactggttttctttccagaga
gcggaacaggcgaggaaaagtagtcccttctcggcgattctgcggagggatctccgtggggcggtgaacgccgatgattatataaggacg
cgccggtccctatcagtgatagagatctccctatcagtgatagagagtgtggcacagctagttccgtcgcagccgggatttgggtcgcggtt
cttgtttgtggatcgctgtgatcgtcacttgacgaacgcgcagccgccATGGCCGCTGATGGCTATCTGCCTGATT
GGCTGGAAGATAACCTGAGTGAGGGCATCCGGGAATGGTGGGCCCTGAAGCctggagcc
ccgaagcccaaagccaaccagcaaaagcaggacgacggccggggtctggtgcttcctggctacaagtacctcggacccttcaacggact
cgacaagggggagcccgtcaacgcggcggacgcagcggccctcgagcacgacaaggcctacgaccagcagctgcaggcgggtgac
aatccgtacctgcggtataaccacgccgacgccgagtttcaggagcgtctgcaagaagatacgtcttttgggggcaacctcgggcgagca
gtcttccaggccaagaagcgggttctcgaacctctcggtctggttgaggaaggcgctaagacggctcctggaaagaagaggccggtagA
ACCTAGCCCCCAGCGATCTCCAGATTCTTCTACCGGCATCGGCAAAAAGGGCCAAC
AGCCTGCGAGAAAGCGGCTGAACTTCGGCCAGACCGGAGACAGCGAGAGCGTGCCC
GACCCTCAGCCTCTGGGCGAGCCCCCTGCCGCCCCTTCTGGCGTGGGCCCCAACACC
ATGGCCGCCGGCGGCGGAGCCCCTATGGCCGACAACAACGAGGGCGCCGACGGCGT
GGGAAGCAGCAGCGGCAATTGGCACTGCGACTCCACCTGGCTGGGCGACAGAGTGA
TCACCACATCTACAAGAACCTGGGCCCTGCCCACCTACAACAATCACCTGTACAAGC
AGATCAGCAATGGAACCTCTGGAGGCGCCACCAACGATAACACCTACTTTGGCTAC
AGCACCCCTTGGGGATATTTCGACTTCAACCGGTTCCACTGTCACTTCAGCCCTCGG
GACTGGCAGCGTCTCATTAACAATAACTGGGGCTTCAGACCAAAGAGACTGAGCTT
CAAGCTGTTCAACATCCAGGTGAAGGAAGTGACCCAAAACGAGGGCACCAAGACCA
TCGCCAATAACCTCACCTCCACAATCCAAGTGTTCACCGATTCCGAGTACCAGCTGC
CTTACGTGCTGGGCTCCGCCCACCAGGGCTGCCTGCCACCATTCCCCGCCGACGTGT
TCATGATCCCTCAGTACGGCTACCTGACACTGAACAACGGATCTCAAGCAGTGGGC
AGAAGCTCCTTCTACTGTCTGGAGTACTTCCCTAGCCAGATGCTGAGGACAGGCAAC
AATTTCCAGTTCACCTACACATTCGAGGACGTCCCTTTTCACAGCTCTTATGCCCATA
GCCAGAGCCTGGATAGACTGATGAACCCTCTGATCGACCAGTATCTGTATTACCTGA
GCAGAACGCAAACAACAGGCGGCACCGCTAATACCCAGACCCTGGGTTTCAGCCAA
GGCGGCCCTAACACAATGGCCAATCAGGCCAAAAACTGGCTTCCTGGCCCCTGCTA
CAGACAGCAAAGAGTGAGCACAACCACCGGCCAGAACAACAACTCTAACTTCGCCT
GGACAGCCGGCACCAAATACCACCTGAACGGCAGAAACAGCCTGGCCAACCCTGGA
ATCGCTATGGCTACACACAAGGACGATGAGGAACGGTTCTTCCCCAGCAACGGCAT
CCTGATCTTCGGCAAGCAGAACGCCGCTCGCGACAACGCCGACTACAGCGACGTGA
TGCTGACCAGCGAGGAAGAAATCAAGACAACAAACCCGGTCGCCACCGAGGAATA
CGGCATTGTGGCCGATAACCTGCAGCAGCAGAATACCGCCCCTCAGATCGGCACCG
TGAACTCCCAGGGAGCCCTGCCTGGCATGGTGTGGCAGAACAGAGATGTGTACCTG
CAGGGCCCTATCTGGGCCAAGATCCCCCACACCGACGGAAATTTCCATCCAAGCCCT
CTGATGGGCGGATTTGGCCTGAAGCACCCCCCTCCACAGATTCTCATCAAAAACACA
CCTGTGCCAGCCGACCCTCCCACAACATTTAATCAGTCTAAGCTGAATAGCTTCATC
ACCCAGTACAGCACCGGCCAGGTGTCCGTGGAAATCGAGTGGGAGCTGCAGAAAGA
GAACAGCAAGAGATGGAACCCTGAGATCCAGTACACCAGCAACTACTACAAAAGCA

CAAGCGTGGACTTCGCCGTTAATACTGAGGGCGTCTACTCAGAGCCTCGGCCCATCG
GCACTCGGTACCTGACCAGAAACCTGaatggctaggatccggccggcctgcaggtgtcctcacaggaacgaagt
ccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaattt
cacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactcatcaatgtatcttattgactggattgagggacagccccccc
caaagcccccagggatgtaattacgtccctcccccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctcccccg
catccccgagccggcagcgtgcggggacagcccgggcacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctcttt
gagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctt
ctataggattgaggtcagagcGACATTGATTATTGACTAGTTATTAATAGTAATCAATTACGGGG
TCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGC
CCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTT
CCCATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGG
TAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATT
GACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGG
GACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGC
GGTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAA
GTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGAACCAAAATCAACGGGACT
TTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTAC
GGTGGGAGGTCTATATAAGCAGAGCTCTCCCTATCAGTGATAGAGATCTCCCTATCA
GTGATAGAGATCGTCGACGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGC
CATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTatcccaattctg
atgcgccggtgatcagatcaaaaacttcagccaggtacggcgctaagacggctcctggaaagaagaggccggtagagccatcaccccag
cgttctccagactcctctacgggcatcggcaagaaaggccaacagcccgccagaaaaagactcaattttggtcagactggcgactcagagt
cagttccagaccctcaacctctcggagaacctccagcagcgccctctggtgtgggacctaatacaatggctgcaggcggtggcgcaccaa
tggcagacaataacgaaggcgccgacggagtgggtagttcctcgggaaattggcattgcgattccacatggctgggcgacagagtcatca
ccaccagcacccgaacctgggccctgcccacctacaacaaccacctctacaagcaaatctccaacgggacatcgggaggagccaccaa
cgacaacacctacttcggctacagcacccccctgggggtattttgactttaacagattccactgccacttttcaccacgtgactggcagcgactc
atcaacaacaactggggattccggcccaagagactcagcttcaagctcttcaacatccaggtcaaggaggtcacgcagaatgaaggcacc
aagaccatcgccaataacctcaccagcaccatccaggtgtttacggactcggagtaccagctgccgtacgttctcggctctgcccaccagg
gctgcctgcctccgttcccggcggacgtgttcatgattccccagtacggctacctaacactcaacaacggtagtcaggccgtgggacgctc
ctccttctactgcctggaatactttccttcgcagatgctgagaaccggcaacaacttccagtttacttacaccttcgaggacgtgcctttccac
agcagctacgcccacagccagagcttggaccggctgatgaatcctctgattgaccagtacctgtactacttgtctcggactcaaacaacagga
ggcacggcaaatacgcagactctgggcttcagccaaggtgggcctaatacaatggccaatcaggcaaagaactggctgccaggaccctg
ttaccgccaacaacgcgtctcaacgacaaccgggcaaaacaacaatagcaactttgcctggactgctgggaccaaataccatctgaatgga
agaaattcattggctaatcctggcatcgctatggcaacacacaaagacgacgaggagcgtttttttcccagtaacgggatcctgatttttggca
aacaaaatgctgccagagacaatgcggattacagcgatgtcatgctcaccagcgaggaagaaatcaaaaccactaaccctgtggctacag
aggaatacggtatcgtggcagataacttgcagcagcaaaacacggctcctcaaattggaactgtcaacagccagggggccttacccggtat
ggtctggcagaaccgggacgtgtacctgcagggtcccatctgggccaagattcctcacacggacggcaacttccacccgtctccgctgat
gggcggctttggcctgaaacatcctccgcctcagatcctgatcaagaacacgcctgtacctgcggatcctccgaccaccttcaaccagtcaa
agctgaactctttcatcacgcaatacagcaccggacaggtcagcgtggaaattgaatgggagctgcagaaggaaaacagcaagcgctgg
aaccccgagatccagtacacctccaactactacaaatctacaagtgtggactttgctgttaatacagaaggcgtgtactctgaaccccgcccc
attggcacccgttacctcacccgtaatctgtaaactagtttgcttgttaatcaataaaccgtttaattcgtttcagttgaGCGGCCGTCGA
GTCTAGAGGGCCCGTTTAAACCCGCTGATCAGCCTCGACTGTGCCTTCTAGTTGCCA
GCCATCTGTTGTTTGCCCCTCCCCCGTGCCTTCCTTGACCCTGGAAGGTGCCACTCCC
ACTGTCCTTTCCTAATAAAATGAGGAAATTGCATCGCATTGTCTGAGTAGGTGTCAT
TCTATTCTGGGGGGTGGGGTGGGGCAGGACAGCAAGGGGGAGGATTGGGAAGACA
ATAGCAGGCATGCTGGGGATGCGGTGGGCTCTATGGCTTCTGAGGCGGAAAGAACC
AGCTGGGGCTCTAGGGGGTATCCCCGGCGCGCCGAATTCGTTAACAAGCTtTAATTAa
CGCgtATAcCTAggATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAA
GAAACAGTGGCAGCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCC
ATTGGTATGGCTTTTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCG
AGAAGCGTTCAGAGGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCC
GCACGCTGCCGGCTCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGG
GCGGCTCGCTGCTGCCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTT
GGGGGGGGGCTGTCCCTGGCCTCCAAGGCCAGCTTCCCACAATAAGTTGGGTGAATT

TTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATGGGAATTCTGT
GGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcAAACGCCAGCAAC
GCGGCCTTTTTACGGTTCCTGGCCTTTTGCTGGCCTTTTGCTCACATGTCCTGCAGGC
AGCTGCGCGCTCGCTCGCTCACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCG
ACCTTTGGTCGCCCGGCCTCAGTGAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAA
CTCCATCACTAGGGGTTCCTGCGGCCGCACGCGTGGAGCTAGTTATTAATAGTAATC
AATTACGGGGTCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACATAACTTAC
GGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGTCAATAAT
GACGTATGTTCCCATAGTAACGTCAATAGGGACTTTCCATTGACGTCAATGGGTGGA
GTATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTAC
GCCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACAT
GACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACC
ATGGTGATGCGGTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGG
GGATTTCCAAGTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGCACCAAAATC
AACGGGACTTTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGT
AGGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTCGTTTAGTGAACCGTCAGAT
CGCCTGGAGACGCCATCCACGCTGTTTTGACCTCCATAGAAGACACCGGGACCGATC
CAGCCTCCGCGGATTCGAATCCCGGCCGGGAACGGTGCATTGGAACGCGGATTCCC
CGTGCCAAGAGTGACGTAAGTACCGCCTATAGAGTCTATAGGCCCACAAAAAATGC
TTTCTTCTTTTAATATACTTTTTTGTTTATCTTATTTCTAATACTTTCCCTAATCTCTTT
CTTTCAGGGCAATAATGATACAATGTATCATGCCTCTTTGCACCATTCTAAAGAATA
ACAGTGATAATTTCTGGGTTAAGGCAATAGCAATATTTCTGCATATAAATATTTCTG
CATATAAATTGTAACTGATGTAAGAGGTTTCATATTGCTAATAGCAGCTACAATCCA
GCTACCATTCTGCTTTTATTTTATGGTTGGGATAAGGCTGGATTATTCTGAGTCCAAG
CTAGGCCCTTTTGCTAATCATGTTCATACCTCTTATCTTCCTCCCACAGCTCCTGGGC
AACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTGGGATTCGAACAT
CGATTGAATTCTGAATGGTGAGCAAGGGCGAGGAGCTGTTCACCGGGGTGGTGCCC
ATCCTGGTCGAGCTGGACGGCGACGTAAACGGCCACAAGTTCAGCGTGTCCGGCGA
GGGCGAGGGCGATGCCACCTACGGCAAGCTGACCCTGAAGTTCATCTGCACCACCG
GCAAGCTGCCCGTGCCCTGGCCCACCCTCGTGACCACCCTGACCTACGGCGTGCAGT
GCTTCAGCCGCTACCCCGACCACATGAAGCAGCACGACTTCTTCAAGTCCGCCATGC
CCGAAGGCTACGTCCAGGAGCGCACCATCTTCTTCAAGGACGACGGCAACTACAAG
ACCCGCGCCGAGGTGAAGTTCGAGGGCGACACCCTGGTGAACCGCATCGAGCTGAA
GGGCATCGACTTCAAGGAGGACGGCAACATCCTGGGGCACAAGCTGGAGTACAACT
ACAACAGCCACAACGTCTATATCATGGCCGACAAGCAGAAGAACGGCATCAAGGTG
AACTTCAAGATCCGCCACAACATCGAGGACGGCAGCGTGCAGCTCGCCGACCACTA
CCAGCAGAACACCCCCATCGGCGACGGCCCCGTGCTGCTGCCCGACAACCACTACC
TGAGCACCCAGTCCGCCCTGAGCAAAGACCCCAACGAGAAGCGCGATCACATGGTC
CTGCTGGAGTTCGTGACCGCCGCCGGGATCACTCTCGGCATGGACGAGCTGTACAA
GTACTCAGATCTCGAGCTCAAGTAGGGATCCTCTAGAGTCGACCTGCAGAAGCTTGC
CTCGAGCAGCGCTGCTCGAGAGATCTACGGGTGGCATCCCTGTGACCCCTCCCCAGT
GCCTCTCCTGGCCCTGGAAGTTGCCACTCCAGTGCCCACCAGCCTTGTCCTAATAAA
ATTAAGTTGCATCATTTTGTCTGACTAGGTGTCCTTCTATAATATTATGGGGTGGAGG
GGGGTGGTATGGAGCAAGGGGCAAGTTGGGAAGACAACCTGTAGGGCCTGCGGGGT
CTATTGGGAACCAAGCTGGAGTGCAGTGGCACAATCTTGGCTCACTGCAATCTCCGC
CTCCTGGGTTCAAGCGATTCTCCTGCCTCAGCCTCCCGAGTTGTTGGGATTCCAGGC
ATGCATGACCAGGCTCAGCTAATTTTTGTTTTTTTGGTAGAGACGGGGTTTCACCATA
TTGGCCAGGCTGGTCTCCAACTCCTAATCTCAGGTGATCTACCCACCTTGGCCTCCC
AAATTGCTGGGATTACAGGCGTGAACCACTGCTCCCTTCCCTGTCCTTCTGATTTTGT
AGGTAACCACGTGCGGACCGAGCGGCCGCAGGAACCCCTAGTGATGGAGTTGGCCA
CTCCCTCTCTGCGCGCTCGCTCGCTCACTGAGGCCGGGCGACCAAAGGTCGCCCGAC
GCCCGGGCTTTGCCCGGGCGGCCTCAGTGAGCGAGCGAGCGCGCAGCTGCCTGCAG
GGGCGCCTGATGCGGTATTTTCTCCTTACGCATCTGTGCGGTATTTCACACCGCATAC
GTCgtaGCtGATcaATTgGCGCGCCGAATTCGTTAACAAGCTtTAATTAaCGCgtATAcCTA

ggATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGC
AGCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGC
TTTTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCA
GAGGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCG
GCTCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTG
CTGCCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGC
TGTCCCTGGCCTCCAAGGCCAGCTTCCCACAATAAGTTGGGTGAATTTTGGCTCATT
CCTCCTTTCTATAGGATTGAGGTCAGAGCggggttggggttgcgccttttccaaggcagccctgggtttgcgca
gggacgcggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcac
ccggatcttcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgc
cggacgtgacaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccgcg
atgggctgtggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggaggggggtgtggg
gcggtagtgtgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgacc
gaatcaccgacctctctccccagAAGCTCCCGGGAGCTTGTATATCCATTTTCGGATCTGATCAGC
ACGTGTTGACAATTAATCATCGGCATAGTATATCGGCATAGTATAATACGACAAGGT
GAGGAACgccaccATGAAAAAGCCTGAACTCACCGCGACGTCTGTCGAGAAGTTTCTG
ATCGAAAAGTTCGACAGCGTCTCCGACCTGATGCAGCTCTCGGAGGGCGAAGAATC
TCGTGCTTTCAGCTTCGATGTAGGAGGGCGTGGATATGTCCTGCGGGTAAATAGCTG
CGCCGATGGTTTCTACAAAGATCGTTATGTTTATCGGCACTTTGCATCGGCCGCGCTC
CCGATTCCGGAAGTGCTTGACATTGGGGAATTCAGCGAGAGCCTGACCTATTGCATC
TCCCGCCGTGCACAGGGTGTCACGTTGCAAGACCTGCCTGAAACCGAACTGCCCGCT
GTTCTGCAGCCGGTCGCGGAGGCCATGGATGCGATCGCTGCGGCCGATCTTAGCCA
GACGAGCGGGTTCGGCCCATTCGGACCGCAAGGAATCGGTCAATACACTACATGGC
GTGATTTCATATGCGCGATTGCTGATCCCCATGTGTATCACTGGCAAACTGTGATGG
ACGACACCGTCAGTGCGTCCGTCGCGCAGGCTCTCGATGAGCTGATGCTTTGGGCCG
AGGACTGCCCCGAAGTCCGGCACCTCGTGCACGCGGATTTCGGCTCCAACAATGTCC
TGACGGACAATGGCCGCATAACAGCGGTCATTGACTGGAGCGAGGCGATGTTCGGG
GATTCCCAATACGAGGTCGCCAACATCTTCTTCTGGAGGCCGTGGTTGGCTTGTATG
GAGCAGCAGACGCGCTACTTCGAGCGGAGGCATCCGGAGCTTGCAGGATCGCCGCG
GCTCCGGGCGTATATGCTCCGCATTGGTCTTGACCAACTCTATCAGAGCTTGGTTGA
CGGCAATTTCGATGATGCAGCTTGGGCGCAGGGTCGATGCGACGCAATCGTCCGATC
CGGAGCCGGGACTGTCGGGCGTACACAAATCGCCCGCAGAAGCGCGGCCGTCTGGA
CCGATGGCTGTGTAGAAGTACTCGCCGATAGTGGAAACCGACGCCCCAGCACTCGT
CCGAGGGCAAAGGAATAGAGCGCGGGGATCTCATGCTGGAGTTCTTCGCCCACCCC
AACTTGTTTATTGCAGCTTATAATGGTTACAAATAAAGCAATAGCATCACAAATTTC
ACAAATAAAGCATTTTTTTCACTGCATTCTAGTTGTGGTTTGTCCAAACTCATCAATG
TATCTTATCATGTCTGTAGCtGATgtATAcCTAggATCCGGCCGGccTGCAggTGTCCTCA
CAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGGAATTGACT
GGATTCCTTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCCAGGTGTCTG
CAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTGCCACCTTCC
CCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGGGAGCGCCGG
ACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAGGGACGTAA
TTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTCTAGAGCGGCCGCCACCGCGG
TGGAGCTCCAGCTTTTGTTCCCTTTAGTGAGGGTTAATTAGATCTTAATACGACTCAC
TATAGGGCGAATTGGGTACCGGGCCCCCCCTCGAGGTCGACGGTATCGATAAGCTTG
ATATCTATAACAAGAAAATATATATATAATAAGTTATCACGTAAGTAGAACATGAA
ATAACAATATAATTATCGTATGAGTTAAATCTTAAAAGTCACGTAAAAGATAATCAT
GCGTCATTTTGACTCACGCGGTCGTTATAGTTCAAAATCAGTGACACTTACCGCATT
GACAAGCACGCCTCACGGGAGCTCCAAGCGGCGACTGAGATGTCCTAAATGCACAG
CGACGGATTCGCGCTATTTAGAAAGAGAGAGCAATATTTCAAGAATGCATGCGTCA
ATTTTACGCAGACTATCTTTCTAGGGTTAATCTAGCTGCATCAGGATCATATCGTCGG
GTCTTTTTTCCGGCTCAGTCATCGCCCAAGCTGGCGCTATCTGGGCATCGGGGAGGA
AGAAGCCCGTGCCTTTTCCCGCGAGGTTGAAGCGGCATGGAAAGAGTTTGCCGAGG
ATGACTGCTGCTGCATTGACGTTGAGCGAAAACGCACGTTTACCATGATGATTCGGG

AAGGTGTGGCCATGCACGCCTTTAACGGTGAACTGTTCGTTCAGGCCACCTGGGATA
CCAGTTCGTCGCGGCTTTTCCGGACACAGTTCCGGATGGTCAGCCCGAAGCGCATCA
GCAACCCGAACAATACCGGCGACAGCCGGAACTGCCGTGCCGGTGTGCAGATTAAT
GACAGCGGTGCGGCGCTGGGATATTACGTCAGCGAGGACGGGTATCCTGGCTGGAT
GCCGCAGAAATGGACATGGATACCCCGTGAGTTACCCGGCGGGCGCGCTTGGCGTA
ATCATGGTCATAGCTGTTTCCTGTGTGAAATTGTTATCCGCTCACAATTCCACACAAC
ATACGAGCCGGAAGCATAAAGTGTAAAGCCTGGGGTGCCTAATGAGTGAGCTAACT
CACATTAATTGCGTTGCGCTCACTGCCCGCTTTCCAGTCGGGAAACCTGTCGTGCCA
GCTGCATTAATGAATCGGCCAACGCGCGGGGAGAGGCGGTTTGCGTATTGGGCGCT
CTTCCGCTTCCTCGCTCACTGACTCGCTGCGCTCGGTCGTTCGGCTGCGGCGAGCGG
TATCAGCTCACTCAAAGGCGGTAATACGGTTATCCACAGAATCAGGGGATAACGCA
GGAAAGAACATGTGAGCAAAAGGCCAGCAAAAGGCCAGGAACCGTAAAAAGGCCG
CGTTGCTGGCGTTTTTCCATAGGCTCCGCCCCCCTGACGAGCATCACAAAAATCGAC
GCTCAAGTCAGAGGTGGCGAAACCCGACAGGACTATAAAGATACCAGGCGTTTCCC
CCTGGAAGCTCCCTCGTGCGCTCTCCTGTTCCGACCCTGCCGCTTACCGGATACCTGT
CCGCCTTTCTCCCTTCGGGAAGCGTGGCGCTTTCTCATAGCTCACGCTGTAGGTATCT
CAGTTCGGTGTAGGTCGTTCGCTCCAAGCTGGGCTGTGTGCACGAACCCCCCGTTCA
GCCCGACCGCTGCGCCTTATCCGGTAACTATCGTCTTGAGTCCAACCCGGTAAGACA
CGACTTATCGCCACTGGCAGCAGCCACTGGTAACAGGATTAGCAGAGCGAGGTATG
TAGGCGGTGCTACAGAGTTCTTGAAGTGGTGGCCTAACTACGGCTACACTAGAAGG
ACAGTATTTGGTATCTGCGCTCTGCTGAAGCCAGTTACCTTCGGAAAAAGAGTTGGT
AGCTCTTGATCCGGCAAACAAACCACCGCTGGTAGCGGTGGTTTTTTTGTTTGCAAG
CAGCAGATTACGCGCAGAAAAAAAGGATCTCAAGAAGATCCTTTGATCTTTTCTACG
GGGTCTGACGCTCAGTGGAACGAAAACTCACGTTAAGGGATTTTGGTCATGAGATTA
TCAAAAAGGATCTTCACCTAGATCCTTTTAAATTAAAAATGAAGTTTTAAATCAATC
TAAAGTATATATGAGTAAACTTGGTCTGACAGTTACCAATGCTTAATCAGTGAGGCA
CCTATCTCAGCGATCTGTCTATTTCGTTCATCCATAGTTGCCTGACTCCCCGTCGTGT
AGATAACTACGATACGGGAGGGCTTACCATCTGGCCCCAGTGCTGCAATGATACCG
CGAGACCCACGCTCACCGGCTCCAGATTTATCAGCAATAAACCAGCCAGCCGGAAG
GGCCGAGCGCAGAAGTGGTCCTGCAACTTTATCCGCCTCCATCCAGTCTATTAATTG
TTGCCGGGAAGCTAGAGTAAGTAGTTCGCCAGTTAATAGTTTGCGCAACGTTGTTGC
CATTGCTACAGGCATCGTGGTGTCACGCTCGTCGTTTGGTATGGCTTCATTCAGCTCC
GGTTCCCAACGATCAAGGCGAGTTACATGATCCCCCATGTTGTGCAAAAAAGCGGTT
AGCTCCTTCGGTCCTCCGATCGTTGTCAGAAGTAAGTTGGCCGCAGTGTTATCACTC
ATGGTTATGGCAGCACTGCATAATTCTCTTACTGTCATGCCATCCGTAAGATGCTTTT
CTGTGACTGGTGAGTACTCAACCAAGTCATTCTGAGAATAGTGTATGCGGCGACCGA
GTTGCTCTTGCCCGGCGTCAATACGGGATAATACCGCGCCACATAGCAGAACTTTAA
AAGTGCTCATCATTGGAAAACGTTCTTCGGGGCGAAAACTCTCAAGGATCTTACCGC
TGTTGAGATCCAGTTCGATGTAACCCACTCGTGCACCCAACTGATCTTCAGCATCTTT
TACTTTCACCAGCGTTTCTGGGTGAGCAAAAACAGGAAGGCAAAATGCCGCAAAAA
AGGGAATAAGGGCGACACGGAAATGTTGAATACTCAT

(153) The sequence of the PB-iCap8split-ITRGFP-hygro-4 vector is provided below:

(154) TABLE-US-00034 PB-iCap8split-ITRGFP-hygro-4 (17,649 bp) (SEQ ID NO: 33)

actcttcctttttcaatattattgaagcatttatcagggttattgtctcatgagcggatacatatttgaatgtatttagaaaaataaaca
aataggggttccgcgcacatttccccgaaaagtgccacctaaattgtaagcgttaatattttgttaaaattcgcgttaaatttttgttaaatca
gctcattttttaaccaataggccgaaatcggcaaaatcccttataaatcaaaagaatagaccgagatagggttgagtgttgttccagtttggaa
caagagtccactattaaagaacgtggactccaacgtcaaagggcgaaaaaccgtctatcagggcgatggcccactacgtgaaccatcaccctaa
tcaagttttttggggtcgaggtgccgtaaagcactaaatcggaaccctaaagggagcccccgatttagagcttgacggggaaagccggcga
acgtggcgagaaaggaagggaagaaagcgaaaggagcgggcgctagggcgctggcaagtgtagcggtcacgctgcgcgtaaccacc
acacccgccgcgcttaatgcgccgctacagggcgcgtcccattcgccattcaggctgcgcaactgttgggaagggcgatcggtgcgggc
ctcttcgctattacgccagctggcgaaagggggatgtgctgcaaggcgattaagttgggtaacgccagggttttcccagtcacgacgttgta
aaacgacggccagtgagcgcgcctcgttcattcacgtttttgaacccgtggaggacgggcagactcgcggtgcaaatgtgttttacagcgtg
atggagcagatgaagatgctcgacacgctgcagaacacgcagctagattaaccctagaaagataatcatattgtgacgtacgttaaagataa
tcatgcgtaaaattgacgcatgtgttttatcggtctgtatatcgaggtttatttattaatttgaatagatattaagttttattatatttacact

tacatactaataataaattcaacaaacaatttatttatgtttatttatttattaaaaaaaaacaaaaactcaaaatttcttctataaagtaaca
aaacttttatcgaattcctgcagcccgggggatccactagttctagagggacagccccccccaaagcccccagggatgtaattacgtccctcc
cccgctagggggcagcagcgagccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtggggacagcccggg
cacggggaaggtggcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcc
tccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgtgatgggaattctgt
ggaatgtggtgagtagcgggctgctgggctggccggggctttcgtggccgccgggccgctcggtgggacggaagcgtgtggagagaccgc
caagggctgtagtctgggtccgcgagcaaggttgccctgaactgggggttgggggggagcgcagcaaaatggcggctgttcccgagtctt
gaatggaagacgcttgtgaggcgggctgtgaggtcgttgaaacaaggtgggggggcatggtgggcggcaagaacccaaggtcttgaggc
cttcgctaatgcgggaaagctcttattcgggtgagatgggctggggcaccatctggggaccctgacgtgaagtttgtcactgactggagaac
tcggtttgtcgtctgttgcgggggcggcagttatggcggtgccgttgggcagtgcacccgtacctttgggagcgcgcgccctcgtcgtgtcg
tgacgtcacccgttctgttggcttataatgcagggtggggccacctgccggtaggtgtgcggtaggcttttctccgtcgcaggacgcagggtt
cgggcctagggtaggctctcctgaatcgacaggcgccggacctctggtgaggggagggataagtgaggcgtcagtttctttggtcggtttta
tgtacctatcttcttaagtagctgaagctccggttttgaactatgcgctcggggttggcgagtgtgttttgtgaagttttttaggcaccttttg
aaatgtaatcatttgggtcaatatgtaattttcagtgttagactagtaaattgtccgctaaattctggccgtttttggcttttttgttagactg
tcagttagggtgtggaaagtcccgcgatcgctagcgtttaaacttaagcttggtaccgagctcggatccactagtccagtgtggtggaattcct
gcttcgcgatgtacgggccagatatacgcgtggcctccgcgccgggttttggcgcctcccgcgggcgcccccctcctcacggcgagcgctgcca
cgtcagacgaagggcgcagcgagcgtcctgatccttccgcccggacgctcaggacagcggcccgctgctcataagactcggccttagaacccca
gtatcagcagaaggacattttaggacgggacttgggtgactctagggcactggttttctttccagagagcggaacaggcgaggaaaagtag
tcccttctcggcgattctgcggagggatctccgtggggcggtgaacgccgatgattatataaggacgcgccggtccctatcagtgatagag
atctccctatcagtgatagagagtgtggcacagctagttccgtcgcagccgggatttgggtcgcggttcttgtttgtggatcgctgtgatcgtc
acttgacgaacgcgcagccgccatggccgctgatggctatctgcctgattggctggaagataacctgagtgagggcatccgggaatggtg
ggccctgaagcctggagccccgaagcccaaagccaaccagcaaaagcaggacgacggccggggtctggtgcttcctggctacaagtac
ctcggacccttcaacggactcgacaagggggagcccgtcaacgggggacgcagcggccctcgagcacgacaaggcctacgacca
gcagctgcaggcgggtgacaatccgtacctgcggtataaccacgccgacgccgagtttcaggagcgtctgcaagaagatacgtcttttgg
gggcaacctcgggcgagcagtcttccaggccaagaagcgggttctcgaacctctcggtctggttgaggaaggcgctaagacggctcctg
gaaagaagaggccggtagaacctagcccccagcgatctccagattcttctaccggcatcggcaaaaagggccaacagcctgcgagaaa
gcggctgaacttcggccagaccggagacagcgagagcgtgcccgaccctcagcctctgggcgagccccctgccgccccttctggcgtg
ggccccaacaccatggccgccggcggcggagcccctatggccgacaacaacgagggcgccgacggcgtgggaagcagcagcggca
attggcactgcgactccacctggctgggcgacagagtgatcaccacatctacaagaacctgggccctgcccacctacaacaatcacctgta
caagcagatcagcaatggaacctctggaggcgccaccaacgataacacctactttggctacagcaccccttggggatatttcgacttcaacc
ggttccactgtcacttcagccctcgggactggcagcgtctcattaacaataactggggcttcagaccaaagagactgagcttcaagctgttca
acatccaggtgaaggaagtgacccaaaacgagggcaccaagaccatcgccaataacctcacctccacaatccaagtgttcaccgattccg
agtaccagctgccttacgtgctgggctccgcccaccagggctgcctgccaccattccccgccgacgtgttcatgatccctcagtacggctac
ctgacactgaacaacggatctcaagcagtgggcagaagctccttctactgtctggagtacttccctagccagatgctgaggacaggcaaca
atttccagttcacctacacattcgaggacgtcccttttcacagctcttatgcccatagccagagcctggatagactgatgaaccctctgatcga
ccagtatctgtattacctgagcagaacgcaaacaacaggcggcaccgctaatacccagaccctgggtttcagccaaggcggccctaacaca
atggccaatcaggccaaaaactggcttcctggcccctgctacagacagcaaagagtgagcacaaccaccggccagaacaacaactctaa
cttcgcctggacagccggcaccaaataccacctgaacggcagaaacagcctggccaaccctggaatcgctatggctacacacaaggacg
atgaggaacggttcttccccagcaacggcatcctgatcttcggcaagcagaacgccgctcgcgacaacgccgactacagcgacgtgatgc
tgaccagcgaggaagaaatcaagacaacaaacccggtcgccaccgaggaatacggcattgtggccgataacctgcagcagcagaatac
cgcccctcagatcggcaccgtgaactcccagggagccctgcctggcatggtgtggcagaacagagatgtgtacctgcagggccctatctg
ggccaagatcccccacaccgacggaaatttccatccaagccctctgatgggggatttggcctgaagcaccccccctccacagattctcatc
aaaaacacacctgtgccagccgaccctcccacaacatttaatcagtctaagctgaatagcttcatcacccagtacagcaccggccaggtgtc
cgtggaaatcgagtgggagctgcagaaagagaacagcaagagatggaaccctgagatccagtacaccagcaactactacaaaagcaca
agcgtggacttcgccgttaatactgagggcgtctactcagagcctcggcccatcggcactcggtacctgaccagaaacctgaatggctagg
atccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaaaacttgtttattgca
gcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagttgtggtttgtccaaactca
tcaatgtatcttattgactggattgagggacagccccccccaaagcccccagggatgtaattacgtccctcccccgctagggggcagcagcga
gccgcccggggctccgctccggtccggcgctccccccgcatccccgagccggcagcgtgcggggacagcccgggcacggggaaggtg
gcacgggatcgctttcctctgaacgcttctcgctgctctttgagcctgcagacacctgggggggatacggggaaaaggcctccaaggccagc
ttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagcgacattgattattgactagttattaatagta
atcaattacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggtaaatggcccgcctggctgaccgcccaacgac
ccccgcccattgacgtcaataatgacgtatgttcccatagtaacgccaatagggactttccattgacgtcaatgggtggagtatttacggtaaa
ctgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatgacggtaaatggcccgcctggcattatgccca

gtacatgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccatggtgatgcggttttggcagtacatcaat
gggcgtggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggagtttgttttggaaccaaaatcaacgggact
ttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtgggaggtctatataagcagagctctccctatcag
tgatagagatctccctatcagtgatagagatcgtcgacgagctcgtttagtgaaccgtcagatcgcctggagacgccatccacgctgttttgac
ctccatagaagacaccgggaccgatccagcctatcccaattctgatgcgccggtgatcagatcaaaaacttcagccaggtacggcgctaaga
cggctcctggaaagaagaggccggtagagccatcaccccagcgttctccagactcctctacgggcatcggcaagaaaggccaacagcc
cgccagaaaaagactcaattttggtcagactggcgactcagagtcagttccagaccctcaacctctcggagaacctccagcagcgccctct
ggtgtgggacctaatacaatggctgcaggcggtggcgcaccaatggcagacaataacgaaggcgccgacggagtgggtagttcctcgg
gaaattggcattgcgattccacatggctgggcgacagagtcatcaccaccagcacccgaacctgggccctgcccacctacaacaaccacc
tctacaagcaaatctccaacgggacatcgggaggagccaccaacgacaacacctacttcggctacagcaccccctgggggtattttgacttt
aacagattccactgccacttttcaccacgtgactggcagcgactcatcaacaacaactggggattccggcccaagagactcagcttcaagct
cttcaacatccaggtcaaggaggtcacgcagaatgaaggcaccaagaccatcgccaataacctcaccagcaccatccaggtgtttacgga
ctcggagtaccagctgccgtacgttctcggctctgcccaccagggctgcctgcctccgttcccggcggacgtgttcatgattccccagtacg
gctacctaacactcaacaacggtagtcaggccgtgggacgctcctccttctactgcctggaatactttccttcgcagatgctgagaaccggca
acaacttccagtttacttacaccttcgaggacgtgcctttccacagcagctacgcccacagccagagcttggaccggctgatgaatcctctga
ttgaccagtacctgtactacttgtctcggactcaaacaacaggaggcacggcaaatacgcagactctgggcttcagccaaggtgggcctaat
acaatggccaatcaggcaaagaactggctgccaggaccctgttaccgccaacaacgcgtctcaacgacaaccgggcaaaacaacaatag
caactttgcctggactgctgggaccaaataccatctgaatggaagaaattcattggctaatcctggcatcgctatggcaacacacaaagacg
acgaggagcgtttttttcccagtaacgggatcctgatttttggcaaacaaaatgctgccagagacaatgcggattacagcgatgtcatgctca
ccagcgaggaagaaatcaaaaccactaaccctgtggctacagaggaatacggtatcgtggcagataacttgcagcagcaaaacacggct
cctcaaattggaactgtcaacagccagggggccttacccggtatggtctggcagaaccgggacgtgtacctgcagggtcccatctgggcc
aagattcctcacacggacggcaacttccacccgtctccgctgatgggcggctttggcctgaaacatcctccgcctcagatcctgatcaagaa
cacgcctgtacctgcggatcctccgaccaccttcaaccagtcaaagctgaactctttcatcacgcaatacagcaccggacaggtcagcgtg
gaaattgaatgggagctgcagaaggaaaacagcaagcgctggaaccccgagatccagtacacctccaactactacaaatctacaagtgtg
gactttgctgttaatacagaaggcgtgtactctgaaccccgccccattggcacccgttacctcacccgtaatctgtaaactagtttgcttgtta
atcaataaaccgtttaattcgtttcagttgagcggccgtcgagtctagagggcccgtttaaacccgctgatcagcctcgactgtgccttctagt
tgccagccatctgttgtttgcccctcccccgtgccttccttgaccctggaaggtgccactcccactgtcctttcctaataaaatgaggaaattg
catcgcattgtctgagtaggtgtcattctattctggggggtggggtggggcaggacagcaagggggaggattgggaagacaatagcaggcatgc
tggggatgcggtgggctctatggcttctgaggcggaaagaaccagctggggctctagggggtatccccggcgcgccgaattcgttaacaa
gctttaattaacgcgtatacctaggatccggccggcctgcaggtgtcctcacaggaacgaagtccctaaagaaacagtggcagccaggttt
agccccggaattgactggattcctttttagggcccattggtatggctttttccccgtatccccccaggtgtctgcaggctcaaagagcagcga
gaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgctgccggctcggggatgcggggggagc
gccggaccggagcggagccccgggcggctcgctgctgccccctagcgggggagggacgtaattacatccctgggggctttgggggggg
ggctgtccctggcctccaaggccagcttcccacaataagttgggtgaattttggctcattcctcctttctataggattgaggtcagagctttgt
gatgggaattctgtggaatgtgtgtcagttagggtgtggaaagtcccgcgatcgctagcaaacgccagcaacgcggcctttttacggttcctgg
ccttttgctggccttttgctcacatgtcctgcaggcagctgcgcgctcgctcgctcactgaggccgcccgggcaaagcccgggcgtcgggcg
acctttggtcgcccggcctcagtgagcgagcgagcgcgcagagagggagtggccaactccatcactaggggttcctgcggccgcacgc
gtggagctagttattaatagtaatcaattacggggtcattagttcatagcccatatatggagttccgcgttacataacttacggtaaatggccc
gcctggctgaccgcccaacgacccccgcccattgacgtcaataatgacgtatgttcccatagtaacgtcaatagggactttccattgacgtcaa
tgggtggagtatttacggtaaactgcccacttggcagtacatcaagtgtatcatatgccaagtacgccccctattgacgtcaatgacggtaaat
ggcccgcctggcattatgcccagtacatgaccttatgggactttcctacttggcagtacatctacgtattagtcatcgctattaccatggtgat
gcggttttggcagtacatcaatgggcgtggatagcggtttgactcacggggatttccaagtctccaccccattgacgtcaatgggagtttgttt
tgcaccaaaatcaacgggactttccaaaatgtcgtaacaactccgccccattgacgcaaatgggcggtaggcgtgtacggtgggaggtctatat
aagcagagctcgtttagtgaaccgtcagatcgcctggagacgccatccacgctgttttgacctccatagaagacaccgggaccgatccagc
ctccgcggattcgaatcccggccgggaacggtgcattggaacgcggattccccgtgccaagagtgacgtaagtaccgcctatagagtctat
aggcccacaaaaaatgctttcttcttttaatatactttttttgtttatcttatttctaatactttccctaatctctttctttcagggcaataatg
atacaatgtatcatgcctctttgcaccattctaaagaataacagtgataatttctgggttaaggcaatagcaatatttctgcatataaatattt
ctgcatataaattgtaactgatgtaagaggtttcatattgctaatagcagctacaatccagctaccattctgcttttattttatggttgggata
aggctggattattctgagtccaagctaggcccttttgctaatcatgttcatacctcttatcttcctcccacagctcctgggcaacgtgctggtc
tgtgtgctggcccatcactttggcaaagaattgggattcgaacatcgattgaattctgaatggtgagcaagggcgaggagctgttcaccggggt
ggtgcccatcctggtcgagctggacggcgacgtaaacggccacaagttcagcgtgtccggcgagggcgagggcgatgccacctacggcaagctg
accctgaagttcatctgcaccaccggcaagctgcccgtgccctggcccaccctcgtgaccaccctgacctacggcgtgcagtgcttcagccgct
accccgaccacatgaagcagcacgacttcttcaagtccgccatgcccgaaggctacgtccaggagcgcaccatcttcttcaaggacgacggcaa
ctacaagacccgcgccgaggtgaagttcgagggcgacaccctggtgaaccgcatcgagctgaagggcatcgacttcaaggaggacggcaacat

cctggggcacaagctggagtacaactacaacagccacaacgtctatatcatggccgacaagcagaagaacggcatcaaggtgaacttcaa
gatccgccacaacatcgaggacggcagcgtgcagctcgccgaccactaccagcagaacacccccatcggcgacggccccgtgctgctg
cccgacaaccactacctgagcacccagtccgccctgagcaaagaccccaacgagaagcgcgatcacatggtcctgctggagttcgtgac
cgccgccgggatcactctcggcatggacgagctgtacaagtactcagatctcgagctcaagtagggatcctctagagtcgacctgcagaa
gcttgcctcgagcagcgctgctcgagagatctacgggtggcatccctgtgacccctccccagtgcctctcctggccctggaagttgccactc
cagtgcccaccagccttgtcctaataaaattaagttgcatcattttgtctgactaggtgtccttctataatattatggggtggagggggggtggt
atggagcaaggggcaagttgggaagacaacctgtagggcctgcggggtctattgggaaccaagctggagtgcagtggcacaatcttggctc
actgcaatctccgcctcctgggttcaagcgattctcctgcctcagcctcccgagttgttgggattccaggcatgcatgaccaggctcagctaat
ttttgtttttttggtagagacggggtttcaccatattggccaggctggtctccaactcctaatctcaggtgatctacccaccttggcctcccaa
attgctgggattacaggcgtgaaccactgctcccttccctgtccttctgattttgtaggtaaccacgtgcggaccgagcggccgcaggaacccc
tagtgatggagttggccactccctctctgcgcgctcgctcgctcactgaggccgggcgaccaaaggtcgcccgacgcccgggctttgcccg
ggcggcctcagtgagcgagcgagcgcgcagctgcctgcaggggcgcctgatgcggtattttctccttacgcatctgtgcggtatttcacac
cgcatacgtcgtagctgatcaattggcgcgccgaattcgttaacaagctttaattaacgcgtatacctaggatccggccggcctgcaggtgtc
ctcacaggaacgaagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctttttagggcccattggtatggcttt
ttccccgtatccccccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccg
ggctgtccccgcacgctgccggctcggggatgcggggggagcgccggaccggagcggagccccgggcggctcgctgctgcccccta
gcgggggagggacgtaattacatccctgggggctttgggggggggctgtccctggcctccaaggccagcttcccacaataagttgggtga
attttggctcattcctcctttctataggattgaggtcagagcggggttggggttgcgccttttccaaggcagccctgggtttgcgcagggacgc
ggctgctctgggcgtggttccgggaaacgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcacccggatct
tcgccgctacccttgtgggccccccggcgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgccggacgtg
acaaacggaagccgcacgtctcactagtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccgcgatgggctgt
ggccaatagcggctgctcagcagggcgcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtggggcggtagtg
tgggccctgttcctgcccgcgcggtgttccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgaccgaatcaccg
acctctctccccagaagctcccgggagcttgtatatccattttcggatctgatcagcacgtgttgacaattaatcatcggcatagtatatcggc
atagtataatacgacaaggtgaggaacgccaccatgaaaaagcctgaactcaccgcgacgtctgtcgagaagtttctgatcgaaaagttcga
cagcgtctccgacctgatgcagctctcggagggcgaagaatctcgtgctttcagcttcgatgtaggagggcgtggatatgtcctgcgggtaa
atagctgcgccgatggtttctacaaagatcgttatgtttatcggcactttgcatcggccgcgctcccgattccggaagtgcttgacattgggga
attcagcgagagcctgacctattgcatctcccgccgtgcacagggtgtcacgttgcaagacctgcctgaaaccgaactgcccgctgttctgc
agccggtcgcggaggccatggatgcgatcgctgcggccgatcttagccagacgagcgggttcggcccattcggaccgcaaggaatcgg
tcaatacactacatggcgtgatttcatatgcgcgattgctgatccccatgtgtatcactggcaaactgtgatggacgacaccgtcagtgcgtcc
gtcgcgcaggctctcgatgagctgatgctttgggccgaggactgccccgaagtccggcacctcgtgcacgcggatttcggctccaacaat
gtcctgacggacaatggccgcataacagcggtcattgactggagcgaggcgatgttcggggattcccaatacgaggtcgccaacatcttct
tctggaggccgtggttggcttgtatggagcagcagacgcgctacttcgagcggaggcatccggagcttgcaggatcgccgcggctccgg
gcgtatatgctccgcattggtcttgaccaactctatcagagcttggttgacggcaatttcgatgatgcagcttgggcgcagggtcgatgcgac
gcaatcgtccgatccggagccgggactgtcgggcgtacacaaatcgcccgcagaagcgcggccgtctggaccgatggctgtgtagaagt
actcgccgatagtggaaaccgacgccccagcactcgtccgagggcaaaggaatagagcgcggggatctcatgctggagttcttcgccca
ccccaacttgtttattgcagcttataatggttacaaataaagcaatagcatcacaaatttcacaaataaagcatttttttcactgcattctagt
tgtggtttgtccaaactcatcaatgtatcttatcatgtctgtagctgatgtatacctaggatccggccggcctgcaggtgtcctcacaggaacg
aagtccctaaagaaacagtggcagccaggtttagccccggaattgactggattcctttttagggcccattggtatggctttttccccgtatccc
cccaggtgtctgcaggctcaaagagcagcgagaagcgttcagaggaaagcgatcccgtgccaccttccccgtgcccgggctgtccccgcacgct
gccggctcggggatgcggggggagcgccggaccggagcggagccccgggcggctcgctgctgcccccctagcgggggagggacgta
attacatccctgggggctttgggggggggctgtccctctagagcggccgccaccgcggtggagctccagcttttgttccctttagtgagggtt
aattagatcttaatacgactcactatagggcgaattgggtaccgggccccccctcgaggtcgacggtatcgataagcttgatatctataacaa
gaaaatatatatatataataagttatcacgtaagtagaacatgaaataacaatataattatcgtatgagttaaatcttaaaagtcacgtaaaagat
aatcatgcgtcattttgactcacgcggtcgttatagttcaaaatcagtgacacttaccgcattgacaagcacgcctcacgggagctccaagcgg
cgactgagatgtcctaaatgcacagcgacggattcgcgctatttagaaagagagagcaatatttcaagaatgcatgcgtcaattttacgcagac
tatctttctagggttaatctagctgcatcaggatcatatcgtcgggtctttttttccggctcagtcatcgcccaagctggcgctatctgggcatc
ggggaggaagaagcccgtgccttttcccgcgaggttgaagcggcatggaaagagtttgccgaggatgactgctgctgcattgacgttgagcg
aaaacgcacgtttaccatgatgattcgggaaggtgtggccatgcacgcctttaacggtgaactgttcgttcaggccacctgggataccagttc
gtcgcggcttttccggacacagttccggatggtcagcccgaagcgcatcagcaacccgaacaataccggcgacagccggaactgccgtg
ccggtgtgcagattaatgacagcggtgcggcgctgggatattacgtcagcgaggacgggtatcctggctggatgccgcagaaatggacat
ggataccccgtgagttacccggcgggcgcgcttggcgtaatcatggtcatagctgtttcctgtgtgaaattgttatccgctcacaattccacac
aacatacgagccggaagcataaagtgtaaagcctggggtgcctaatgagtgagctaactcacattaattgcgttgcgctcactgcccgctttc
cagtcgggaaacctgtcgtgccagctgcattaatgaatcggccaacgcgcggggagaggcggtttgcgtattgggcgctcttccgcttcct

cgctcactgactcgctgcgctcggtcgttcggctgcggcgagcggtatcagctcactcaaaggcggtaatacggttatccacagaatcagg
ggataacgcaggaaagaacatgtgagcaaaaggccagcaaaaggccaggaaccgtaaaaaggccgcgttgctggcgtttttccataggc
tccgcccccctgacgagcatcacaaaaatcgacgctcaagtcagaggtggcgaaacccgacaggactataaagataccaggcgtttcccc
ctggaagctccctcgtgcgctctcctgttccgaccctgccgcttaccggatacctgtccgcctttctcccttcgggaagcgtggcgctttctca
tagctcacgctgtaggtatctcagttcggtgtaggtcgttcgctccaagctgggctgtgtgcacgaaccccccgttcagcccgaccgctgcgc
cttatccggtaactatcgtcttgagtccaacccggtaagacacgacttatcgccactggcagcagccactggtaacaggattagcagagcga
ggtatgtaggcggtgctacagagttcttgaagtggtggcctaactacggctacactagaaggacagtatttggtatctgcgctctgctgaagc
cagttaccttcggaaaaagagttggtagctcttgatccggcaaacaaaccaccgctggtagcggtggtttttttgtttgcaagcagcagattac
gcgcagaaaaaaaggatctcaagaagatcctttgatcttttctacggggtctgacgctcagtggaacgaaaactcacgttaagggattttggt
catgagattatcaaaaaggatcttcacctagatccttttaaattaaaaatgaagttttaaatcaatctaaagtatatatgagtaaacttggtct
gacagttaccaatgcttaatcagtgaggcacctatctcagcgatctgtctatttcgttcatccatagttgcctgactccccgtgtgtagataac
tacgatacgggagggcttaccatctggccccagtgctgcaatgataccgcgagacccacgctcaccggctccagatttatcagcaataaaccag
ccagccggaagggccgagcgcagaagtggtcctgcaactttatccgcctccatccagtctattaattgttgccgggaagctagagtaagtagtt
cgccagttaatagtttgcgcaacgttgttgccattgctacaggcatcgtggtgtcacgctcgtcgtttggtatggcttcattcagctccggttc
ccaacgatcaaggcgagttacatgatcccccatgttgtgcaaaaaagcggttagctccttcggtcctccgatcgttgtcagaagtaagttggcc
gcagtgttatcactcatggttatggcagcactgcataattctcttactgtcatgccatccgtaagatgcttttctgtgactggtgagtactcaa
ccaagtcattctgagaatagtgtatgcggcgaccgagttgctcttgcccggcgtcaatacgggataataccgcgccacatagcagaactttaaa
agtgctcatcattggaaaacgttcttcggggcgaaaactctcaaggatcttaccgctgttgagatccagttcgatgtaacccactcgtgcaccc
aactgatcttcagcatcttttactttcaccagcgtttctgggtgagcaaaaacaggaaggcaaaatgccgcaaaaaagggaataagggcgacac
ggaaatgttgaatactcat

(155) The sequence of the pPBBG-iCap8cd-ITRGFP-hygro vector is provided below:

(156) TABLE-US-00035 pPBBG-iCap8cd-ITRGFP-hygro (14,262 bp) (SEQ ID NO: 34)

ACTCTTCCTTTTTCAATATTATTGAAGCATTTATCAGGGTTATTGTCTCATGA
GCGGATACATATTTGAATGTATTTAGAAAAATAAACAAATAGGGGTTCCGCGCACA
TTTCCCCGAAAAGTGCCACCTAAATTGTAAGCGTTAATATTTTGTTAAAATTCGCGTT
AAATTTTTGTTAAATCAGCTCATTTTTTAACCAATAGGCCGAAATCGGCAAAATCCC
TTATAAATCAAAAGAATAGACCGAGATAGGGTTGAGTGTTGTTCCAGTTTGGAACA
AGAGTCCACTATTAAAGAACGTGGACTCCAACGTCAAAGGGCGAAAAACCGTCTAT
CAGGGCGATGGCCCACTACGTGAACCATCACCCTAATCAAGTTTTTTGGGGTCGAGG
TGCCGTAAAGCACTAAATCGGAACCCTAAAGGGAGCCCCCGATTTAGAGCTTGACG
GGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGGAAGAAAGCGAAAGGAGCGGG
CGCTAGGGCGCTGGCAAGTGTAGCGGTCACGCTGCGCGTAACCACCACACCCGCCG
CGCTTAATGCGCCGCTACAGGGCGCGTCCCATTCGCCATTCAGGCTGCGCAACTGTT
GGGAAGGGCGATCGGTGCGGGCCTCTTCGCTATTACGCCAGCTGGCGAAAGGGGGA
TGTGCTGCAAGGCGATTAAGTTGGGTAACGCCAGGGTTTTCCCAGTCACGACGTTGT
AAAACGACGGCCAGTGAGCGCGCCTCGTTCATTCACGTTTTTGAACCCGTGGAGGAC
GGGCAGACTCGCGGTGCAAATGTGTTTTACAGCGTGATGGAGCAGATGAAGATGCT
CGACACGCTGCAGAACACGCAGCTAGATTAACCCTAGAAAGATAATCATATTGTGA
CGTACGTTAAAGATAATCATGCGTAAAATTGACGCATGTGTTTTATCGGTCTGTATA
TCGAGGTTTATTTATTAATTTGAATAGATATTAAGTTTTATTATATTTACACTTACAT
ACTAATAATAAATTCAACAAACAATTTATTTATGTTTATTTATTTATTAAAAAAAAA
CAAAAACTCAAAATTTCTTCTATAAAGTAACAAAACTTTTATCGAATTCCTGCAGCC
CGGGGGATCCACTAGTTCTAGAGGGACAGCCCCCCCCCAAAGCCCCCAGGGATGTA
ATTACGTCCCTCCCCCGCTAGGGGGCAGCAGCGAGCCGCCCGGGGCTCCGCTCCGGT
CCGGCGCTCCCCCCGCATCCCCGAGCCGGCAGCGTGCGGGGACAGCCCGGGCACGG
GGAAGGTGGCACGGGATCGCTTTCCTCTGAACGCTTCTCGCTGCTCTTTGAGCCTGC
AGACACCTGGGGGGATACGGGGAAAAGGCCTCCAAGGCCAGCTTCCCACAATAAGT
TGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGGTCAGAGCTTTGTGATG
GGAATTCTGTGGAATGTGTGTCAGTTAGGGTGTGGAAAGTCCCgcGATCgcTAGcGTTT
AAACTTAAGCTTGGTACCGAGCTCGGATCCACTAGTCCAGTGTGGTGGAATTCCTGC
TTCGCGATGTACGGGCCAGATATACGCGTTGACATTGATTATTGACTAGTTATTAAT
AGTAATCAATTACGGGGTCATTAGTTCATAGCCCATATATGGAGTTCCGCGTTACAT
AACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGT
CAATAATGACGTATGTTCCCATAGTAACGCCAATAGGGACTTTCCATTGACGTCAAT

GGGTGGAGTATTTACGGTAAACTGCCCACTTGGCAGTACATCAAGTGTATCATATGC
CAAGTACGCCCCCTATTGACGTCAATGACGGTAAATGGCCCGCCTGGCATTATGCCC
AGTACATGACCTTATGGGACTTTCCTACTTGGCAGTACATCTACGTATTAGTCATCGC
TATTACCATGGTGATGCGGTTTTGGCAGTACATCAATGGGCGTGGATAGCGGTTTGA
CTCACGGGGATTTCCAAGTCTCCACCCCATTGACGTCAATGGGAGTTTGTTTTGGAA
CCAAAATCAACGGGACTTTCCAAAATGTCGTAACAACTCCGCCCCATTGACGCAAAT
GGGCGGTAGGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTCTCCCTATCAGTG
ATAGAGATCTCCCTATCAGTGATAGAGATCGTCGACGAGCTCGTTTAagttgcgcagccatcg
acgtcagacgcggaagcttcgatcaactacgcagacaggtaAGTaaacaaatgttctcgtcacgtgggcatgaatctgatgctgtttccct
gcagacaatgcgagagaatgaatcagaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaatctc
aacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgggaaaggtgccagacgcttgcactgcctgcgatct
ggtcaatgtggatttggatgactgcatctttgaacaataaatgatttaaatcaggtatggctgccgatggttatcttccagattggctcgagga
caacctctctgagggcattcgcgagtggtgggcgctgaaacctggagccccgaagcccaaagccaaccagcaaaagcaggacgacggc
cggggtctggtgcttcctggctacaagtacctcggacccttcaacggactcgacaagggggagcccgtcaacgcggcggacgcagcgg
ccctcgagcacgacaaggcctacgaccagcagctgcagggggtgacaatccgtacctgcggtataaccacgccgacgccgagtttcag
gagcgtctgcaagaagatacgtcttttgggggcaacctcgggcgagcagtcttccaggccaagaagcgggttctcgaacctctcggtctgg
ttgaggaaggcgctaagacggctcctggaaagaagaggccggtagagccatcaccccagcgttctccagactcctctacgggcatcggc
aagaaaggccaacagcccgccagaaaaagactcaattttggtcagactggcgactcagagtcagttccagaccctcaacctctggagaa
cctccagcagcgccctctggtgtgggacctaatacaatggctgcaggcggtggcgcaccaatggcagacaataacgaaggcgccgacg
gagtgggtagttcctcgggaaattggcattgcgattccacatggctgggcgacagagtcatcaccaccagcacccgaacctgggccctgc
ccacctacaacaaccacctctacaagcaaatctccaacgggacatcgggaggagccaccaacgacaacacctacttcggctacagcacc
ccctgggggtattttgactttaacagattccactgccacttttcaccacgtgactggcagcgactcatcaacaacaactggggattccggccc
aagagactcagcttcaagctcttcaacatccaggtcaaggaggtcacgcagaatgaaggcaccaagaccatcgccaataacctcaccagc
accatccaggtgtttacggactcggagtaccagctgccgtacgttctcggctctgcccaccagggctgcctgcctccgttcccggcggacgt
gttcatgattccccagtacggctacctaacactcaacaacggtagtcaggccgtgggacgctcctccttctactgcctggaatactttccttcg
cagatgctgagaaccggcaacaacttccagtttacttacaccttcgaggacgtgcctttccacagcagctacgcccacagccagagcttgga
ccggctgatgaatcctctgattgaccagtacctgtactacttgtctcggactcaaacaacaggaggcacggcaaatacgcagactctgggct
tcagccaaggtgggcctaatacaatggccaatcaggcaaagaactggctgccaggaccctgttaccgccaacaacgcgtctcaacgacaa
ccgggcaaaacaacaatagcaactttgcctggactgctgggaccaaataccatctgaatggaagaaattcattggctaatcctggcatcgct
atggcaacacacaaagacgacgaggagcgtttttttcccagtaacgggatcctgatttttggcaaacaaaatgctgccagagacaatgcgga
ttacagcgatgtcatgctcaccagcgaggaagaaatcaaaaccactaaccctgtggctacagaggaatacggtatcgtggcagataacttg
cagcagcaaaacacggctcctcaaattggaactgtcaacagccagggggccttacccggtatggtctggcagaaccgggacgtgtacctg
cagggtcccatctgggccaagattcctcacacggacggcaacttccacccgtctccgctgatgggcggctttggcctgaaacatcctccgc
ctcagatcctgatcaagaacacgcctgtacctgcggatcctccgaccaccttcaaccagtcaaagctgaactctttcatcacgcaatacagca
ccggacaggtcagcgtggaaattgaatgggagctgcagaaggaaaacagcaagcgctggaaccccgagatccagtacacctccaacta
ctacaaatctacaagtgtggactttgctgttaatacagaaggcgtgtactctgaaccccgccccattggcacccgttacctcacccgtaatctg
taaactagtttgcttgttaatcaataaaccgtttaattcgtttcagttgaGCGGCCGTCGAGTCTAGAGGGCCCGTTTA
AACCCGCTGATCAGCCTCGACTGTGCCTTCTAGTTGCCAGCCATCTGTTGTTTGCCCC
TCCCCCGTGCCTTCCTTGACCCTGGAAGGTGCCACTCCCACTGTCCTTTCCTAATAAA
ATGAGGAAATTGCATCGCATTGTCTGAGTAGGTGTCATTCTATTCTGGGGGGTGGGG
TGGGGCAGGACAGCAAGGGGGAGGATTGGGAAGACAATAGCAGGCATGCTGGGGA
TGCGGTGGGCTCTATGGCTTCTGAGGCGGAAAGAACCAGCTGGGGCTCTAGGGGGT
ATCCCCGGCGCGCCGAATTCGTTAACAAGCTtTAATTAaCGCgtATAcCTAggATCCGGC
CGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGT
TTAGCCCCGGAATTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTTTTTCCCC
GTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAG
CGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGG
ATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCT
AGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTGG
CCTCCAAGGCCAGCTTCCCACAATAAGTTGGGTGAATTTTGGCTCATTCCTCCTTTCT
ATAGGATTGAGGTCAGAGCTTTGTGATGGGAATTCTGTGGAATGTGTGTCAGTTAGG
GTGTGGAAAGTCCCgcGATCgcTAGcAAACGCCAGCAACGCGGCCTTTTTACGGTTCC
TGGCCTTTTGCTGGCCTTTTGCTCACATGTCCTGCAGGCAGCTGCGCGCTCGCTCGCT
CACTGAGGCCGCCCGGGCAAAGCCCGGGCGTCGGGCGACCTTTGGTCGCCCGGCCT

CAGTGAGCGAGCGAGCGCGCAGAGAGGGAGTGGCCAACTCCATCACTAGGGGTTCC
TGCGGCCGCACGCGTGGAGCTAGTTATTAATAGTAATCAATTACGGGGTCATTAGTT
CATAGCCCATATATGGAGTTCCGCGTTACATAACTTACGGTAAATGGCCCGCCTGGC
TGACCGCCCAACGACCCCCGCCCATTGACGTCAATAATGACGTATGTTCCCATAGTA
ACGTCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAAACTGCC
CACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACGTCAAT
GACGGTAAATGGCCCGCCTGGCATTATGCCCAGTACATGACCTTATGGGACTTTCCT
ACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTGATGCGGTTTTGGC
AGTACATCAATGGGCGTGGATAGCGGTTTGACTCACGGGGATTTCCAAGTCTCCACC
CCATTGACGTCAATGGGAGTTTGTTTTGCACCAAAATCAACGGGACTTTCCAAAATG
TCGTAACAACTCCGCCCCATTGACGCAAATGGGCGGTAGGCGTGTACGGTGGGAGG
TCTATATAAGCAGAGCTCGTTTAGTGAACCGTCAGATCGCCTGGAGACGCCATCCAC
GCTGTTTTGACCTCCATAGAAGACACCGGGACCGATCCAGCCTCCGCGGATTCGAAT
CCCGGCCGGGAACGGTGCATTGGAACGCGGATTCCCCGTGCCAAGAGTGACGTAAG
TACCGCCTATAGAGTCTATAGGCCCACAAAAAATGCTTTCTTCTTTTAATATACTTTT
TTGTTTATCTTATTTCTAATACTTTCCCTAATCTCTTTCTTTCAGGGCAATAATGATAC
AATGTATCATGCCTCTTTGCACCATTCTAAAGAATAACAGTGATAATTTCTGGGTTA
AGGCAATAGCAATATTTCTGCATATAAATATTTCTGCATATAAATTGTAACTGATGT
AAGAGGTTTCATATTGCTAATAGCAGCTACAATCCAGCTACCATTCTGCTTTTATTTT
ATGGTTGGGATAAGGCTGGATTATTCTGAGTCCAAGCTAGGCCCTTTTGCTAATCAT
GTTCATACCTCTTATCTTCCTCCCACAGCTCCTGGGCAACGTGCTGGTCTGTGTGCTG
GCCCATCACTTTGGCAAAGAATTGGGATTCGAACATCGATTGAATTCTGAATGGTGA
GCAAGGGCGAGGAGCTGTTCACCGGGGTGGTGCCCATCCTGGTCGAGCTGGACGGC
GACGTAAACGGCCACAAGTTCAGCGTGTCCGGCGAGGGCGAGGGCGATGCCACCTA
CGGCAAGCTGACCCTGAAGTTCATCTGCACCACCGGCAAGCTGCCCGTGCCCTGGCC
CACCCTCGTGACCACCCTGACCTACGGCGTGCAGTGCTTCAGCCGCTACCCCGACCA
CATGAAGCAGCACGACTTCTTCAAGTCCGCCATGCCCGAAGGCTACGTCCAGGAGC
GCACCATCTTCTTCAAGGACGACGGCAACTACAAGACCCGCGCCGAGGTGAAGTTC
GAGGGCGACACCCTGGTGAACCGCATCGAGCTGAAGGGCATCGACTTCAAGGAGGA
CGGCAACATCCTGGGGCACAAGCTGGAGTACAACTACAACAGCCACAACGTCTATA
TCATGGCCGACAAGCAGAAGAACGGCATCAAGGTGAACTTCAAGATCCGCCACAAC
ATCGAGGACGGCAGCGTGCAGCTCGCCGACCACTACCAGCAGAACACCCCCATCGG
CGACGGCCCCGTGCTGCTGCCCGACAACCACTACCTGAGCACCCAGTCCGCCCTGAG
CAAAGACCCCAACGAGAAGCGCGATCACATGGTCCTGCTGGAGTTCGTGACCGCCG
CCGGGATCACTCTCGGCATGGACGAGCTGTACAAGTACTCAGATCTCGAGCTCAAGT
AGGGATCCTCTAGAGTCGACCTGCAGAAGCTTGCCTCGAGCAGCGCTGCTCGAGAG
ATCTACGGGTGGCATCCCTGTGACCCCTCCCCAGTGCCTCTCCTGGCCCTGGAAGTT
GCCACTCCAGTGCCCACCAGCCTTGTCCTAATAAAATTAAGTTGCATCATTTTGTCTG
ACTAGGTGTCCTTCTATAATATTATGGGGTGGAGGGGGGTGGTATGGAGCAAGGGG
CAAGTTGGGAAGACAACCTGTAGGGCCTGCGGGGTCTATTGGGAACCAAGCTGGAG
TGCAGTGGCACAATCTTGGCTCACTGCAATCTCCGCCTCCTGGGTTCAAGCGATTCT
CCTGCCTCAGCCTCCCGAGTTGTTGGGATTCCAGGCATGCATGACCAGGCTCAGCTA
ATTTTTGTTTTTTTGGTAGAGACGGGGTTTCACCATATTGGCCAGGCTGGTCTCCAAC
TCCTAATCTCAGGTGATCTACCCACCTTGGCCTCCCAAATTGCTGGGATTACAGGCG
TGAACCACTGCTCCCTTCCCTGTCCTTCTGATTTTGTAGGTAACCACGTGCGGACCGA
GCGGCCGCAGGAACCCCTAGTGATGGAGTTGGCCACTCCCTCTCTGCGCGCTCGCTC
GCTCACTGAGGCCGGGCGACCAAAGGTCGCCCGACGCCCGGGCTTTGCCCGGGCGG
CCTCAGTGAGCGAGCGAGCGCGCAGCTGCCTGCAGGGGCGCCTGATGCGGTATTTT
CTCCTTACGCATCTGTGCGGTATTTCACACCGCATACGTCgtaGCtGATcaATTgGCGCG
CCGAATTCGTTAACAAGCTtTAATTAaCGCgtATAcCTAggATCCGGCCGGccTGCAggTG
TCCTCACAGGAACGAAGTCCCTAAAGAAACAGTGGCAGCCAGGTTTAGCCCCGGAA
TTGACTGGATTCCTTTTTTAGGGCCCATTGGTATGGCTTTTTCCCCGTATCCCCCCAG
GTGTCTGCAGGCTCAAAGAGCAGCGAGAAGCGTTCAGAGGAAAGCGATCCCGTGCC
ACCTTCCCCGTGCCCGGGCTGTCCCCGCACGCTGCCGGCTCGGGGATGCGGGGGGA

GCGCCGGACCGGAGCGGAGCCCCGGGCGGCTCGCTGCTGCCCCCTAGCGGGGGAGG
GACGTAATTACATCCCTGGGGGCTTTGGGGGGGGGCTGTCCCTGGCCTCCAAGGCCA
GCTTCCCACAATAAGTTGGGTGAATTTTGGCTCATTCCTCCTTTCTATAGGATTGAGG
TCAGAGCggggttggggttgcgccttttccaaggcagccctgggtttgcgcagggacgcggctgctctgggcgtggttccgggaaa
cgcagcggcgccgaccctgggtctcgcacattcttcacgtccgttcgcagcgtcacccggatcttcgccgctacccttgtgggccccccgg
cgacgcttcctgctccgcccctaagtcgggaaggttccttgcggttcgcggcgtgccggacgtgacaaacggaagccgcacgtctcacta
gtaccctcgcagacggacagcgccagggagcaatggcagcgcgccgaccgcgatgggctgtggccaatagcggctgctcagcagggc
gcgccgagagcagcggccgggaaggggcggtgcgggaggcggggtgtggggcggtagtgtgggccctgttcctgcccgcgcggtgt
tccgcattctgcaagcctccggagcgcacgtcggcagtcggctccctcgttgaccgaatcaccgacctctctccccagAAGCTCCC
GGGAGCTTGTATATCCATTTTCGGATCTGATCAGCACGTGTTGACAATTAATCATCG
GCATAGTATATCGGCATAGTATAATACGACAAGGTGAGGAACgccaccATGAAAAAGC
CTGAACTCACCGCGACGTCTGTCGAGAAGTTTCTGATCGAAAAGTTCGACAGCGTCT
CCGACCTGATGCAGCTCTCGGAGGGCGAAGAATCTCGTGCTTTCAGCTTCGATGTAG
GAGGGCGTGGATATGTCCTGCGGGTAAATAGCTGCGCCGATGGTTTCTACAAAGATC
GTTATGTTTATCGGCACTTTGCATCGGCCGCGCTCCCGATTCCGGAAGTGCTTGACAT
TGGGGAATTCAGCGAGAGCCTGACCTATTGCATCTCCCGCCGTGCACAGGGTGTCAC
GTTGCAAGACCTGCCTGAAACCGAACTGCCCGCTGTTCTGCAGCCGGTCGCGGAGG
CCATGGATGCGATCGCTGCGGCCGATCTTAGCCAGACGAGCGGGTTCGGCCCATTCG
GACCGCAAGGAATCGGTCAATACACTACATGGCGTGATTTCATATGCGCGATTGCTG
ATCCCCATGTGTATCACTGGCAAACTGTGATGGACGACACCGTCAGTGCGTCCGTCG
CGCAGGCTCTCGATGAGCTGATGCTTTGGGCCGAGGACTGCCCCGAAGTCCGGCAC
CTCGTGCACGCGGATTTCGGCTCCAACAATGTCCTGACGGACAATGGCCGCATAACA
GCGGTCATTGACTGGAGCGAGGCGATGTTCGGGGATTCCCAATACGAGGTCGCCAA
CATCTTCTTCTGGAGGCCGTGGTTGGCTTGTATGGAGCAGCAGACGCGCTACTTCGA
GCGGAGGCATCCGGAGCTTGCAGGATCGCCGCGGCTCCGGGCGTATATGCTCCGCA
TTGGTCTTGACCAACTCTATCAGAGCTTGGTTGACGGCAATTTCGATGATGCAGCTT
GGGCGCAGGGTCGATGCGACGCAATCGTCCGATCCGGAGCCGGGACTGTCGGGCGT
ACACAAATCGCCCGCAGAAGCGCGGCCGTCTGGACCGATGGCTGTGTAGAAGTACT
CGCCGATAGTGGAAACCGACGCCCCAGCACTCGTCCGAGGGCAAAGGAATAGAGCG
CGGGGATCTCATGCTGGAGTTCTTCGCCCACCCCAACTTGTTTATTGCAGCTTATAAT
GGTTACAAATAAAGCAATAGCATCACAAATTTCACAAATAAAGCATTTTTTTCACTG
CATTCTAGTTGTGGTTTGTCCAAACTCATCAATGTATCTTATCATGTCTGTAGCtGATg
tATAcCTAggATCCGGCCGGccTGCAggTGTCCTCACAGGAACGAAGTCCCTAAAGAAA
CAGTGGCAGCCAGGTTTAGCCCCGGAATTGACTGGATTCCTTTTTAGGGCCCATTGG
TATGGCTTTTTCCCCGTATCCCCCCAGGTGTCTGCAGGCTCAAAGAGCAGCGAGAAG
CGTTCAGAGGAAAGCGATCCCGTGCCACCTTCCCCGTGCCCGGGCTGTCCCCGCACG
CTGCCGGCTCGGGGATGCGGGGGGAGCGCCGGACCGGAGCGGAGCCCCGGGCGGC
TCGCTGCTGCCCCCTAGCGGGGGAGGGACGTAATTACATCCCTGGGGGCTTTGGGGG
GGGGCTGTCCCTCTAGAGCGGCCGCCACCGCGGTGGAGCTCCAGCTTTTGTTCCCTT
TAGTGAGGGTTAATTAGATCTTAATACGACTCACTATAGGGCGAATTGGGTACCGGG
CCCCCCCTCGAGGTCGACGGTATCGATAAGCTTGATATCTATAACAAGAAAATATAT
ATATAATAAGTTATCACGTAAGTAGAACATGAAATAACAATATAATTATCGTATGAG
TTAAATCTTAAAAGTCACGTAAAAGATAATCATGCGTCATTTTGACTCACGCGGTCG
TTATAGTTCAAAATCAGTGACACTTACCGCATTGACAAGCACGCCTCACGGGAGCTC
CAAGCGGCGACTGAGATGTCCTAAATGCACAGCGACGGATTCGCGCTATTTAGAAA
GAGAGAGCAATATTTCAAGAATGCATGCGTCAATTTTACGCAGACTATCTTTCTAGG
GTTAATCTAGCTGCATCAGGATCATATCGTCGGGTCTTTTTTCCGGCTCAGTCATCGC
CCAAGCTGGCGCTATCTGGGCATCGGGGAGGAAGAAGCCCGTGCCTTTTCCCGCGA
GGTTGAAGCGGCATGGAAAGAGTTTGCCGAGGATGACTGCTGCTGCATTGACGTTG
AGCGAAAACGCACGTTTACCATGATGATTCGGGAAGGTGTGGCCATGCACGCCTTTA
ACGGTGAACTGTTCGTTCAGGCCACCTGGGATACCAGTTCGTCGCGGCTTTTCCGGA
CACAGTTCCGGATGGTCAGCCCGAAGCGCATCAGCAACCCGAACAATACCGGCGAC
AGCCGGAACTGCCGTGCCGGTGTGCAGATTAATGACAGCGGTGCGGCGCTGGGATA
TTACGTCAGCGAGGACGGGTATCCTGGCTGGATGCCGCAGAAATGGACATGGATAC

CCCGTGAGTTACCCGGCGGGCGCGCTTGGCGTAATCATGGTCATAGCTGTTTCCTGT
GTGAAATTGTTATCCGCTCACAATTCCACACAACATACGAGCCGGAAGCATAAAGT
GTAAAGCCTGGGGTGCCTAATGAGTGAGCTAACTCACATTAATTGCGTTGCGCTCAC
TGCCCGCTTTCCAGTCGGGAAACCTGTCGTGCCAGCTGCATTAATGAATCGGCCAAC
GCGCGGGGAGAGGCGGTTTGCGTATTGGGCGCTCTTCCGCTTCCTCGCTCACTGACT
CGCTGCGCTCGGTCGTTCGGCTGCGGCGAGCGGTATCAGCTCACTCAAAGGCGGTA
ATACGGTTATCCACAGAATCAGGGGATAACGCAGGAAAGAACATGTGAGCAAAAG
GCCAGCAAAAGGCCAGGAACCGTAAAAAGGCCGCGTTGCTGGCGTTTTTCCATAGG
CTCCGCCCCCCTGACGAGCATCACAAAAATCGACGCTCAAGTCAGAGGTGGCGAAA
CCCGACAGGACTATAAAGATACCAGGCGTTTCCCCCTGGAAGCTCCCTCGTGCGCTC
TCCTGTTCCGACCCTGCCGCTTACCGGATACCTGTCCGCCTTTCTCCCTTCGGGAAGC
GTGGCGCTTTCTCATAGCTCACGCTGTAGGTATCTCAGTTCGGTGTAGGTCGTTCGCT
CCAAGCTGGGCTGTGTGCACGAACCCCCCGTTCAGCCCGACCGCTGCGCCTTATCCG
GTAACTATCGTCTTGAGTCCAACCCGGTAAGACACGACTTATCGCCACTGGCAGCAG
CCACTGGTAACAGGATTAGCAGAGCGAGGTATGTAGGCGGTGCTACAGAGTTCTTG
AAGTGGTGGCCTAACTACGGCTACACTAGAAGGACAGTATTTGGTATCTGCGCTCTG
CTGAAGCCAGTTACCTTCGGAAAAAGAGTTGGTAGCTCTTGATCCGGCAAACAAAC
CACCGCTGGTAGCGGTGGTTTTTTTGTTTGCAAGCAGCAGATTACGCGCAGAAAAAA
AGGATCTCAAGAAGATCCTTTGATCTTTTCTACGGGGTCTGACGCTCAGTGGAACGA
AAACTCACGTTAAGGGATTTTGGTCATGAGATTATCAAAAAGGATCTTCACCTAGAT
CCTTTTAAATTAAAAATGAAGTTTTAAATCAATCTAAAGTATATATGAGTAAACTTG
GTCTGACAGTTACCAATGCTTAATCAGTGAGGCACCTATCTCAGCGATCTGTCTATTT
CGTTCATCCATAGTTGCCTGACTCCCCGTCGTGTAGATAACTACGATACGGGAGGGC
TTACCATCTGGCCCCAGTGCTGCAATGATACCGCGAGACCCACGCTCACCGGCTCCA
GATTTATCAGCAATAAACCAGCCAGCCGGAAGGGCCGAGCGCAGAAGTGGTCCTGC
AACTTTATCCGCCTCCATCCAGTCTATTAATTGTTGCCGGGAAGCTAGAGTAAGTAG
TTCGCCAGTTAATAGTTTGCGCAACGTTGTTGCCATTGCTACAGGCATCGTGGTGTC
ACGCTCGTCGTTTGGTATGGCTTCATTCAGCTCCGGTTCCCAACGATCAAGGCGAGT
TACATGATCCCCCATGTTGTGCAAAAAAGCGGTTAGCTCCTTCGGTCCTCCGATCGT
TGTCAGAAGTAAGTTGGCCGCAGTGTTATCACTCATGGTTATGGCAGCACTGCATAA
TTCTCTTACTGTCATGCCATCCGTAAGATGCTTTTCTGTGACTGGTGAGTACTCAACC
AAGTCATTCTGAGAATAGTGTATGCGGCGACCGAGTTGCTCTTGCCCGGCGTCAATA
CGGGATAATACCGCGCCACATAGCAGAACTTTAAAAGTGCTCATCATTGGAAAACG
TTCTTCGGGGCGAAAACTCTCAAGGATCTTACCGCTGTTGAGATCCAGTTCGATGTA
ACCCACTCGTGCACCCAACTGATCTTCAGCATCTTTTACTTTCACCAGCGTTTCTGGG
TGAGCAAAAACAGGAAGGCAAAATGCCGCAAAAAAGGGAATAAGGGCGACACGGA
AATGTTGAATACTCAT

Example 5: Performance of AAV Stable Producer Cell Lines

(157) To generate stable AAV producer cell lines, one transposon from each group of iHelper, iRep and iCap-GOI was used to integrate into the genome of stable HEK293-TetR-KRAB C20 cell lines through a “cut and paste” mechanism by the co-transfection of these three transposon plasmids with mRNA encoding super piggyBac transposase (SPB-100). After transfection, three antibiotics including puromycin, blasticidin and hygromycin were added into the cell culture medium to select the transposon integrated cells. After stable cell pool generation and screening, one highly productive cell pool was selected to monitor the long-term stability of cells for AAV productivity. The suspension pool was left untreated or induced by doxycycline from cell passage 5 to passage 25 over three months. As shown in FIG. **7**, the productivity of polyclonal pool of AAV producer cell line is stable for 25 passages upon induction by Dox. AAV production without Dox induction showed extremely low basal/leaky levels of AAVs.

(158) To evaluate the performance of AAV stable producer cell lines, multiple single cell clones were isolated by single cell printing and expanded for testing. First, the viral protein expression was evaluated by western blot analysis. Top three clonal cells (Clone A, B and C) were uninduced or induced by doxycycline for AAV production. Whole cell lysates were harvested three days post induction and prepared for the Western blot analysis. The whole cell lysate produced from a triple plasmid transfection process was used as positive control. The results show the robust induction of protein expression of helper, Rep and Cap

genes in these clonal AAV producer cell lines, at the level similar or higher than the control samples, while showing undetectable level of leakiness protein expression in uninduced cells (See FIG. **8**).
(159) The AAV productivity of the top three clones expressing Green Fluorescent Protein (GFP), a gene of interest, was examined in 3 Liter single-use bioreactors. The AAV genomic titer by ddPCR titer assay showed strong induction of AAV production in all clones in the range from 4.3×10.sup.10 vg/mL to 2.8×10.sup.11 vg/mL on day 5 harvest, which was comparable or higher than the AAV productivity using conventional triple transfection method (See FIG. **9**).
(160) The productivity of Clones A, B and C expressing GFP was further compared to three representative clonal cells (Clone 1, Clone 2 and Clone 3) uninduced or induced by doxycycline. Clone 1, Clone 2 and Clone 3 comprise nucleic acid sequences disclosed in U.S. Pat. No. 11,739,347 B2 that were designed using E2A and E4 genes under control of an inducible CMV promoter, VA RNA under control of an H1 promoter, and Rep and Cap genes under control of a native promoter. The AAV productivity of Clones A, B and C was significantly higher than Clone 1, Clone 2 and Clone 3 after Dox induction (See FIG. **10**).
(161) The functionality of the AAV produced from the bioreactors was tested by the cell-based infectivity and potency assay. The purified AAV was serially diluted and transduced to HT1080 cells in 96-well plates. The percentage of GFP-positive cells from transduced AAV was quantified by flow cytometry analysis and used to calculate the transduction unit (TU). The normalized viral genome per TU result suggests that the AAV produced from AAV producer cell lines are functional in infecting cells and expressing a gene of interest, such as GFP (See FIG. **11**).
(162) It will be readily apparent to one of ordinary skill in the relevant arts that other suitable modifications and adaptations to the methods and applications described herein can be made without departing from the scope of any of the embodiments.
(163) It is to be understood that while certain embodiments have been illustrated and described herein, the claims are not to be limited to the specific forms or arrangement of parts described and shown. In the specification, there have been disclosed illustrative embodiments and, although specific terms are employed, they are used in a generic and descriptive sense only and not for purposes of limitation. Modifications and variations of the embodiments are possible in light of the above teachings. It is therefore to be understood that the embodiments may be practiced otherwise than as specifically described.
(164) All references cited herein, including patents, patent applications, papers, textbooks and the like, and the references cited therein, to the extent that they are not already, are hereby incorporated herein by reference in their entirety.

Claims

1. An isolated nucleic acid molecule operably comprising: a. a first transposon-specific inverted terminal repeat (ITR); b. a first inducible promoter comprising a native E2A promoter and two tetracycline operator sequences (a first TetO.sub.2 sequence); c. an E2A coding sequence under control of the first inducible promoter; d. a viral associated (VA) non-coding RNA under control of a second inducible promoter comprising a native VA promoter or a constitutive promoter comprising a native VA promoter; e. a third inducible promoter comprising a promoter native to an E4 coding sequence and a second TetO.sub.2 sequence; f. the E4 coding sequence under control of the third inducible promoter; g. an antibiotic resistance gene; and h. a second ITR, wherein the isolated nucleic acid molecule does not include a Rep gene and/or Cap gene.
2. The nucleic acid molecule of claim 1, wherein the nucleic acid molecule further comprises a core insulator sequence inserted between the E4 coding sequence and the antibiotic resistance gene.
3. The nucleic acid molecule of claim 1, further comprising a core insulator sequence inserted between the first ITR and the first inducible promoter.